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Scholz

(54) **EPHA2 ANTIBODIES**

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(21) Appl. No.: **18/854,466**

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(86) PCT No.: **PCT/US2023/065395**

§ 371 (c)(1),
(2) Date: **Oct. 4, 2024**

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Publication Classification

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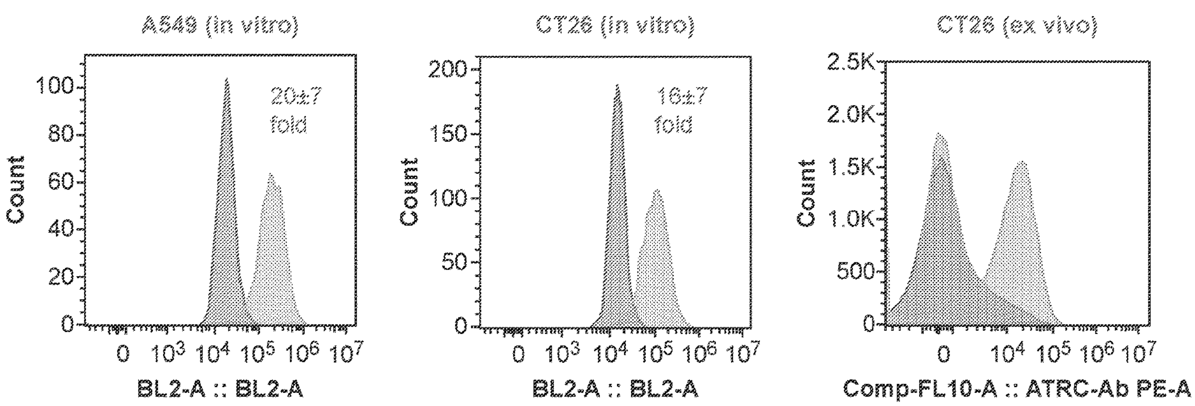
C07K 16/28 (2006.01)

(52) **U.S. Cl.**
CPC ***A61K 47/68031*** (2023.08); ***A61K 9/0019*** (2013.01); ***A61K 47/6879*** (2017.08); ***A61K 47/6889*** (2017.08); ***A61P 35/00*** (2018.01); ***C07K 16/2809*** (2013.01); ***C07K 16/2818*** (2013.01); ***C07K 16/2866*** (2013.01); ***C07K 16/2878*** (2013.01)

(57) **ABSTRACT**

Provided herein immunoconjugate comprising an antibody that binds to ephrin receptor A2 (EphA2) and a cytotoxic agent. The cytotoxic agent is an auristatin analogue, and the antibody is conjugated to the auristatin analogue. These immunoconjugates comprising EphA2 antibodies bind preferentially to tumor tissue compared to normal tissue, and they can be used in methods of inducing an immune response and methods of inhibiting tumor cell growth.

Specification includes a Sequence Listing.



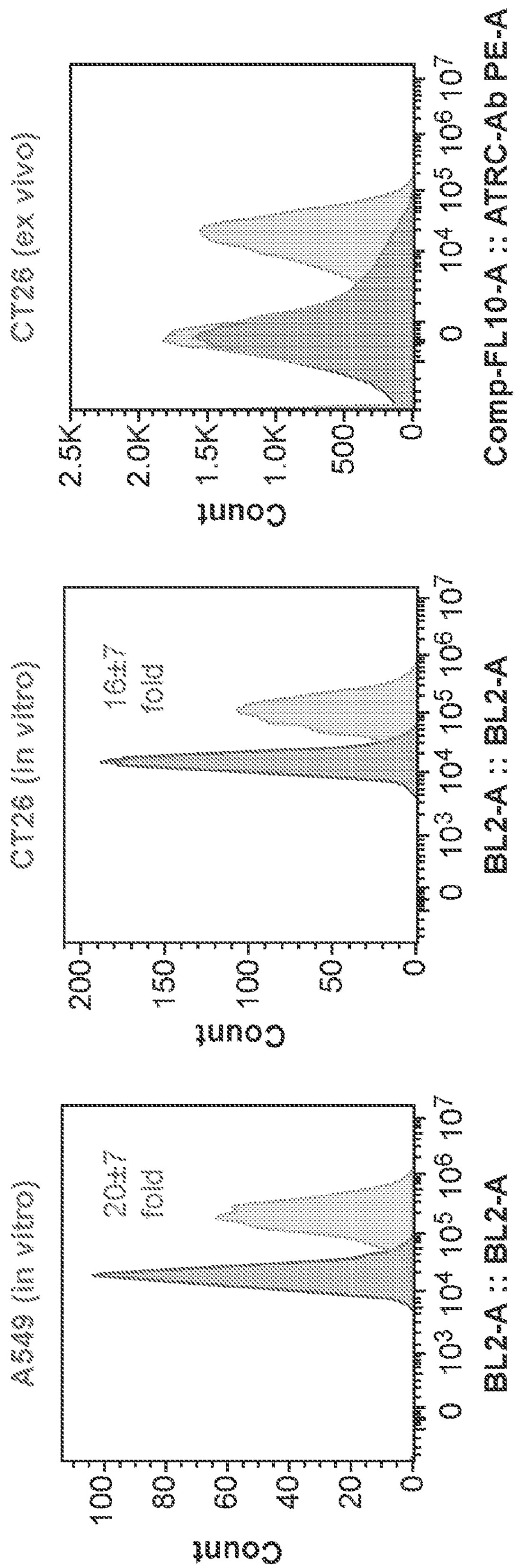

A549 (in vitro)
20±7
fold
Count
BL2-A :: BL2-A
CT26 (in vitro)
16±7
fold
Count
BL2-A :: BL2-A
CT26 (ex vivo)
Count
Comp-FL10-A :: ATRC-Ab PE-A

FIG. 1

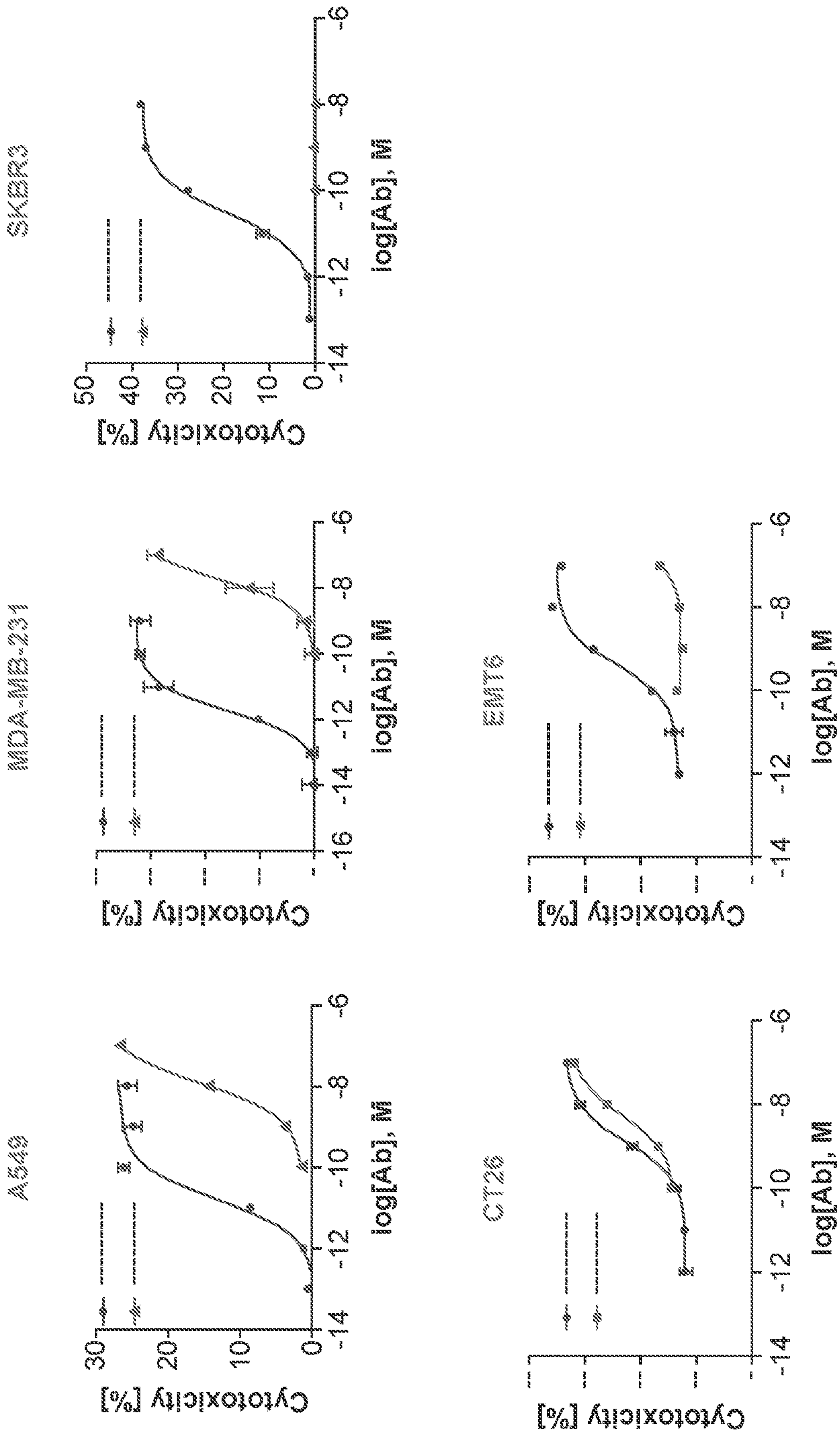


FIG. 2A

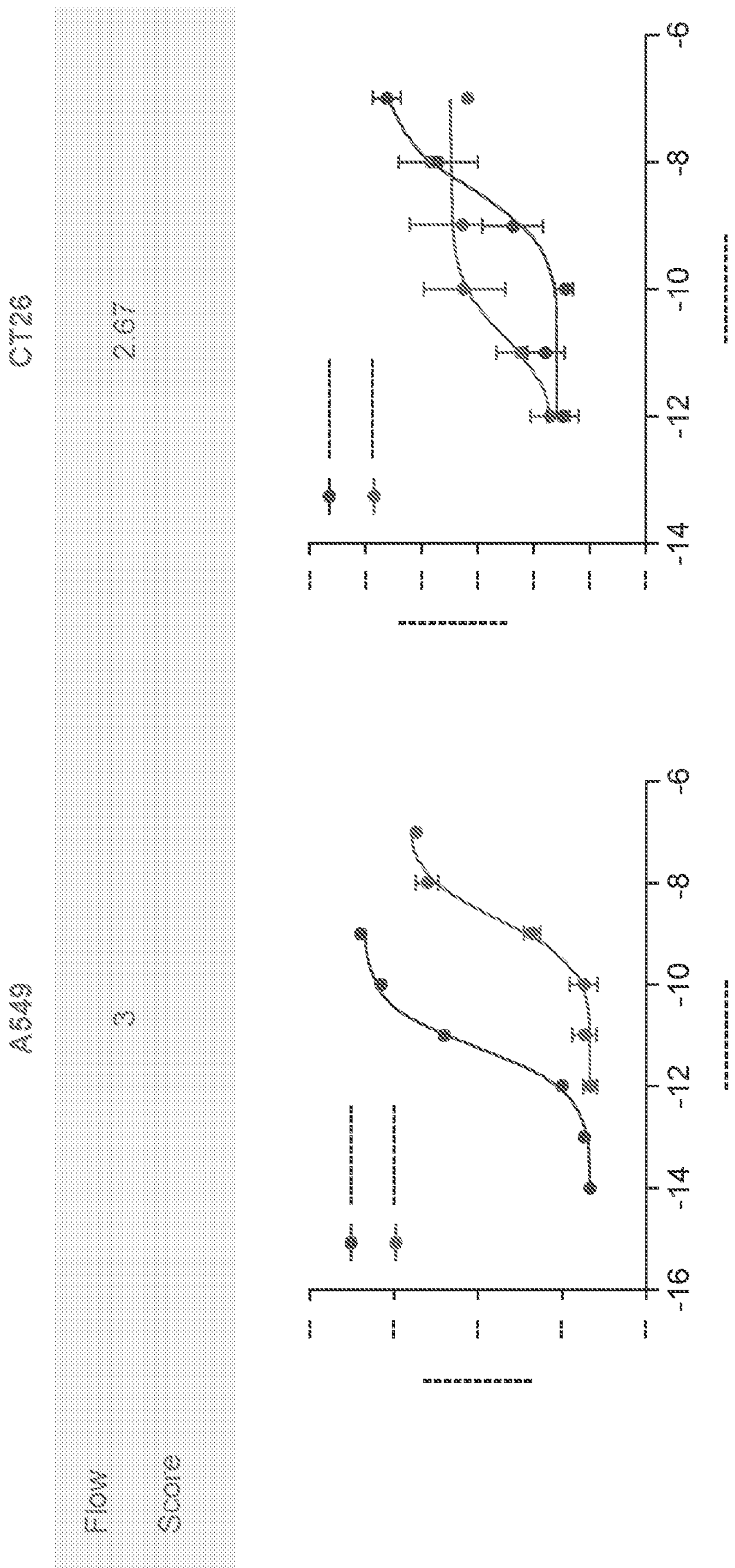
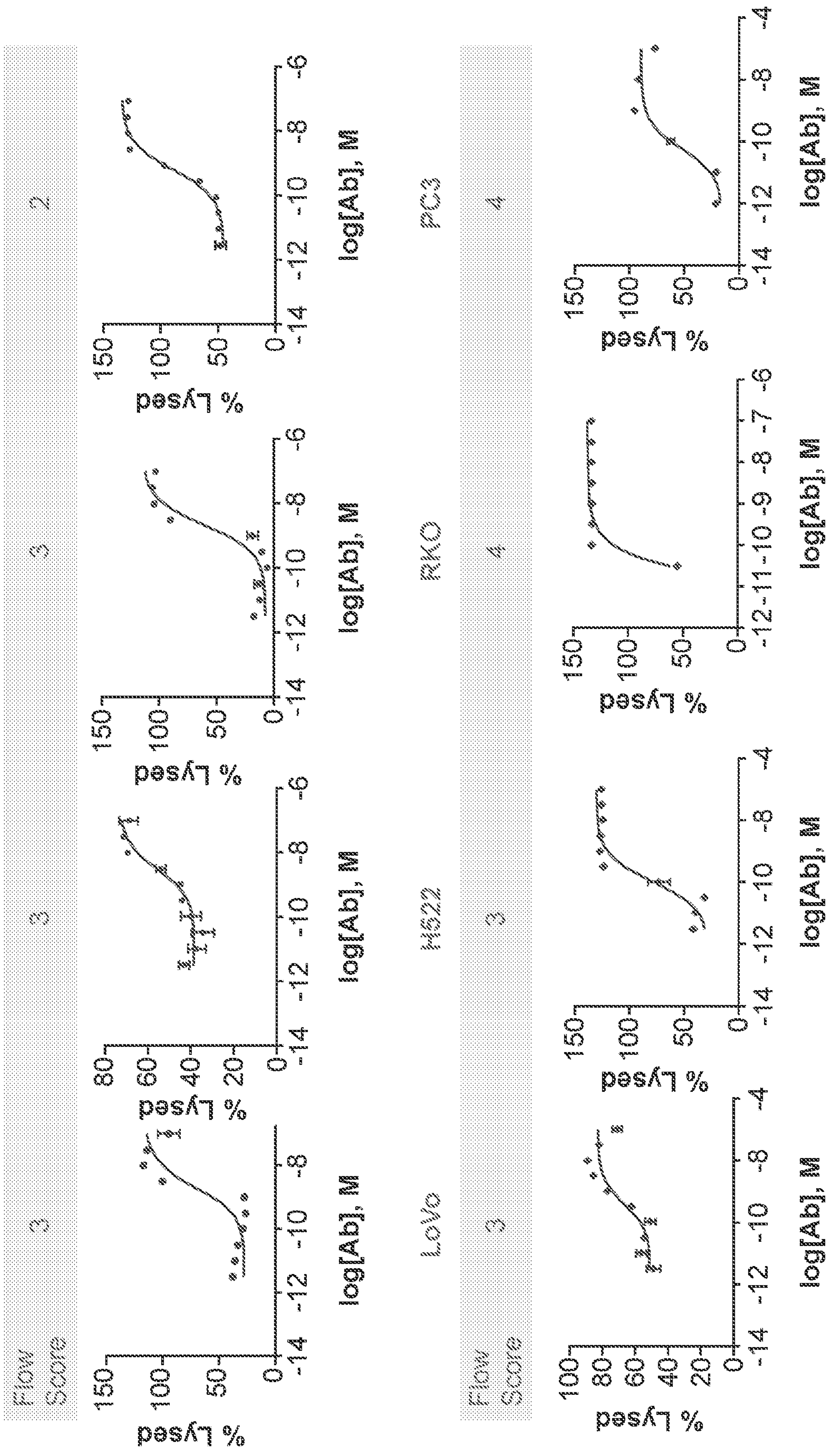


FIG. 2B



Flow Score
A549
3
SKOV3
3
MDA-MB-231
3
A375
2
LoVo
3
H522
3
RKO
4
PC3
4
% Lysed
log[Ab], M

FIG. 3

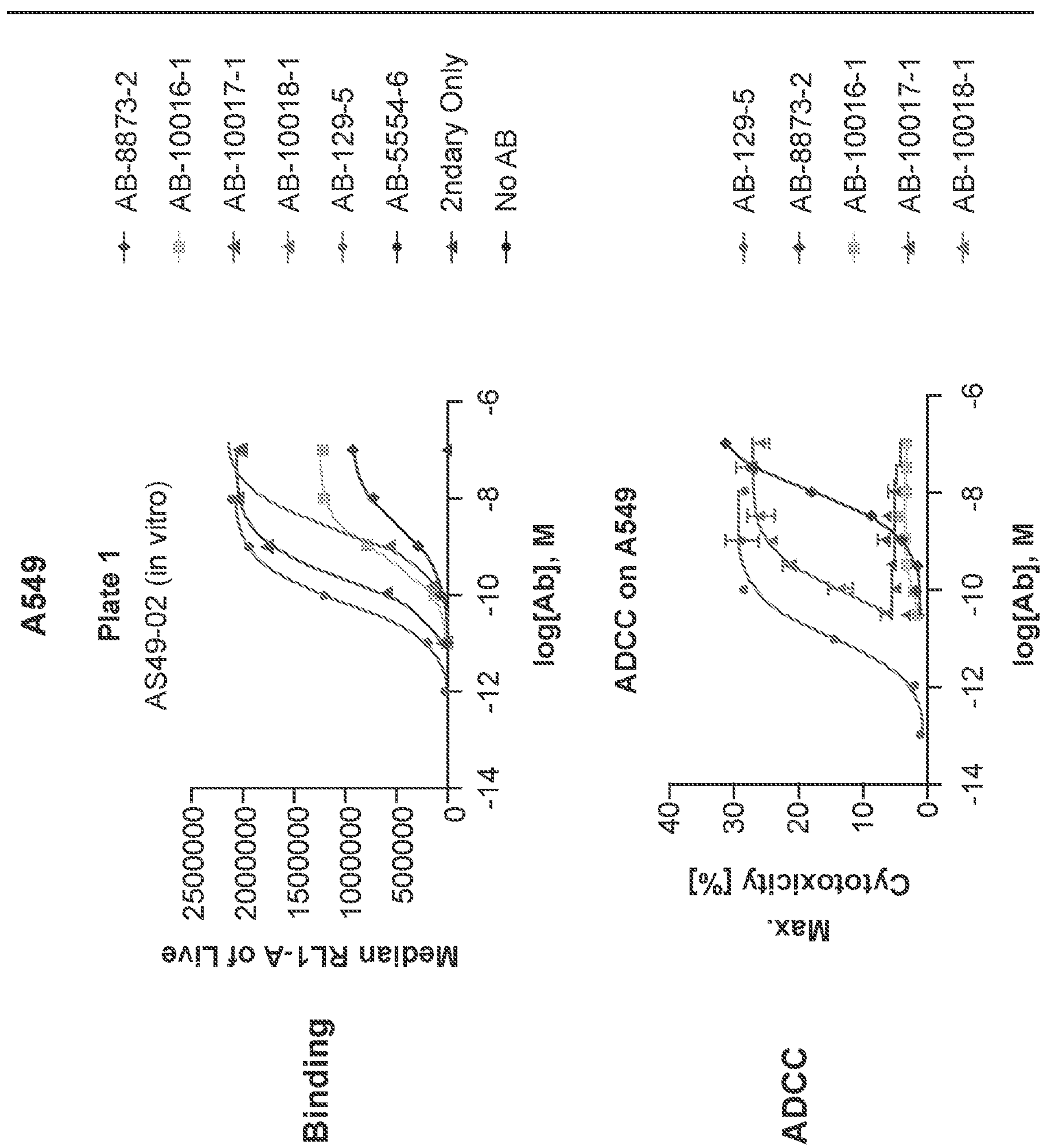
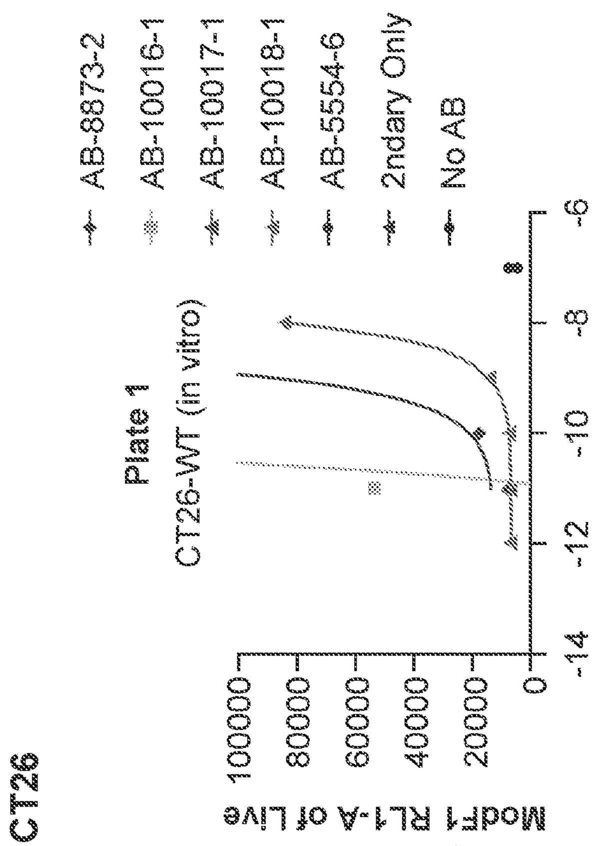

A549
Binding
Plate 1
AS49-02 (in vitro)
Median RL1-A of Live
2500000
2000000
1500000
1000000
500000
0
-14
-12
-10
-8
-6
log[Ab], M
AB-8873-2
AB-10016-1
AB-10017-1
AB-10018-1
AB-129-5
AB-5554-6
2ndary Only
No AB
ADCC
ADCC on A549
Max.
Cytotoxicity [%]
40
30
20
10
0
-14
-12
-10
-8
-6
log[Ab], M
AB-129-5
AB-8873-2
AB-10016-1
AB-10017-1
AB-10018-1

FIG. 4


CT26
Plate 1
CT26-WT (in vitro)
ModFI RL1-A of Live
log[Ab], M
AB-8873-2
AB-10016-1
AB-10017-1
AB-10018-1
AB-5554-6
2ndary Only
No AB

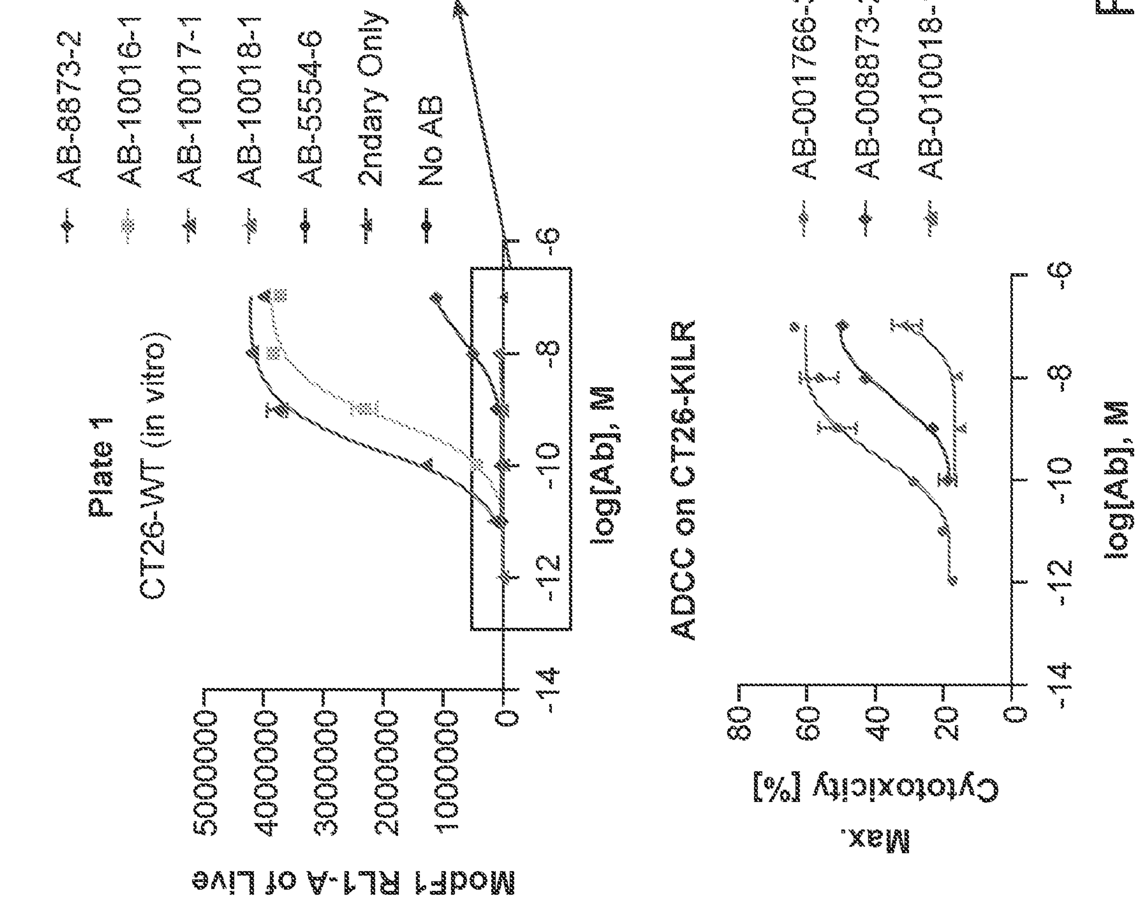


Plate 1
CT26-WT (in vitro)
ModFI RL1-A of Live
log[Ab], M
AB-8873-2
AB-10016-1
AB-10017-1
AB-10018-1
AB-5554-6
2ndary Only
No AB
ADCC on CT26-KILR
Max. Cytotoxicity [%]
log[Ab], M
AB-001766-3
AB-008873-2
AB-010018-1

FIG. 4 Contd...

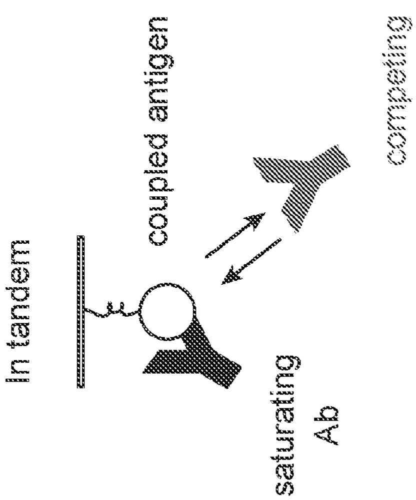

In tandem
coupled antigen
saturating Ab
competing Ab

FIG. 5A

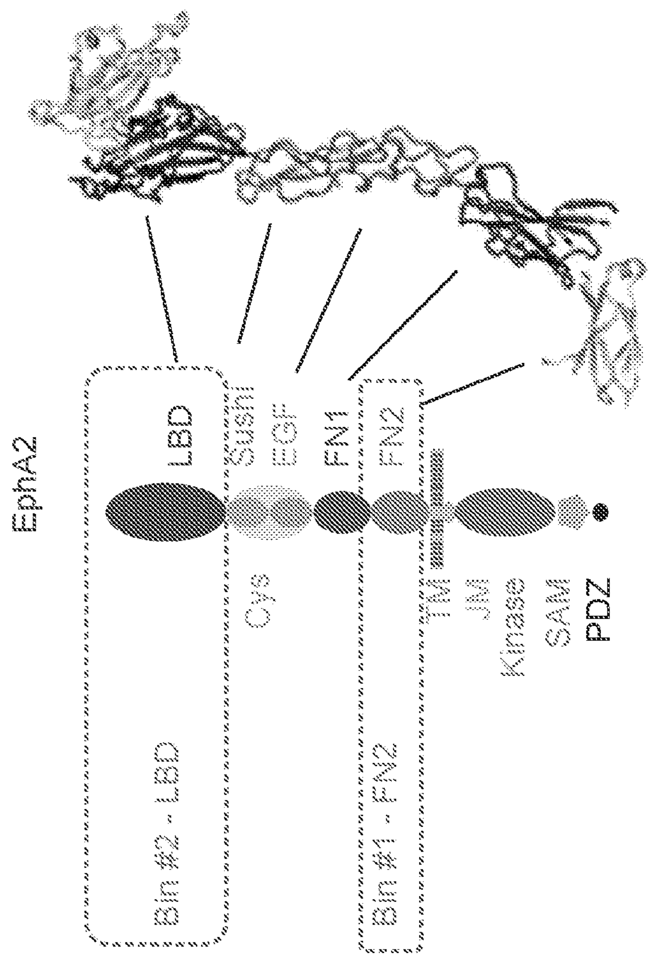

EphA2
Bin #2 - LBD
Bin #1 - FN2
LBD
Sushi
EGF
FN1
FN2
Cys
TM
JM
Kinase
SAM
PDZ

FIG. 5B

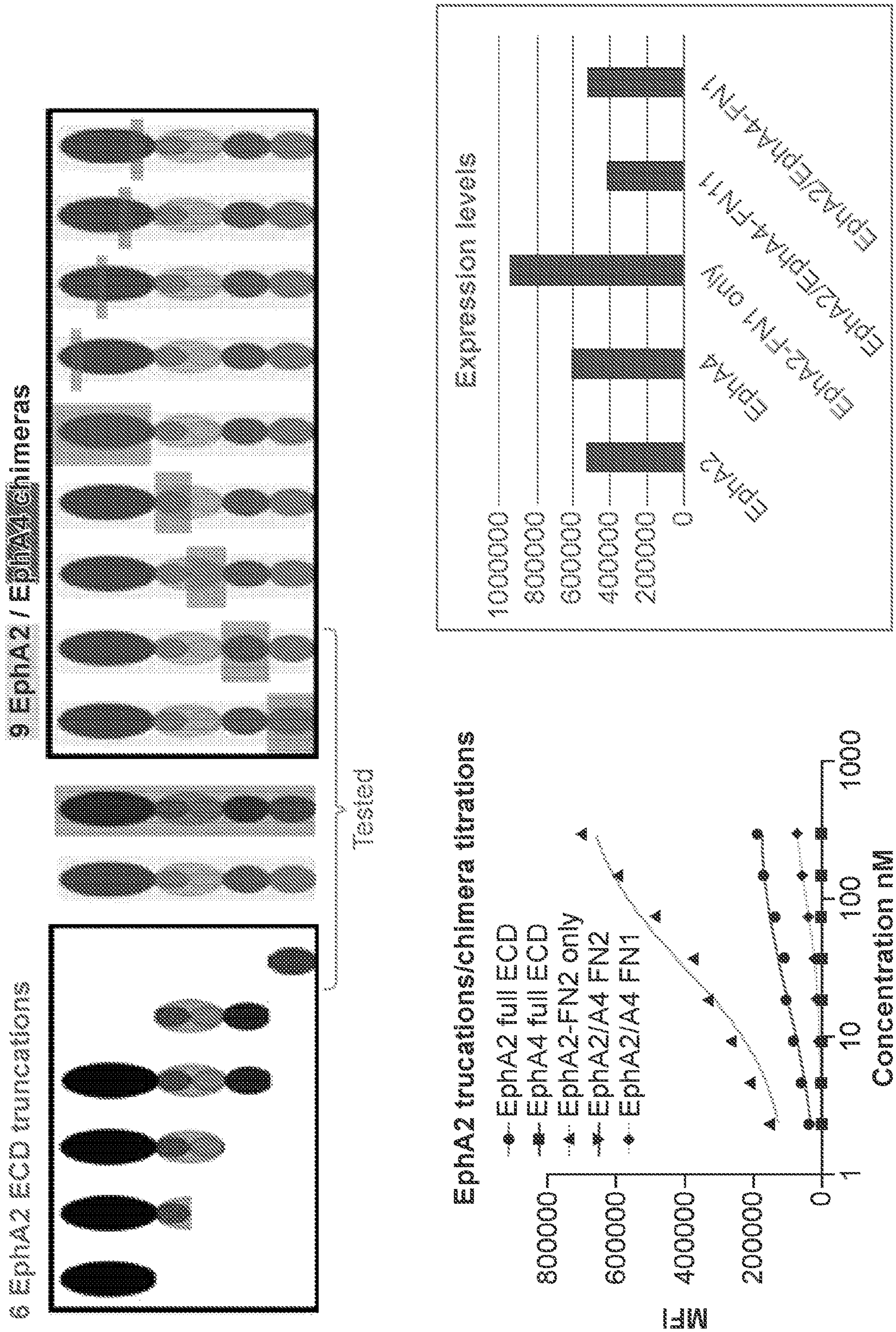

6 EphA2 ECD truncations
Tested
9 EphA2 / EphA4 chimeras
EphA2 trucations/chimera titrations
EphA2 full ECD
EphA4 full ECD
EphA2-FN2 only
EphA2/A4 FN2
EphA2/A4 FN1
MFI
800000
600000
400000
200000
0
1
10
100
1000
Concentration nM
Expression levels
1000000
800000
600000
400000
200000
0
EphA2
EphA4
EphA2-FN1 only
EphA2/EphA4-FN1/1
EphA2/EphA4-FN1

FIG. 6

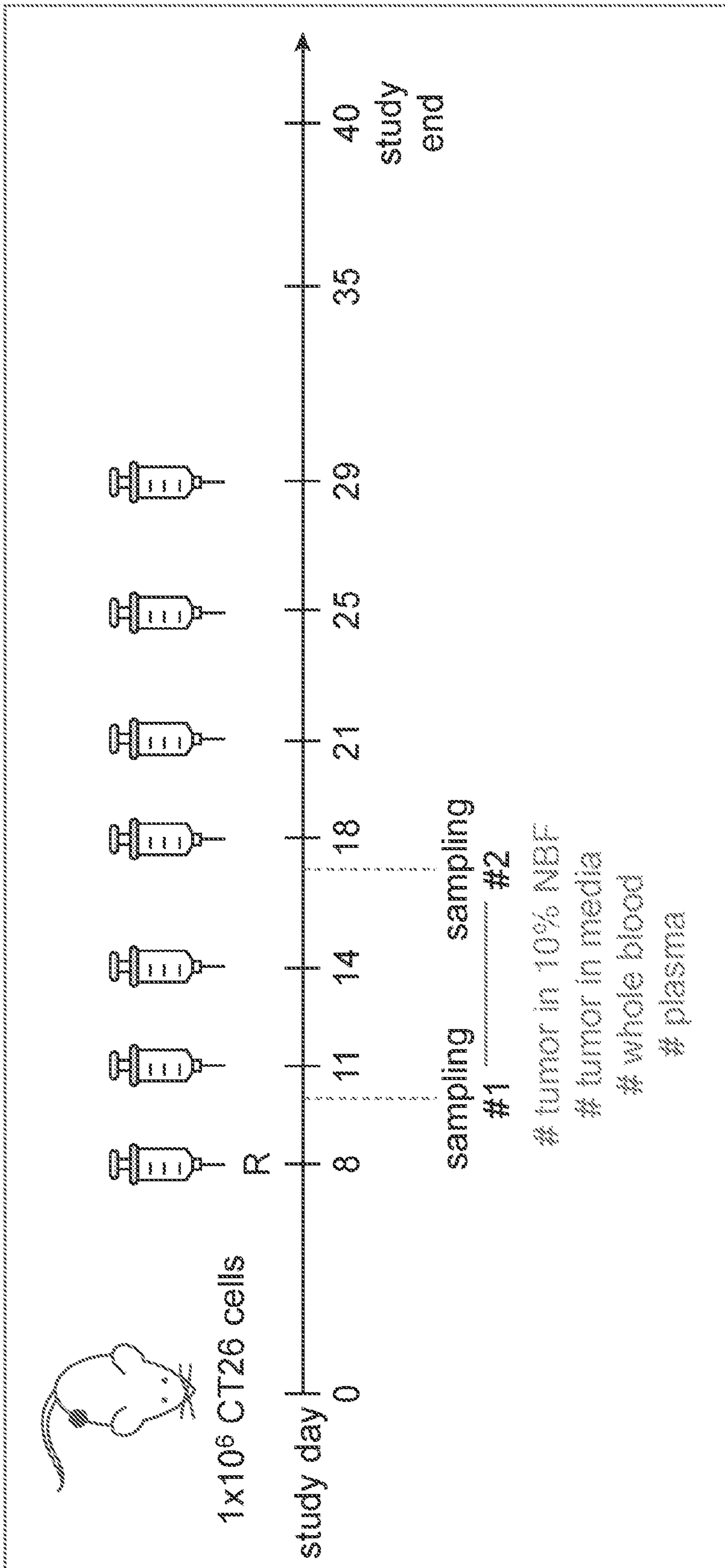

1x10^6 CT26 cells
study day
0
R
8
11
14
18
21
25
29
35
40
study end
sampling #1
sampling #2
tumor in 10% NBF
tumor in media
whole blood
plasma

FIG. 7

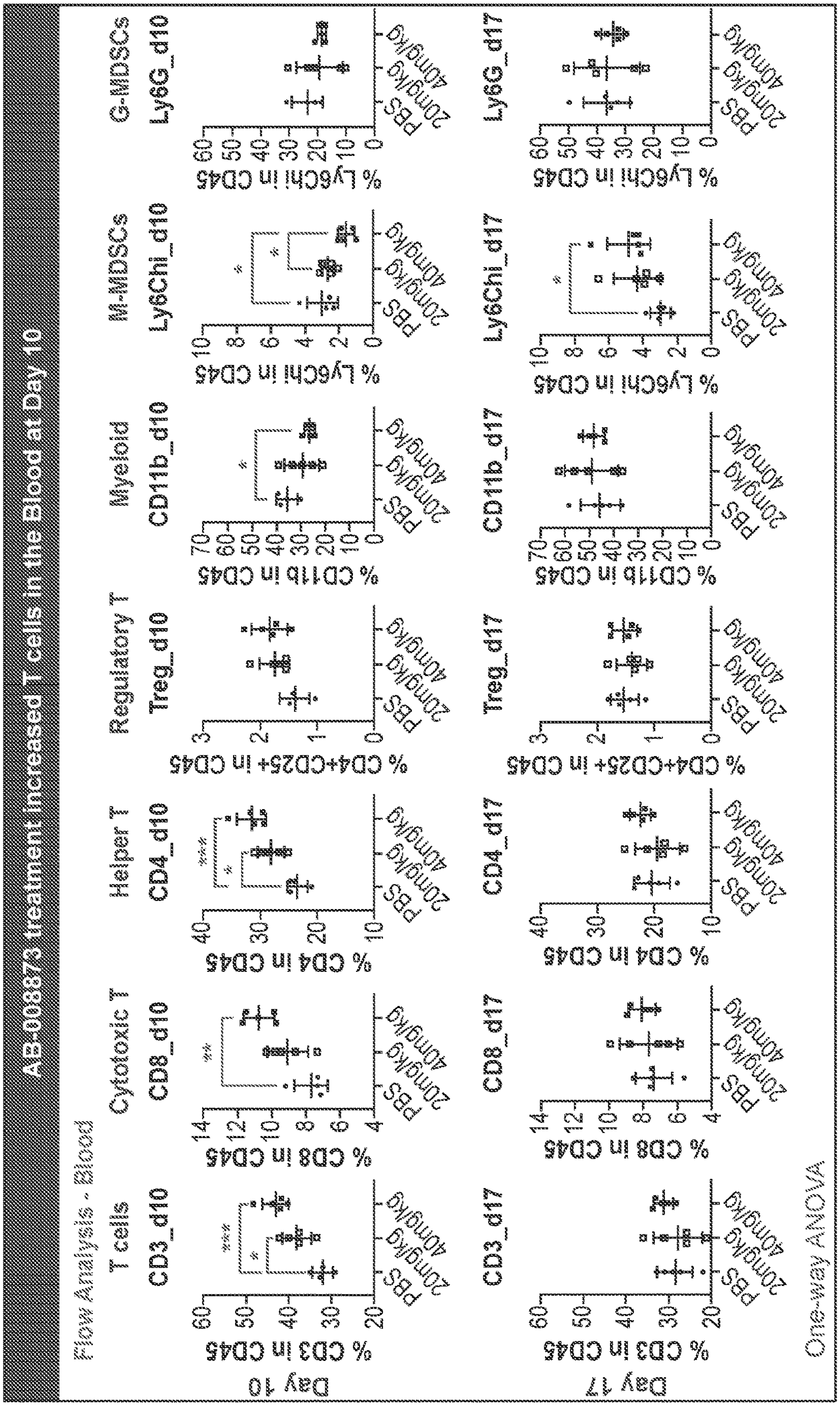
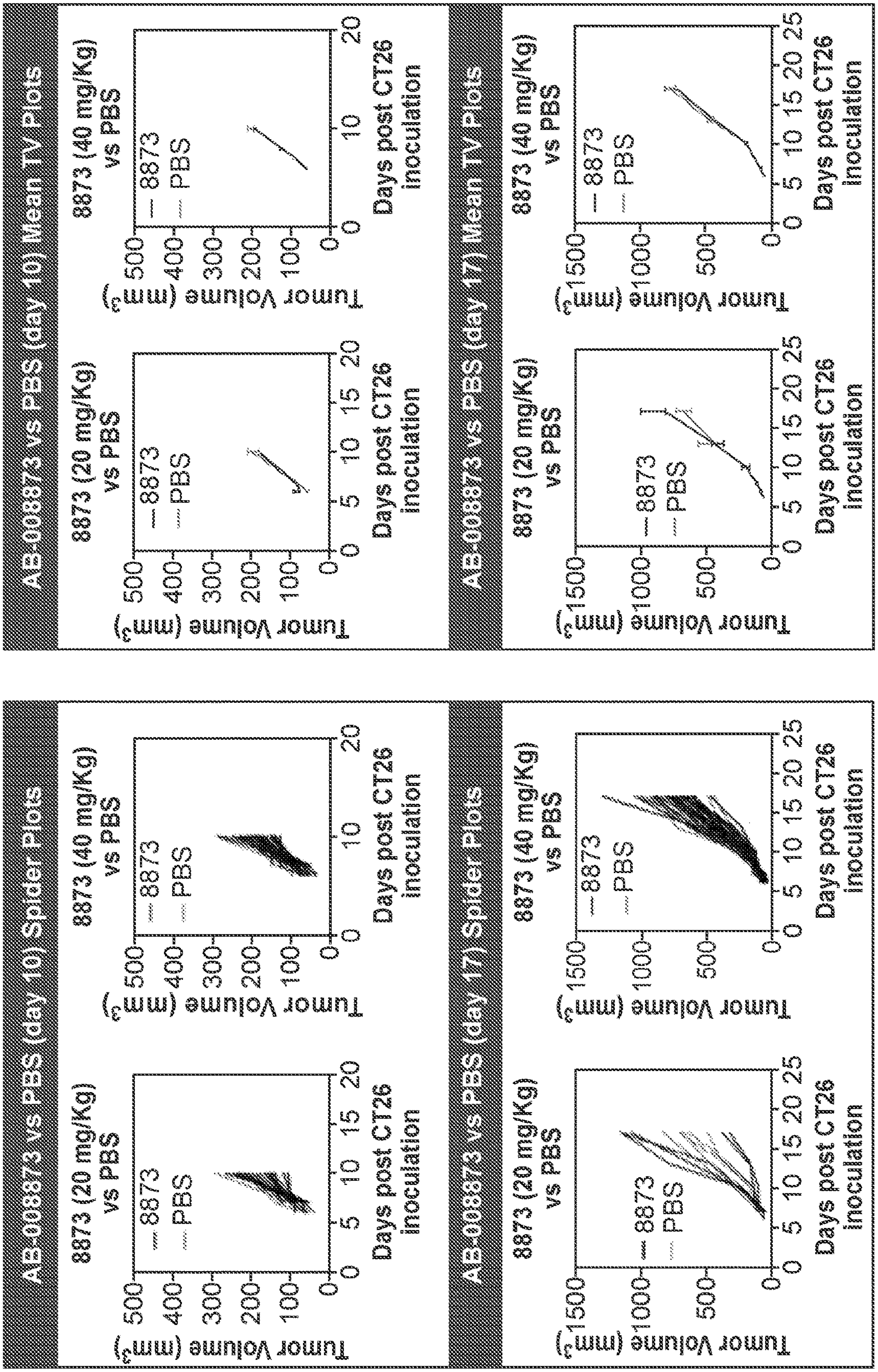

AB-008873 treatment increased T cells in the Blood at Day 10
Flow Analysis - Blood
Day 10
Day 17
T cells
Cytotoxic T
Helper T
Regulatory T
Myeloid
M-MDSCs
G-MDSCs
CD3_d10
CD8_d10
CD4_d10
Treg_d10
CD11b_d10
Ly6Chi_d10
Ly6G_d10
CD3_d17
CD8_d17
CD4_d17
Treg_d17
CD11b_d17
Ly6Chi_d17
Ly6G_d17
% CD3 in CD45
% CD8 in CD45
% CD4 in CD45
% CD4+CD25+ in CD45
% CD11b in CD45
% Ly6Chi in CD45
PBS
20mg/kg
40mg/kg
One-way ANOVA

FIG. 8



AB-008873 vs PBS (day 10) Spider Plots
8873 (20 mg/Kg) vs PBS
8873 (40 mg/Kg) vs PBS
AB-008873 vs PBS (day 17) Spider Plots
AB-008873 vs PBS (day 10) Mean TV Plots
AB-008873 vs PBS (day 17) Mean TV Plots
8873
PBS
Tumor Volume (mm³)
Days post CT26 inoculation

FIG. 9

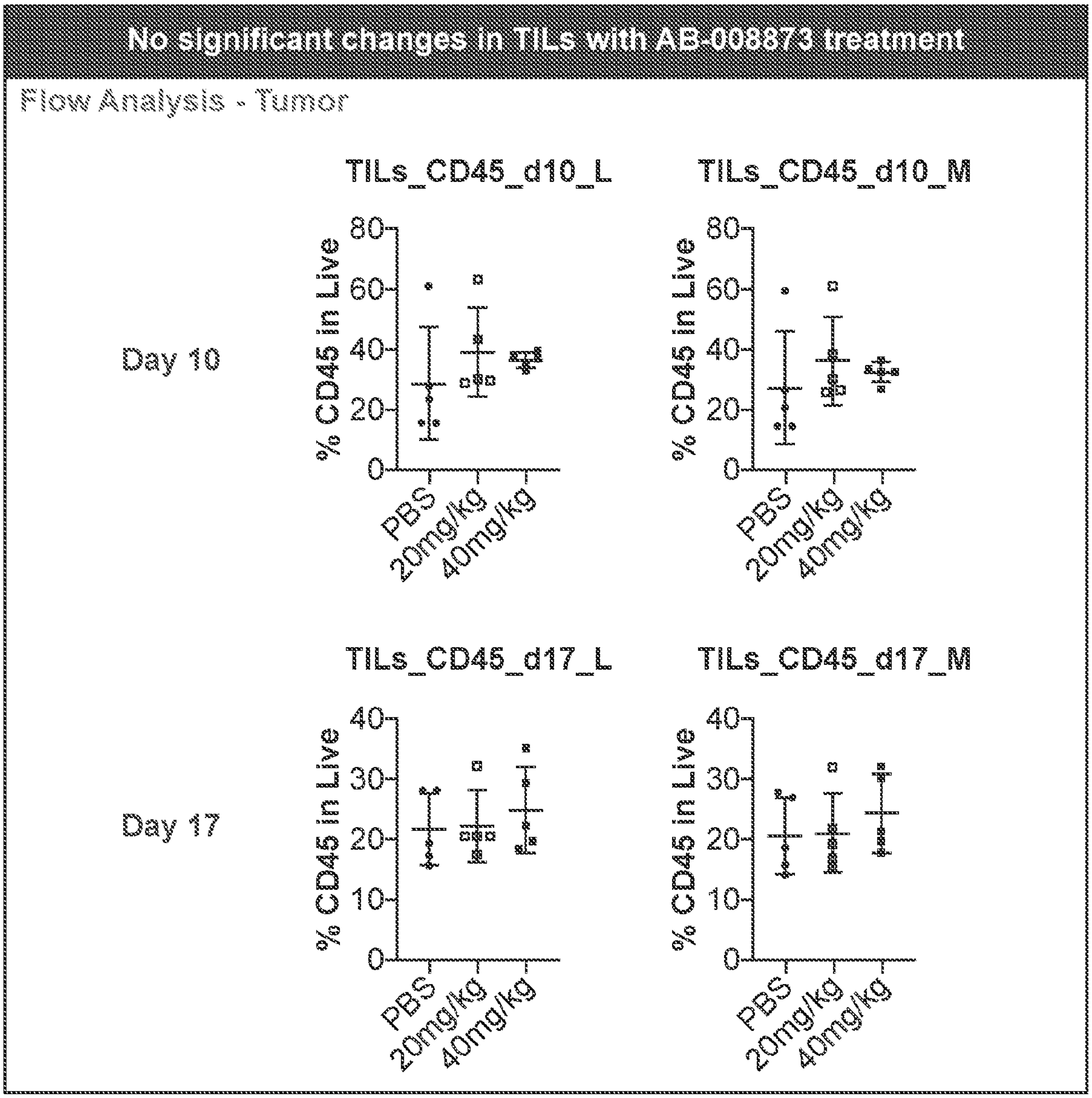

No significant changes in TILs with AB-008873 treatment
Flow Analysis - Tumor
Day 10
Day 17
TILs_CD45_d10_L
TILs_CD45_d10_M
TILs_CD45_d17_L
TILs_CD45_d17_M
% CD45 in Live
80
60
40
30
20
10
0
PBS
20mg/kg
40mg/kg

FIG. 10

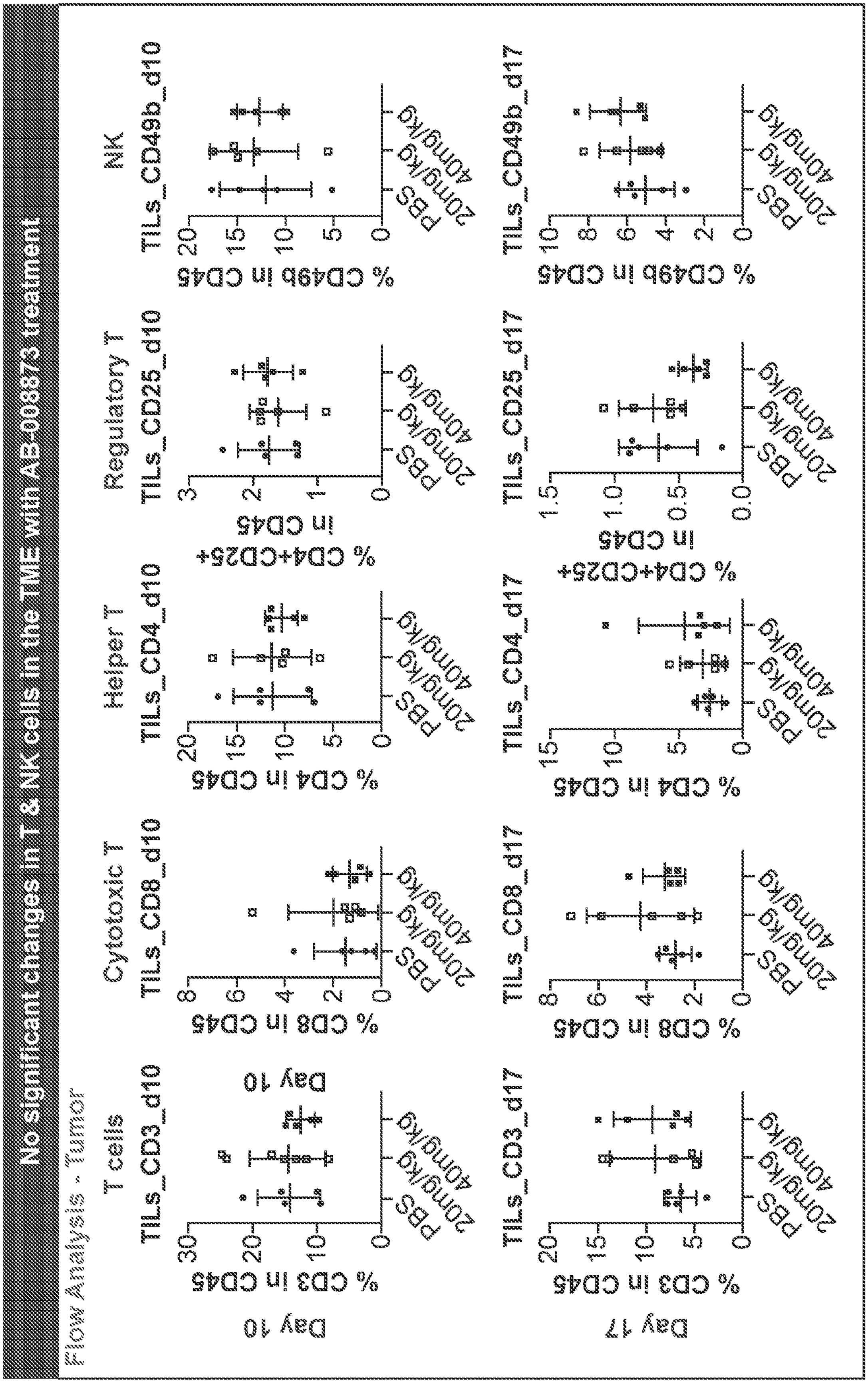

No significant changes in T & NK cells in the TME with AB-008873 treatment
Flow Analysis - Tumor
T cells
Cytotoxic T
Helper T
Regulatory T
NK
Day 10
Day 17
TILs_CD3_d10
TILs_CD8_d10
TILs_CD4_d10
TILs_CD25_d10
TILs_CD49b_d10
TILs_CD3_d17
TILs_CD8_d17
TILs_CD4_d17
TILs_CD25_d17
TILs_CD49b_d17
% CD3 in CD45
% CD8 in CD45
% CD4 in CD45
% CD4+CD25+ in CD45
% CD49b in CD45
PBS
20mg/kg
40mg/kg

FIG. 11

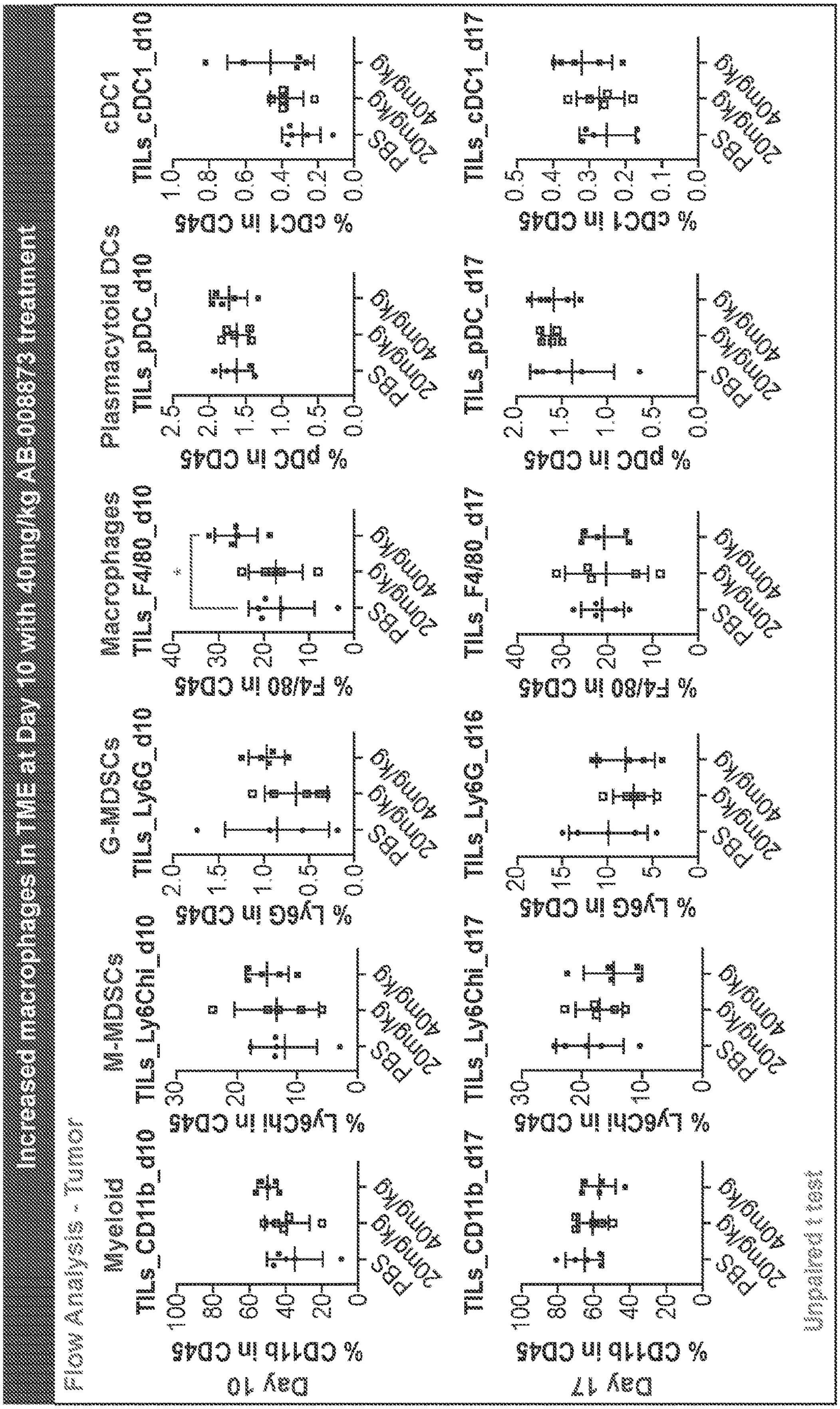

Increased macrophages in TME at Day 10 with 40mg/kg AB-008873 treatment
Flow Analysis - Tumor
Day 10
Day 17
Myeloid
M-MDSCs
G-MDSCs
Macrophages
Plasmacytoid DCs
cDC1
TILs_CD11b_d10
TILs_Ly6Chi_d10
TILs_Ly6G_d10
TILs_F4/80_d10
TILs_pDC_d10
TILs_cDC1_d10
TILs_CD11b_d17
TILs_Ly6Chi_d17
TILs_Ly6G_d16
TILs_F4/80_d17
TILs_pDC_d17
TILs_cDC1_d17
% CD11b in CD45
% Ly6Chi in CD45
% Ly6G in CD45
% F4/80 in CD45
% pDC in CD45
% cDC1 in CD45
PBS
20mg/kg
40mg/kg
Unpaired t test

FIG. 12

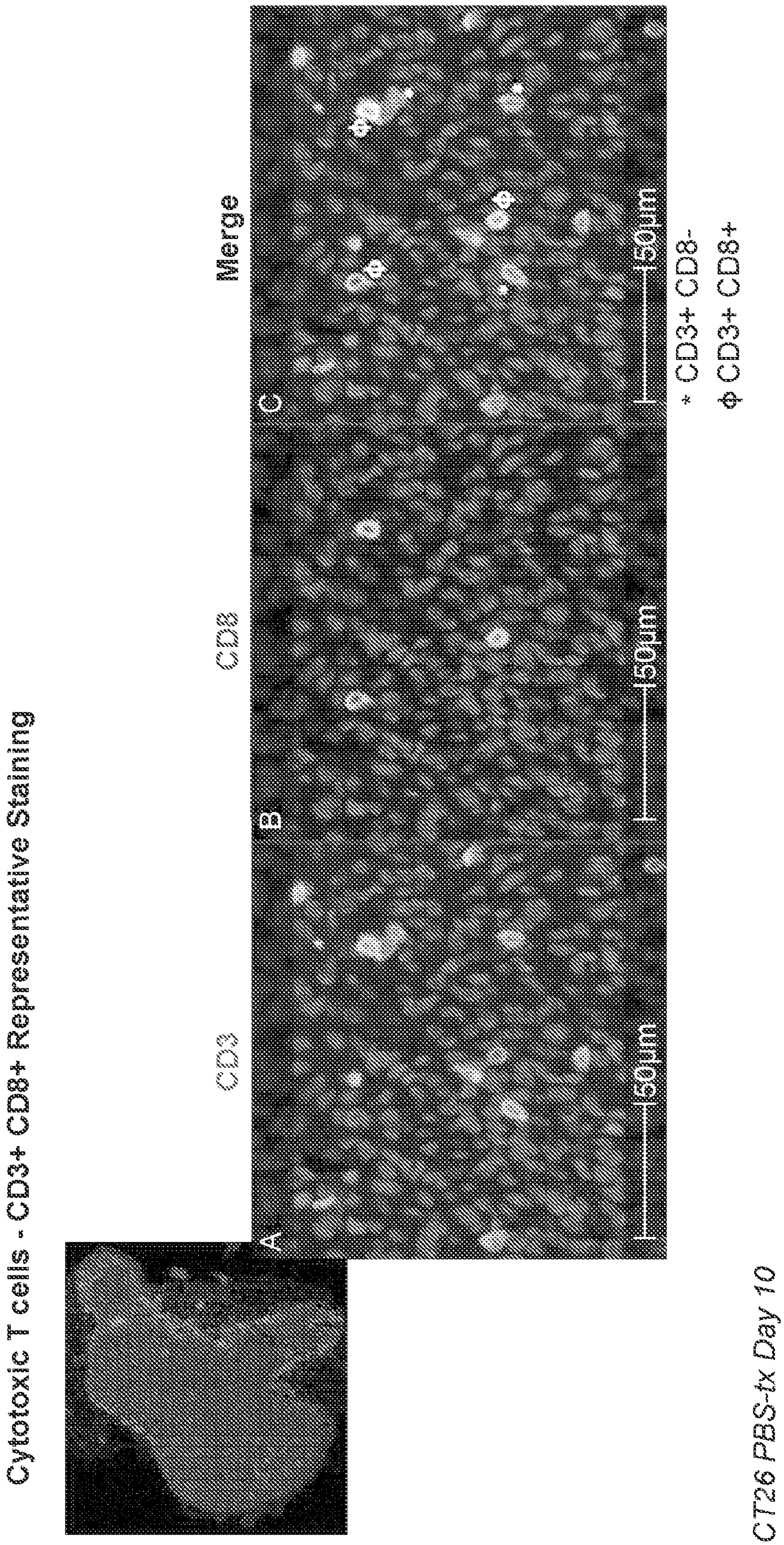
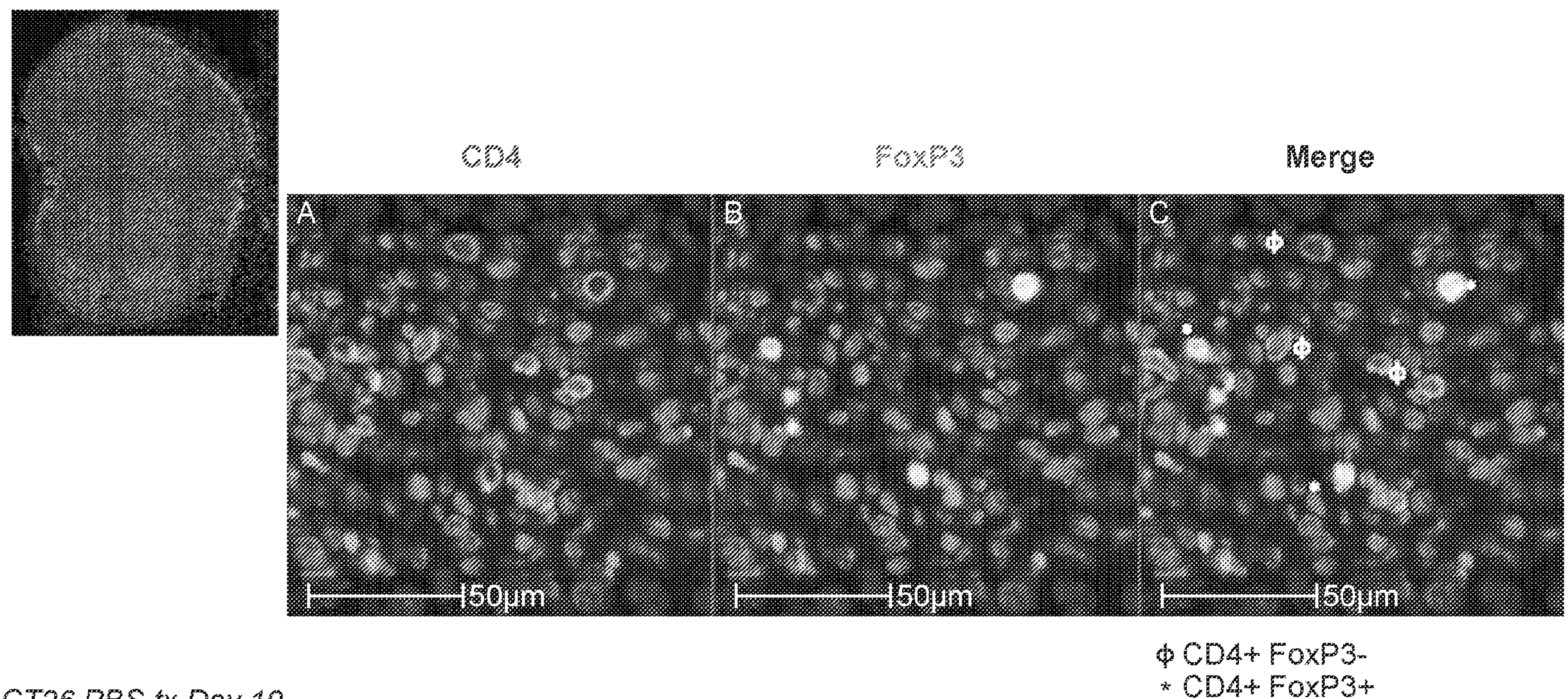

Cytotoxic T cells - CD3+ CD8+ Representative Staining
A
CD3
50µm
B
CD8
50µm
C
Merge
50µm
* CD3+ CD8-
ɸ CD3+ CD8+
CT26 PBS-tx Day 10

FIG. 13

Regulatory T cells - CD4+ FoxP3+ Representative Staining



CT26 PBS-tx Day 10

FIG. 14

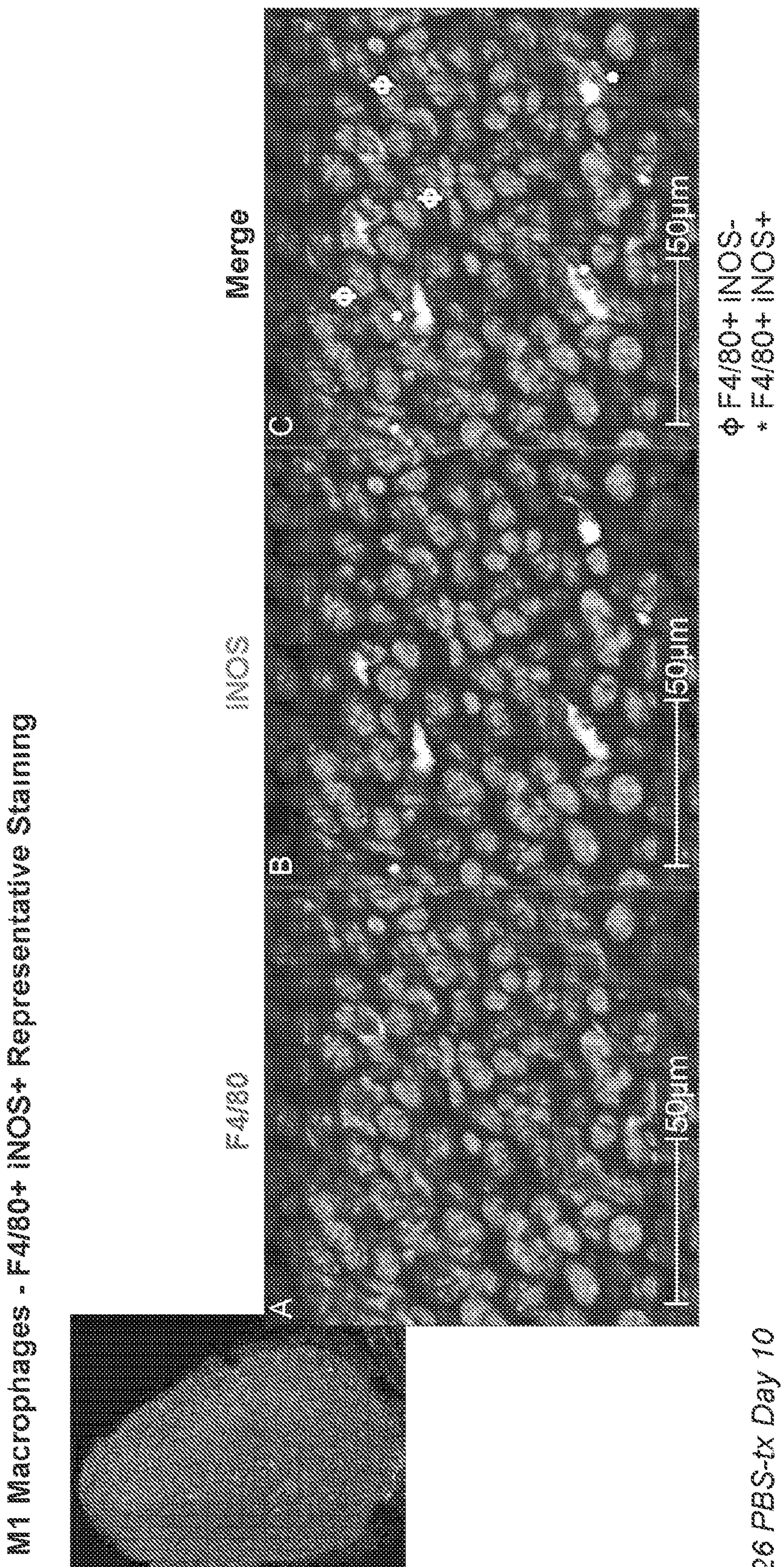
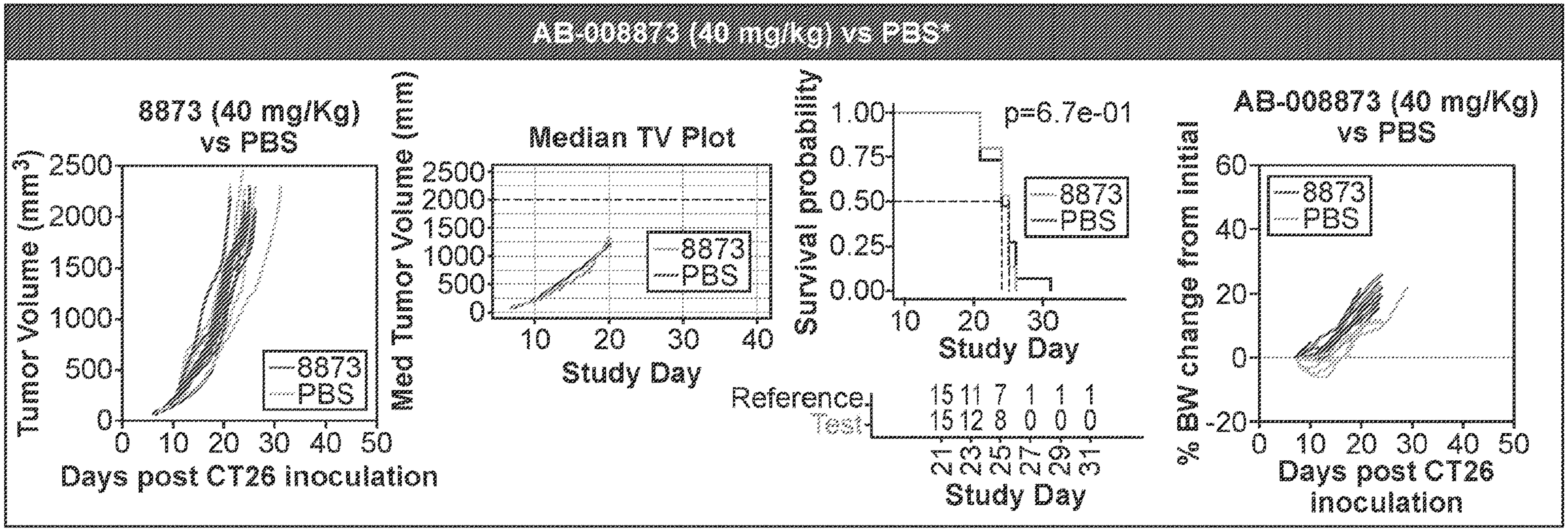

M1 Macrophages - F4/80+ iNOS+ Representative Staining
A
F4/80
50μm
B
iNOS
50μm
C
Merge
50μm
ф F4/80+ iNOS-
* F4/80+ iNOS+
CT26 PBS-tx Day 10

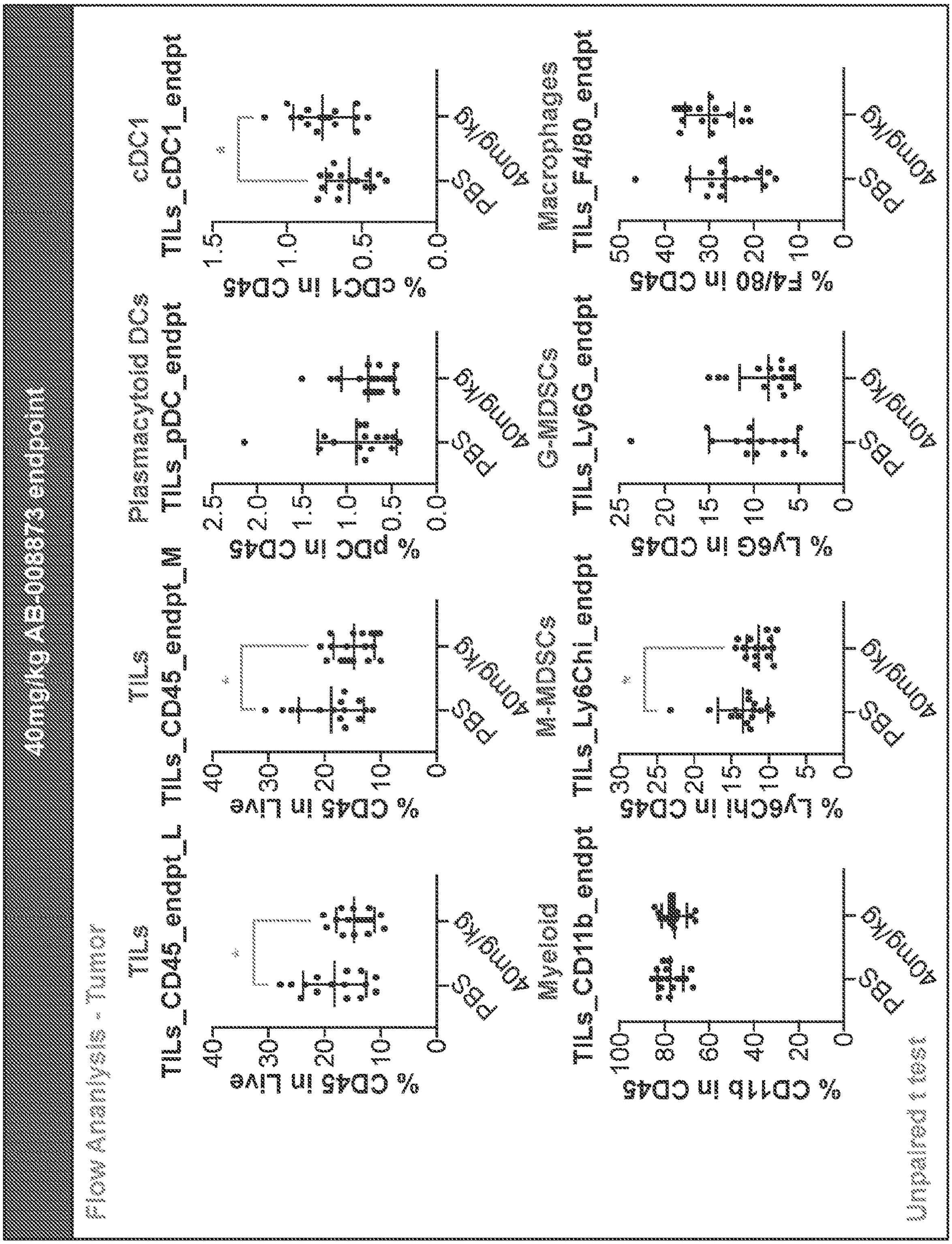
FIG. 15



*TGD study arms. Mice received 7 cycles (b.i.w.) of treatment and were euthanized when tumor burden endpoint criteria were met.

Group	Dosw (mpk)	n	Ref. Grp	Survival p-value	Median Survival	Reference Median Survival	NAAC p-value	NAAC effect size	NGRM p-value	NGRM effect size
8873	40	15	PBS	6.68E-01	25	24	3.72E-01	0.0071	5.33E-01	-0.0014

FIG. 16



40mg/kg AB-008873 endpoint
Flow Ananlysis - Tumor
TILs
TILs_CD45_endpt_L
% CD45 in Live
TILs
TILs_CD45_endpt_M
% CD45 in Live
Plasmacytoid DCs
TILs_pDC_endpt
% pDC in CD45
cDC1
TILs_cDC1_endpt
% cDC1 in CD45
Myeloid
TILs_CD11b_endpt
% CD11b in CD45
M-MDSCs
TILs_Ly6Chi_endpt
% Ly6Chi in CD45
G-MDSCs
TILs_Ly6G_endpt
% Ly6G in CD45
Macrophages
TILs_F4/80_endpt
% F4/80 in CD45
PBS
40mg/kg
Unpaired t test

FIG. 17

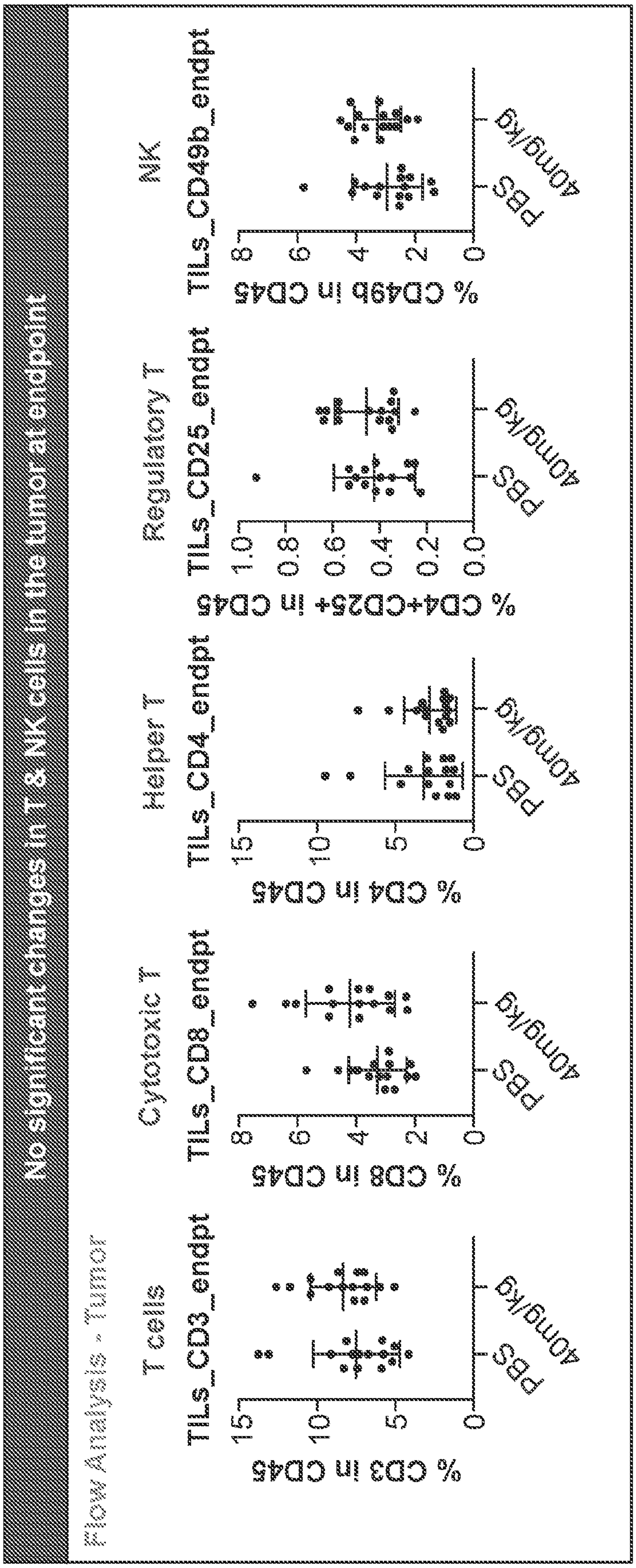

No significant changes in T & NK cells in the tumor at endpoint
Flow Analysis - Tumor
T cells
TILs_CD3_endpt
% CD3 in CD45
Cytotoxic T
TILs_CD8_endpt
% CD8 in CD45
Helper T
TILs_CD4_endpt
% CD4 in CD45
Regulatory T
TILs_CD25_endpt
% CD4+CD25+ in CD45
NK
TILs_CD49b_endpt
% CD49b in CD45
PBS
40mg/kg

FIG. 18

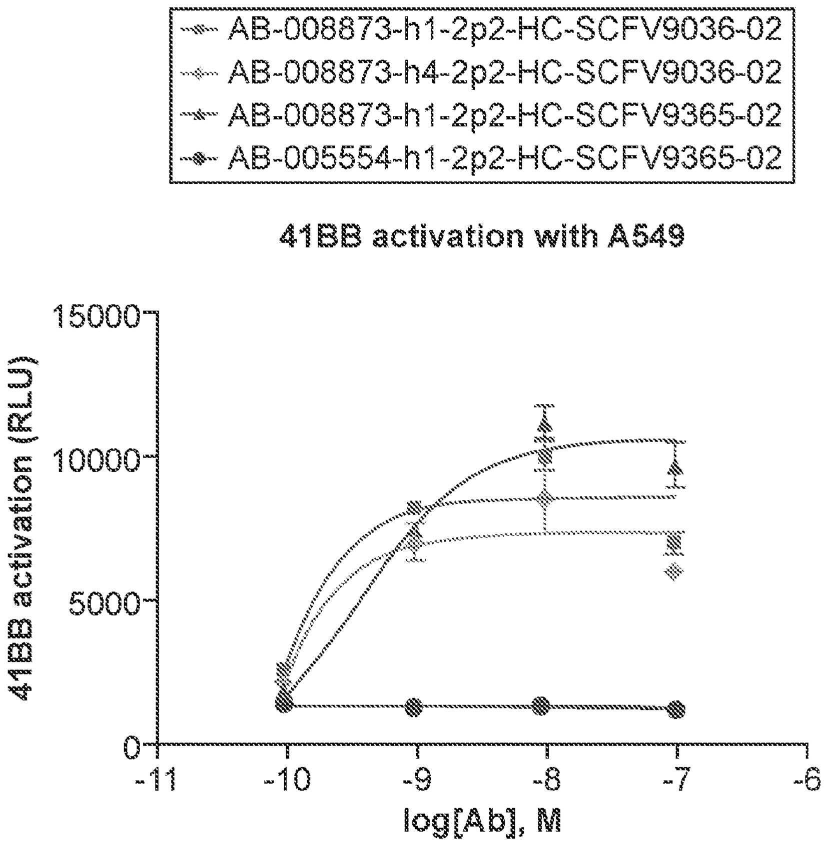

AB-008873-h1-2p2-HC-SCFV9036-02
AB-008873-h4-2p2-HC-SCFV9036-02
AB-008873-h1-2p2-HC-SCFV9365-02
AB-005554-h1-2p2-HC-SCFV9365-02
41BB activation with A549
15000
10000
5000
0
41BB activation (RLU)
-11
-10
-9
-8
-7
-6
log[Ab], M

FIG. 19A

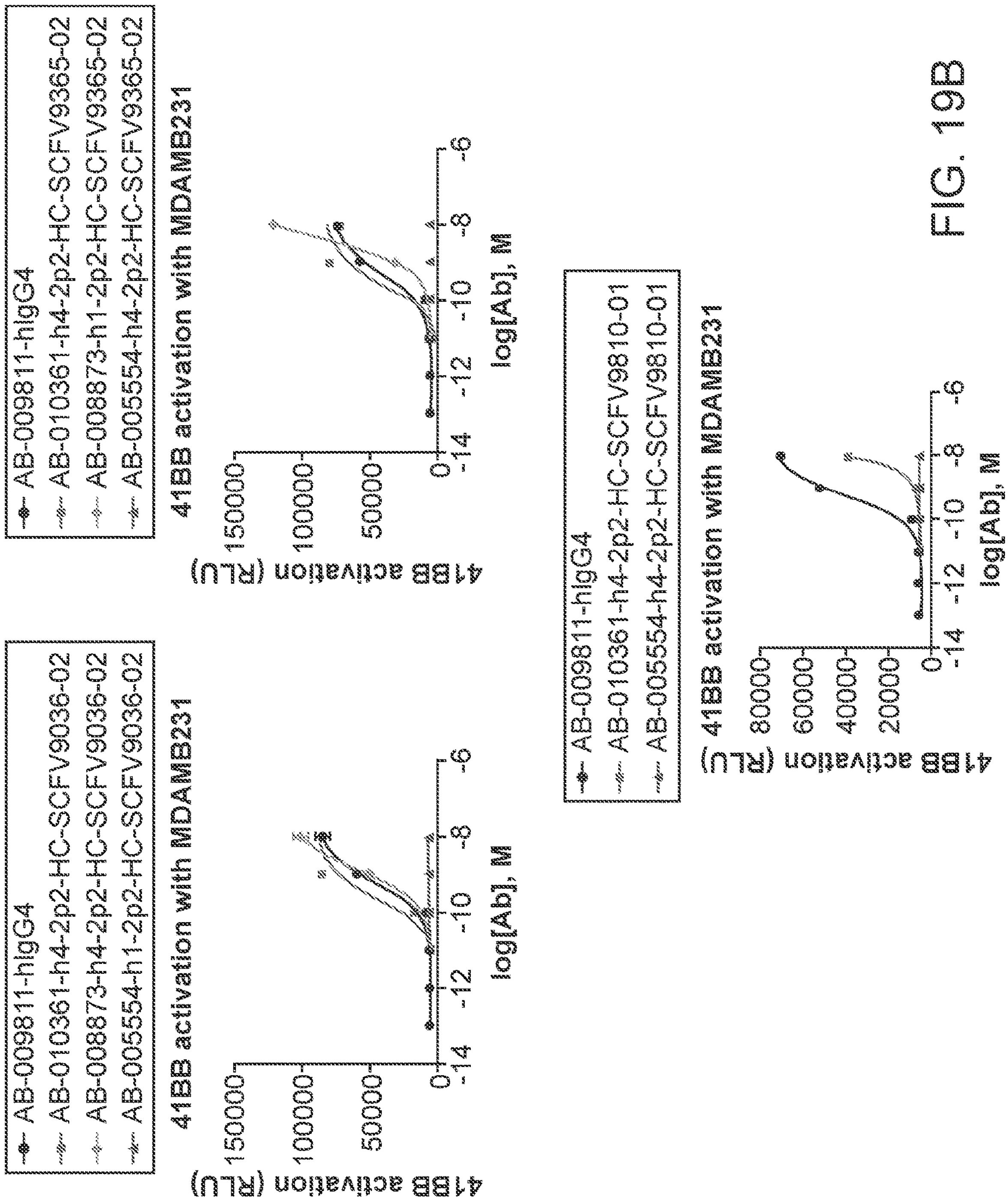

41BB activation with MDAMB231
41BB activation (RLU)
log[Ab], M
AB-009811-hIgG4
AB-010361-h4-2p2-HC-SCFV9036-02
AB-008873-h4-2p2-HC-SCFV9036-02
AB-005554-h1-2p2-HC-SCFV9036-02
41BB activation with MDAMB231
41BB activation (RLU)
log[Ab], M
AB-009811-hIgG4
AB-010361-h4-2p2-HC-SCFV9365-02
AB-008873-h1-2p2-HC-SCFV9365-02
AB-005554-h4-2p2-HC-SCFV9365-02
41BB activation with MDAMB231
41BB activation (RLU)
log[Ab], M
AB-009811-hIgG4
AB-010361-h4-2p2-HC-SCFV9810-01
AB-005554-h4-2p2-HC-SCFV9810-01

FIG. 19B

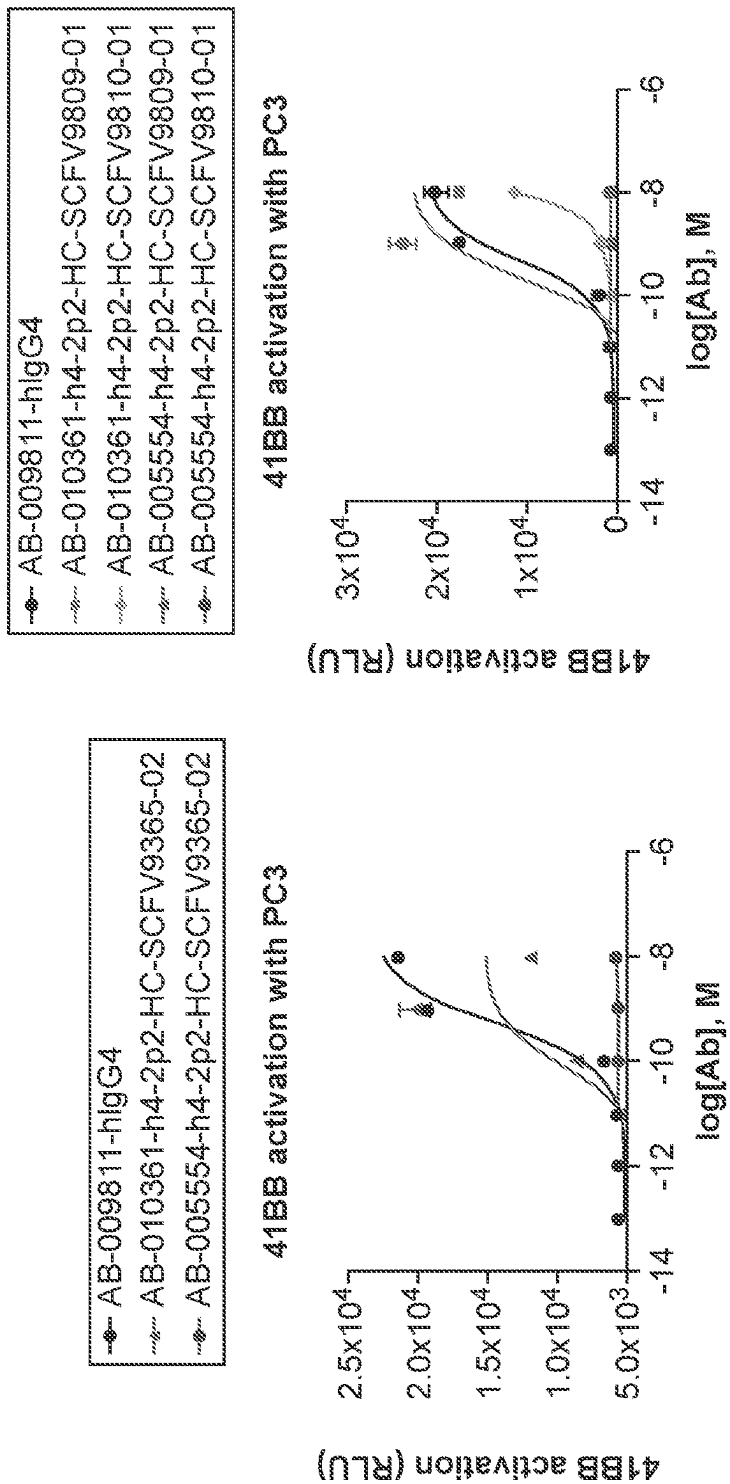

AB-009811-hIgG4
AB-010361-h4-2p2-HC-SCFV9809-01
AB-010361-h4-2p2-HC-SCFV9810-01
AB-005554-h4-2p2-HC-SCFV9809-01
AB-005554-h4-2p2-HC-SCFV9810-01
41BB activation with PC3
41BB activation (RLU)
3x10^4
2x10^4
1x10^4
0
-14
-12
-10
-8
-6
log[Ab], M
AB-009811-hIgG4
AB-010361-h4-2p2-HC-SCFV9365-02
AB-005554-h4-2p2-HC-SCFV9365-02
41BB activation with PC3
41BB activation (RLU)
2.5x10^4
2.0x10^4
1.5x10^4
1.0x10^4
5.0x10^3
-14
-12
-10
-8
-6
log[Ab], M

FIG. 19C

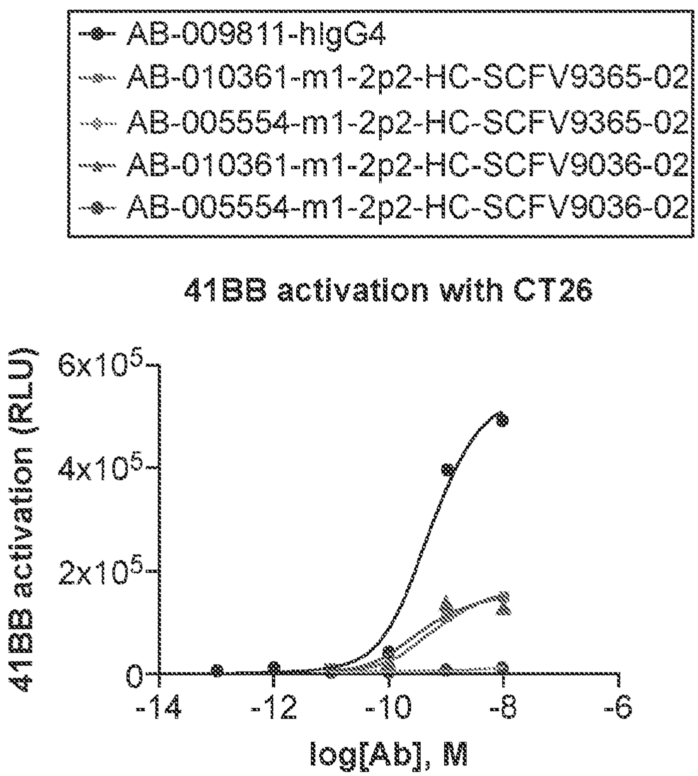

AB-009811-hIgG4
AB-010361-m1-2p2-HC-SCFV9365-02
AB-005554-m1-2p2-HC-SCFV9365-02
AB-010361-m1-2p2-HC-SCFV9036-02
AB-005554-m1-2p2-HC-SCFV9036-02
41BB activation with CT26
41BB activation (RLU)
6x10^5
4x10^5
2x10^5
0
-14
-12
-10
-8
-6
log[Ab], M

FIG. 19D

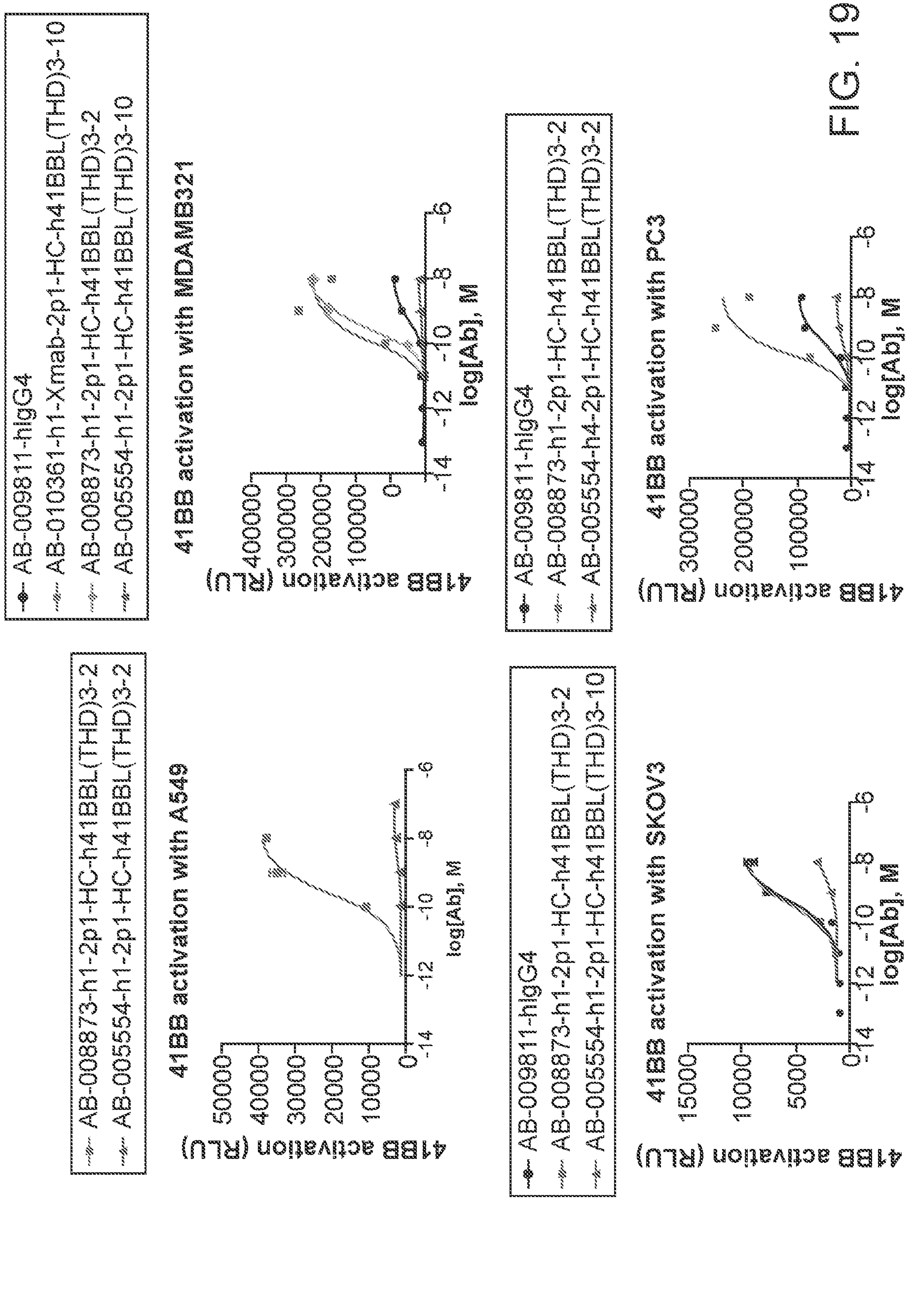
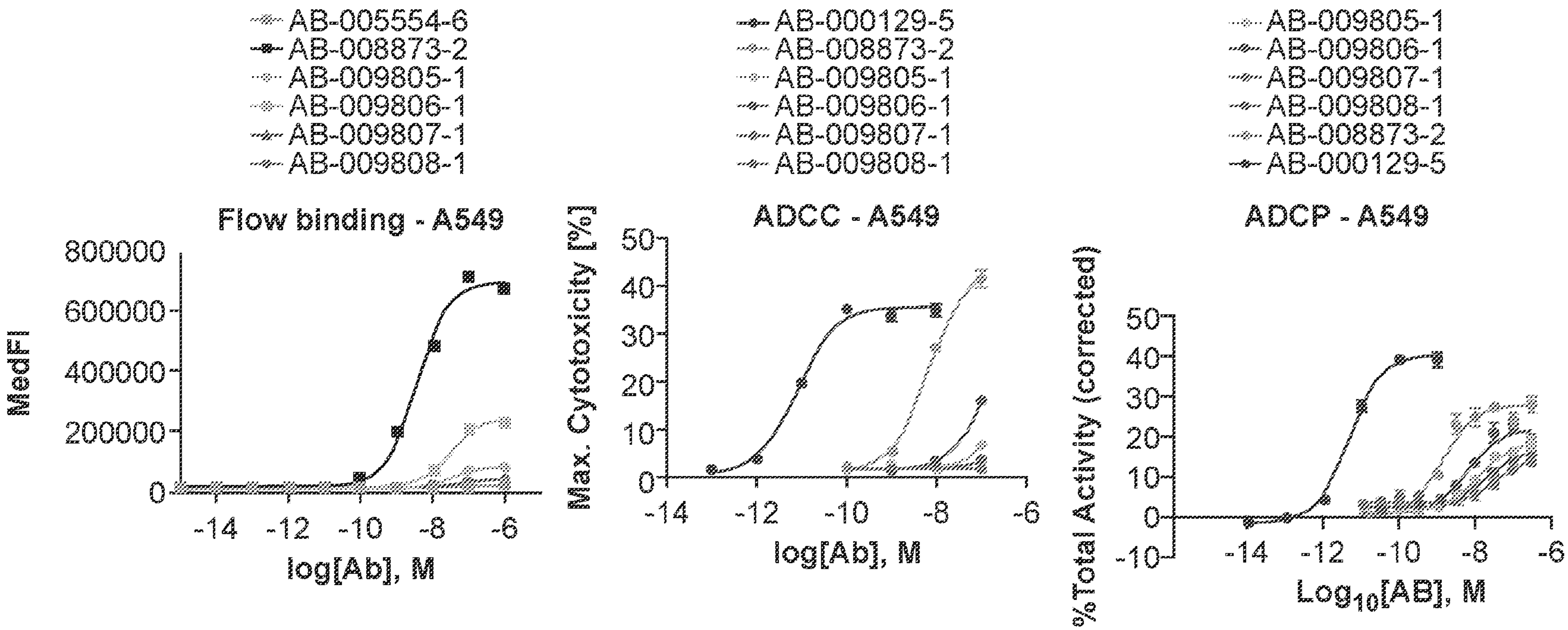

41BB activation with A549
41BB activation (RLU)
log[Ab], M
AB-008873-h1-2p1-HC-h41BBL(THD)3-2
AB-005554-h1-2p1-HC-h41BBL(THD)3-2
41BB activation with MDAMB321
AB-009811-hIgG4
AB-010361-h1-Xmab-2p1-HC-h41BBL(THD)3-10
AB-008873-h1-2p1-HC-h41BBL(THD)3-2
AB-005554-h1-2p1-HC-h41BBL(THD)3-10
41BB activation with SKOV3
AB-009811-hIgG4
AB-008873-h1-2p1-HC-h41BBL(THD)3-2
AB-005554-h1-2p1-HC-h41BBL(THD)3-10
41BB activation with PC3
AB-009811-hIgG4
AB-008873-h1-2p1-HC-h41BBL(THD)3-2
AB-005554-h4-2p1-HC-h41BBL(THD)3-2

FIG. 19E



Ab	Flow Delta activity (MedFI)	ADCC Delta activity (%)	ADCP Delta activity (%)	# Germline deviations	N-linked glyco
AB-008873	680293	39.45	27.43	27	N
AB-009805	75111	5.06	17.3	22	Y
AB-009806	230730	14.47	17.87	21	Y
AB-009807	12737	0.4	11.47	22	Y
AB-009808	29470	1.31	12.83	20	Y

FIG. 20

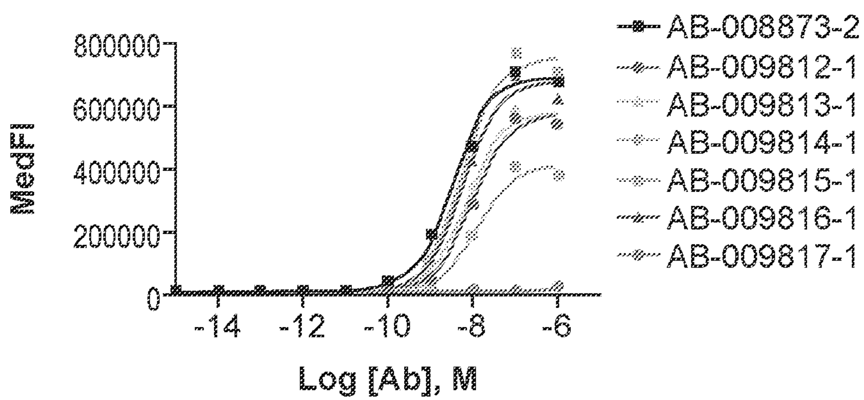


FIG. 21A

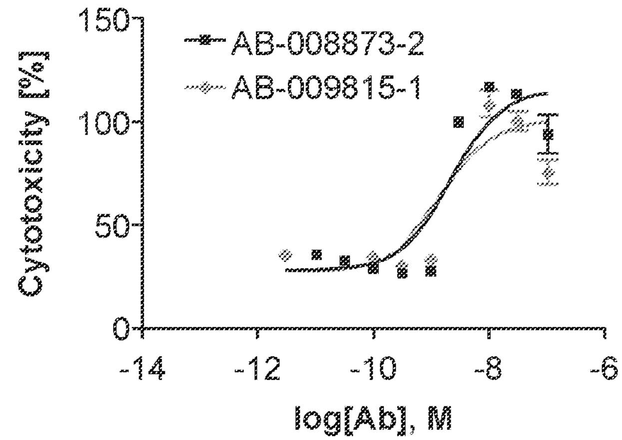


FIG. 21B

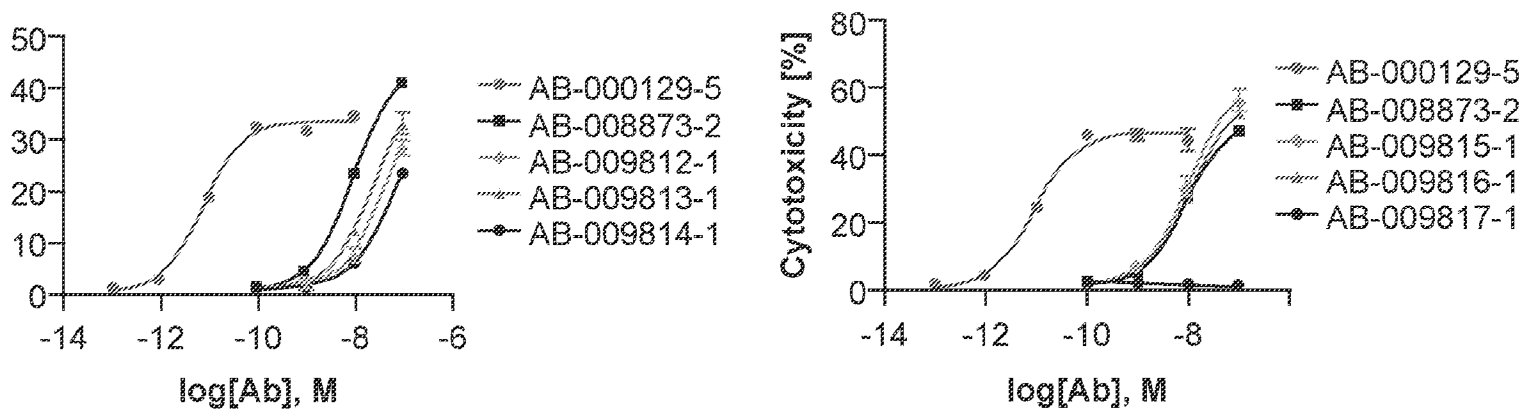


FIG. 21C

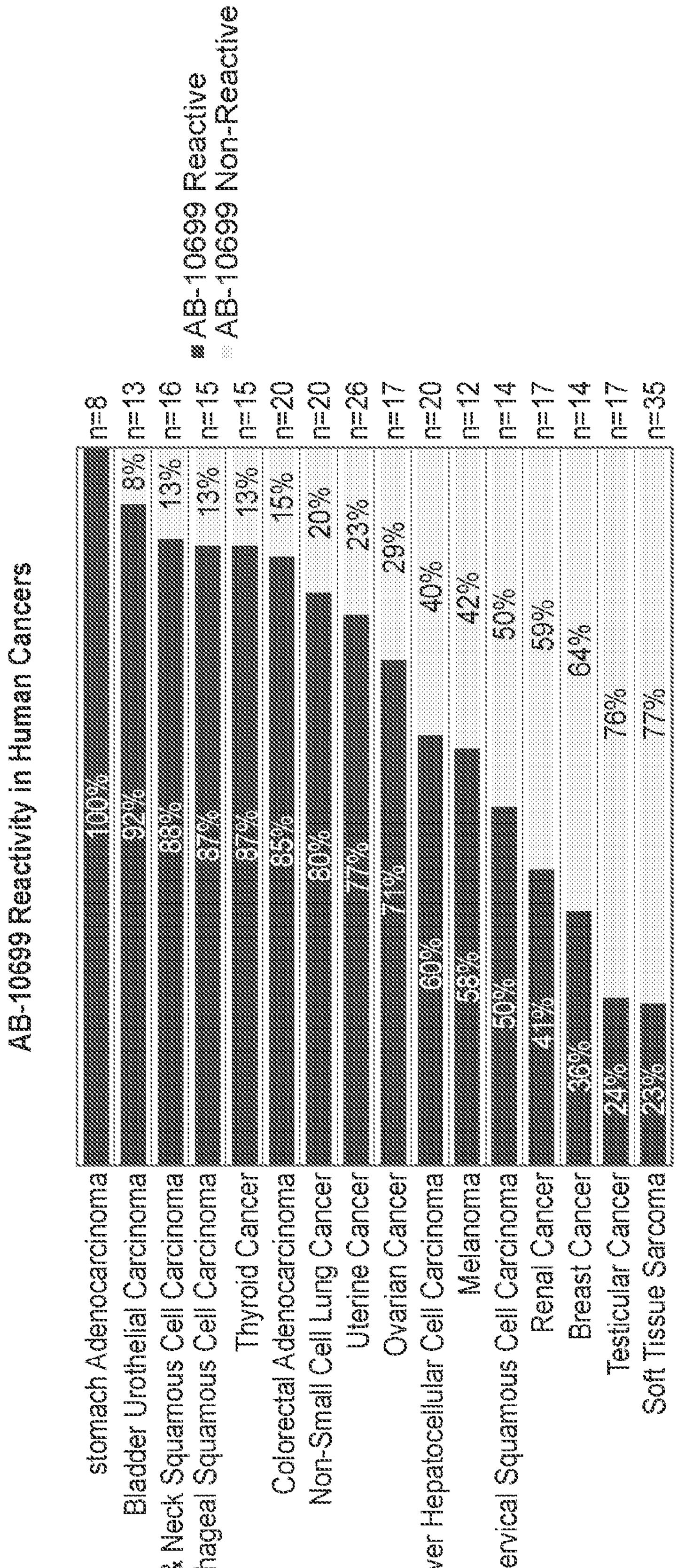

AB-10699 Reactivity in Human Cancers
AB-10699 Reactive
AB-10699 Non-Reactive
stomach Adenocarcinoma 100% n=8
Bladder Urothelial Carcinoma 92% 8% n=13
Head & Neck Squamous Cell Carcinoma 88% 13% n=16
Esophageal Squamous Cell Carcinoma 87% 13% n=15
Thyroid Cancer 87% 13% n=15
Colorectal Adenocarcinoma 85% 15% n=20
Non-Small Cell Lung Cancer 80% 20% n=20
Uterine Cancer 77% 23% n=26
Ovarian Cancer 71% 29% n=17
Liver Hepatocellular Carcinoma 60% 40% n=20
Melanoma 58% 42% n=12
Cervical Squamous Cell Carcinoma 50% 50% n=14
Renal Cancer 41% 59% n=17
Breast Cancer 36% 64% n=14
Testicular Cancer 24% 76% n=17
Soft Tissue Sarcoma 23% 77% n=35

FIG. 22

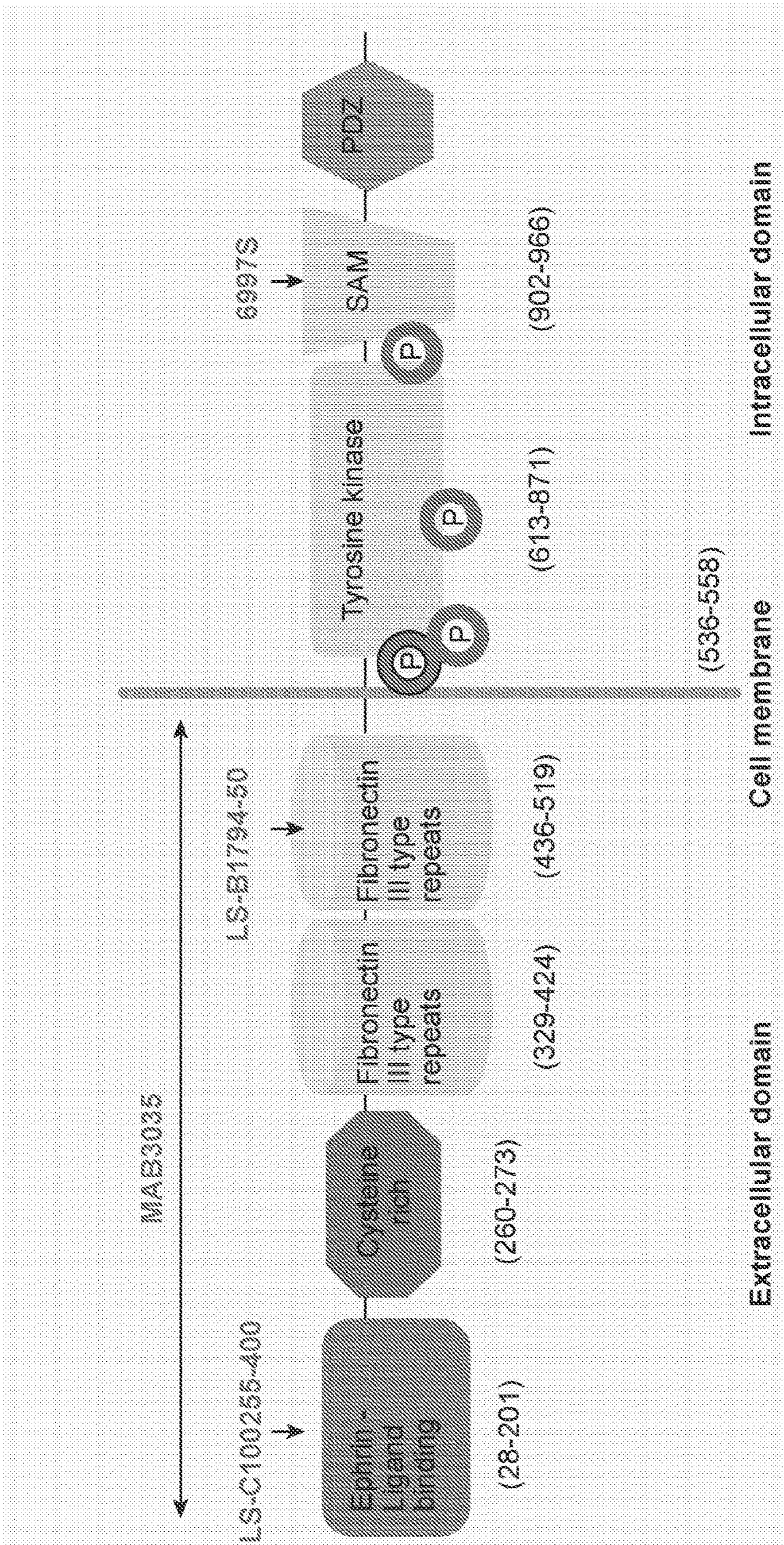

LS-C100255-400
Ephrin-Ligand binding
(28-201)
MAB3035
Cysteine rich
(260-273)
Fibronectin III type repeats
(329-424)
LS-B1794-50
Fibronectin III type repeats
(436-519)
Extracellular domain
Cell membrane
(536-558)
P
P
Tyrosine kinase
P
(613-871)
P
6997S
SAM
(902-966)
PDZ
Intracellular domain

FIG. 23

IHC score of 8873 and variants on 21 types of FF human tumor and TAT

- compared with clinical candidate EphA2 antibodies and commercial EphA2 Abs

Ab Info			NSCLC-SCL	NSCLC-Adeno	RCC-clear cell	RCC-papillary	MELA	ESO	OVA	TNBC
8873 and variants	AB-009873		1	2	1	1	2	2	2	1
	AB-009812		1	2	1	1	2	2	2	1
	AB-009813		1	2	1	1	2	2	2	1
	AB-009815		1	2	1	1	2	2	2	1
	AB-009816		1	2	1	1	2	2	2	1
Clinical candidate EphA2 antibodies	AB-010018		2	2	1	1	3	2	2	2
	AB-010016		1	1	1	1	1.5	1	1	1
	AB-010017		1	2	1	1	2	1	1	1
Commercial anti-EphA2 Abs	Extracellular	MAB3035 (Gin-25-Asn534)	2	2	1	1	3	2	2	1
		LS-C16249-100(extracelluiar)	2	2	1	1	3	2	2	1
		LS-C100255-400(AA 30-60)	1	1	1	1	2	1	1	1
		LS-81794-50(AA 450-500)	1	2	1	2	2	1	1	1
		Ab73254 (extracellular)	2	2	1	1	2	1	1	1
	Cytoplasmic	69975 (around Arg007)	1	1	1	1	2	2	1	1

Ab Info			% overlap of positive cancer types between 8873 and other Abs	
8873 variants	AB-009873			
	AB-009812		91%	
	AB-009813		91%	
	AB-009815		100%	
	AB-009816		100%	
Clinical candidate EphA2 antibodies	AB-010018		82%	
	AB-010016		45%	
	AB-010017		55%	
Commercial anti-EphA2 Abs	Extracellular	MAB3035 (Gin-25-Asn534)	73%	
		LS-C16249-100(extracellular)	91%	
		LS-C100255-400(AA 30-60)	27%	
		LS-81794-50(AA 450-500)	36%	
		Ab73254 (extracellular)	27%	
	Cytoplasmic	69975 (around Arg007)	45%	

FIG. 24A

IHC score												
CR+ BRC	Her2+ DAC	CRC	PANC	PROC	STS	URD	ANC	HNC	GAC	UCEC-endometioid	UCEC-clear cell	UCEC-papillary
1	2	1	1	1	2	1	2	2	2	1	2	2
1	2	1	1	1	2	1	2	2	2	1	1	2
1	1	1	1	1	2	1	2	2	2	1	2	2
1	2	1	1	1	2	1	2	2	2	1	2	2
1	2	1	1	1	2	1	2	2	2	1	2	2
2	2	1	1	1	2	1	2	1	1	2	2	2
1	1	1	1	1	1	1	2	1	1	1	2	2
1	1	2	1	1	1	1	2	2	1	2	2	3
1	1	1	2	1	1	1	2	2	1	2	2	2
1	2	1	1	1	2	1	2	2	1	2	2	3
1	1	1	1	1	1	1	1	1	1	1	2	2
1	1	1	1	1	1	1	1	1	1	1	2	2
1	1	1	1	1	1	1	1	1	1	1	2	1
1	1	1	1	1	1	1	2	1	1	1	2	2

% overlap of positive cancer types between DS-8895 and other Abs
69%
62%
62%
69%
69%
31%
46%
69%
85%
23%
31%
31%
38%

- Same concentration, exposure time and LUT for 8873 and variants and clinical candidate EphA2 Abs
- Utilized manufacture recommendations for concentration for commercial antibodies
- Commercial antibodies are from different species and use different isotype controls

FIG. 24A Contd...

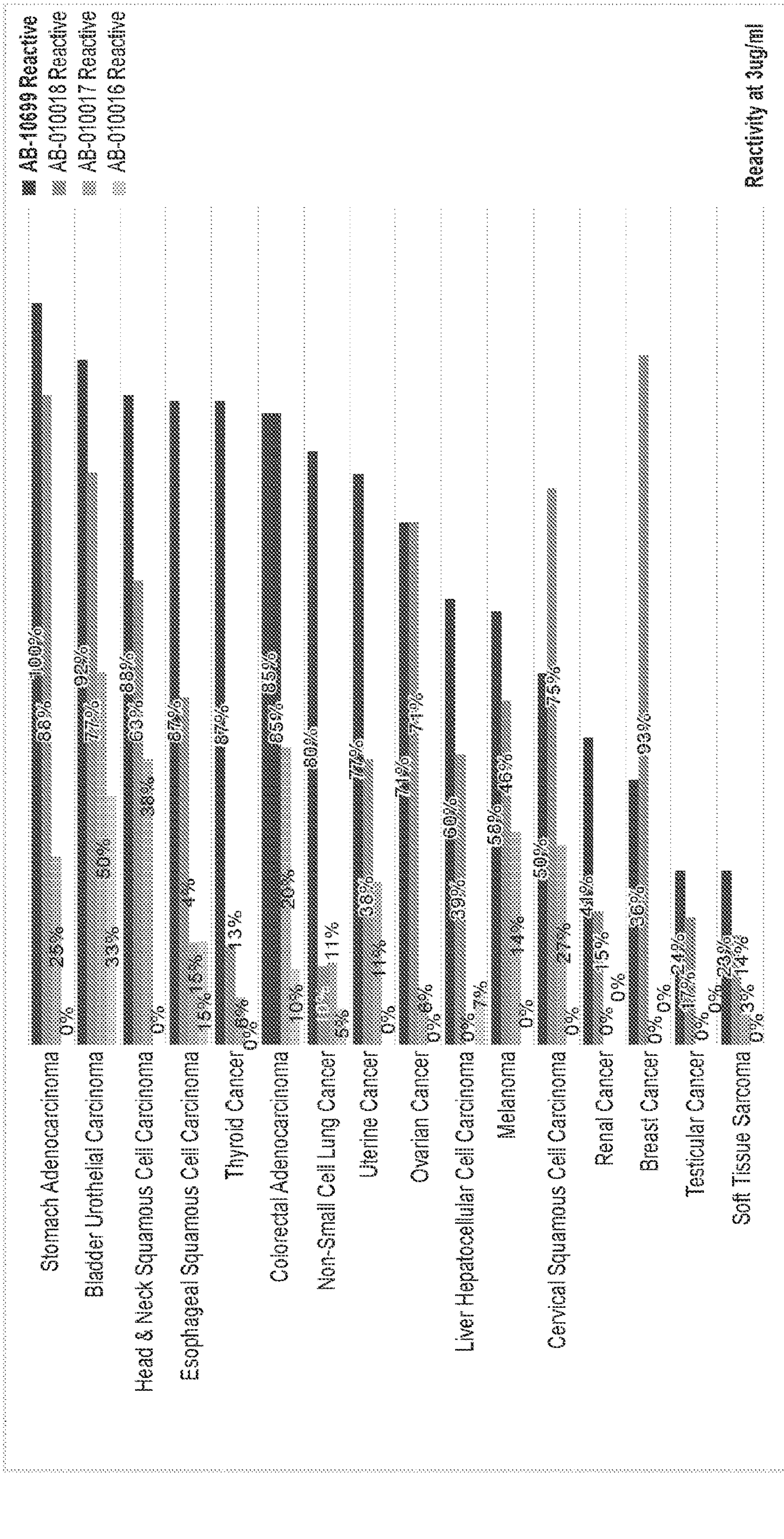
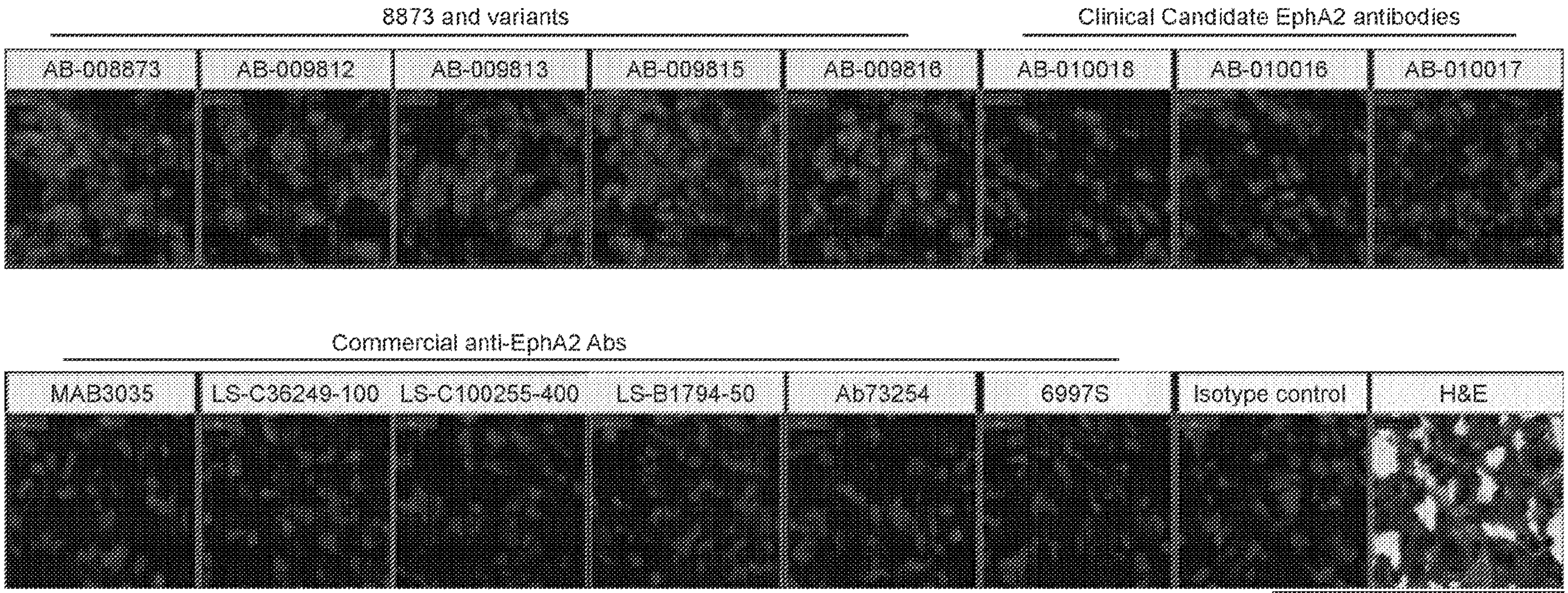
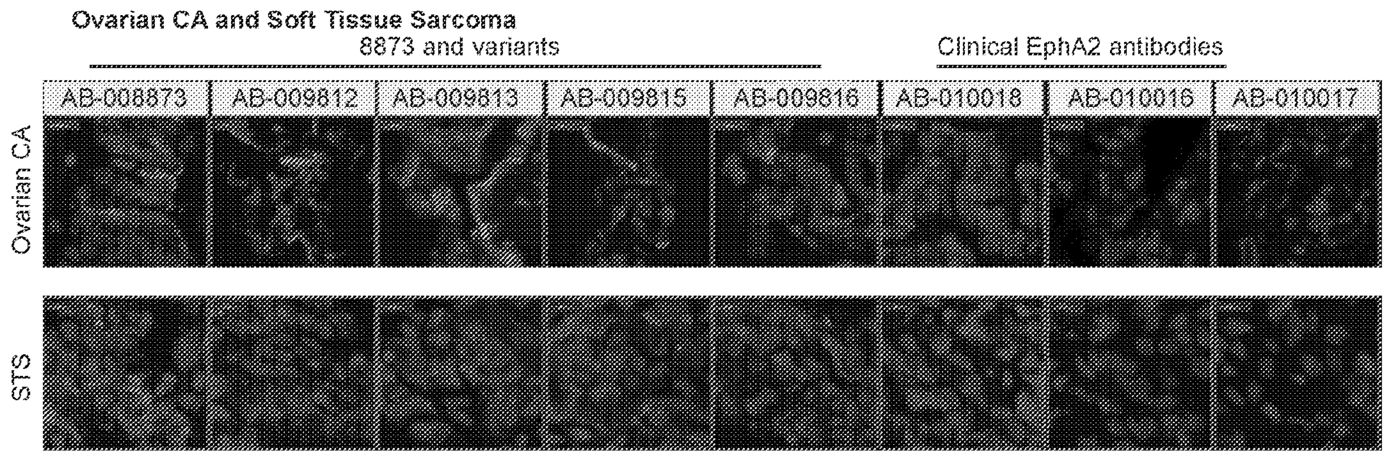

AB-10699 Reactive
AB-010018 Reactive
AB-010017 Reactive
AB-010016 Reactive
Reactivity at 3ug/ml
Stomach Adenocarcinoma
Bladder Urothelial Carcinoma
Head & Neck Squamous Cell Carcinoma
Esophageal Squamous Cell Carcinoma
Thyroid Cancer
Colorectal Adenocarcinoma
Non-Small Cell Lung Cancer
Uterine Cancer
Ovarian Cancer
Liver Hepatocellular Cell Carcinoma
Melanoma
Cervical Squamous Cell Carcinoma
Renal Cancer
Breast Cancer
Testicular Cancer
Soft Tissue Sarcoma

FIG. 24B



Antibody	GAC
AB-008873	2
AB-009812	2
	2
AB-009815	2
AB-009816	2
AB-010018	1
AB-010016	1
AB-010017	1
MAB3035	1
LS-C36249-100	1
LS-C100255-400	1
LS-B1794-50	1
Ab73254	1
6997S	1

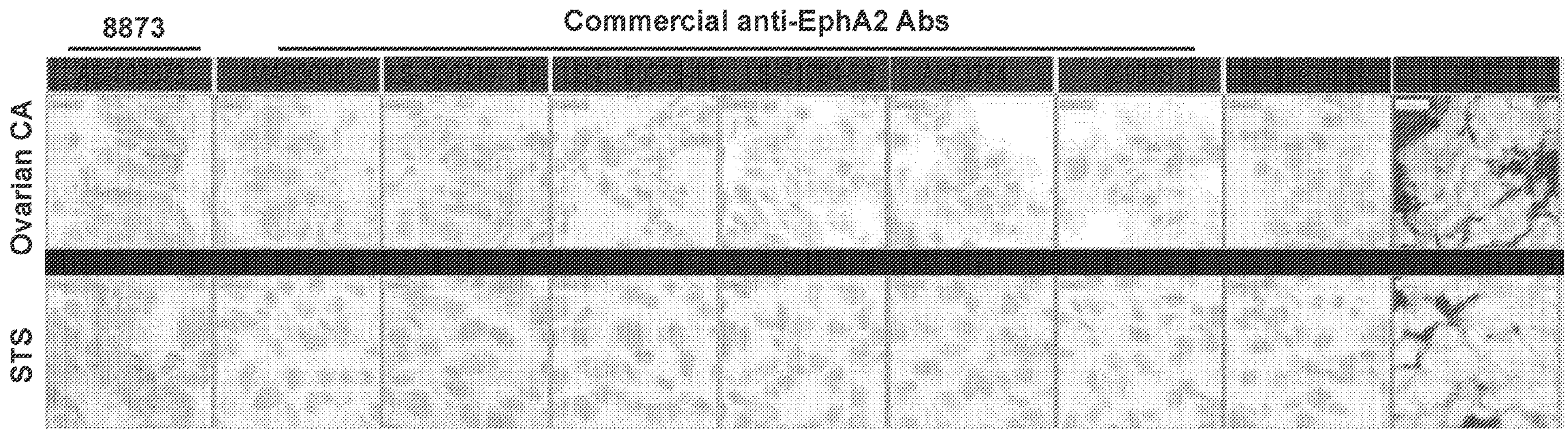
FIG. 25



Antibody	OVA	STS
AB-008873	2	2
AB-009812	2	2
AB-009813	2	2
AB-009815	2	2
AB-009816	2	2
AB-010018	2	2
AB-010016	1	1
AB-010017	1	1

FIG 26

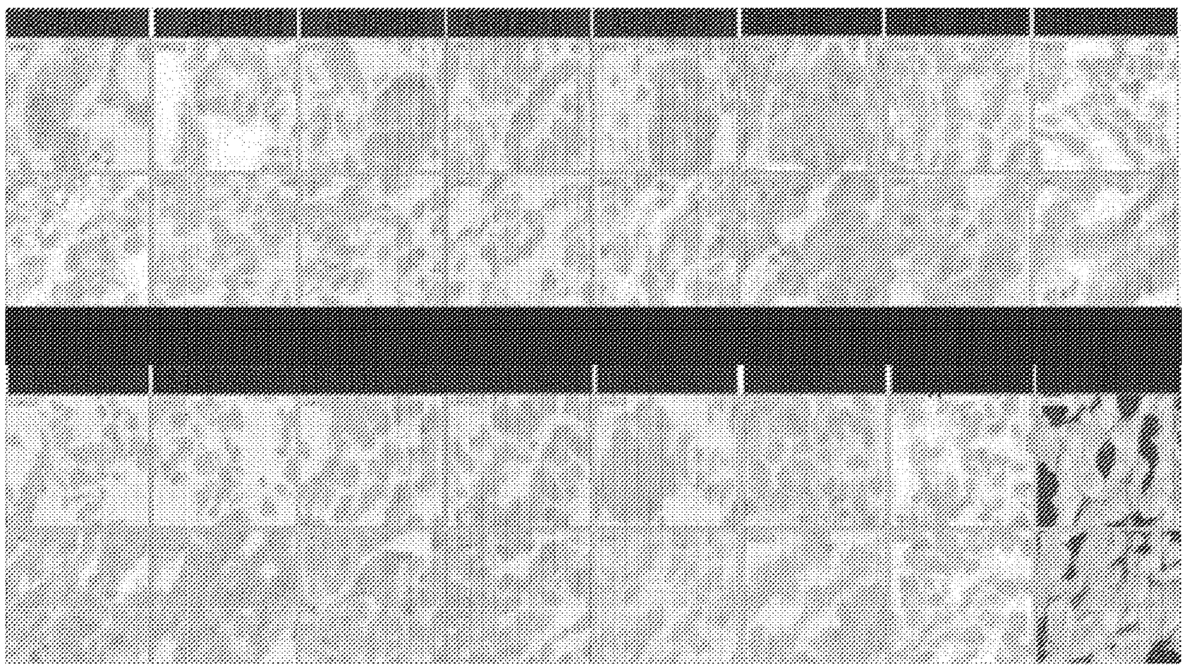
Ovarian CA and Soft Tissue Sarcoma



Antibody	OVA	STS
AB-008873	2	2
MAB3035	2	1
LS-C36249-100	2	2
LS-C100255-400	1	1
LS-B1794-50	1	1
Ab73254	1	1
6997S	1	1

- 8873 and variants show tumor selectivity in OVA and STS
- Only LS-C36249-100 commercial anti-EphA2 Ab shows weak and diffused signals in OVA and STS

FIG. 27

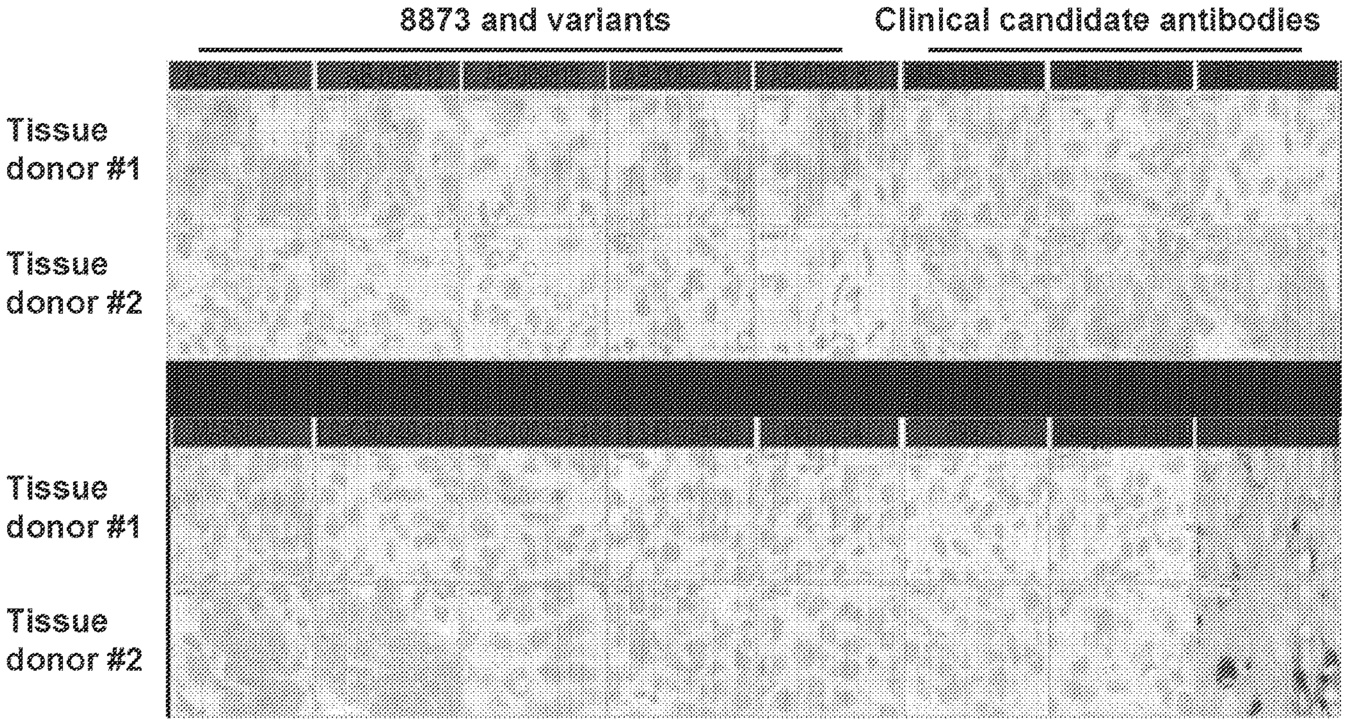


- 8873, variants, clinical candidate EphA2 Abs and commercial anti-EphA2 Abs show tumor selectivity in different tissue donors in Uterine CA
- Differences in donor tissue binding suggest possible differential epitope avilability and/or processing between donors -possibly unique epitope recognition by 8873 paratope

Antibody	Clear cell UCEC
AB-008873	2
AB-009812	1
AB-009813	2
AB-009815	2
AB-009816	2
AB-010018	2
AB-010016	2
AB-010017	2
MAB3035	2
LS-C36249-100	2
LS-C100255-400	2
LS-B1794-50	2
Ab73254	2
6997S	2

FIG. 28

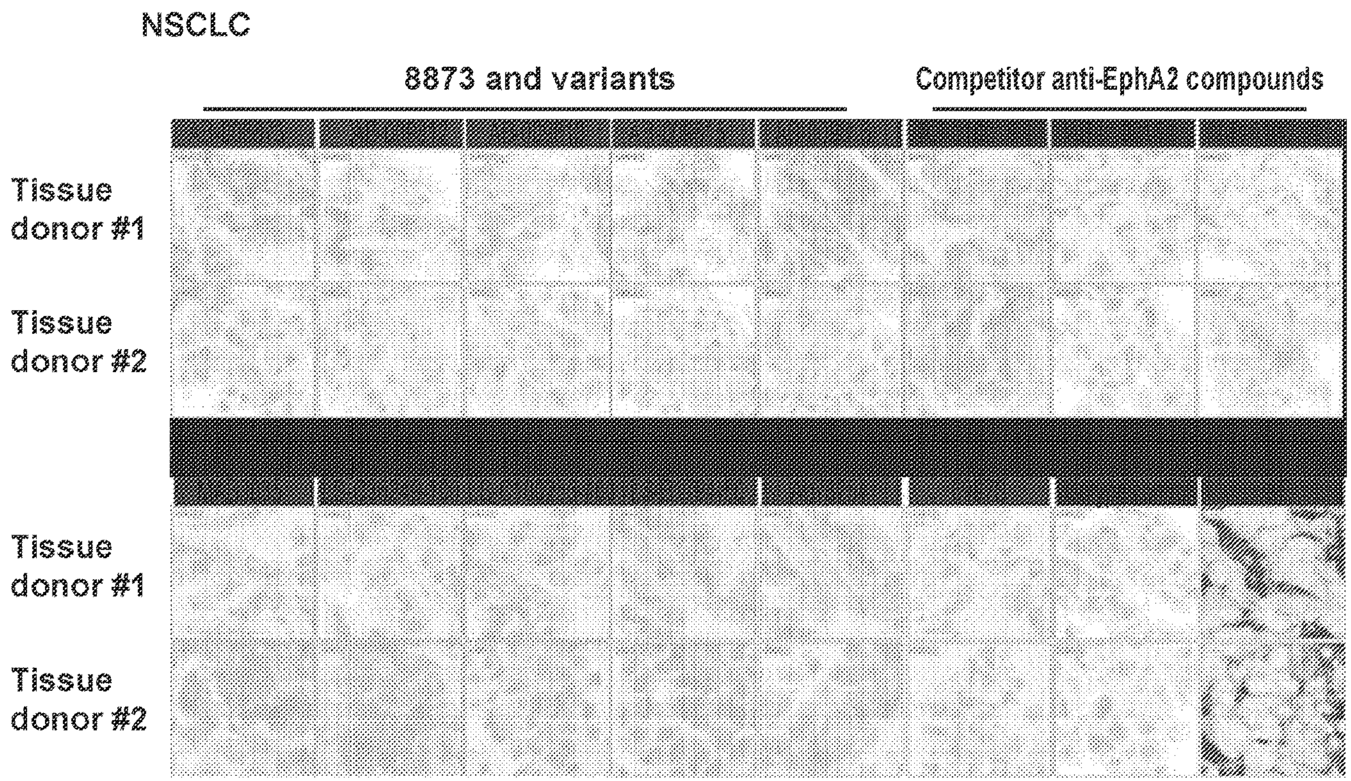
Head and neck CA



- 8873, variants, clinical candidate antibodies and commercial anti-EphA2 Abs show tumor selectivity in different tissue donors in Head and neck CA
- Differences in donor tissue binding suggest possible differential epitope avilability and/or processing between donors -possibly unique epitope recognition by 8873 paratope

Antibody	MNC
AB-008873	2
AB-009812	2
AB-009813	2
AB-009815	2
AB-009816	2
AB-010018	1
AB-010016	2
AB-010017	2
MAB3035	2
LS-C36249-100	2
LS-C100255-400	1
LS-B1794-50	1
Ab73254	1
6997S	1

FIG. 29



- 8873, variants, clinical candidate EphA2 antibodies and commercial anti-EphA2 Abs show tumor selectivity in different tissue donors in Lung Adenocarcinoma
- Differences in donor tissue binding suggest possible differential epitope avilability and/or processing between donors - possibly unique epitope recognition by 8873 paratope

Antibody	NSCLC-Adeno
AB-008873	2
AB-009812	2
AB-009813	2
AB-009815	2
AB-009816	2
AB-010018	2
AB-010016	1
AB-010017	2
MAB3035	2
LS-C36249-100	2
LS-C100255-400	1
LS-B1794-50	2
Ab73254	2
6997S	1

FIG. 30

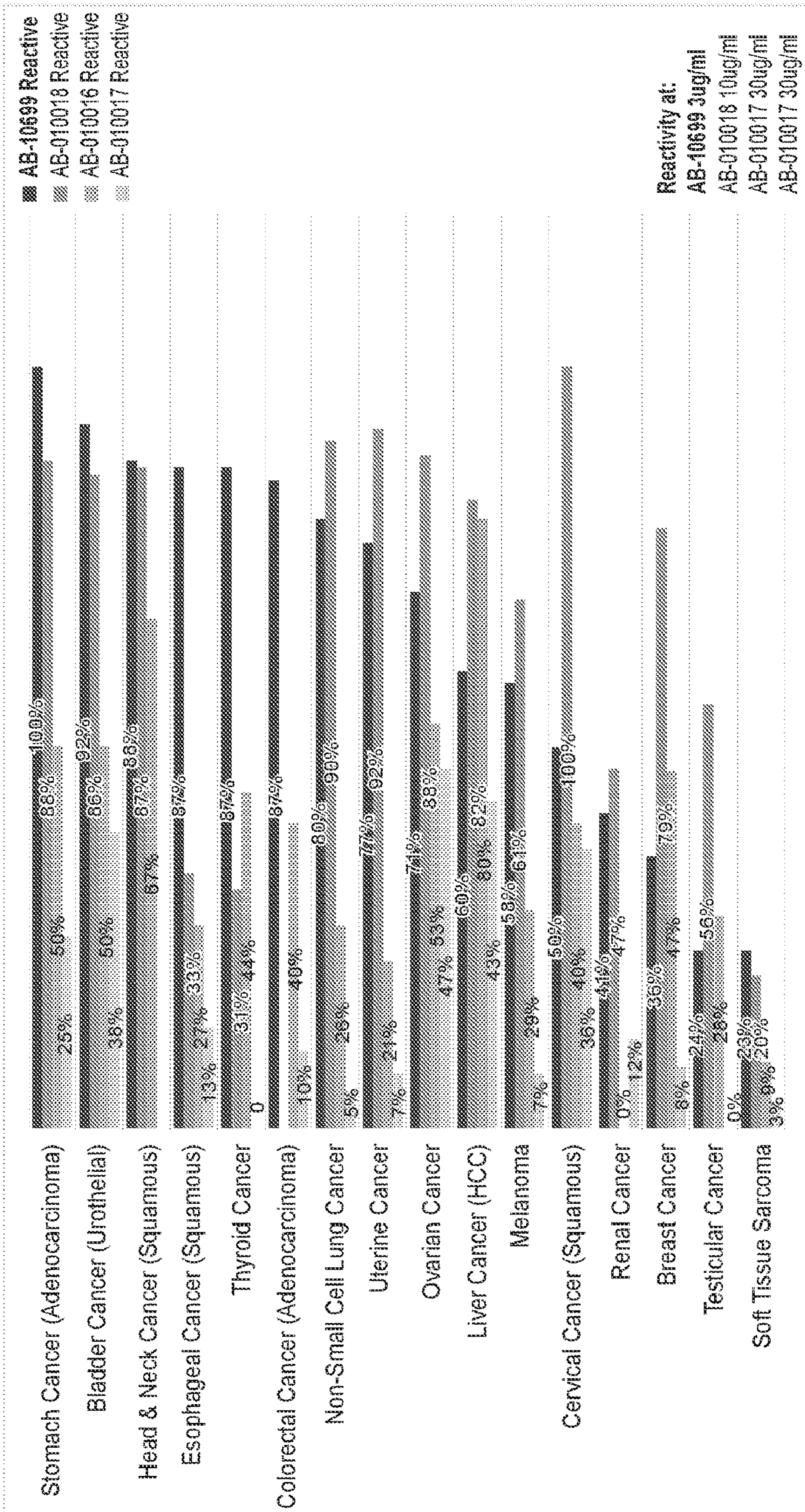

AB-10699 Reactive
AB-010018 Reactive
AB-010016 Reactive
AB-010017 Reactive
Reactivity at:
AB-10699 3ug/ml
AB-010018 10ug/ml
AB-010017 30ug/ml
AB-010017 30ug/ml
Stomach Cancer (Adenocarcinoma)
Bladder Cancer (Urothelial)
Head & Neck Cancer (Squamous)
Esophageal Cancer (Squamous)
Thyroid Cancer
Colorectal Cancer (Adenocarcinoma)
Non-Small Cell Lung Cancer
Uterine Cancer
Ovarian Cancer
Liver Cancer (HCC)
Melanoma
Cervical Cancer (Squamous)
Renal Cancer
Breast Cancer
Testicular Cancer
Soft Tissue Sarcoma

FIG. 31

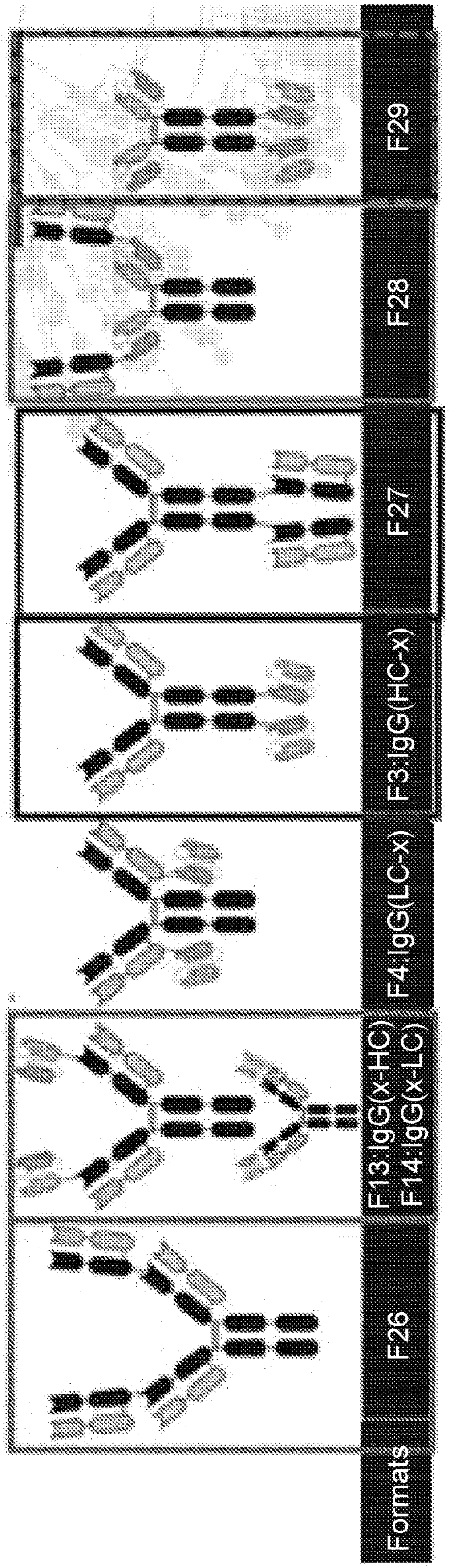
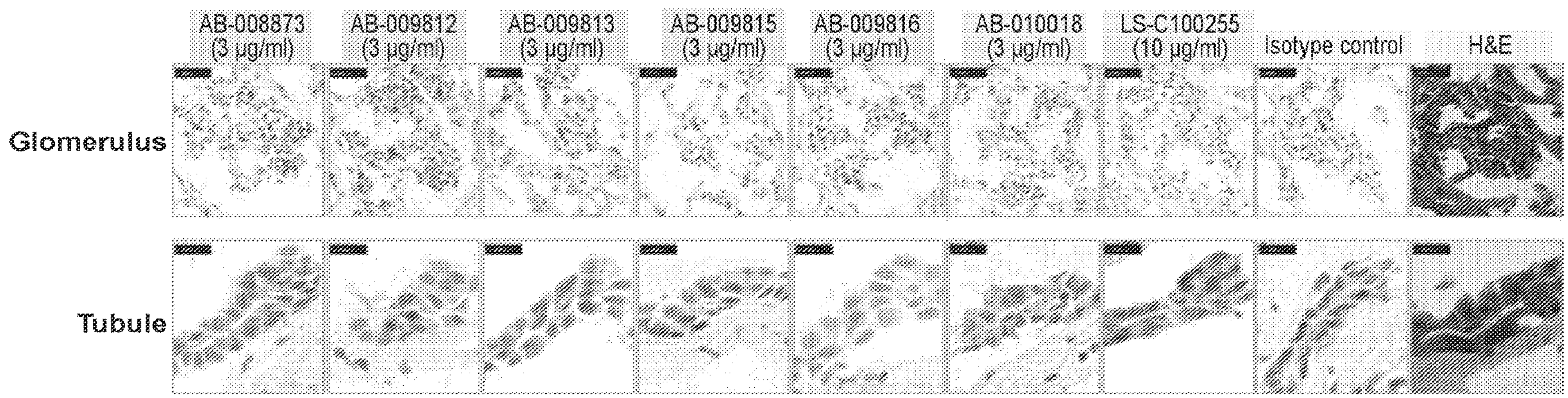

Formats
F26
F13:IgG(x-HC)
F14:IgG(x-LC)
F4:IgG(LC-x)
F3:IgG(HC-x)
F27
F28
F29

FIG. 32

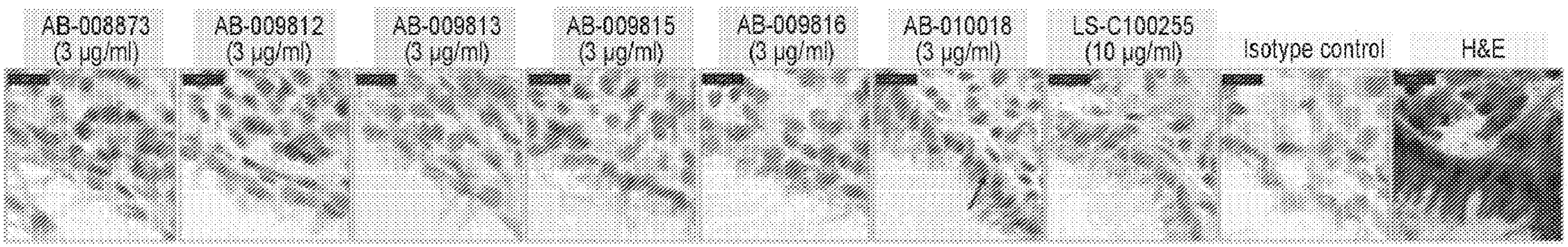
Normal Kidney



- DS-8895 and anti-EphA2Ab show more signals in tubules

FIG. 33

Normal Stomach



- DS-8895 shows more signals in epithelium

FIG. 34

Comparison of AB-008873 ADC activity with engineered variant AB-009815 (H76PT Mutation)

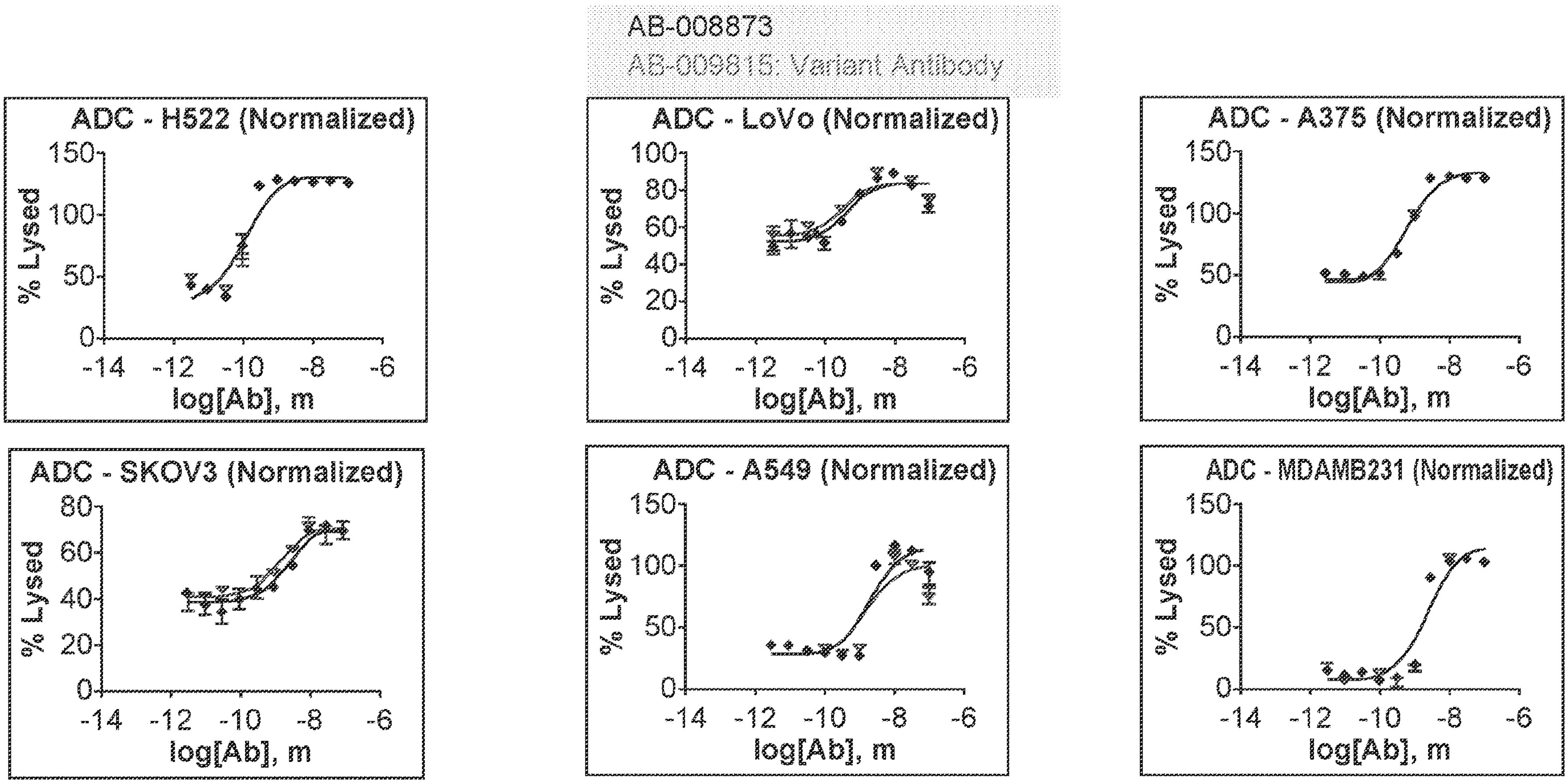


FIG. 35

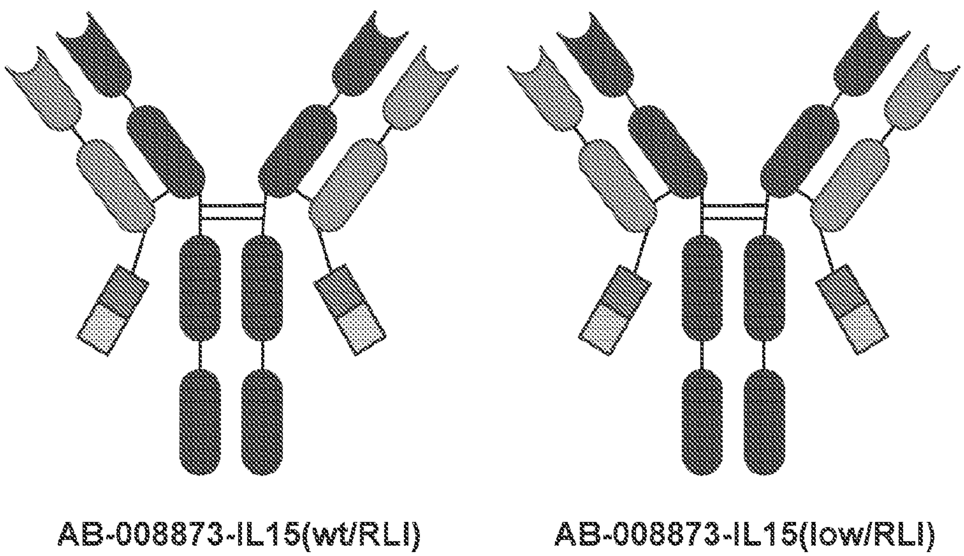


FIG. 36

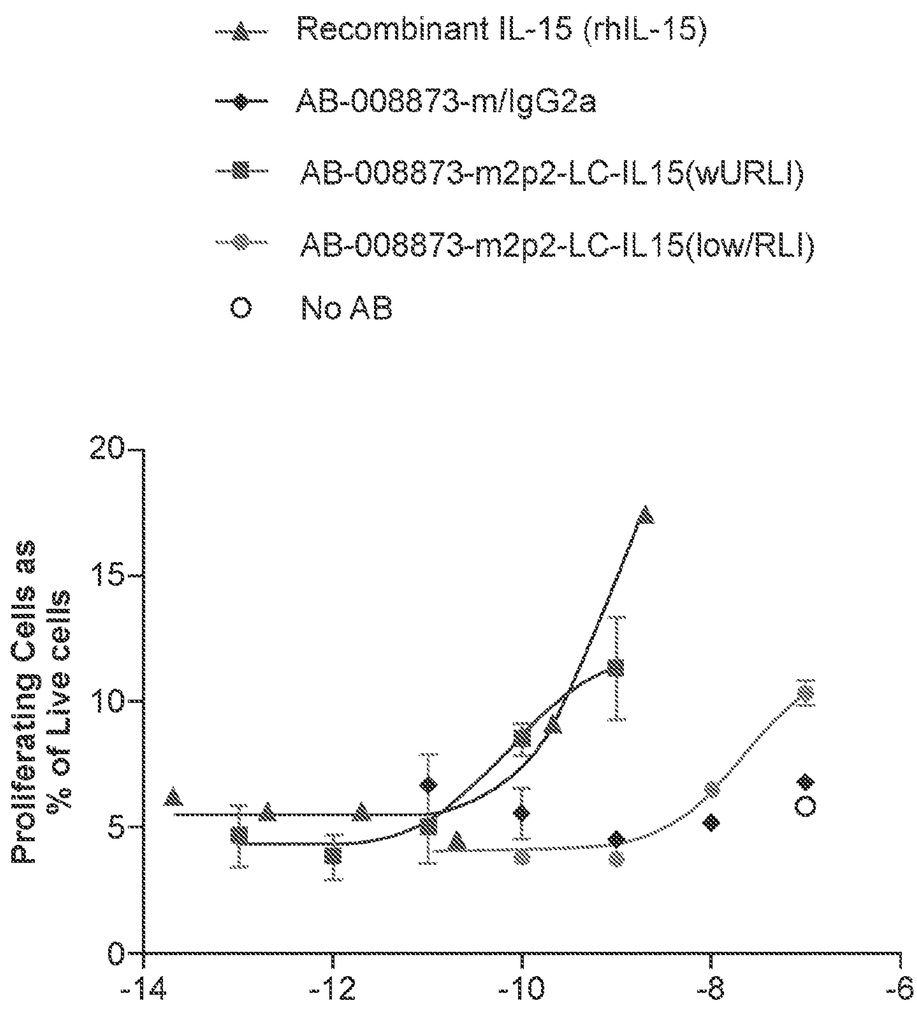

Recombinant IL-15 (rhIL-15)
AB-008873-m/IgG2a
AB-008873-m2p2-LC-IL15(wURLI)
AB-008873-m2p2-LC-IL15(low/RLI)
No AB
Proliferating Cells as % of Live cells
20
15
10
5
0
-14
-12
-10
-8
-6
log[Ab], M

FIG. 37

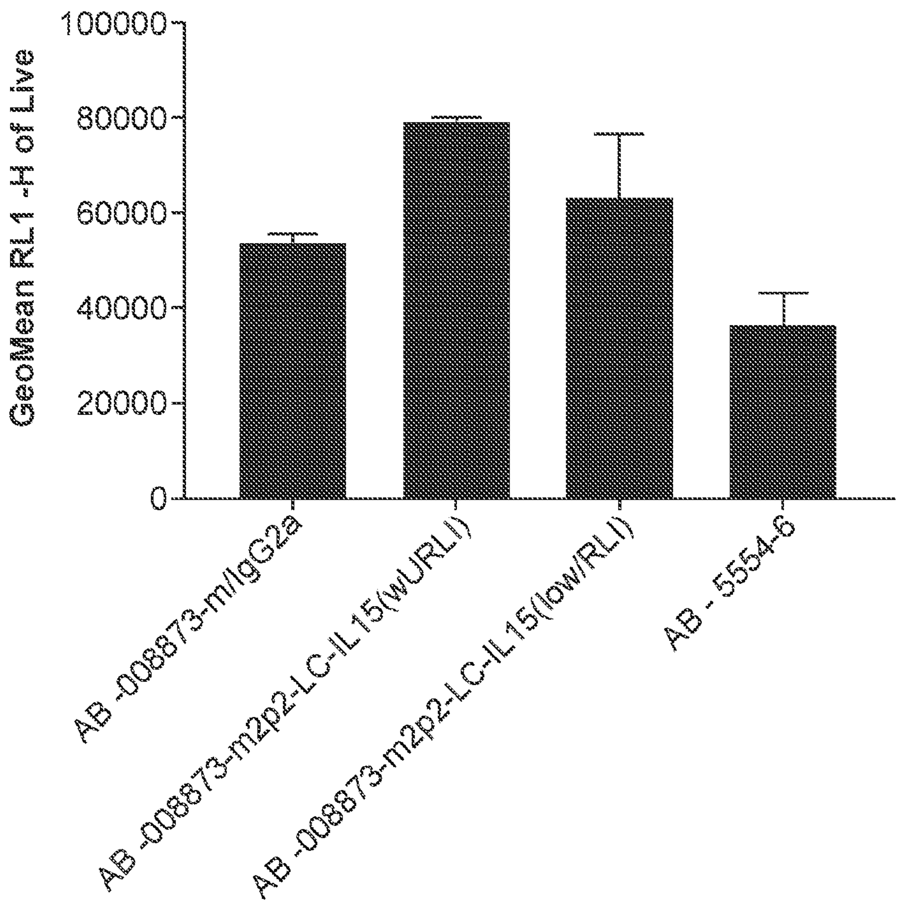

GeoMean RL1 -H of Live
100000
80000
60000
40000
20000
0
AB -008873-m/IgG2a
AB -008873-m2p2-LC-IL15(wURLI)
AB -008873-m2p2-LC-IL15(low/RLI)
AB - 5554-6

FIG. 38

VH

IMGT																			
Kabat																			
Kabat #s	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
AB-008873	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009805	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009806	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009807	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009808	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009812	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009813	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009814	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009815	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009816	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S
AB-009817	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S

FIG. 39A

							IMGT																			
											KABAT															
20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	L	S	D	Y	H	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	G	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
L	T	C	A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E

FIG. 39A Contd...

IMGT					IMGT																				
Kabat											KABAT														
Kabat #s	47	48	49	50	51	52	52A	52B	52C	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68
AB-008873	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T
AB-009805	W	I	G	E	I	N	H	S	G	S	T	N	Y	N	-	-	-	P	S	L	K	S	R	V	T
AB-009806	W	I	G	E	I	N	H	S	G	S	T	N	Y	N	-	-	-	P	S	L	K	S	R	V	T
AB-009807	W	I	G	E	V	N	H	S	G	S	T	S	Y	N	-	-	-	P	S	L	K	S	R	V	T
AB-009808	W	I	G	E	I	N	H	S	G	S	T	N	Y	N	-	-	-	P	S	L	K	S	R	V	T
AB-009812	W	I	G	E	V	N	H	S	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T
AB-009813	W	I	G	E	V	N	H	A	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T
AB-009814	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	-	-	-	P	S	L	K	S	R	V	T
AB-009815	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T
AB-009816	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T
AB-009817	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S	R	V	T

FIG. 39A Contd...

69	70	71	72	73	74	75	76	77	78	79	80	81	82	82A	82B	82C	83	84	85	86	87	88
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	S	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	T	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	A	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A

FIG. 39A Contd...

IMGT					IMGT														
Kabat							KABAT												
Kabat #s	89	90	91	92	93	94	95	96	97	98	99	100	100A	100B	100C	100D	100E	100F	100G
AB-008873	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009805	V	Y	Y	C	A	K	P	F	R	P	H	C	T	N	G	V	C	Y	S
AB-009806	V	Y	Y	C	A	K	P	F	R	P	H	C	T	N	G	V	C	Y	S
AB-009807	V	Y	Y	C	A	K	P	F	R	P	H	C	T	N	G	V	C	Y	S
AB-009808	V	Y	Y	C	A	K	P	F	R	P	H	C	T	N	G	V	C	Y	S
AB-009812	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009813	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009814	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009815	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009816	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S
AB-009817	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	A	V	C	Y	S

FIG. 39A Contd...

100H	100I	100J	100K	101	102	103	104	105	106	107	108	109	110	111	112	113
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	A	T	V	S	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	S	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	S	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	S	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
G	D	A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S

FIG. 39A Contd...

VL

IMGT																										IMGT	
Kabat																											
Kabat #s	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
AB-008873	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009805	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009806	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009807	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009808	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009812	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009813	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009814	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009815	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009816	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I
AB-009817	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N	I

FIG. 39B

KABAT																			
29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	T	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V

FIG. 39B Contd...

IMGT		IMGT																								
Kabat		KABAT																								
Kabat #s	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
AB-008873	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009805	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009806	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009807	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009808	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009812	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009813	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009814	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009815	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009816	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T
AB-009817	Y	D	D	S	D	R	P	S	G	I	P	E	Q	F	S	G	S	N	S	G	N	T	A	T	L	T

FIG. 39B Contd...

														IMGT																				
														KABAT																				
75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	95A	95H	96	97	98	99	100	101	102	103	104	105	106	107
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
I	S	R	V	E	A	G	D	E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L

FIG. 39B Contd...

VH CDRs

IMGT																						
Kabat																						
Kabat #s	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
AB-008873	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-009815	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-010148	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-010357	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-010361	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-010363	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C
AB-010699	Q	V	Q	L	Q	Q	W	G	A	G	L	L	K	P	S	E	T	L	S	L	T	C

FIG. 39C

			IMGT																				
								KABAT															
23	24	25	26	27	28	29	30	31	32	33	34	35	36	33	38	39	40	41	42	43	44	45	46
A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	D	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	G	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	G	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E
A	V	Y	G	G	S	F	S	G	Y	Y	W	S	W	I	R	Q	P	P	G	K	G	L	E

FIG. 39C Contd...

IMGT					IMGT																
Kabat				KABAT																	
Kabat #s	47	48	49	50	51	52	52A	52B	52C	53	54	55	56	57	58	59	60	61	62	63	64	65
AB-008873	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-009815	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-010148	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-010357	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-010361	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-010363	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S
AB-010699	W	I	G	E	V	N	H	R	G	S	I	N	Y	N	N	Y	N	P	S	L	K	S

FIG. 39C Contd...

66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	82A	82B	82C	83	84	85	86	87	88
R	V	T	I	S	V	D	P	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	T	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A
R	V	T	I	S	V	D	T	S	K	N	Q	F	S	L	K	L	S	S	V	T	A	A	D	T	A

FIG. 39C Contd...

IMGT					IMGT																
Kabat							KABAT														
Kabat #s	89	90	91	92	93	94	95	96	97	98	99	100	100A	100B	100C	100D	100E	100F	100G	100H	100I
AB-008873	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-009815	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-010148	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-010357	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-010361	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-010363	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D
AB-010699	V	Y	Y	C	A	K	P	L	R	P	H	C	T	N	G	V	C	Y	S	G	D

FIG. 39C Contd...

100J	100K	101	102	103	104	105	106	107	108	109	110	111	112	113
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S
A	F	D	I	W	G	Q	G	T	M	V	T	V	A	S

FIG. 39C Contd...

VL-CDRs

IMGT																											
Kabat																											
Kabat #s	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
AB-008873	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-009815	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-010148	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-010357	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-010361	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-010363	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N
AB-010699	S	Y	V	L	T	Q	P	P	S	-	V	S	V	A	P	G	Q	T	A	R	I	T	C	G	G	N	N

FIG. 39D

IMGT																				
KABAT																				
28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V
I	G	S	K	N	V	H	W	Y	Q	Q	K	P	G	Q	A	P	V	L	V	V

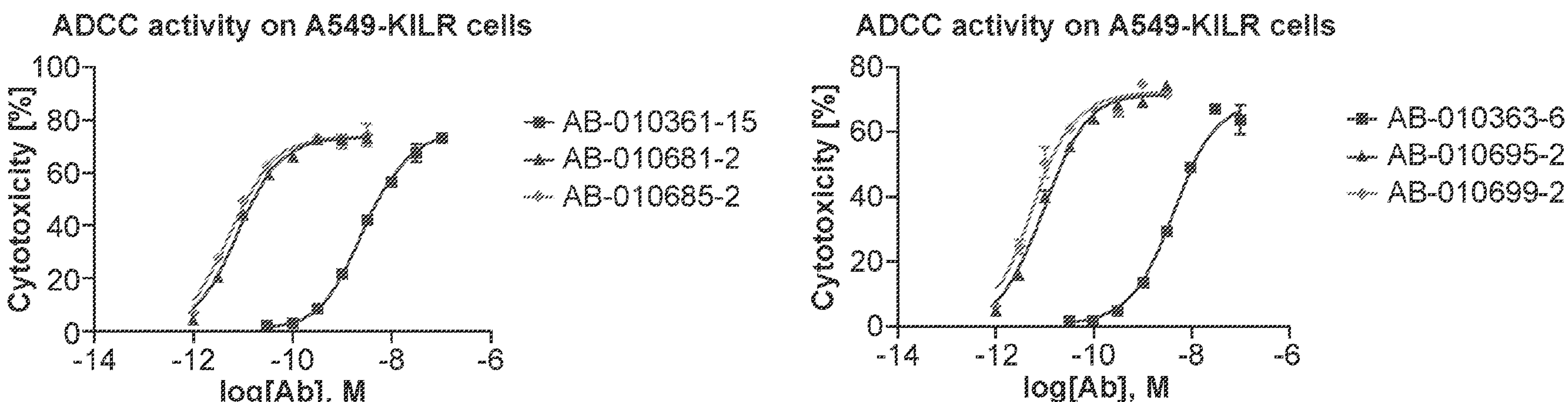
FIG. 39D Contd...

IMGT		IMGT																																
Kabat					KABAT																													
Kabat #s	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82
AB-008873	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-009815	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-010148	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-010357	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-010361	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-010363	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D
AB-010699	Y	D	D	S	D	R	P	S	G	I	P	I	Q	F	S	G	S	N	S	G	M	T	A	T	L	T	I	S	R	V	E	A	G	D

FIG. 39D Contd...

						IMGT																				
												KABAT														
83	84	85	86	87	88	89	90	91	92	93	94	95	95A	95B	96	97	98	99	100	101	102	103	104	105	106	107
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	L	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	V	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	V	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	S	S	S	D	H	V	V	F	G	G	G	T	K	L	T	V	L
E	A	D	Y	Y	C	Q	V	W	D	H	S	S	D	H	V	V	F	G	G	G	T	K	L	T	V	L

FIG. 39D Contd...



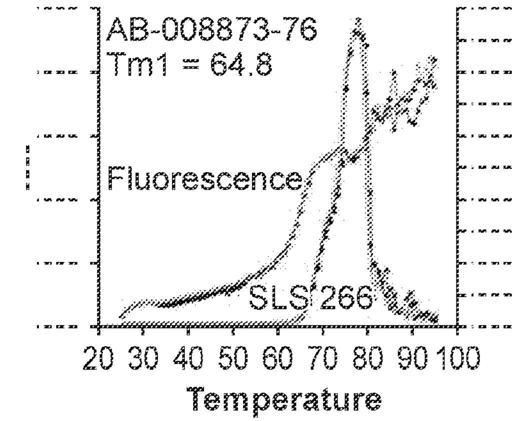
	EC50
AB-010361-15	2.625e-009
AB-010681-2	7.901e-012
AB-010685-2	5.689e-012

	EC50
AB-010363-6	4.246e-009
AB-010695-2	9.614e-012
AB-010699-2	5.786e-012

FIG. 40

AB-008873:
Tm1 = 64.8

Mutations	10357 Average Tm1 (°C)	10361 Average Tm1 (°C)	10363 Average Tm1 (°C)
Parent	64.8	64.7	68.0
H107TI_L23GR	66.7	65.5	66.5
L23GR_L29SY	64.7	65.5	66.4
L23GR_L30KM	64.5	64.7	67.0
L23GR_L92SH	64.4	64.1	65.8
H107TI_L29SY	64.3	63.6	65.8
L29SY_L30KM	64.5	63.9	65.7
L29SY_L92SH	64.0	64.4	65.9
H107TI_L30KM	64.5	64.4	66.8
L30KM_L92SH	64.5	64.2	66.5
H107TI_L92SH	65.0	64.4	66.9
H107TI_L29SY_L92SH	64.9	64.2	66.4
L93SR	64.7	65.2	66.8
L93SE	65.3	64.9	67.7
L29SY_L93SE	64.9	64.5	66.7



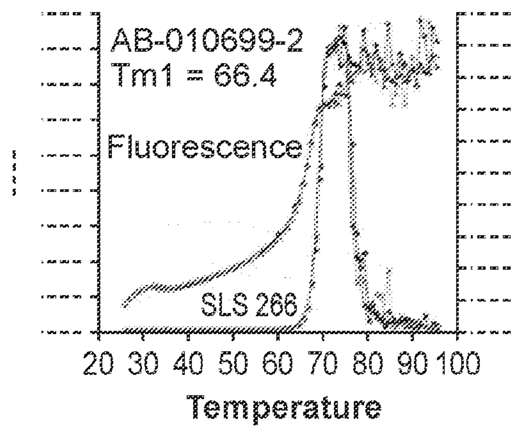
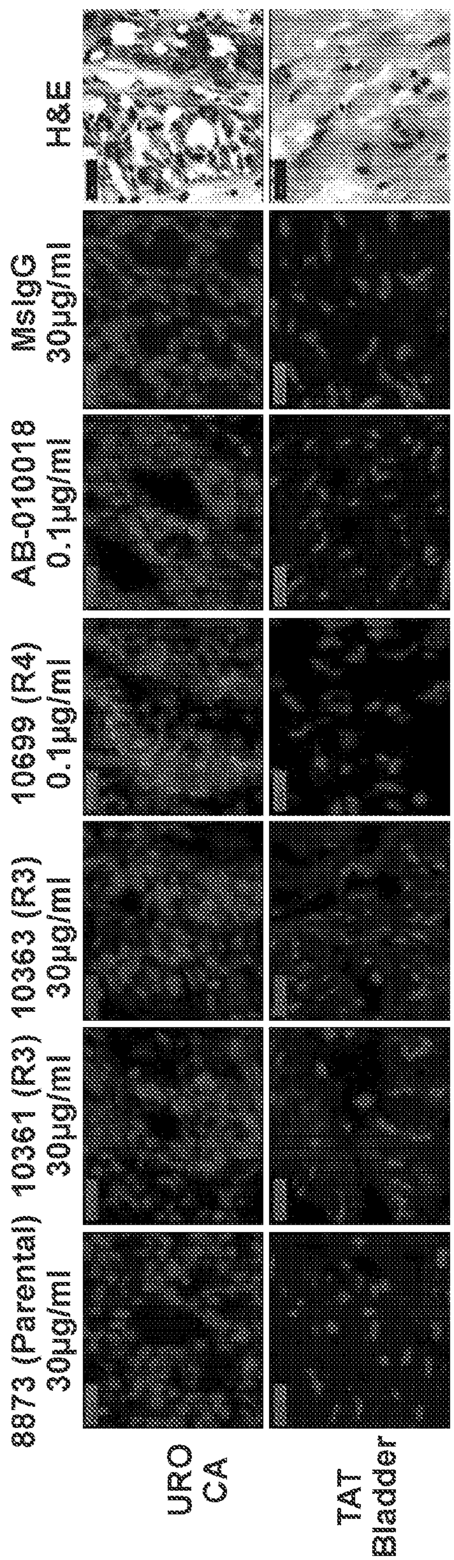
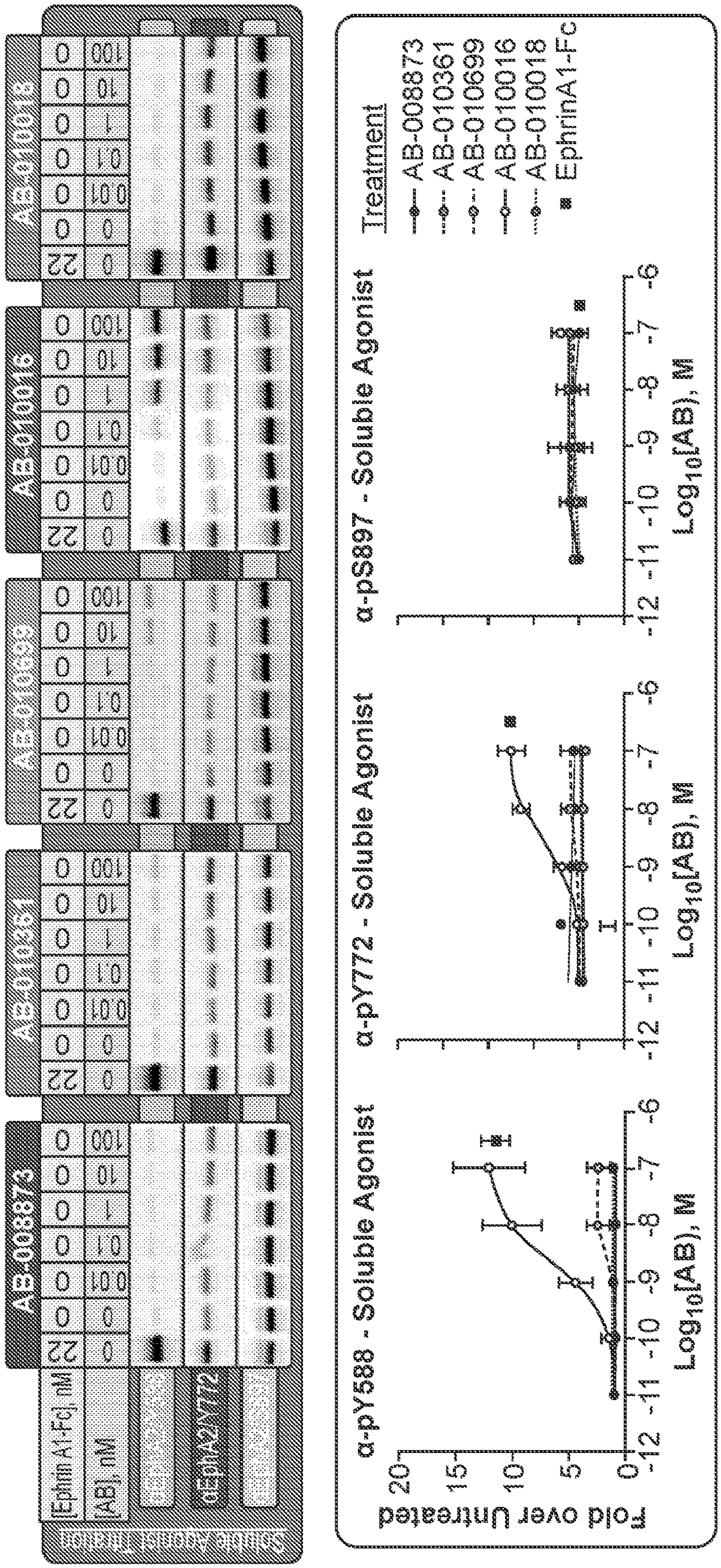


FIG. 41



H&E
MsIgG 30µg/ml
AB-010018 0.1µg/ml
10699 (R4) 0.1µg/ml
10363 (R3) 30µg/ml
10361 (R3) 30µg/ml
8873 (Parental) 30µg/ml
URO CA
TAT Bladder

FIG. 42



Soluble Agonist Titration
[Ephrin A1-Fc], nM
[AB], nM
AB-008873
AB-010361
AB-010699
AB-010016
AB-010018
α-pY588 - Soluble Agonist
α-pY772 - Soluble Agonist
α-pS897 - Soluble Agonist
Fold over Untreated
Log10[AB], M
Treatment
AB-008873
AB-010361
AB-010699
AB-010016
AB-010018
EphrinA1-Fc

FIG. 43A

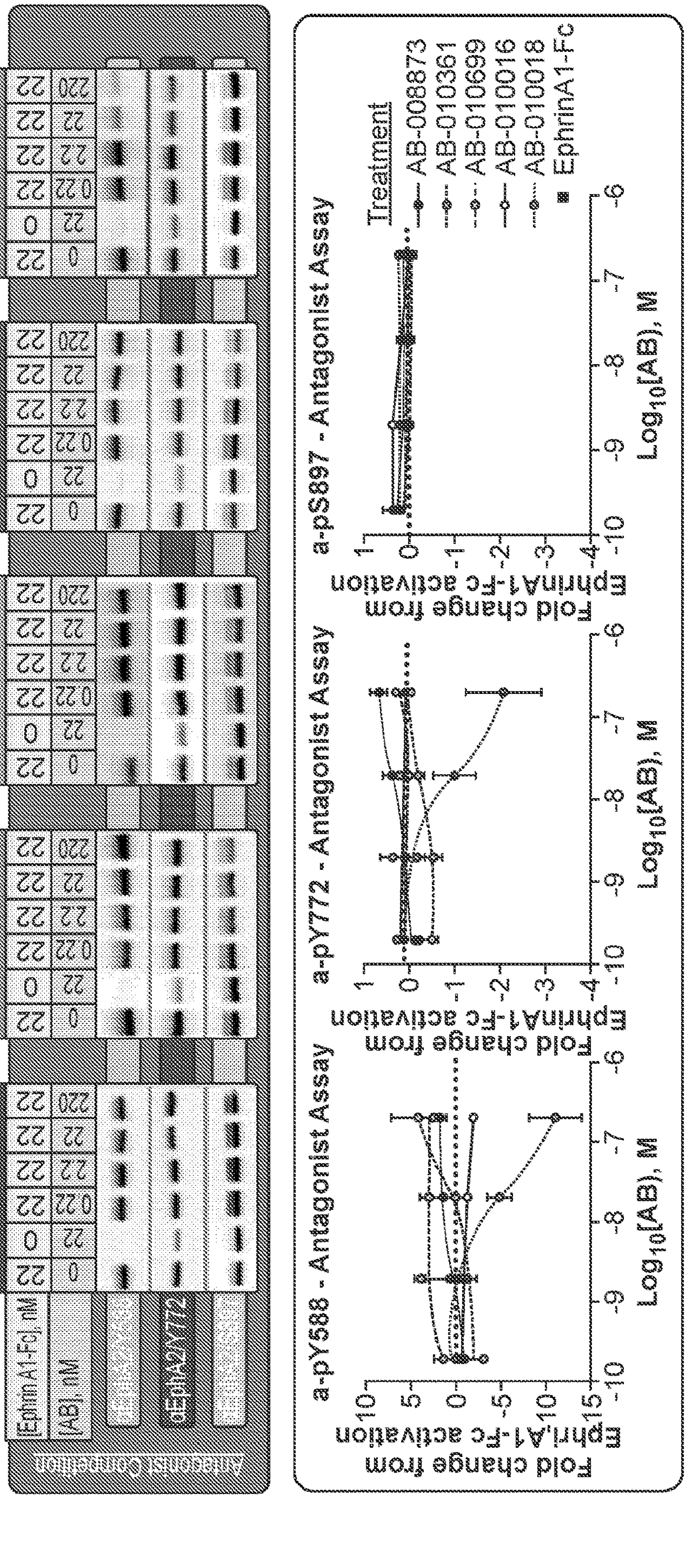

Antagonist Competition
[EphrinA1-Fc], nM
[AB], nM
dEphA2Y772
AB-008873
AB-010361
AB-010699
AB-010016
AB-010018
a-pY588 - Antagonist Assay
a-pY772 - Antagonist Assay
a-pS897 - Antagonist Assay
Fold change from EphrinA1-Fc activation
Log10[AB], M
Treatment
AB-008873
AB-010361
AB-010699
AB-010016
AB-010018
EphrinA1-Fc

FIG. 43B

Fine Epitope Mapping by Yeast Display

Select for mutations that are well-displayed but negatively affect binding

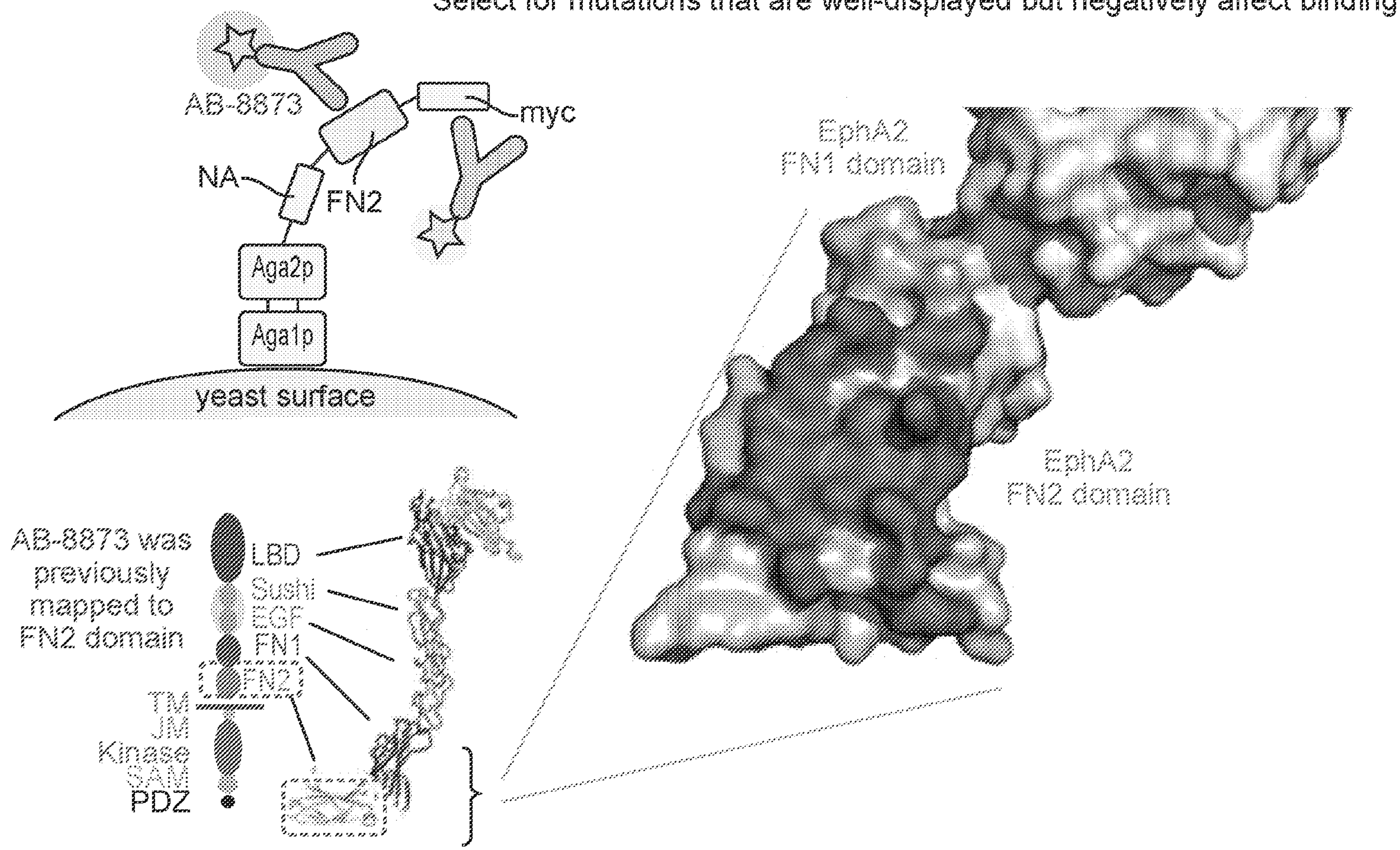
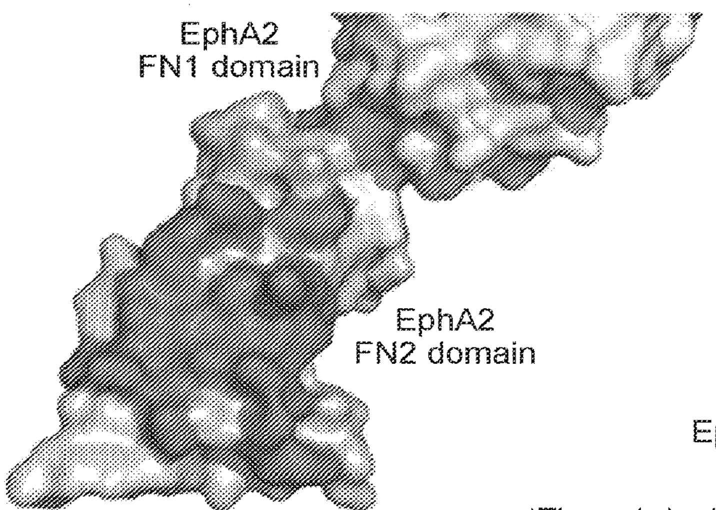


FIG. 44

Epitope is Conformational, and Not Conserved in the Eph Family



EphA2	TEPPK...VRLEGRST	KKGDS	TYLVQVQ	AGSKVHEFQIL
EphA1	AESLSGLSLRLVKKEP	LNQCE	TYIVRVR	PFSPDHEFRIS
EphA3	AAPSPVLTIKKDRTSR	KQEQE	IYVFQIR	TNSRKFEFEIS
EphA4	AAPSSIALVQAKEVTR	KDQNE	SYVFHVR	DFSEPLEVTIN
EphA5	AAPSPVTNVKKGKIAK	K.DQE	VYVFQIR	VFSRRFEFEIT
EphA6	DAPSLIGMMRKDWASQ	KEHEC	KYVFHIR	GYSQKFEFEIG
EphA7	AAPSQVSCVMKERVLQ	KDQRD	VYVFQIR	NYSPRLDVAIL
EphA8	AAPSQVVVIRQERAGQ	KDKEM	RYVFQVR	RFSQAMEVEIG
EphA10	GAPWEEGEIRRDRVEP	KGQSE	RYVFQIR	SFNPSIEVQIL
EphB1	AAPSTVPIMHQVSATM	KEHNE	VYVVQVR	KFSGKMCFQIL
EphB2	AAPSAVSIMHQVSRTV	KELSE	IYVFQVR	RYSGKMYFQIN
EphB3	AAPSEVPTLRLHSSSG	K..SF	RYVVQVR	QYSRPARFFIT
EphB4	EVPPAVSDIRVTRSSP	KGAEG	SYLVQVR	PFGQEHKSQIQ
EphB6	EVPSAVPVVHQVSRAS	QAEDE	IYGFQVR	PYGGKVYFQIL
EphA2 #	439------------447	476-447	504-506	519-----525

Adapted from doi: 10.1038/nsmb.1782

*The red shading of sequence indicates degree of conservation as calculated with ESPRIPT

FIG. 45

Epitope is conserved across tox species (mouse, rat, cyno)

All FN2 Divergence	446	455	460	461	479	485	488	496	500	529	533	534
EphA2_FL(1-976)	G	S	P	P	S	R	E	D	D	P	G	N
cyEphA2_FL(1-976)	G	S	P	P	S	R	E	D	D	P	G	S
muEphA2_FL(1-977)	D	T	V	S	A	R	E	D	D	T	A	N
raEphA2_FL(1-977)	D	A	V	P	A	H	D	G	G	T	A	T

Epitope	439	441	443	444	447	476	477	504	506	519	520	521	522	523	524	525
EphA2_FL(1-976)	P	K	R	L	R	K	G	L	Q	S	K	V	H	E	F	Q
muEphA2_FL(1-977)	P	K	R	L	R	K	G	L	Q	S	K	V	H	E	F	Q
cyEphA2_FL(1 -976)	P	K	R	L	R	K	G	L	Q	S	K	V	H	E	F	Q
raEphA2_FL(1-977)	P	K	R	L	R	K	G	L	Q	S	K	V	H	E	F	Q

FIG. 46

Co-Crystal Demonstrates HCDR3 Re-Arrangement vs Antibody Alone

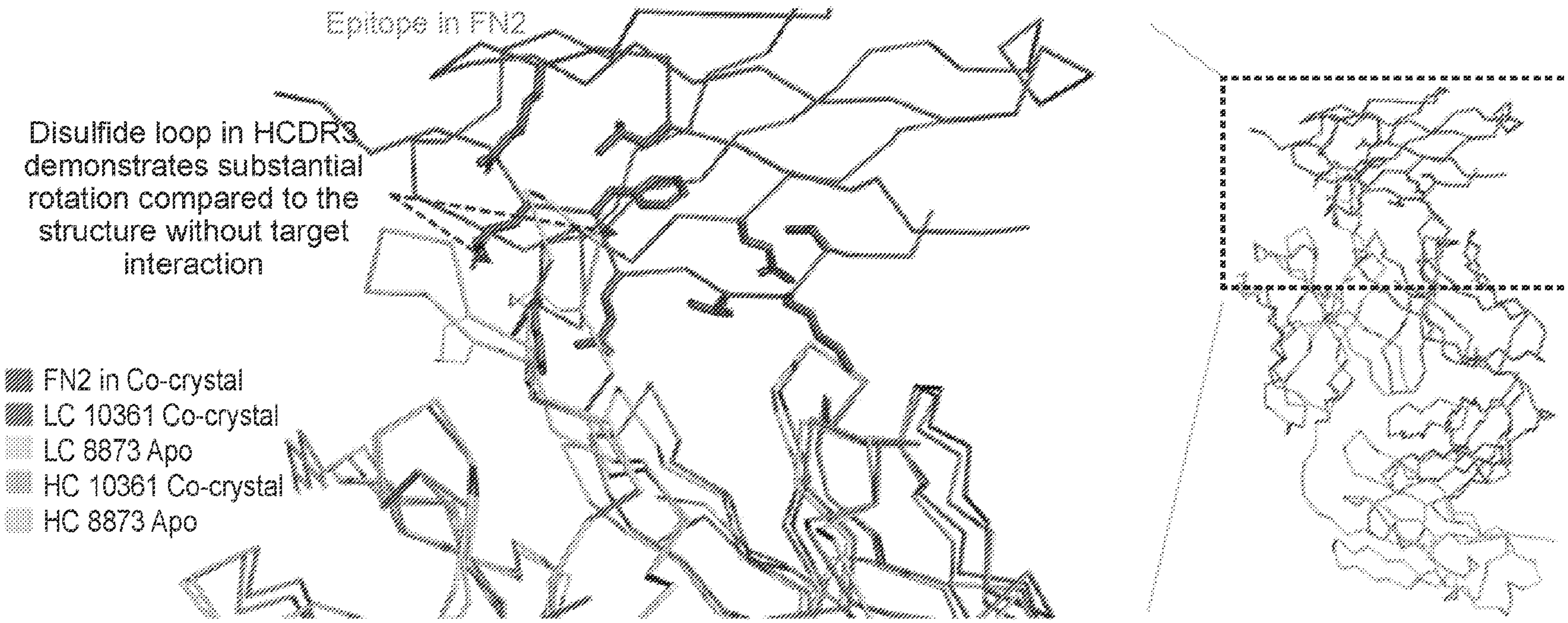


FIG. 47

EphA2-IL15

In Vivo TGI/R with *Unoptimized* 8873 Fv

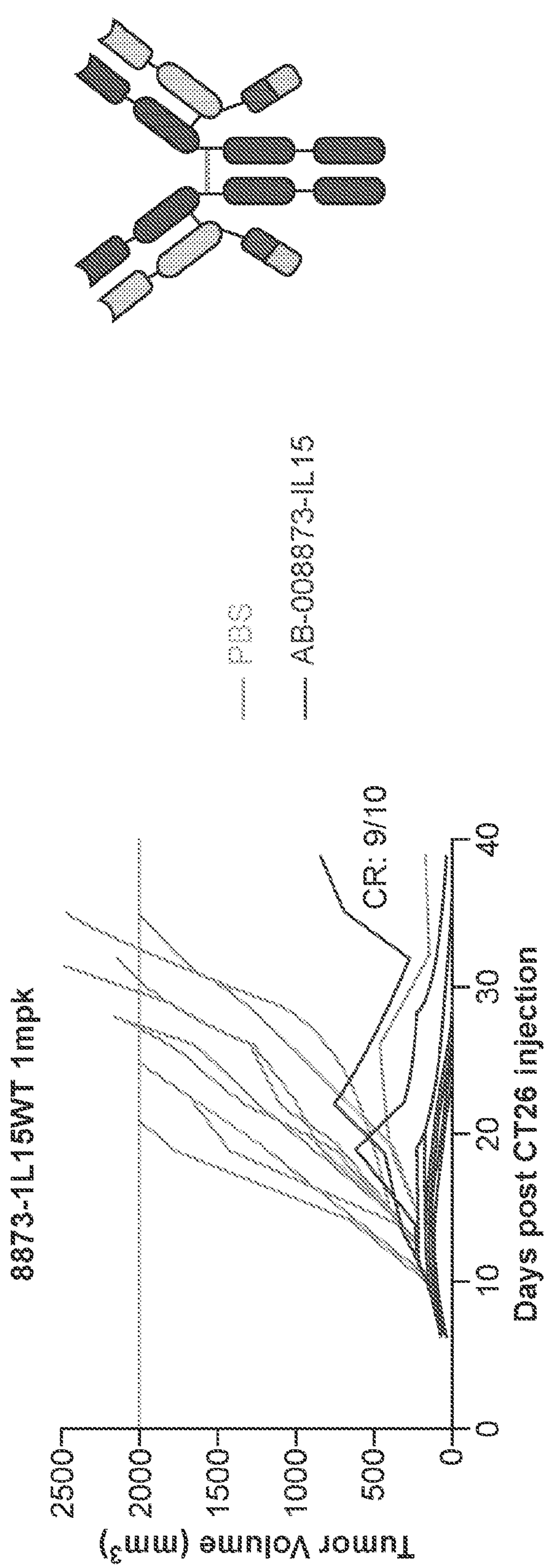
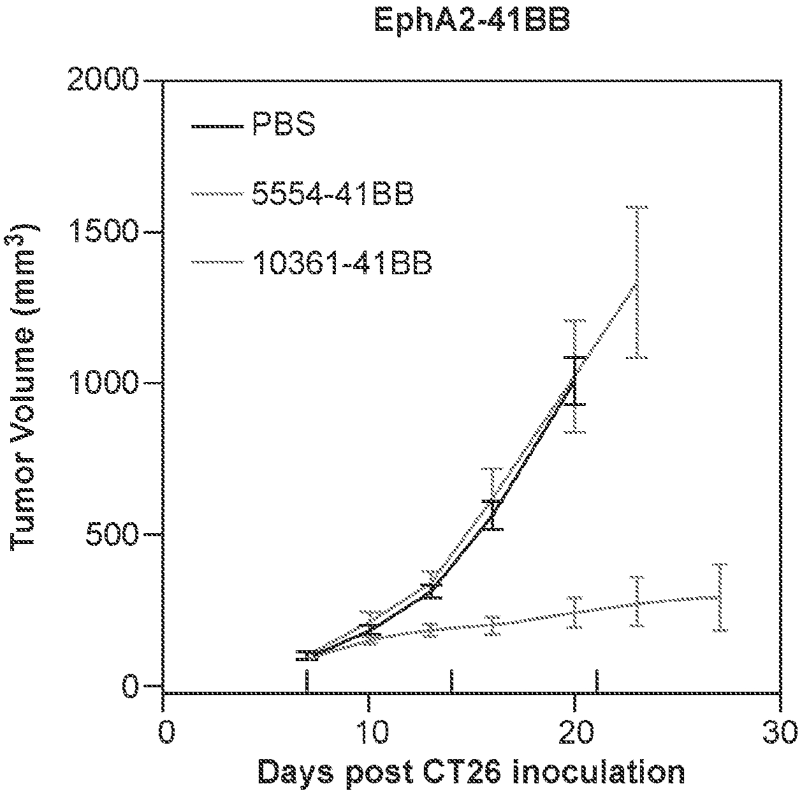
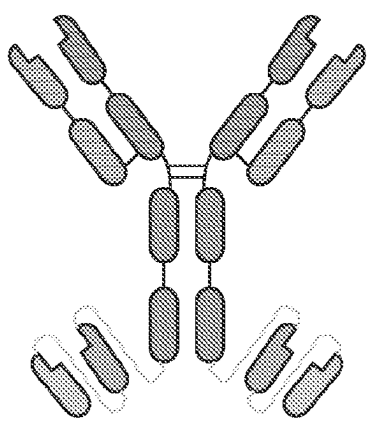


FIG. 48



anti-EphA2 10361



anti-4-1BB 9365

FIG. 49

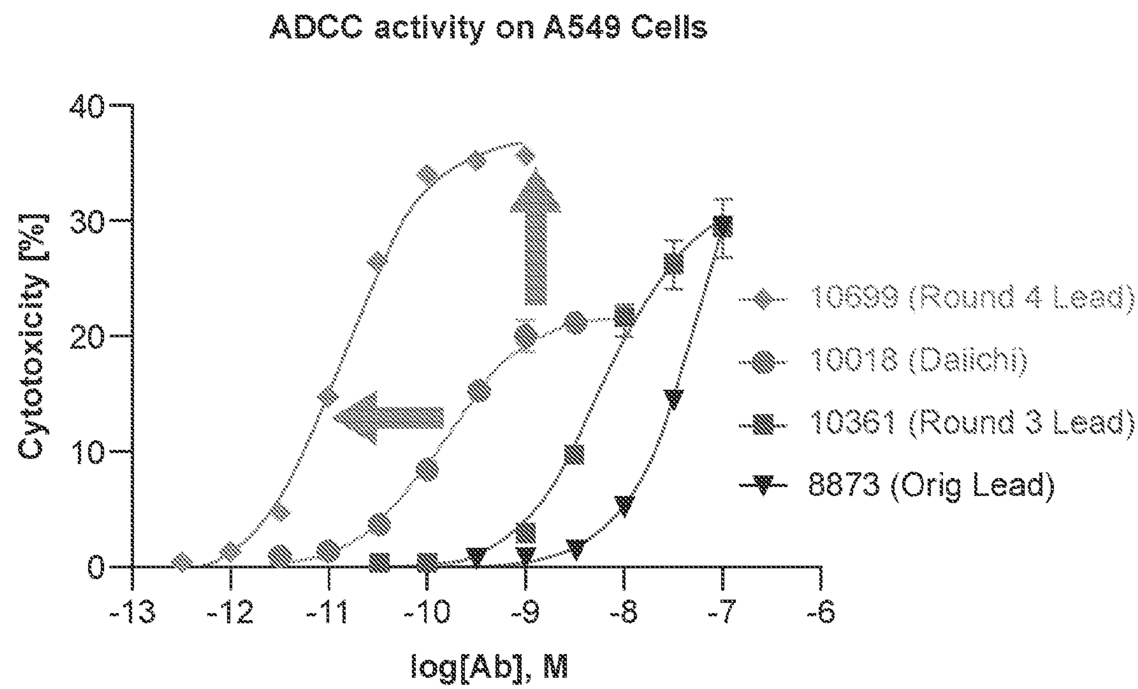


FIG. 50A

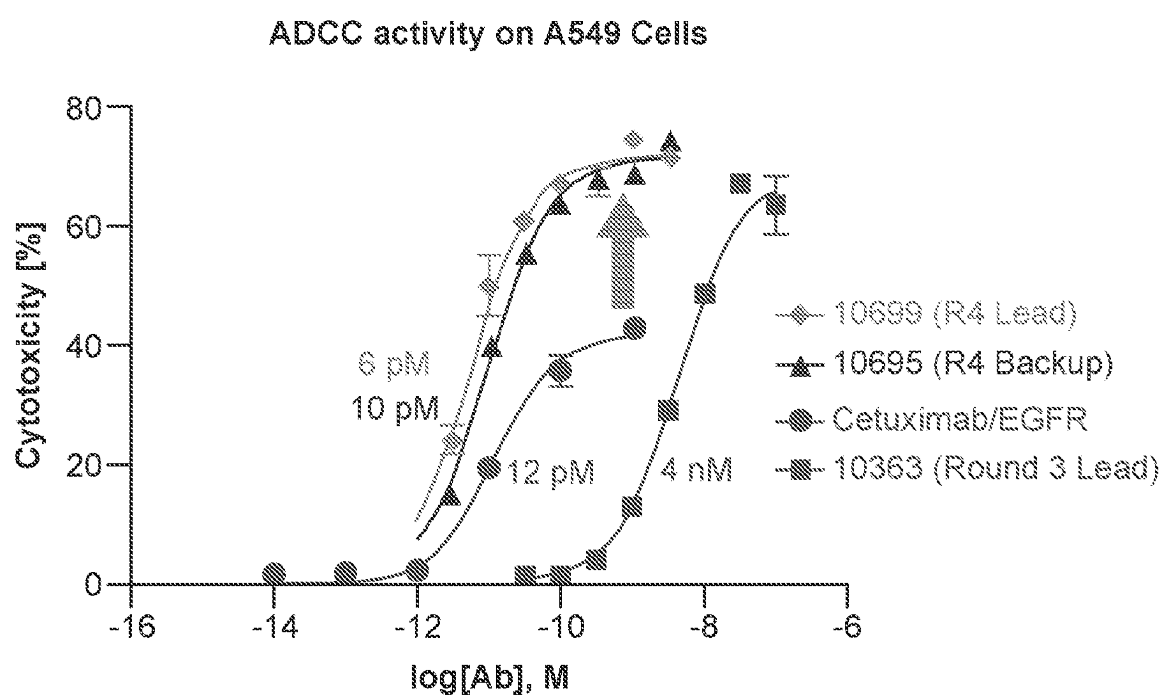


FIG. 50B

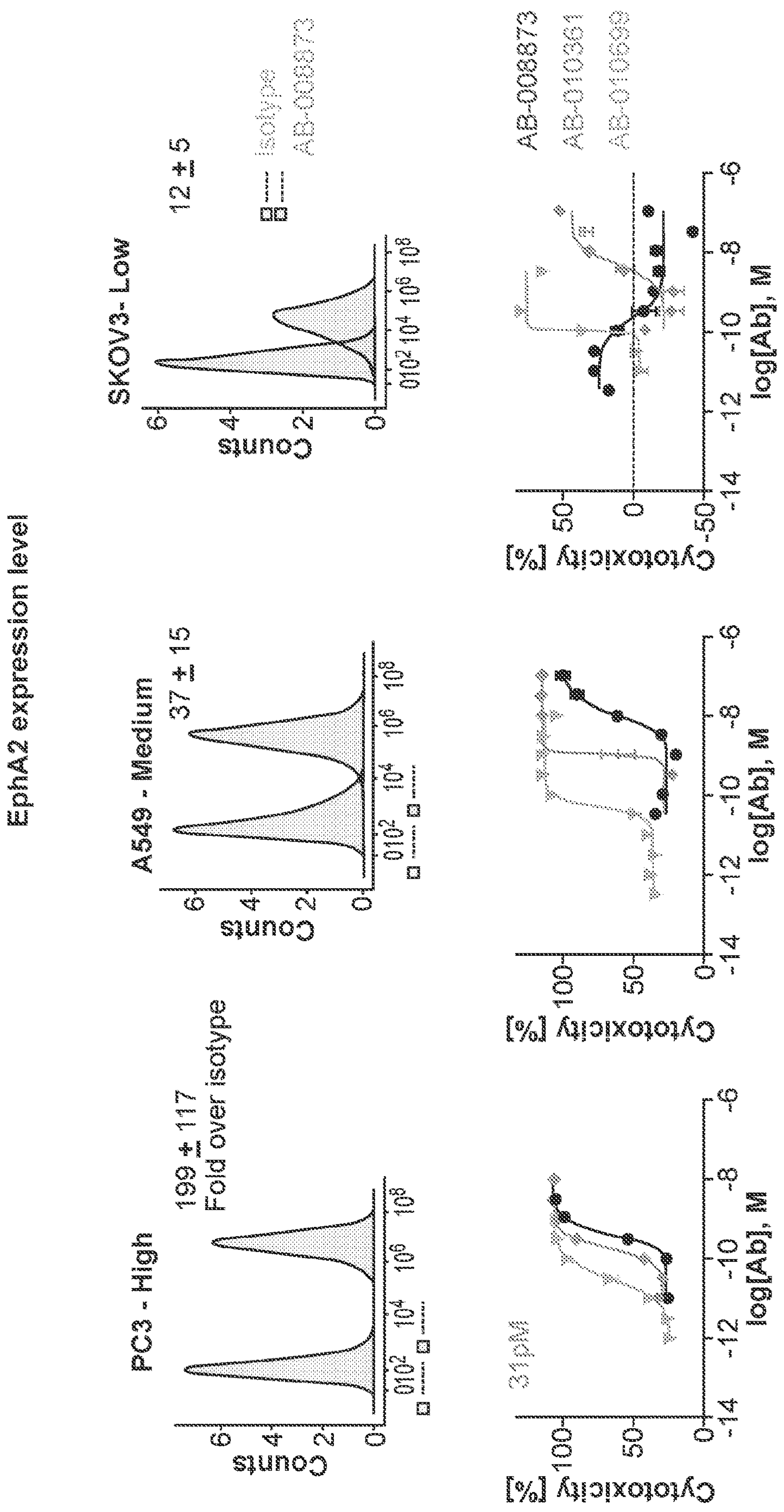
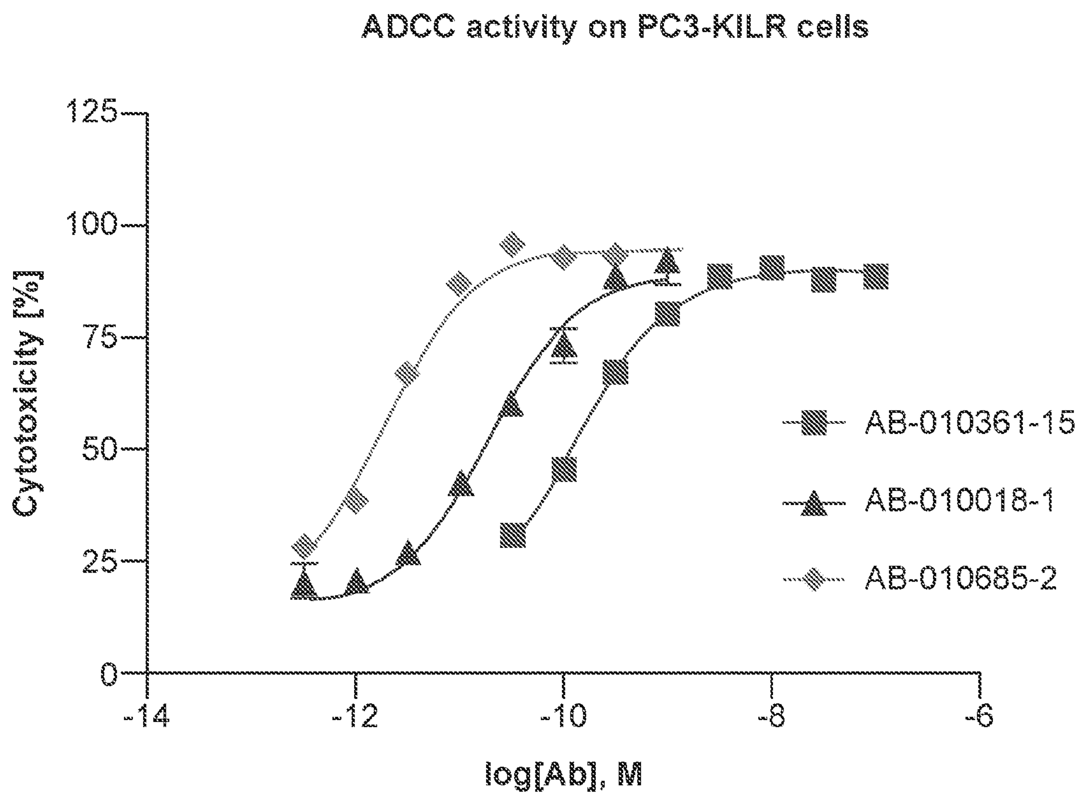

EphA2 expression level
PC3 - High
199 ± 117
Fold over isotype
A549 - Medium
37 ± 15
SKOV3- Low
12 ± 5
isotype
AB-008873
Counts
Cytotoxicity [%]
log[Ab], M
31pM
AB-008873
AB-010361
AB-010699

FIG. 51



	EC50
AB-010361-15	1.326e-010
AB-010018-1	1.841e-011
AB-010685-2	1.813e-012

FIG. 52

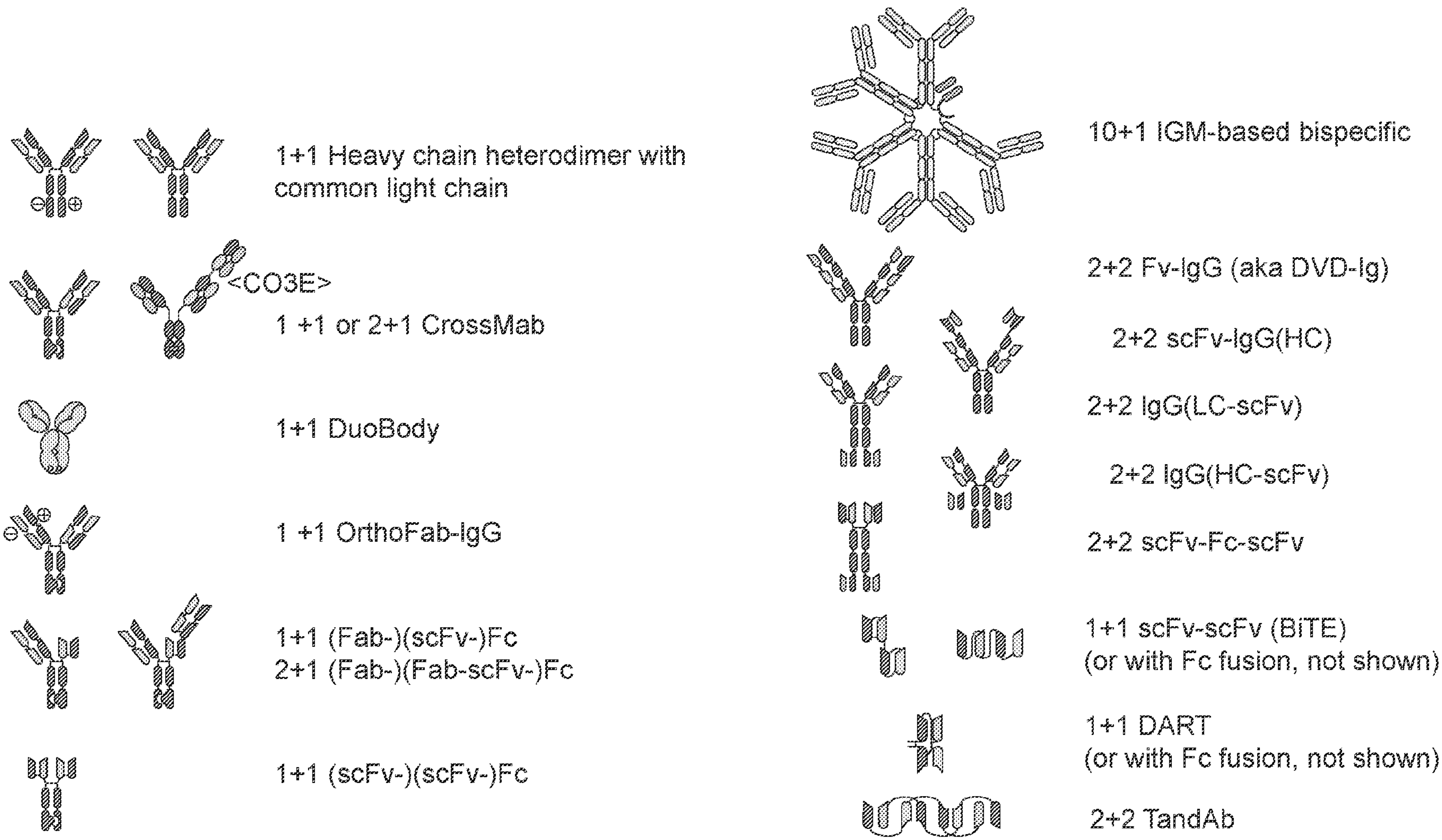

1+1 Heavy chain heterodimer with common light chain
<CO3E>
1 +1 or 2+1 CrossMab
1+1 DuoBody
1 +1 OrthoFab-IgG
1+1 (Fab-)(scFv-)Fc
2+1 (Fab-)(Fab-scFv-)Fc
1+1 (scFv-)(scFv-)Fc
10+1 IGM-based bispecific
2+2 Fv-IgG (aka DVD-Ig)
2+2 scFv-IgG(HC)
2+2 IgG(LC-scFv)
2+2 IgG(HC-scFv)
2+2 scFv-Fc-scFv
1+1 scFv-scFv (BiTE)
(or with Fc fusion, not shown)
1+1 DART
(or with Fc fusion, not shown)
2+2 TandAb

FIG. 53

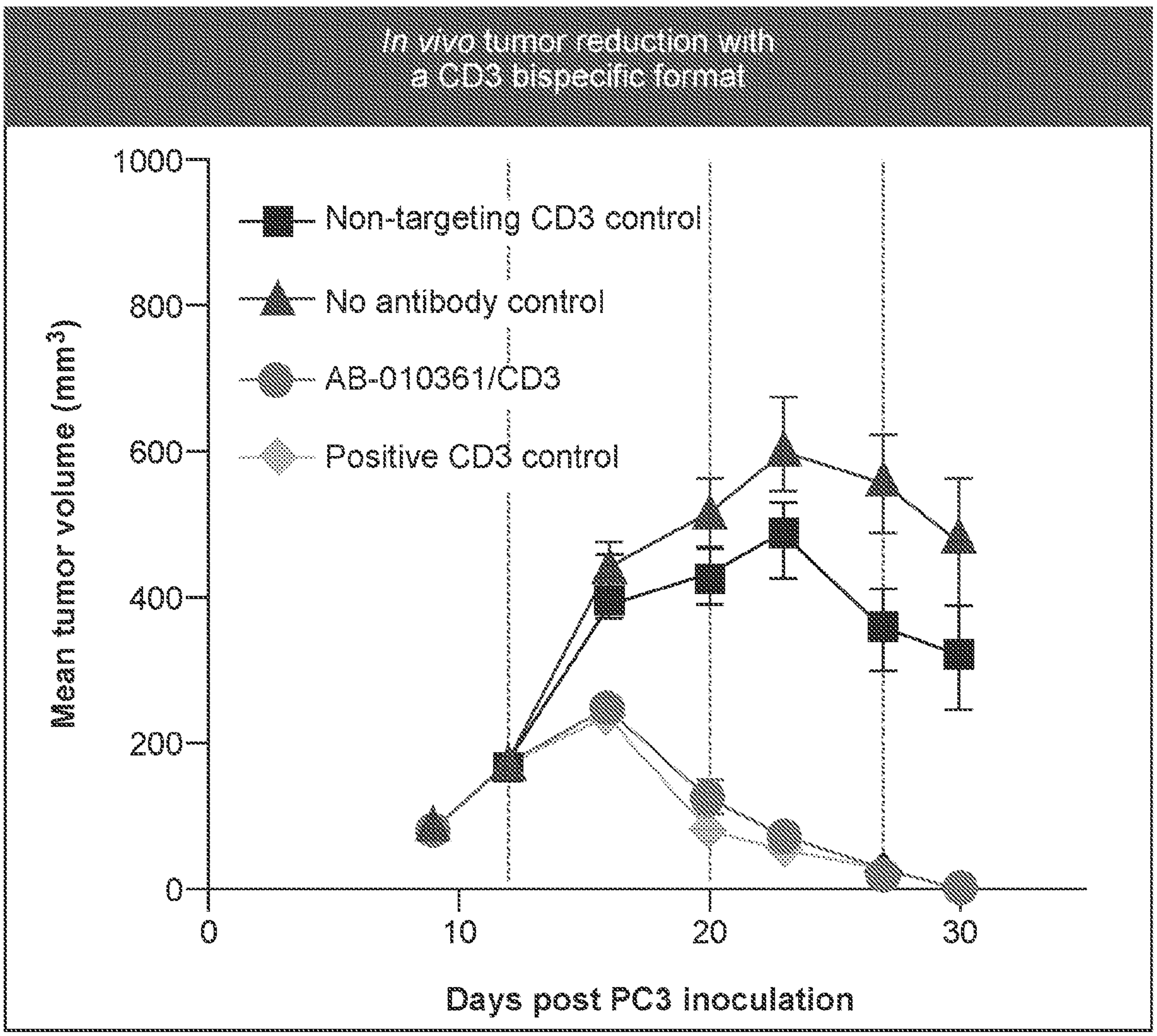
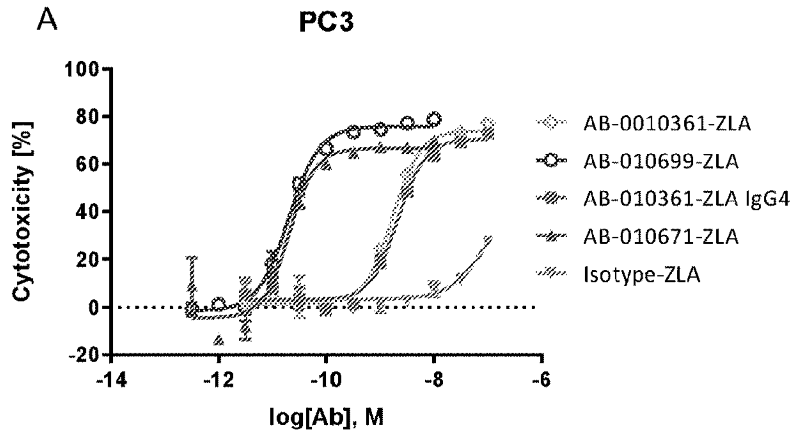
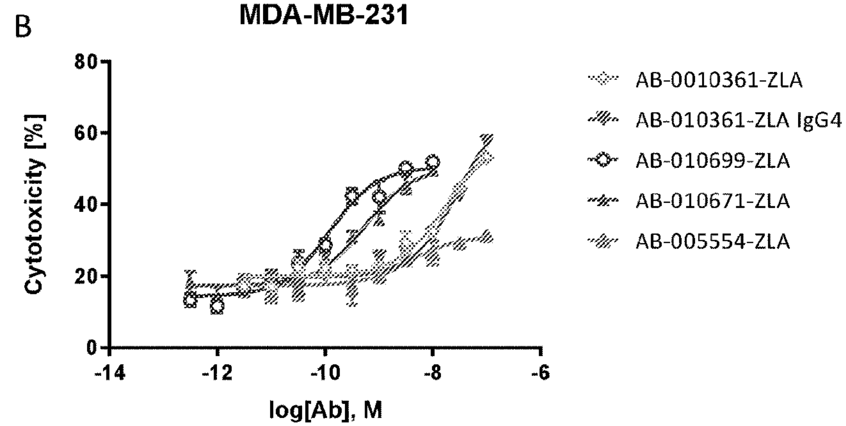
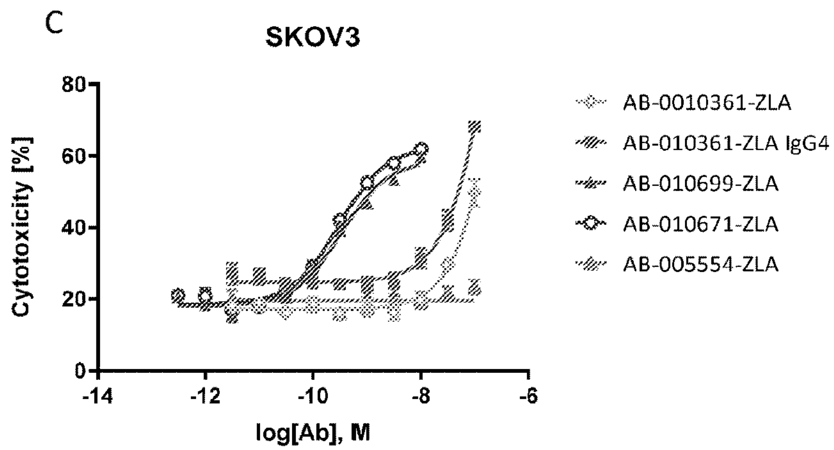

In vivo tumor reduction with
a CD3 bispecific format
Non-targeting CD3 control
No antibody control
AB-010361/CD3
Positive CD3 control
Mean tumor volume (mm3)
1000
800
600
400
200
0
0
10
20
30
Days post PC3 inoculation

FIG. 54



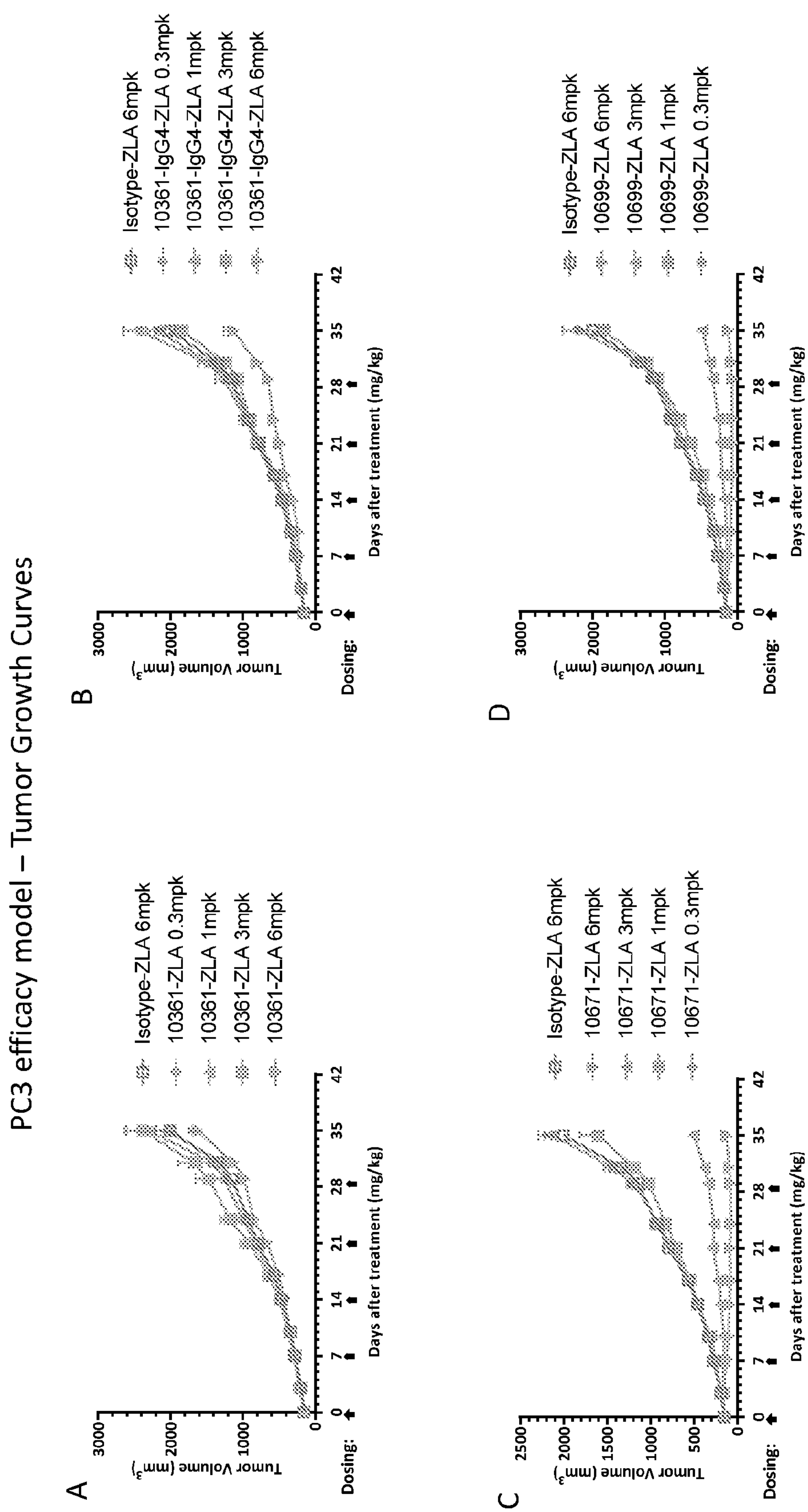




D

EC50 [M]			
	PC3	MDA-MB-231	SKOV3
AB-010361-ZLA	1.708E-09	1.696E-08	>2.5E-07
AB-010361-IgG4-ZLA	2.122E-09	3.14E-08	>2.5E-07
AB-010699-ZLA	2.028E-11	1.216E-10	3.357E-10
AB-010671-ZLA	2.086E-11	5.432E-10	3.058E-10

FIG. 55



PC3 efficacy model – Tumor Growth Curves
A
Tumor Volume (mm³)
Days after treatment (mg/kg)
Dosing:
Isotype-ZLA 6mpk
10361-ZLA 0.3mpk
10361-ZLA 1mpk
10361-ZLA 3mpk
10361-ZLA 6mpk
B
Tumor Volume (mm³)
Days after treatment (mg/kg)
Dosing:
Isotype-ZLA 6mpk
10361-IgG4-ZLA 0.3mpk
10361-IgG4-ZLA 1mpk
10361-IgG4-ZLA 3mpk
10361-IgG4-ZLA 6mpk
C
Tumor Volume (mm³)
Days after treatment (mg/kg)
Dosing:
Isotype-ZLA 6mpk
10671-ZLA 6mpk
10671-ZLA 3mpk
10671-ZLA 1mpk
10671-ZLA 0.3mpk
D
Tumor Volume (mm³)
Days after treatment (mg/kg)
Dosing:
Isotype-ZLA 6mpk
10699-ZLA 6mpk
10699-ZLA 3mpk
10699-ZLA 1mpk
10699-ZLA 0.3mpk

FIG. 56

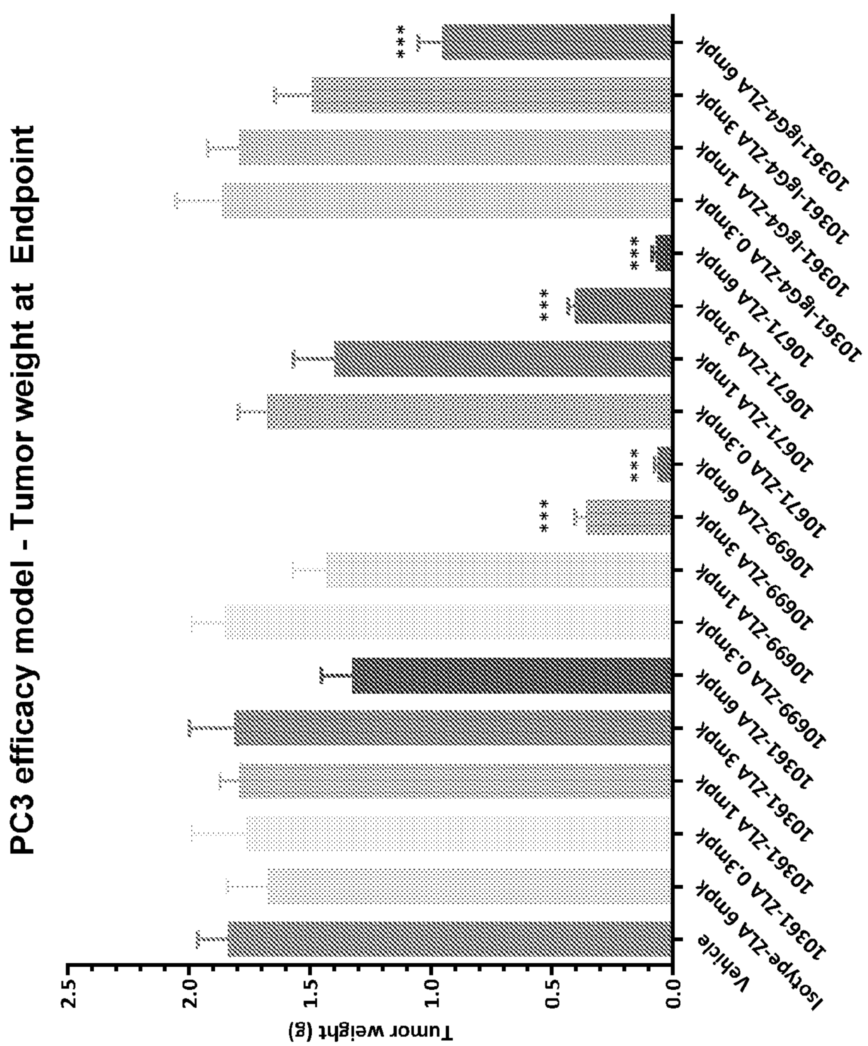

PC3 efficacy model - Tumor weight at Endpoint
Tumor weight (g)
0.0
0.5
1.0
1.5
2.0
2.5
Vehicle
Isotype-ZLA 6mpk
10361-ZLA 0.3mpk
10361-ZLA 1mpk
10361-ZLA 3mpk
10361-ZLA 6mpk
10699-ZLA 0.3mpk
10699-ZLA 1mpk
10699-ZLA 3mpk
10699-ZLA 6mpk
10671-ZLA 0.3mpk
10671-ZLA 1mpk
10671-ZLA 3mpk
10671-ZLA 6mpk
10361-IgG4-ZLA 0.3mpk
10361-IgG4-ZLA 1mpk
10361-IgG4-ZLA 3mpk
10361-IgG4-ZLA 6mpk
***
***
***
***
***

FIG. 57

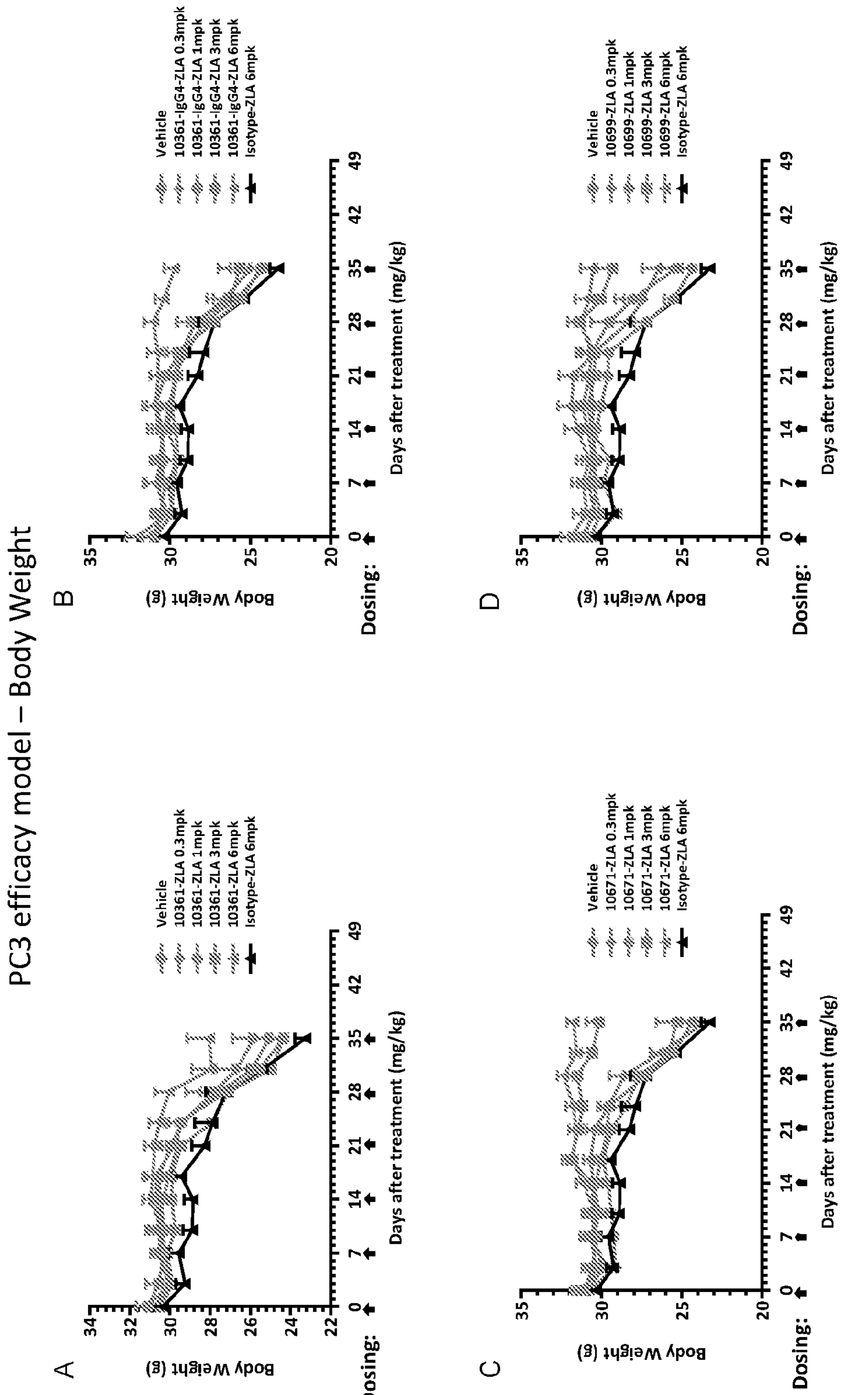
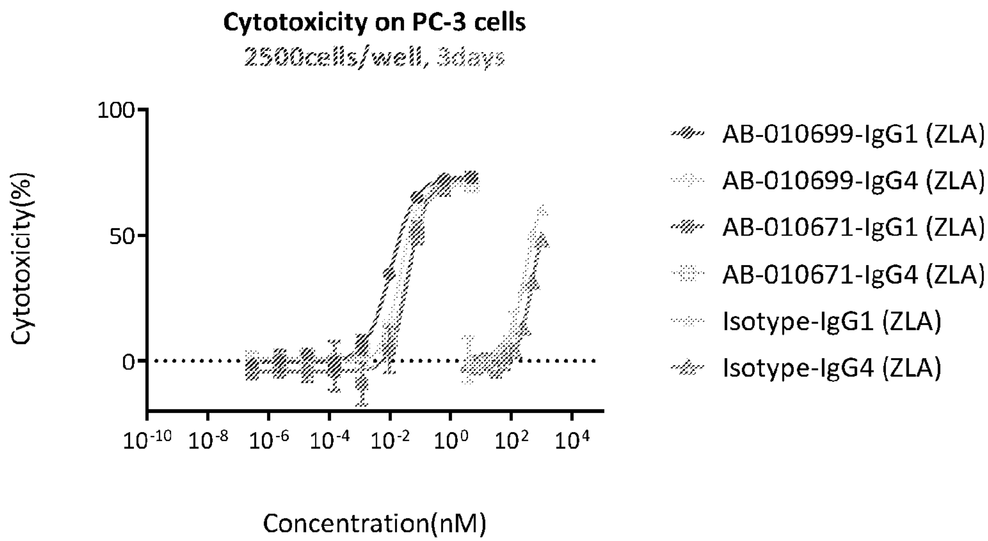

PC3 efficacy model – Body Weight
A
Body Weight (g)
Days after treatment (mg/kg)
Dosing:
Vehicle
10361-ZLA 0.3mpk
10361-ZLA 1mpk
10361-ZLA 3mpk
10361-ZLA 6mpk
Isotype-ZLA 6mpk
B
Body Weight (g)
Days after treatment (mg/kg)
Dosing:
Vehicle
10361-IgG4-ZLA 0.3mpk
10361-IgG4-ZLA 1mpk
10361-IgG4-ZLA 3mpk
10361-IgG4-ZLA 6mpk
Isotype-ZLA 6mpk
C
Body Weight (g)
Days after treatment (mg/kg)
Dosing:
Vehicle
10671-ZLA 0.3mpk
10671-ZLA 1mpk
10671-ZLA 3mpk
10671-ZLA 6mpk
Isotype-ZLA 6mpk
D
Body Weight (g)
Days after treatment (mg/kg)
Dosing:
Vehicle
10699-ZLA 0.3mpk
10699-ZLA 1mpk
10699-ZLA 3mpk
10699-ZLA 6mpk
Isotype-ZLA 6mpk

FIG. 58



A

ADCs(nM)	PC-3, 3 Days	
	IC50 (nM)	Max Cytotoxicity (%)
AB-010699-IgG1 (ZLA)	0.011	74
AB-010699-IgG4 (ZLA)	0.023	72
AB-010671-IgG1 (ZLA)	0.041	74
AB-010671-IgG4 (ZLA)	0.041	70
Isotype-IgG1 (ZLA)	303	62
Isotype-IgG4 (ZLA)	477	48

B

FIG. 59

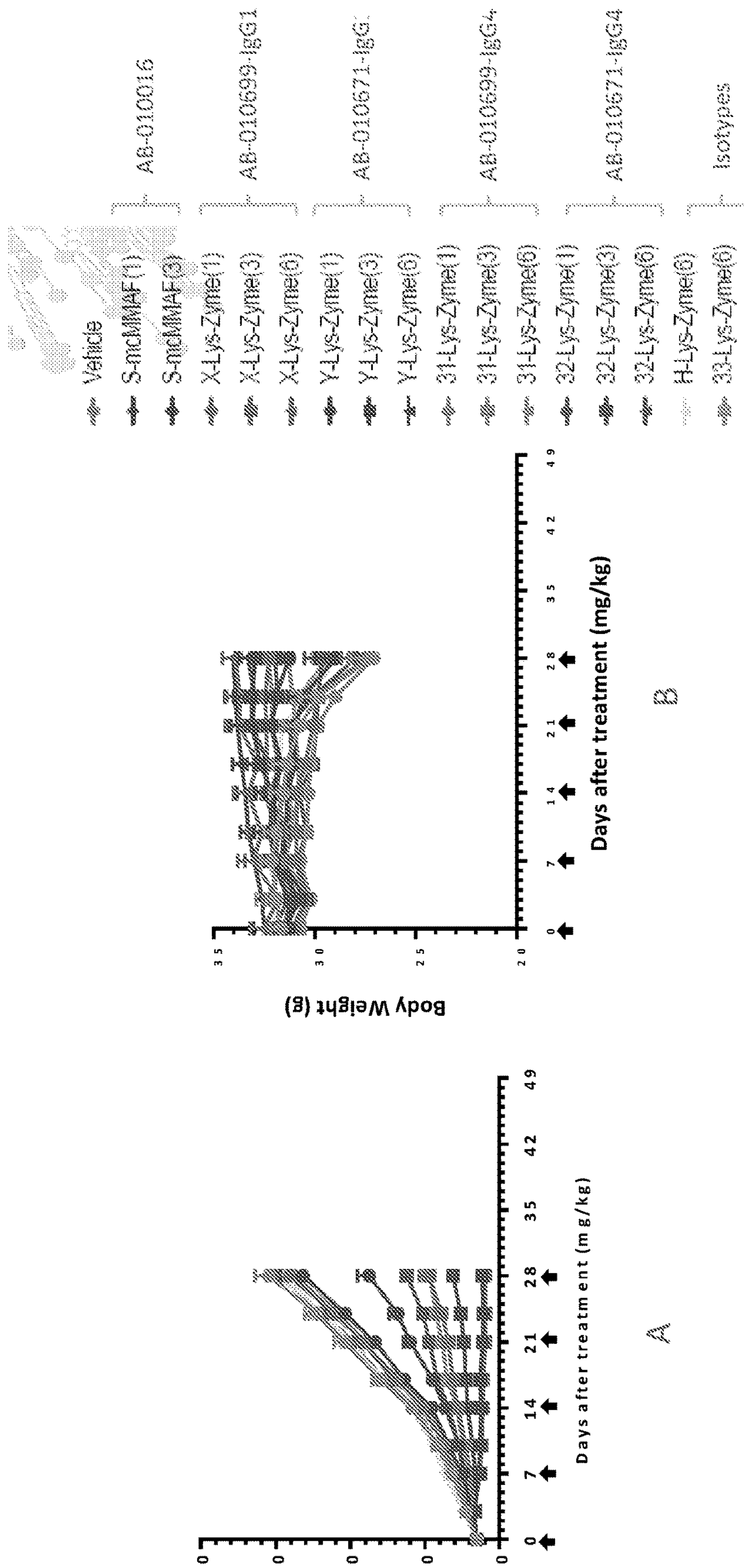

A
Tumor Volume (mm³)
2000
1500
1000
500
0
0
7
14
21
28
35
42
49
Days after treatment (mg/kg)
B
Body Weight (g)
35
30
25
20
Days after treatment (mg/kg)
Vehicle
S-mcMMAF(1)
S-mcMMAF(3)
AB-010016
X-Lys-Zyme(1)
X-Lys-Zyme(3)
X-Lys-Zyme(6)
AB-010699-IgG1
Y-Lys-Zyme(1)
Y-Lys-Zyme(3)
Y-Lys-Zyme(6)
31-Lys-Zyme(1)
31-Lys-Zyme(3)
31-Lys-Zyme(6)
AB-010699-IgG4
32-Lys-Zyme(1)
32-Lys-Zyme(3)
32-Lys-Zyme(6)
AB-010671-IgG4
H-Lys-Zyme(6)
33-Lys-Zyme(6)
Isotypes

FIG. 60

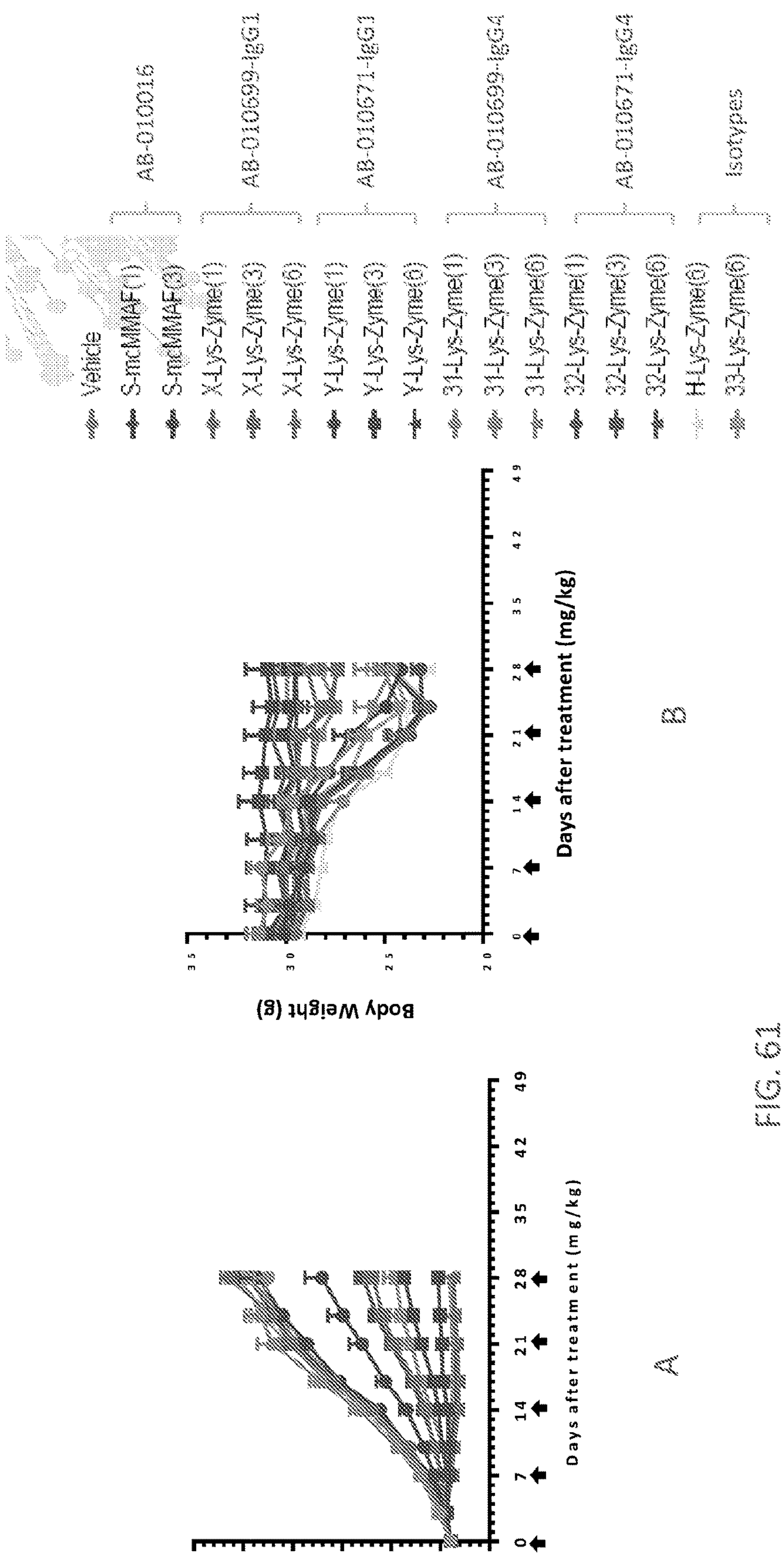

Tumor Volume (mm³)
3000
2500
2000
1500
1000
500
0
0
7
14
21
28
35
42
49
Days after treatment (mg/kg)
A
Body Weight (g)
35
30
25
20
Days after treatment (mg/kg)
B
Vehicle
S-mcMMAF(1)
S-mcMMAF(3)
AB-010016
X-Lys-Zyme(1)
X-Lys-Zyme(3)
X-Lys-Zyme(6)
AB-010699-IgG1
Y-Lys-Zyme(1)
Y-Lys-Zyme(3)
Y-Lys-Zyme(6)
AB-010671-IgG1
31-Lys-Zyme(1)
31-Lys-Zyme(3)
31-Lys-Zyme(6)
AB-010699-IgG4
32-Lys-Zyme(1)
32-Lys-Zyme(3)
32-Lys-Zyme(6)
AB-010671-IgG4
H-Lys-Zyme(6)
33-Lys-Zyme(6)
Isotypes

FIG. 61

EPHA2 ANTIBODIES

RELATED APPLICATIONS

[0001] This application claims priority to U.S. Application No. 63/362,476, filed on Apr. 5, 2022, and U.S. Application No. 63/374,797, filed on Sep. 7, 2022. The entire content of each provisional application is herein incorporated by reference for all purposes.

REFERENCE TO SEQUENCE LISTING

[0002] The Sequence Listing .xml file, identified as 097519-1374172-2901 PCT, is 11,32,370 bytes in size and was created on Mar. 28, 2023. The Sequence Listing is hereby incorporated by reference in its entirety.

FIELD OF CANCER THERAPEUTICS

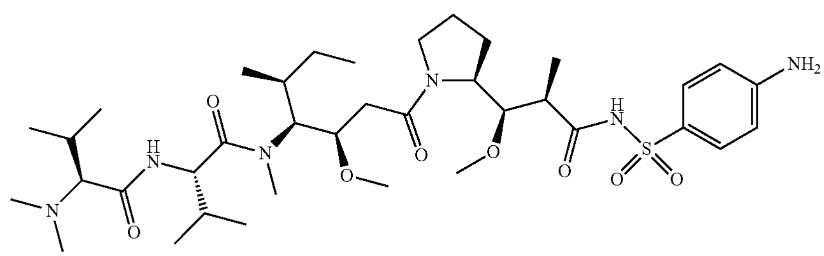
[0003] This application relates to therapeutic antibodies for the treatment of various cancers.

BACKGROUND

[0004] EphA2 is a tyrosine kinase and belongs to the family of Ephrin receptors. Like other Ephrin receptors, it is a single-pass membrane protein with large extracellular N-termini. EphA2 is typically involved in cell-cell repulsion or adhesion processes. EphA2 is also known to promote angiogenesis. EphA2 is overexpressed in tumor tissues compared to normal adult tissues, indicating its potential application in cancer treatment. Although several EphA2 antibodies have been developed, some of which are being assessed in clinical trials, these antibodies have the shortcomings of high toxicity and insufficient efficacy. They are unable to meet the needs of cancer patients.

BRIEF SUMMARY

[0005] Provided herein is an immunoconjugate comprising an antibody that binds to ephrin receptor A2 (EphA2) and a cytotoxic agent, wherein the cytotoxic agent is an auristatin analogue, and wherein the antibody is conjugated to the auristatin analogue. In some embodiments, the auristatin analogue has a formula of:



[0006] In some embodiments, the linker has a formula of

$$\#\text{—}Z\text{—}[Str]_s\text{—}AA_1\text{—}[AA_2]_m\text{—}[X]_o\text{—}\% \tag{V}$$

[0007] wherein:

[0008] Z is a linking group that joins the linker to a target group on the antibody;

[0009] Str is a stretcher:

[0010] AA_1 and AA_2 are each independently an amino acid, wherein AA_1-$[AA_2]_r$ forms a protease cleavage site;

[0011] X is a self-immolative group;

[0012] s is 0 or 1;

[0013] m is 1, 2, 3 or 4;

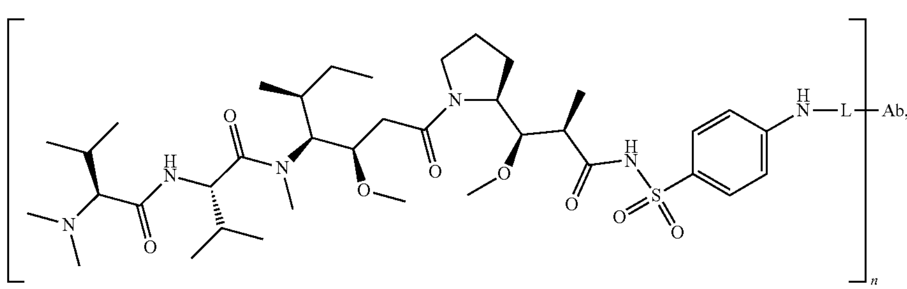
[0014] o is 0, 1 or 2;

[0015] # is the point of attachment to the antibody, and

[0016] % is the point of attachment to the auristatin analogue.

[0017] In some embodiments, the immunoconjugate has a formula of:

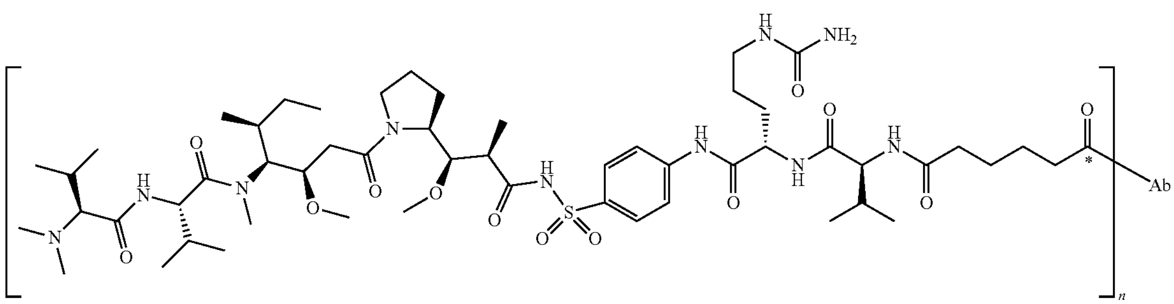
(I)



wherein: L is a cleavable linker; n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and Ab is the antibody.

[0018] In some embodiments, the immunoconjugate has a formula of:

(II)



wherein n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and Ab is the antibody.

[0019] In some embodiments, the antibody of the immunoconjugate binds to an epitope of EphA2 located in the FN2 domain of EphA2, wherein the FN2 domain has the sequence of

```
                                         (SEQ ID NO: 95)
TEPPKVRLEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVR
RTEGFSVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTLSPEGSG
N.
```

[0020] In some embodiments, the epitope comprises at least one, at least two, at least three, at least four, at least five, or at least six, at least seven, at least eight, or at least nine, or all ten of amino acid residues Leu444, Arg447, Lys476, Gln506, Ser519, Lys520, Val521, Glu523, Phe524, and Gln525 of SEQ ID NO: 94.

[0021] In some embodiments, the antibody of the immunoconjugate has one or more properties of: a) activates the ephrin A1-EphA2 signaling axis at least 95%, 90%, 85%, or 80% less than ephrin-A1 activates the ephrin A1-EphA2 signaling axis, b) binds preferentially to a tumor tissue than normal tissue, c) has an EC_{50} for activation of the ephrinA1-EphA2 signaling axis that is at least 10-, 20-, 50-, 100-, 200-, 500-fold less potent than the ADCC EC_{50} of the antibody; and/or d) has ADCP activity, and c) has ADCC activity.

[0022] In some embodiments, the antibody of the immunoconjugate disclosed herein binds to a tumor tissue preferentially compared to normal tissue.

[0023] In some embodiments, the antibody of the immunoconjugate comprises a heavy chain variable region comprising: an HCDR1 of any one of SEQ ID NOS:1-11 and 201-265 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence; an HCDR2 of any one of SEQ ID NOS:12-22 and 266-330 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; and an HCDR3 of any one of SEQ ID NOS:23-33 and 331-395 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence; a light chain variable region comprising: an LCDR1 of any one of SEQ ID NOS:34-44 and 396-460 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; an LCDR2 of any one of SEQ ID NOS:45-55 and 461-525 or a variant thereof in which 1, 2, or 3 amino acid is substituted relative to the sequence; and an LCDR3 of any one of SEQ ID NOS:56-66 and 526-590 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence.

[0024] In some embodiments, the antibody of the immunoconjugate comprises a heavy chain variable region comprising: an HCDR1 comprising a sequence (SEQ ID NO:1) or a variant HCDR1 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence; an HCDR2 comprising a sequence (SEQ ID NO:12), or a variant HCDR2 in which 1, 2, 3, 4, or 5 amino acids substituted relative to the sequence; and an HCDR3 comprising a sequence (SEQ ID NO:23), or a variant HCDR3 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence. The antibody further comprises a light chain variable region comprising: an LCDR1 comprising a sequence (SEQ ID NO:34) or a variant LCDR1 in which 1, 2, 3, 4, or 5 amino acids is substituted relative to the sequence; an LCDR2 comprising a sequence (SEQ ID NO:45), or variant LCDR2 in which 1, 2, or 3 amino acids is substituted relative to the sequence; and an LCDR3 comprising a sequence (SEQ ID NO:56), or a variant LCDR3 in which 1, 2, 3, 4, or 5 amino acids is substituted relative to the sequence.

[0025] In some embodiments, the antibody of the immunoconjugate comprises: (a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769); (b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNNYNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 770); (c) an HCDR3 sequence of $AKPX_1RPHCX_2NGVCX_3SGDAFDI$, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S (SEQ ID NO: 771); (d) an LCDR1 sequence of $X_1GNNIGX_2X_3X_4VH$, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or S (SEQ TD NO: 772); (e) an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence of QVWDX1 X2SDHX3V, where X1 is H or S, X2 is E, R, or S, and X3 is L or V (SEQ ID NO: 773).

[0026] In some embodiments, the antibody of the immunoconjugate comprises: (a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, and X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769); (b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 774); (c) an HCDR3 sequence of

AKPX_1RPHCX_2NGVCX_3SGDAFDI, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S (SEQ ID NO: 771); (d) an LCDR1 sequence of X_1GNNIGX_2 X_3 X_4VH, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or S (SEQ ID NO: 772); (e) an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence of QVWDX1 X2SDHX3V, where X1 is H or S, X2 is E, R, or S, and X3 is L or V (SEQ ID NO: 773).

[0027] In some embodiments, the antibody comprises: (a) an HCDR1 sequence of GGS$X_1$$X_2DYX_3$WS where X_1 is F or L, X_2 is S or N, and X_3 is Y or H (SEQ ID NO: 775); (b) an HCDR2 sequence of EX1NHX2 GS X_3 X_4YNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 775); or an HCDR2 sequence of EX_1NHX_2 GS X_3 X_4YNNYNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 N or S(SEQ ID NO: 777); and (c) an HCDR3 sequence of AKP X_1RPHCTNGVCX2SGDAFDI, where X_1 is L or F and X_2 is Y or S (SEQ ID NO: 778); (d) an LCDR1 sequence of GGNNIGX_1KNVH, where X_1 is S or T (SEQ ID NO: 779); (e) an LCDR2 sequence DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence QVWDSSSDHLV (SEQ ID NO: 56).

[0028] In some embodiments, the antibody of the immunoconjugate comprises a V_H comprising an amino acid sequence having at least 95% identity to a V_H in Table 8; and a V_L comprising an amino sequence having at 95% identity to a V_L in Table 8. In some embodiments, the V_H comprises an amino acid sequence having at least 95% identity to (SEQ ID NO: 67); and the V_L comprises an amino sequence having at 95% identity to (SEQ ID NO: 78). In some embodiments, the V_H comprises an amino acid sequence having at least 95% identity to a V_H of any one of SEQ ID NOS:67-77 and 591-655. The V_L comprises an amino sequence having at 95% identity to a V_L of any one of SEQ ID NOS:78-88 and 656-720.

[0029] In some embodiments, the antibody of the immunoconjugate comprises: a V_H region comprising amino acid sequence SEQ ID NO:67 and a V_L region comprising amino acid sequence SEQ ID NO:78; a V_H region comprising amino acid sequence SEQ ID NO:68 and a V_L region comprising amino acid sequence SEQ ID NO:79; a V_H region comprising amino acid sequence SEQ ID NO:69 and a VL region comprising amino acid sequence SEQ ID NO:80; a V_H region comprising amino acid sequence SEQ ID NO:70 and a VL region comprising amino acid sequence SEQ ID NO:81; a V_H region comprising amino acid sequence SEQ ID NO:71 and a V_L region comprising amino acid sequence SEQ TD NO:82; a V_H region comprising amino acid sequence SEQ TD NO:72 and a V_L region comprising amino acid sequence SEQ ID NO:83; a V_H region comprising amino acid sequence SEQ ID NO:73 and a V_L region comprising amino acid sequence SEQ ID NO:84; a V_H region comprising amino acid sequence SEQ ID NO:74 and a V_L region comprising amino acid sequence SEQ ID NO:85; a V_H region comprising amino acid sequence SEQ ID NO:75 and a V_L region comprising amino acid sequence SEQ ID NO:86; or a V_H region comprising amino acid sequence SEQ ID NO:76 and a V_L region comprising amino acid sequence SEQ ID NO:87; a V_H region comprising amino acid sequence SEQ ID NO:77 and a V_L region comprising amino acid sequence SEQ ID NO:88, a V_H region comprising amino acid sequence SEQ ID NO:652 and a V_L region comprising amino acid sequence SEQ ID NO:717, or a V_H region comprising amino acid sequence SEQ ID NO:607 and a V_L region comprising amino acid sequence SEQ ID NO:672.

[0030] In some embodiments, antibody of the immunoconjugate comprises: (1) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein: (a) the V_H region has at least 70% identity to SEQ ID NO:67; and comprises a CDR1 of SEQ ID NO:1, or the CDR1 of SEQ ID NO:1 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:12, or the CDR2 of SEQ ID NO:12 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:23 or the CDR3 of SEQ ID NO:23 in which 1, 2, 3, 4, or 5 are substituted; and (b) the V_L region has at least 70% identity to SEQ ID NO: 78, and comprises a CDR1 of SEQ ID NO:34 or the CDR1 of SEQ ID NO:34 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:45, or the CDR2 of SEQ ID NO:45 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:56 or the CDR3 of SEQ ID NO:56 in which 1, 2, 3, 4, or 5 are substituted; (2) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein: (a) the V_H region has at least 70% identity to SEQ ID NO:652; and comprises a CDR1 of SEQ ID NO:262, or the CDR1 of SEQ ID NO:262 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:327, or the CDR2 of SEQ ID NO:327 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:392 or the CDR3 of SEQ ID NO:392 in which 1, 2, 3, 4, or 5 are substituted; and (b) the VL region has at least 70% identity to SEQ ID NO: 717, and comprises a CDR1 of SEQ TD NO:457 or the CDR1 of SEQ TD NO:457 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:522, or the CDR2 of SEQ ID NO:522 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:587 or the CDR3 of SEQ ID NO:587 in which 1, 2, 3, 4, or 5 are substituted; or (3) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein: (a) the V_H region has at least 70% identity to SEQ ID NO:607; and comprises a CDR1 of SEQ ID NO:217, or the CDR1 of SEQ ID NO:217 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:282, or the CDR2 of SEQ ID NO:282 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:347 or the CDR3 of SEQ ID NO:347 in which 1, 2, 3, 4, or 5 are substituted; and (b) the V_L region has at least 70% identity to SEQ ID NO: 672, and comprises a CDR1 of SEQ ID NO:412 or the CDR1 of SEQ ID NO:412 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:477, or the CDR2 of SEQ ID NO:477 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:542 or the CDR3 of SEQ ID NO:542 in which 1, 2, 3, 4, or 5 are substituted.

[0031] The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises: a heavy chain variable (VH) region and a light chain variable (VL) region, (i) wherein the VH region has at least 70% identity to SEQ ID NO:67 and comprises (a) an HCDR1 sequence of GGSX1X2DYX3WS where X1 is F or L, X2 is S or N, and X3 is Y or H (SEQ ID NO: 775); (b) an HCDR2 sequence of EX1NHX2 CS X3 X4YNPSLKS, where X1 is I or V, X2 is R or S, X3 is I or T, and X4 is N or S (SEQ ID NO: 776); or an HCDR2 sequence of EX1NHX2 GS X3 X4YNNYNPSLKS, where X1 is I or V, X2 is R or S, X3 is I or T, and X4 N or S (SEQ ID NO: 777); and (c) an HCDR3 sequence of AKP X1RPHCTNGVCX2SGDAFDI, where X1 is L or F and X2 is Y or S (SEQ ID NO: 778); and (ii) wherein the VL region has at least 70% identity to SEQ ID NO:78 and comprises (d) an LCDR1 sequence of GGNNIGX1KNVH, where X1 is S or T (SEQ ID NO: 779); (e) an LCDR2 sequence DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence QVWDSSSDHLV (SEQ ID NO: 56).

[0032] In some embodiments, the antibody of the immunoconjugate comprises an HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702, or a variant thereof in which at least one, two, three, four, five, or all six of the CDRs contain 1 or 2 amino acid substitutions compared to the corresponding CDR.

[0033] In some embodiments, the antibody of the immunoconjugate comprises a heavy chain variable region and a light chain variable region of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702, or a variant thereof, wherein the variant comprises a heavy chain variable region having a sequence that is at least 95% identical to that of the corresponding heavy chain variable region and a light chain variable region having a sequence that is at least 95% identical to the corresponding light chain variable region.

[0034] In some embodiments, at least 1 or 2 of the substitutions referenced above are conservative substitutions and at least 50% of the substitutions are conservative substitutions. In some embodiments, all the substitutions are conservative substitutions.

[0035] In some embodiments, the antibody of the immunoconjugate comprises a V_H region of any one of SEQ ID NO 67-77 and 591-655, and/or a V_L region of any one of SEQ ID NO: 78-88 and 656-720, or an antibody comprising a V_H region with at least 80% identity to any one of SEQ ID NO 67-77 and 591-655 and a V_L region having at least 80% identity to any one of SEQ ID NO: 78-88 and 656-720, with variations to the corresponding V_H or V_L regions present only in Framework regions.

[0036] In some embodiments, the antibody of the immunoconjugate comprises an HCDR1 of any one of SEQ ID NOS: 1-11, and 201-265, an HCDR2 of any one of SEQ ID NOS:12-22 and 266-330, an HCDR3 of any one of SEQ ID NOS:23-33 and 331-395, an LCDR1 of any one of SEQ ID NOS:34-44 and 396-460, an LCDR2 of any one of SEQ ID NOS:45-55 and 461-525, an LCDR3 of any one of SEQ ID NOS:56-66 and 526-590, wherein the FW regions in the V_H region are at least 80% identical to the FW regions present in the V_H region of SEQ ID NO 67-77 and 591-655, and wherein the FW regions in the V_L region are at least 80% identical to the FW regions present in the V_L region of SEQ ID NO: 78-88 and 656-720.

[0037] In some embodiments, the antibody of the immunoconjugate is a non-natural antibody.

[0038] In some embodiments, the antibody is a bispecific antibody comprising two antigen binding fragments, one binding to EphA2, and the other binds to a different antigen. In some embodiments, the different antigen is 4-1bb or CD3.

[0039] In some embodiment, the immunoconjugate comprising the antibody disclosed herein further comprises one or more of an IL-15 receptor agonist, a TGFβ trap, a TLR agonist, or a 4-1BB ligand (4-1BBL).

[0040] Also provided in this disclosure is a pharmaceutical composition comprising any of the immunoconjugates disclosed herein and a pharmaceutically acceptable carrier.

[0041] Also provided in this disclosure is a method of inducing an immune response and/or treating cancer, the method comprising administering any one of the immunoconjugates disclosed herein. In some embodiments, the immune response comprises an antibody-dependent cellular cytotoxicity (ADCC) and/or antibody dependent cellular phagocytosis (ADCP). In some embodiments, the antibody is administered intravenously.

[0042] Also provided in this disclosure is a method of treating a cancer patient having a tumor overexpressing EphA2, the method comprising administering any one of the immunoconjugates disclosed herein to the patient. In some embodiments, the cancer is a gastric cancer, ovarian cancer, soft tissue sarcoma, lung cancer, head and neck cancer, uterine cancer, breast cancer, colorectal cancer, esophageal cancer, stomach cancer, kidney cancer, melanoma, liver cancer, bladder cancer, or testicular cancer. In some embodiments, the cancer is non-small cell lung cancer, triple-negative breast cancer, colorectal cancer, ovarian cancer or melanoma. In some embodiments, the method further comprises administering chemotherapy and/or radiation therapy. In some embodiments, the method comprises administering an agent that targets an immunological checkpoint antigen. In some embodiments, the agent is a monoclonal antibody. In some embodiments, the monoclonal antibody blocks PD-1 ligand binding to PD-1. In some embodiments, the monoclonal antibody is an anti-PD-1 antibody.

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] FIG. **1** shows surface binding profile of AB-008873 in A549 cells and CT26 cells (in vitro and ex vivo).

[0044] FIG. **2**A and FIG. **2**B shows results of assessing the ADCC activity (FIG. **2**A) and ADCP activity (FIG. **2**B) of AB-008873 in vitro functional assays.

[0045] FIG. **3** shows the ADC activity of AB-008873 on in vitro cell lines.

[0046] FIG. **4** compares the binding activity and functional activity of AB-008873 and other EphA2 antibodies on mouse and human cells.

[0047] FIG. **5**A illustrates a method of octet in tandem binning for epitope mapping. FIG. **5**B illustrates the epitopes in the EphA2 domains recognized by EphA2 antibodies.

[0048] FIG. **6** shows the results from a yeast display that demonstrates the binding of AB-008873 to FN2 of EphA2.

[0049] FIG. **7** shows the design of an in vivo study to assess the effect of the AB-008873 treatment.

[0050] FIG. **8** shows that AB-008873 treatment increased T cells in the blood.

[0051] FIG. **9** shows the effects on tumor growth of treatment of AB-008873 on mice inoculated with CT26 cells.

[0052] FIG. **10** shows that AB-008873 treatment did not result in significant changes in TIL number.

[0053] FIG. **11** shows that AB-008873 treatment did not significantly change T & NK cell numbers in the TME.

[0054] FIG. **12** shows increased macrophage numbers in TME at Day 10 with 40 mg/kg AB-008873 treatment.

[0055] FIG. **13** shows representative CD3+ CD8+ staining of cytotoxic T cells.

[0056] FIG. **14** shows representative CD4+ FoxP3+ staining of regulatory T cells.

[0057] FIG. **15** shows representative F4/80+ iNOS+ staining on M1 macrophages.

[0058] FIG. **16** shows the effect of 40 mg/kg AB-00 8873 administered on tumor growth and body weight in mice inoculated with CT26 cells (murine colon carcinoma cells).

[0059] FIG. **17** shows flow cytometry analysis of total tumor-infiltrating leukocytes (TIL) and myeloid cells from tumors in mice inoculated with CT26 cells. The mice were treated with 40 mg/kg AB-008873 or PBS.

[0060] FIG. **18** shows flow cytometry analysis of lymphoid cells from tumors in mice inoculated with CT26 cells. The mice were treated with 40 mg/kg AB-008873 or PBS.

[0061] FIGS. **19**A, **19**B, **19**C, **19**D, and **19**E show results of activation of 4-1BB by EphA2-4-1BB bispecific antibodies in tumor cell lines.

[0062] FIG. **20** shows the functional activity of AB-008873 sibling antibodies (antibodies that are derived from the same lineage as AB-008873)

[0063] FIG. **21**A-C show the functional activity of engineered AB-008873 variants: flow binding, ADC activity, and ADCC activity, respectively.

[0064] FIG. **22** shows the reactivity of AB-010699 across a panel of human cancer samples.

[0065] FIG. **23** shows the binding sites of various commercial EphA2 antibodies.

[0066] FIG. **24** shows the reactivity of EphA2 antibodies across a panel of human cancer samples.

[0067] FIG. **25** shows immunofluorescence study results demonstrating a possible unique epitope of AB-008873 in gastric cancer.

[0068] FIG. **26** compares EphA2 antibodies in binding ovarian cancer and soft tissue sarcoma cells.

[0069] FIG. **27** compares AB-008873 with commercial anti-EphA2 antibodies in binding to ovarian cancer and soft tissue sarcoma cells.

[0070] FIG. **28** compares the binding pattern of EphA2 antibodies in tumor tissues from different uterine cancer donors.

[0071] FIG. **29** shows the binding pattern of EphA2 antibodies in tumor tissues from different head and neck cancer donors.

[0072] FIG. **30** shows the binding pattern of EphA2 antibodies in tumor tissues from different Non-small Cell Lung cancer (NSCLC) donors.

[0073] FIG. **31** shows the reactivity of EphA2 antibodies across a panel of additional human cancer samples.

[0074] FIG. **32** shows schematics of the formats of multivalent antibodies, including the format of the Fc regions.

[0075] FIG. **33** shows the binding of AB-008873, variants thereof, AB-010018 and commercial anti-EphA2 antibodies to normal kidney.

[0076] FIG. **34** shows the binding of AB-008873 and its variants, AB-010018, and a commercial anti-EphA2 antibody to a normal stomach.

[0077] FIG. **35** shows a comparison of the ADC activity of AB-008873 with AB-009815 (comprising an H76PT mutation relative to AB-008873). This mutation removes a medium-risk DP proteolytic cleavage site that could cause AB degradation in the serum, decreasing half-life after injection in mice/humans. The results showed that the ADC activity of AB-009815 is substantially similar to that of AB-008873 in H522, LoVo, A375, SKOV3, A549, and MDAMB231 cells.

[0078] FIG. **36** is a schematic illustration of AB-008873-IL15 fusions.

[0079] FIG. **37** shows that AB-008873-IL15 multiMabs treatment resulted in dose-dependent increases in human NK cell proliferation.

[0080] FIG. **38** shows the flow cytometry analysis of the binding of the AB-008873-IL15 multiMabs to tumor cells using flow cytometry.

[0081] FIG. **39**A-D show alignments and CDR designations for various EphA2 antibodies. FIG. **39**A-D discloses SEQ ID NOS 67-88, 67, 75, 598, 603, 607, 609, 652, 78, 86, 663, 668, 672, 674, and 717, respectively, in order of appearance.

[0082] FIG. **40** compares the ADCC activity of several of anti-EphA2 antibodies AB-010361 ("AB-010361-15"), AB-010681 ("AB-010681-2"), and AB-010685 ("AB-010681-2") on A549 cells.

[0083] FIG. **41** shows the results of thermostability analysis of the anti-EphA2 antibodies AB-008873 and AB-010699.

[0084] FIG. **42** shows immunofluorescence staining results of AB-008873 variants as compared to AB-008873, and AB-010018.

[0085] FIG. **43**A and FIG. **43**B show the phosphorylation of Y588 and S897 on EphA2 by anti-EphA2 antibodies AB-008873, AB-010361. AB-010699, AB-010016, AB-010018, and AB-010019 in agonist (FIG. **43**A) and antagonist (FIG. **43**B) assay.

[0086] FIG. **44** shows identification of AB-008873 epitope residues by yeast display. The epitope residues identified by this method include Pro439, Lys441, Arg443, Leu444, Arg447, Lys476, Gly477, Leu504, Gln506, Ser519, Lys520, Val521, His522, Glu523, Phe524, and Gln525.

[0087] FIG. **45** shows the alignment of the epitope residues across the EphA/B family members. The results show that the residues are divergent across the EphA/B family and the epitope is conformational. FIG. **45** discloses SEQ ID NOS 783-827 and 829-838, respectively, in order of appearance.

[0088] FIG. **46** shows the alignment of the epitope residues across multiple species: human, cynomolgus monkey ("cyno"), mouse, and rat. FIG. **46** discloses SEQ ID NOS 839-841 and 828, respectively, in order of appearance.

[0089] FIG. **47** illustrates the co-crystallization configuration of the EphA2 protein and the AB-008873 antibody. It indicates that the HCDRs changes conformation upon binding to the EphA2 protein: the disulfide loop in HCDR3 shows substantial rotation as compared to the structure that is not in contact with EphA2.

[0090] FIG. **48** shows measurements of tumor growth in mice inoculated with CT26 cells upon treatment of the AB-008873-IL15.

[0091] FIG. **49** shows measurements of tumor growth in mice inoculated with CT26 cells upon treatment of a bispecific antibody comprising scFv of the anti-4-1BB antibody 9365 and AB-010361.

[0092] FIG. **50**A compares the cytotoxicity of AB-008873, AB-010361, AB-010699, and AB-010018 on A549 cells. FIG. **50**B compares the cytotoxicity of AB-010699, AB-010695, AB-010363, AB-010699, and Cetuximab on A549 cells.

[0093] FIG. **51** compares the cytotoxicity of AB-008873, AB-010361, and AB-010699 in an ADC assay.

[0094] FIG. **52** compares the cytotoxicity of AB-010361 ("AB-010361-15"), AB-010018 ("AB-010018-1"), and AB-010685 ("AB-010685-2") on PC3-KILR cells in an ADCC assay.

[0095] FIG. **53** shows representative anti-CD3 bispecific antibody constructs.

[0096] FIG. **54** shows measurements of tumor growth of NSG-DKO mice carrying PC3 tumor cells upon treatment of a bispecific antibody comprised of AB-010361 and an anti-CD3 binding arm.

[0097] FIG. **55**A-D show cytotoxicity results of antibody-drug conjugates, each comprising an EphA2 antibody, AB-010361 (IgG1), AB-010361 (IgG4), AB-10699 (IgG1), or AB-010671 (IgG1), directly conjugated to a ZymeLink Auristatin (ZLA) payload. The cytotoxicity on SKOV3, MDAM1B231, and PC3 cells are shown in FIG. **55**A-C and EC_{50} values are reported in FIG. **55**D.

[0098] FIG. **56**A-D show measurements of tumor growth of NPG mice carrying PC3 tumor cells upon treatment of antibody-drug conjugates each comprising an EphA2 antibody and a ZLA payload.

[0099] FIG. **57** shows endpoint tumor weight analysis of NPG mice carrying tumors established from PC3 cells upon treatment of ZLA-conjugated EphA2 antibodies.

[0100] FIG. **58**A-**58**D show the body weight analysis of NPG mice that carrying tumors established from PC3 cells upon treatment of ZLA-conjugated EphA2 antibodies.

[0101] FIG. **59**A shows cytotoxicity results on PC3 cells of AB-10699-IgG1, AB-10699-IgG4, AB-010671-IgG1, and AB-010671-IgG4 directly conjugated to a ZymeLink Auristatin (ZLA) payload. The IC_{50} values of the study are reported in FIG. **59**B. IgG1 and IgG4 isotype antibodies are included as controls.

[0102] FIG. **60**A shows measurements of tumor growth of NPG mice carrying PC3 tumor cells measuring ~150 mm^3 upon administration of AB-10699-IgG1, AB-10699-IgG4, AB-010671-IgG1, and AB-010671-IgG4 directly conjugated to a ZymeLink Auristatin (ZLA). FIG. **60**B shows the body weight of the mice over the course of the study. IgG1 and IgG4 isotype antibodies are included as controls.

[0103] FIG. **61**A shows measurements of tumor growth of NPG mice carrying PC3 tumor cells measuring ~400 mm^3 upon administration of AB-10699-IgG1, AB-10699-IgG4, AB-010671-IgG1, and AB-010671-IgG4 directly conjugated to a ZymeLink Auristatin (ZLA). FIG. **61**B shows the body weight of the mice over the course of the study. IgG1 and IgG4 isotype antibodies are included as controls.

DETAILED DESCRIPTION

[0104] This application incorporates the entire disclosure of PCT publication WO 2022/187710 by reference for all purposes.

[0105] As used in herein, the singular forms "a," "an," and "the" include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to "an antibody" optionally includes a combination of two or more such molecules, and the like.

[0106] The term "about," as used herein, refers to the usual error range for the respective value readily known to the skilled person in this technical field, for example ±20%, ±10%, or ±5%, are within the intended meaning of the recited value.

[0107] As used herein, the term "antibody" means an isolated or recombinant binding agent that comprises the necessary variable region sequences to specifically bind an antigenic epitope. Therefore, an "antibody" as used herein is any form of antibody of any class or subclass or fragment thereof that exhibits the desired biological activity, e.g., binding a specific target antigen. Thus, it is used in the broadest sense and specifically covers a monoclonal antibody (including full-length monoclonal antibodies), human antibodies, chimeric antibodies, nanobodies, diabodies, multispecific antibodies (e.g., bispecific antibodies), and antibody fragments including but not limited to scFv, Fab, and the like so long as they exhibit the desired biological activity.

[0108] "Antibody fragments" comprise a portion of an intact antibody, for example, the antigen-binding or variable region of the intact antibody. Examples of antibody fragments include Fab, Fab', $F(ab')_2$, and Fv fragments; diabodies; linear antibodies (e.g., Zapata et al., Protein Eng. 8(10): 1057-1062 (1995)); single-chain antibody molecules (e.g., scFv); and multispecific antibodies formed from antibody fragments. Papain digestion of antibodies produces two identical antigen-binding fragments, called "Fab" fragments, each with a single antigen-binding site, and a residual "Fc" fragment, a designation reflecting the ability to crystallize readily. Pepsin treatment yields an $F(ab')_2$ fragment with two antigen combining sites and is still capable of cross-linking antigen.

[0109] As used herein, the term "targets a tumor," or "tumor-targeting," with respect to an antibody, refers to an antibody that binds preferentially to a tumor tissue than normal tissue. In some embodiments, the normal tissue is the tissue that is adjacent to the tumor, referred to as tumor-adjacent tissue or TAT. In some embodiments, a tumor targeting antibody also decreases the rate of tumor growth, tumor size, invasion, and/or metastasis, via direct or indirect effects on tumor cells.

[0110] As used herein, "V-region" refers to an antibody variable region domain comprising the segments of Framework 1, CDR1, Framework 2, CDR2, and Framework 3, including CDR3 and Framework 4. The heavy chain V-region, V_H, is a consequence of rearrangement of a V-gene (HV), a D-gene (HD), and a J-gene (HJ), in what is termed V(D)J recombination during B-cell differentiation. The light chain V-region, V_L, is a consequence of rearrangement of a V-gene (LV) and a J-gene (LJ).

[0111] As used herein, "complementarity-determining region (CDR)" refers to the three hypervariable regions (HVRs) in each chain that interrupt the four "framework" regions established by the light and heavy chain variable regions. The CDRs are the primary contributors to binding to an epitope of an antigen. The CDRs of each chain are referred to as CDR1, CDR2, and CDR3, numbered sequentially starting from the N-terminus, and are also identified by the chain in which the CDR is located. Thus, for example, a V_H CDR3 (HCDR3) is in the variable domain of the heavy chain of the antibody in which it is found, whereas a V_L CDR3 (LCDR3) is the CDR3 from the variable domain of the light chain of the antibody in which it is found. The term "CDR" is used interchangeably with "HVR" in this application when referring to CDR sequences.

[0112] The amino acid sequences of the CDRs and framework regions can be determined using various well-known definitions in the art, e.g., Kabat, Chothia, international ImMunoGeneTics database (IMGT), and AbM (see, e.g., Chothia & Lesk, 1987, Canonical structures for the hypervariable regions of immunoglobulins. J. Mol. Biol. 196, 901-917; Chothia C. et al., 1989, Conformations of immunoglobulin hypervariable regions. Nature 342, 877-883; Chothia C. et al., 1992, structural repertoire of the human V_H segments J. Mol. Biol. 227, 799-817; Al-Lazikani et al., J. Mol. Biol. 1997, 273(4)). Definitions of antigen combining sites are also described in the following: Ruiz et al., IMGT, the international ImMunoGeneTics database. Nucleic Acids Res., 28, 219-221 (2000); and Lefranc, M.-P. IMGT, the international ImMunoGeneTics database. Nucleic Acids Res. January 1; 29(1):207-9 (2001); MacCallum et al, Antibody-antigen interactions: Contact analysis and binding site topography, J. Mol. Biol., 262 (5), 732-745 (1996); and Martin et al, Proc. Natl Acad. Sci. USA, 86, 9268-9272 (1989); Martin et al, Methods Enzymol., 203, 121-153, (1991); Pedersen et al, Immunomethods, 1, 126, (1992); and Rees et al, In Sternberg M. J. E. (ed.), Protein Structure Prediction. Oxford University Press, Oxford, 141-172 1996). Reference to CDRs as determined by Kabat numbering are based, for example, on Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institute of Health, Bethesda, MD (1991)). Chothia CDRs are determined as defined by Chothia (see, e.g., Chothia and Lesk J. Mol. Biol. 196:901-917 (1987)).

[0113] CDRs as shown in Tables 6 and 7 are defined by IMGT and Kabat. The V_H CDRs as listed in Table 6, are defined as follows: HCDR1 is defined by combining Kabat and IMGT; HCDR2 is defined by Kabat; and the HCDR3 is defined by IMGT. The V_L CDRs as listed in Table 7 are defined by Kabat. FIG. **39**A-**39**B shows alignment of EphA2 antibody V_H and V_L sequences with CDRs designated by Kabat and IMGT. As known in the art, numbering and placement of the CDRs can differ depending on the numbering system employed. It is understood that disclosure of a variable heavy and/or variable light sequence includes the disclosure of the associated CDRs, regardless of the numbering system employed.

[0114] A "nonlinear epitope" or "conformational epitope" comprises noncontiguous polypeptides (or amino acids) within the antigenic protein to which an antibody specific to the epitope binds. A conformational epitope is typically formed by a three-dimensional interaction of amino acids in the epitope that may not necessarily be contained in a single stretch of amino acids.

[0115] An "Fc region" refers to the constant region of an antibody excluding the first constant region immunoglobulin domain. Thus, e.g., for human immunoglobulins, "Fc" refers to the last two constant region immunoglobulin domains of IgA, IgD, and IgG, and the last three constant region immunoglobulin domains of IgE and IgM, and the flexible hinge N-terminal to these domains. For IgA and IgM Fc may include the J chain. For IgG, Fc comprises immunoglobulin domains Cγ2 and Cγ3 and the hinge between Cγ1 and Cγ2. It is understood in the art that the boundaries of the Fc region may vary, however, the human IgG heavy chain Fc region is usually defined to comprise residues C226 or P230 to its carboxyl-terminus, using the numbering according to the EU index as in Kabat et al. (1991, NIH Publication 91-3242, National Technical Information Service, Springfield, Va.). The term "Fc region" may refer to this region in isolation or this region in the context of an antibody or antibody fragment. "Fc region" includes naturally occurring allelic variants of the Fc region as well as modified Fc regions, e.g., that are modified to modulate effector function or other properties such as pharmacokinetics, stability, or production properties of an antibody. Fc regions also include variants that do not exhibit alterations in biological function. For example, one or more amino acids can be deleted from the N-terminus or C-terminus of the Fc region of an immunoglobulin without substantial loss of biological function. Such variants can be selected according to general rules known in the art to have minimal effect on activity (see, e.g., Bowie et al., *Science* 247:306-1310, 1990). For example, for IgG4 antibodies, a single amino acid substitution (S228P according to Kabat numbering; designated IgG4Pro) may be introduced to abolish the heterogeneity observed in recombinant IgG4 antibody (see, e.g., Angal et al., *Mol Immunol* 30:105-108, 1993).

[0116] An "EC_{50}" as used herein in the context of a binding or functional assay, refers to the half maximal effective concentration, which is the concentration of an antibody that induces a response (readouts can include but are not limited to fluorescence or luminescence signals) halfway between the baseline and maximum after a specified exposure time. In some embodiments, the "fold over EC_{50}" is determined by dividing the EC_{50} of a reference antibody by the EC_{50} of the test antibody.

[0117] The term "equilibrium dissociation constant" abbreviated (K_D), refers to the dissociation rate constant (k_d, time^{-1}) divided by the association rate constant (k_a, time^{-1} M^{-1}). Equilibrium dissociation constants can be measured using any method. Thus, in some embodiments antibodies of the present disclosure have a K_D of less than about 50 nM, typically less than about 25 nM, or less than 10 nM, e.g., less than about 5 nM or than about 1 nM and often less than about 10 nM as determined by surface plasmon resonance analysis using a biosensor system such as a Biacore® system performed at 37° C. In some embodiments, an antibody of the present disclosure has a K_D of less than 5×10^{-5} M, less than 10^{-5} M, less than 5×10^{-6}M, less than 10^{-6}M, less than 5×10^{-7} M, less than 10^{-7} M, less than 5×10^{-8}M, less than 10^{-8} M, less than 5×10^{-9} M, less than 10^{-9} M, less than 5×10^{-10} M, less than 10^{-10} M, less than 5×10^{-11} M, less than 10^{-11} M, less than 5×10^{-12} M, less than 10^{-12} M, less than 5×10^{-13} M, less than 10^{-13} M, less than 5×10^{-14} M, less than 10^{-14} M, less than 5×10^{-15} M, or less than 10^{-15} M or lower as measured as a bivalent antibody. In the context of the present disclosure, an "improved" K_D refers to a lower K_D. In some embodiments, an antibody of the present disclosure has a K_D of less than 5×10^{-5} M, less than 10^{-5} M, less than 5×10^{-6}M, less than 10^{-6} M, less than 5×10^{-7} M, less than 10^{-7} M, less than 5×10^{-8} M, less than 10^{-8} M, less than 5×10^{-9} M, less than 10^{-9} M, less than 5×10^{-10} M, less than 10^{-10} M, less than 5×10^{-11} M, less than 10^{-11} M, less than 5×10^{-12} M, less than 10^{-12} M, less than 5×10^{-13} M, less than 10^{-13} M, less than 5×10^{-14} M, less than 10^{-14} M, less than 5×10^{-15} M, or less than 10^{-15} M or lower as measured as a monovalent antibody, such as a monovalent Fab. In some embodiments, an EphA2 antibody of the present disclosure has K_D less than 100 pM, e.g., or less than 75 pM, e.g., in the range of 1 to 100 pM, when measured by surface plasmon resonance analysis using a biosensor system such as a Biacore® system performed at 37° C. In some embodiments, an EphA2 antibody of the present disclosure has K_D of greater than 100 pM, e.g., in the range of 100-1000 pM or 500-1000 pM when measured by surface plasmon resonance analysis using a biosensor system such as a Biacore® system performed at 37° C. The term "EphA2 antibody" is used interchangeably with "anti-EphA2 antibody" in this application.

[0118] The term "monovalent molecule" as used herein refers to a molecule with one antigen-binding site, e.g., a Fab or scFv.

[0119] The term "bivalent molecule" as used herein refers to a molecule with two antigen-binding sites. In some embodiments, a bivalent molecule of the present disclosure is a bivalent antibody or a bivalent fragment thereof. In some embodiments, a bivalent molecule of the present disclosure is a bivalent antibody. In some embodiments, a bivalent molecule of the present disclosure is an IgG. In general, monoclonal antibodies have a bivalent basic structure. IgG and IgE have only one bivalent unit, while IgA and IgM consist of multiple bivalent units (2 and 5, respectively) and thus have higher valencies. This bivalency increases the avidity of antibodies for antigens.

[0120] The terms "monovalent binding" or "monovalently binds to" as used herein refer to the binding of one antigen-binding site to its antigen.

[0121] The terms "bivalent binding" or "bivalently binds to" as used herein refer to the binding of both antigen-binding sites of a bivalent molecule to its antigen. In some embodiments, both antigen-binding sites of a bivalent molecule share the same antigen specificity.

[0122] The term "valency" as used herein refers to the number of different binding sites of an antibody for an antigen. A monovalent antibody comprises one binding site for an antigen. A bivalent antibody comprises two binding sites for the same antigen. A multivalent antibody comprises two or more binding sites for the same antigen. A trivalent antibody comprises three binding sites for the same antigen. A tetravalent antibody comprises four binding sites for the same antigen.

[0123] The term "avidity" as used herein in the context of antibody binding to an antigen refers to the combined binding strength of multiple binding sites of the antibody. Thus, "bivalent avidity" refers to the combined strength of two binding sites.

[0124] The terms "identical" or percent "identity," in the context of two or more polypeptide sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues that are the same (e.g., at least 70%, at least 75%, at least 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or higher) identity over a specified region, e.g., the length of the two sequences, when compared and aligned for maximum correspondence over a comparison window or designated region. Alignment for purposes of determining percent amino acid sequence identity can be performed in various methods, including those using publicly available computer software such as BLAST. BLAST-2, ALIGN or Megalign (DNASTAR) software. Examples of algorithms that are suitable for determining percent sequence identity and sequence similarity the BLAST 2.0 algorithms, which are described in Altschul et al., Nuc. Acids Res. 25:3389-3402 (1977) and Altschul et al., J. Mol. Biol. 215:403-410 (1990). Thus, for purposes of this invention, BLAST 2.0 can be used with the default parameters to determine percent sequence identity.

[0125] The terms "corresponding to," "determined with reference to," or "numbered with reference to" when used in the context of the identification of a given amino acid residue in a polypeptide sequence, refers to the position of the residue of a specified reference sequence when the given amino acid sequence is maximally aligned and compared to the reference sequence. The polypeptide that is aligned to the reference sequence need not be the same length as the reference sequence.

[0126] A "conservative" substitution as used herein refers to a substitution of an amino acid such that charge, polarity, hydropathy (hydrophobic, neutral, or hydrophilic), and/or size of the side group chain is maintained. Illustrative sets of amino acids that may be substituted for one another include (i) positively-charged amino acids Lys and Arg; and His at pH of about 6; (ii) negatively charged amino acids Glu and Asp; (iii) aromatic amino acids Phe. Tyr and Trp; (iv) nitrogen ring amino acids His and Trp; (v) aliphatic hydrophobic amino acids Ala, Val, Leu and Ile; (vi) hydrophobic sulfur-containing amino acids Met and Cys, which are not as hydrophobic as Val, Leu, and Ile; (vii) small polar uncharged amino acids Ser, Thr, Asp, and Asn (viii) small hydrophobic or neutral amino acids Gly, Ala, and Pro; (ix) amide-comprising amino acids Asn and Gln; and (xi) beta-branched amino acids Thr, Val, and Ile. Reference to the charge of an amino acid in this paragraph refers to the charge at pH 6-7.

[0127] The terms "nucleic acid" and "polynucleotide" are used interchangeably and as used herein refer to both sense and anti-sense strands of RNA, cDNA, genomic DNA, and synthetic forms and mixed polymers of the above. In particular embodiments, a nucleotide refers to a ribonucleotide, deoxynucleotide or a modified form of either type of nucleotide, or combinations thereof. The terms also include, but is not limited to, single- and double-stranded forms of DNA. In addition, a polynucleotide, e.g., a cDNA or mRNA, may include either or both naturally occurring and modified nucleotides linked together by naturally occurring and/or non-naturally occurring nucleotide linkages. The nucleic acid molecules may be modified chemically or biochemically or may contain non-natural or derivatized nucleotide bases, as will be readily appreciated by those of skill in the art. Such modifications include, for example, labels, methylation, substitution of one or more of the naturally occurring nucleotides with an analogue, internucleotide modifications such as uncharged linkages (e.g., methyl phosphonates, phosphotriesters, phosphoramidates, carbamates, and the like), charged linkages (e.g., phosphorothioates, phosphorodithioates, and the like), pendent moieties (e.g., polypeptides), intercalators (e.g., acridine, psoralen, and the like), chelators, alkylators, and modified linkages (e.g., alpha anomeric nucleic acids, and the like). The above term is also intended to include any topological conformation, including single-stranded, double-stranded, partially duplexed, triplex, hairpinned, circular and padlocked conformations. A reference to a nucleic acid sequence encompasses its complement unless otherwise specified. Thus, a reference to a nucleic acid molecule having a particular sequence should be understood to encompass its complementary strand, with its complementary sequence. The term also includes codon-optimized nucleic acids that encode the same polypeptide sequence.

[0128] The term "vector," as used herein, refers to a nucleic acid molecule capable of propagating another nucleic acid to which it is linked. The term includes the vector as a self-replicating nucleic acid structure as well as the vector incorporated into the genome of a host cell into which it has been introduced. A "vector" as used here refers to a recombinant construct in which a nucleic acid sequence of interest is inserted into the vector. Certain vectors can direct the expression of nucleic acids to which they are operatively linked. Such vectors are referred to herein as "expression vectors".

[0129] A "substitution," as used herein, denotes the replacement of one or more amino acids or nucleotides by different amino acids or nucleotides, respectively.

[0130] An "isolated" nucleic acid refers to a nucleic acid molecule that has been separated from a component of its natural environment. An isolated nucleic acid includes a nucleic acid molecule contained in cells that ordinarily contain the nucleic acid molecule, but the nucleic acid molecule is present extrachromosomally or at a chromosomal location that is different from its natural chromosomal location.

[0131] "Isolated nucleic acid encoding an antibody or fragment thereof" refers to one or more nucleic acid molecules encoding antibody heavy or light chains (or fragments thereof), including such nucleic acid molecule(s) in a single vector or separate vectors, and such nucleic acid molecule(s) present at one or more locations in a host cell.

[0132] The terms "host cell," "host cell line," and "host cell culture" are used interchangeably and refer to cells into which exogenous nucleic acid has been introduced, including the progeny of such cells. Thus, a host cell can be a recombinant host cell, a primary transformed cell and progeny derived therefrom without regard to the number of passages.

[0133] A polypeptide "variant," as the term is used herein, is a polypeptide that typically differs from one or more polypeptide sequences specifically disclosed herein in one or more substitutions, deletions, additions and/or insertions.

[0134] The term "cancer cell" or "tumor cell" as used herein refers to a neoplastic cell. The term includes cells from tumors that are benign as well as malignant. Neoplastic transformation is associated with phenotypic changes of the tumor cell relative to the cell type from which it is derived. The changes can include loss of contact inhibition, morphological changes, and unregulated cell growth,

[0135] "Inhibiting growth of a tumor" and "inhibiting growth of a cancer" as used interchangeably and refer to slowing growth and/or reducing the cancer cell burden of a patient who has cancer. "Inhibiting growth of a cancer" thus includes killing cancer cells, as well as decreasing the rate of tumor growth, tumor size, invasion, and/or metastasis by direct or indirect effects on tumor cells.

[0136] As used herein, "therapeutic agent" refers to an agent that will cure, or at least partially arrest the symptoms of a disease and complications associated with the disease when administered to a patient suffering from the disease in a therapeutically effective dose.

[0137] As used herein, the term "a tumor overexpressing EphA2", or "a tumor that overexpresses EphA2", or "a cancer overexpressing EphA2" or "a cancer that overexpresses EphA2", refers to a tumor or cancer that expresses EphA2 or demonstrates EpA2 reactivity at a level that higher than the level of EphA2 expressed or EphA2 reactivity in normal tissue (e.g., tumor adjacent tissues or TAT). In certain embodiments, a tumor or cancer that overexpresses EphA2 is one that expresses EphA2 at a level that is at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 80%, or at 100% higher than the normal tissue (e.g., tumor adjacent tissues or TAT). In certain embodiments, a tumor or cancer that overexpresses EphA2 is one that demonstrates EphA2 reactivity at level that is at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 80%, or at 100% higher than the normal tissue (e.g., tumor adjacent tissues or TAT).

[0138] In some aspects, the disclosure additionally provides methods of identifying subjects who are candidates for treatment with an EphA2 antibody having tumor-targeting effects. Thus, in one embodiment, the invention provides a method of identifying a patient who can benefit from treatment with an EphA2 antibody of the present disclosure. In one embodiment, the patient has tumor that expresses EphA2. In one embodiment, the patient has tumor that overexpresses EphA2. In some embodiments, the tumor sample is from a primary tumor. In alternative embodiments, the tumor sample is a metastatic lesion. Binding of antibody to tumor cells through a binding interaction with the EphA2 can be measured using any assay, such as immunohistochemistry or flow cytometry. In some embodiments, binding of antibody to at least 0.2%, 0.5%, or 1%, or at least 5% or 10%, or at least 20%, 30%, or 50%, of the tumor cells in a sample may be used as a selection criterion for determining a patient to be treated with an EphA2 antibody as described herein. In other embodiments, analysis of components of the blood, e.g., circulating protein levels, is used to identify a patient whose tumor cells are overexpressing EphA2.

[0139] An EphA2 antibody disclosed herein can be used to treat several different cancers. In some embodiments, a cancer patient who can benefit from the treatment of the EphA2 antibody has a cancer expressing EphA2. In some embodiments, a cancer patient who can benefit from the treatment of the EphA2 antibody has a cancer overexpressing EphA2. In some embodiments, the cancer is a carcinoma or a sarcoma.

[0140] As used herein, the term "an antibody binds to an EphA2" or "an antibody binds to EphA2," means that the antibody binds to EphA2 under permissible conditions (e.g., in a suitable buffer), and the detected signal resulted from the binding is at least 2-fold, at least 5-fold, at least 10 fold, at least 20 fold, at least 100 fold, at least 150 fold, or at least 200 fold above a reference level. In some embodiments, the reference level is a detected signal produced by contacting a control antibody with the EphA2, or by contacting the antibody with a control protein.

[0141] As used herein, the term "sibling antibodies" refer to antibodies derived from the same B-cell clonal lineage. In some embodiments, sibling antibodies share sequence and structural properties as well as epitope specificity.

[0142] As used herein, the term "convergent antibodies" refer to that antibodies derived from different B-cell lineages that exhibit similar sequence and structural properties.

[0143] Several anti-EphA2 comparator antibodies are referenced in this disclosure. AB-010018 is a mouse monoclonal antibody precursor of clinical candidate humanized DS-8895a (Daiichi Sankyo Company, Ltd.), as disclosed in U.S. Pat. No. 9,150,657, SEQ ID NO: 35 and 37, Hasegawa J. et al., Cancer Biol Ther. 2016 November; 17(11):1158-1167. doi: 10.1080/15384047.2016.1235663. Epub 2016 Sep. 21; Shitara K. et al., J Immunother Cancer, 2019 Aug. 14; 7(1):219. doi: 10.1186/s40425-019-0679-9)),

[0144] AB-010016 is an anti-EphA2 antibody generated based on clinical candidate Medi-547 (MedImmune, Inc.) It was generated using the V_H and V_L amino acid sequences from Protein Data Bank code 3SKJ, Peng, L., et. al, (2011) J Mol Biol 413: 390-405.

[0145] AB-010017 is an anti-EphA2 antibody generated based on clinical candidate MM-310 (Merrimack, Inc.) It was generated using V_H and V_L amino acid sequences from the scFv of SEQ ID NO: 40 in US20190070113A1; See also, Geddie et al., MAbs. 2017 January; 9(1): 58-67; Kamoun et al., Nat Biomed Eng. 2019 April; 3(4):264-280. doi: 10.1038/s41551-019-0385-4. Epub 2019 Apr. 5.)

Overview

[0146] This application relates to tumor-targeting antibodies that bind to EphA2 and bind preferentially to tumor tissues than normal tissues. In some embodiments, the EphA2 antibody binds to one or more epitopes in the FN2 domain of EphA2. The EphA2 antibodies disclosed herein can target a tumor in a variety of ways, for example, it can mediate antibody-dependent cell-mediated cytotoxicity (ADCC) and/or ADCP to target and kill tumor cells. These antibodies also demonstrated significant cytotoxicity towards tumor cells and show therapeutic potential in treating cancers. Some of the EphA2 antibodies can target both mouse tumor and human tumor cells. The EphA2 antibodies are also useful in detecting tumors suitable for treatment in diagnostic applications.

[0147] As compared to existing clinical candidates and commercial EphA2 antibodies, EphA2 antibodies provided in this disclosure showed a higher binding affinity to various tumors and lower binding to normal tissues (e.g., tumor adjacent tissues) and exhibit differential binding profile on various cancer tumor types. Unlike existing clinical candidates and commercial EphA2 antibodies, the EphA2 antibodies disclosed herein exhibit reduced agonistic or antagonistic effects on the EphA2 receptor. These properties render them ideal for use as cancer therapeutics.

EPHA2

[0148] Ephrin receptor A2 (EphA2) is expressed mainly in proliferating epithelial cells in adults. Under normal conditions, EphA2 interacts with membrane proteins, ephrin A1, on the neighboring cell and induce diverse signaling networks following cell-to-cell contact. For example, EphA2 may decrease in cell-extracellular matrix (ECM) attachment upon phosphorylation.

[0149] EphA2 is overexpressed in many solid tumors including melanoma, glioma, prostate cancer, breast cancer, ovarian cancer, lung cancer, colon cancer, esophageal cancer, gastric cancer cervical cancer, bladder cancer. EphA2 contributes to cell adhesion and angiogenesis, and its expression is also linked to increased malignancy and poor prognosis.

[0150] Human EphA2 has a sequence of

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                                      (SEQ ID NO: 94)
MELQAARACFALLWGCALAAAAAAQGKEVVLLDFAAAGGELGWLTHPYG

KGWDLMQNIMNDMPIYMYSVCNVMSGDQDNWLRTNWVYRGEAERIFIEL

KFTVRDCNSFPGGASSCKETFNLYYAESDLDYGTNFQKRLFTKIDTIAP

DEITVSSDFEARHVKLNVEERSVGPLTRKGFYLAFQDIGACVALLSVRV

YYKKCPELLQGLAHFPETIAGSDAPSLATVAGTCVDHAVVPPGGEEPRM

HCAVDGEWLVPIGQCLCQAGYEKVEDACQACSPGFFKFEASESPCLECP

EHTLPSPEGATSCECEEGFFRAPQDPASMPCTRPPSAPHYLTAVGMGAK

VELRWTPPQDSGGREDIVYSVTCEQCWPESGECGPCEASVRYSEPPHGL

TRTSVTVSDLEPHMNYTFTVEARNGVSGLVTSRSFRTASVSINQTEPPK

VRLEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVRRTEGF

SVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTLSPEGSGNLAVIG

GVAVGVVLLLVLAGVGFFIHRRRKNQRARQSPEDVYFSKSEQLKPLKTY

VDPHTYEDPNQAVLKFTTEIHPSCVTRQKVIGAGEFGEVYKGMLKTSSG

KKEVPVAIKTLKAGYTEKQRVDFLGEAGIMGQFSHHNIIRLEGVISKYK

PMMIITEYMENGALDKFLREKDGEFSVLQLVGMLRGIAAGMKYLANMNY

VHRDLAARNILVNSNLVCKVSDFGLSRVLEDDPEATYTTSGGKIPIRWT

APEAISYRKFTSASDVWSFGIVMWEVMTYGERPYWELSNHEVMKAINDG

FRLPTPMDCPSAIYQLMMQCWQQERARRPKFADIVSILDKLIRAPDSLK

TLADFDPRVSIRLPSTSGSEGVPFRTVSEWLESIKMQQYTEHFMAAGYT

AIEKVVQMTNDDIKRIGVRLPGHQKRIAYSLLGLKDQVNTVGIPI
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[0151] Human EphA2 comprises: a signal peptide having the sequence of:

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                                      (SEQ ID NO: 96)
MELQAARACFALLWGCALAAAAAA;
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a ligand binding domain having the sequence of

(SEQ ID NO: 97)
QGKEVVLLDFAAAGGELGWLTHPYGKGWDLMQNIMNDMPIYMYSVCNVM

SGDQDNWLRTNWVYRGEAERIFIELKFTVRDCNSFPGGASSCKETFNLY

YAESDLDYGTNFQKRLFTKIDTIAPDEITVSSDFEARHVKLNVEERSVG

PLTRKGFYLAFQDIGACVALLSVRVYYK;

a sushi domain having the sequence of

(SEQ ID NO: 98)
KCPELLQGLAHFPETIAGSDAPSLATVAGTCVDHAVVPPGGEEPRMHCA
VDGEWLVPIGQCL;

an EGF domain having the sequence of

(SEQ ID NO: 99)
CQAGYEKVEDACQACSPGFFKFEASESPCLECPEHTLPSPEGATSCECE
EGFFRAPQDPASMPCT;

an FN1 domain having the sequence of

(SEQ ID NO: 100)
RPPSAPHYLTAVGMGAKVELRWTPPQDSGGREDIVYSVTCEQCWPESGE
CGPCEASVRYSEPPHGLTRTSVTVSDLEPHMNYTFTVEARNGVSGLVTS
RSFRTASVSINQ;
and

an FN2 domain having the sequence of

(SEQ ID NO: 95)
TEPPKVRLEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVR
RTEGFSVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTLSPEGSGN

[0152] EPHA2 comprises an ectodomain, an intracellular region, and a single transmembrane helix. The ectodomain of EphA2, consisting of amino acid residues 23-546 using SEQ ID NO: 94 as a reference, comprises a ligand binding domain (LBD), which interacts with ephrin A1, a Sushi domain, an epidermal growth factor like domain, and two fibronectin type III domains (FN1 and FN2). The intracellular region comprises a tyrosine kinase domain and a sterile α-motif domain. The transmembrane helix of the protein is flanked by juxtamembrane linkers, which connects the ectodomain and the intracellular domain. A membrane-binding motif FN2, located within amino acid residues 437-534 of SEQ ID NO: 94, includes positively charged residues, which can recruit negatively charged lipids to the site of membrane-protein interaction. The interactions of FN2 with lipids stabilize the otherwise flexible EphA2 ectodomain in two main conformations relative to the membrane.

[0153] The EphA2 ectodomain is highly conserved across species. The ectodomain of human EphA2 has sequence identity of 90.8%, 90.4%, 98.8%, and 98.8% to mouse, rat, macaque, and cyno, respectively.

EPHA2 Antibodies

[0154] For purpose of this disclosure, an EphA2 antibody may be any one of AB-008873, the siblings and variants thereof, as further described below.

[0155] AB-008873 was discovered in antibody repertoires generated by Immune Repertoire Capture® (IRC™) technology from plasmablast B cells isolated from a non-small cell lung cancer patient with an active anti-tumor immune response after treatment with the anti-PD-1 antibody OPDTVO® (nivolumab) (Bristol Myers Squibb). The IRC™ technology and its use in antibody discovery is well known and disclosed in, e.g., WO 2012148497A2, the entire content of which is herein incorporated by reference. AB-008873 aligns to its closest human germline genes (IGHV4-34*02, IGHD2-8*01, IGHJ3*02, IGLV3-21*02, and IGLJ2*01), and to its four known siblings AB-009805, AB-009806. AB-009807, and AB-009808. Structures of these antibodies are further disclosed below, e.g., in Tables 6-8.

Variants

[0156] In some embodiments, variants of any of the EphA2 antibodies disclosed herein can be generated by introducing mutations to the heavy chain and/or light chain sequences. In some embodiments, the mutation(s) are introduced into one or more of the CDRs of an EphA2 antibody disclosed herein, e.g., AB-008873, AB-010148, or AB-010363. In some embodiments, the mutation(s) are introduced in the framework regions. In some embodiments, an EphA2 antibody provided herein comprises a V_H region of any one of SEQ ID NO 67-77 and 591-655, and/or a V_L region of any one of SEQ ID NO: 78-88 and 656-720, or an antibody comprising a V_H region with at least 80% identity to any one of SEQ ID NO 67-77 and 591-655 and a V_L region having at least 80% identity to any one of SEQ ID NO: 78-88 and 656-720, with variations to the corresponding V_H or V_L regions present only in Framework regions.

[0157] In some embodiments, EphA2 antibody provided herein comprises an HCDR1 of any one of SEQ ID NOS: 1-11, and 201-265, an HCDR2 of any one of SEQ ID NOS:12-22 and 266-330, an HCDR3 of any one of SEQ ID NOS:23-33 and 331-395, an LCDR1 of any one of SEQ ID NOS:34-44 and 396-460, an LCDR2 of any one of SEQ ID NOS:45-55 and 461-525, an LCDR3 of any one of SEQ ID NOS:56-66 and 526-590; and the FW regions in the V_H region are at least 80% identical to the FW regions present in the V_H region of SEQ ID NO 67-77 and 591-655, and wherein the FW regions in the V_L region are at least 80% identical to the FW regions present in the V_L region of SEQ ID NO: 78-88 and 656-720

[0158] In some embodiments, a variant is engineered to be as much like self as possible to minimize immunogenicity. One approach to do so is to identify a close germline sequence and mutate one of the EphA2 antibodies at as many mismatched positions (also known as "germline deviations") to the germline residue type as possible. In some embodiments, a variant (e.g., AB-009812) comprises a mutation H54RS relative to AB-008873 (heavy chain position 54 mutated from arginine to serine). In some embodiments, a variant (e.g., AB-009813) comprises a mutation H54RA relative to AB-008873 (heavy chain position 54 mutated from arginine to alanine). In some embodiments, a variant (e.g., AB-009814) comprises deletion of three residues, H61N-H62Y-H63N, relative to AB-008873. In some embodiments, a variant comprises one or more mutations selected from the group consisting of H54RS, H54RA, and deletions of three residues H61N-H62Y-H63N, relative to the sequence of the heavy chain variable region of AB-008873 (SEQ ID NO: 67). Variant antibodies AB-009812, AB-009813, and AB-009814 are described in Tables 1-3.

[0159] Additional variants antibodies may also be generated by introducing one or more mutations to any one of the AB-008873 and its siblings. In some embodiments, a variant comprises one or more mutations selected from the group consisting of H31DG, L31NS, L97LV, H51VI, H86TS, H129AS, and L60QR relative to AB-009815. Each using the Hv and Lv sequences of the AB-009815 (SEQ ID NO: 75 and 86) as references. For any the mutation disclosed in this application, the name indicates whether the mutation is a heavy chain or light chain, the position in the heavy chain or light chain of the mutation, the amino acid residue at the position before introduction of the mutation, and the amino acid at the position after introduction of the mutation. For example, H129AS, refers to that the alanine at heavy chain position 129 is mutated to a serine. Table 1 shows exemplary variants of AB-009815.

TABLE 1

Exemplary variants of AB-009815

Antibody ID	Mutations relative to AB-009815
AB-010141	H31DG
AB-010142	L31NS
AB-010143	L97LV
AB-010144	H51VI
AB-010145	H86TS
AB-010146	H129AS
AB-010147	L60QR
AB-010148	H51VI; H86TS
AB-010149	H51VI; H129AS
AB-010150	H86TS; H129AS
AB-010151	H51VI, H86TS, H129AS
AB-010152	H51VI, H86TS, H129AS, L60QR

[0160] Further mutations can be introduced to any one of the AB-008873 variants listed in Table 1. For example, one or more mutations are introduced to AB-0010148 and exemplary variants resulted from the further mutagenesis are shown in in Table 2, below.

TABLE 2

Exemplary AB-0010148 variants

Antibody ID	Mutations relative to AB-0010148
AB-010357	L97LV
AB-010358	L60QR
AB-010359	L97LV; L60QR
AB-010360	H31DG
AB-010361	H31DG; L97LV
AB-010362	H31DG; L60QR
AB-010363	H31DG; L60QR; L97LV
AB-010364	H31DG; H54RA; L97LV
AB-010365	H31DG; H54RA; L97LV; L60QR
AB-010366	H31DG; H54RQ; L97LV
AB-010367	H31DG; H54RQ; L97LV; L60QR

[0161] Further mutations can be introduced to any one of the AB-0010148 variants listed in Table 2. For example, one or more mutations are introduced to AB-010357, AB-010361, AB-010363, and exemplary antibodies resulted from the further mutagenesis are shown in in Table 3, 4, 5, respectively.

TABLE 3

Exemplary variants of AB-010357

Antibody ID	Mutations relative to AB-010357
AB-010661	H107TI; L23GR
AB-010662	L23GR; L29SY
AB-010663	L23GR; L30KM
AB-010664	L23GR; L92SH
AB-010665	H107T; L29SY
AB-010666	L29SY; L30KM
AB-010667	L29SY; L92SH
AB-010668	H107TI; L30KM
AB-010669	L30KM; L92SH
AB-010670	H107TI; L92SH
AB-010671	H107TI; L29SY; L92SH
AB-010672	L93SR
AB-010673	L93SE
AB-010674	L29SY; L93SE

TABLE 4

Exemplary variants of AB-010361

Antibody ID	Mutations relative to AB-010361
AB-010675	H107TI; L23GR
AB-010676	L23GR; L29SY
AB-010677	L23GR; L30KM
AB-010678	L23GR; L92SH
AB-010679	H107TI; L29SY
AB-010680	L29SY; L30KM
AB-010681	L29SY; L92SH
AB-010682	H107TI; L30KM
AB-010683	L30KM; L92SH
AB-010684	H107TI; L92SH
AB-010685	H107TI; L29SY; L92SH
AB-010686	L93SR
AB-010687	L93SE
AB-010688	L29SY; L93SE

TABLE 5

shows exemplary variants of AB-010363

Antibody ID	Mutations relative to AB-010363
AB-010689	H107TI; L23GR
AB-010690	L23GR; L29SY
AB-010691	L23GR; L30KM
AB-010692	L23GR; L92SH
AB-010693	H107TI; L29SY
AB-010694	L29SY; L30KM
AB-010695	L29SY; L92SH
AB-010696	H107TI; L30KM
AB-010697	L30KM; L92SH
AB-010698	H107TI; L92SH
AB-010699	H107TI; L29SY; L92SH
AB-010700	L93SR
AB-010701	L93SE
AB-010702	L29SY; L93SE

[0162] Methods of generating variants are further described in the section entitled "ENGINEERING VARIANTS" and Example 2 below.

Structures of the EphA2 Antibodies

[0163] In some embodiments, an EphA2 antibody that binds to EphA2 and comprises a heavy chain variable region comprising: an HCDR1 comprising a sequence (SEQ ID NO:1), or a variant HCDR1 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence; an HCDR2 comprising a sequence (SEQ ID NO:12), or a variant HCDR2 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; and an HCDR3 comprising a sequence (SEQ ID NO:23), or a variant HCDR3 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence. In some embodiments, the antibody comprises a light chain variable region comprising: an LCDR1 comprising a sequence (SEQ ID NO:34), or a variant LCDR1 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; an LCDR2 comprising a sequence (SEQ ID NO:45), or variant LCDR2 in which 1, 2, or 3 amino acid is substituted relative to the sequence; and an LCDR3 comprising a sequence (SEQ ID NO:56), or a variant LCDR3 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence.

[0164] In some embodiments, an antibody that binds to EphA2 comprises: a heavy chain variable region comprising: an HCDR1 of any one of SEQ ID NOS: 1-11 and 201-265 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence; an HCDR2 of any one of SEQ ID NOS: 12-22 and 266-330 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; and an HCDR3 of any one of SEQ ID NOS: 23-33 and 331-395 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence. The antibody also comprises a light chain variable region comprising: an LCDR1 of any one of SEQ ID NOS: 34-44 and 396-460 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; an LCDR2 of any one of SEQ ID NOS: 45-55 and 461-525 or a variant thereof in which 1, 2, or 3 amino acid is substituted relative to the sequence; and an LCDR3 of any one of SEQ ID NOS: 56-66 and 526-590 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence.

[0165] In some embodiments, an antibody that binds to EphA2 comprises (a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, and X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769); (b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNNYNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 770); and (c) an HCDR3 sequence of $AKPX_1RPHC X_2NGVCX_3SGDAFDI$, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S(SEQ ID NO: 771). The antibody also comprises (d) an LCDR1 sequence of $X_1GNNIGX_2 X_3 X_4VH$, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or S (SEQ ID NO: 772); I an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence of QVWDX1 X2SDHX3V, where X1 is H or S, X2 is E, R, or S, and X3 is L or V (SEQ ID NO: 773).

[0166] In some embodiments, an antibody that binds to EphA2 comprises (a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, and X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769); (b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 774); or an HCDR2 sequence of EX_1NHX_2 GS $X_3X_4YNNYNPSLKS$, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 N or S (SEQ ID NO: 777); and (c) an HCDR3 sequence of $AKPX_1RPHCX_2NGVCX_3SGDAFDI$, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S (SEQ ID NO: 771). The antibody also comprises (d) an LCDR1 sequence of $X_1GNNIGX_2 X_3 X_4VH$, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or I (SEQ ID NO: 780) (e) an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and (f) an LCDR3 sequence of QVWDX1 X2SDHX3V, where X1 is H or S, X2 is E, R, or S, and X3 is L or V. (SEQ ID NO: 773).

[0167] In some embodiments, an antibody that binds to EphA2 comprises a V_H comprising an amino acid sequence having at least 95% identity to (SEQ ID NO: 67); and a V_L comprising an amino sequence having at 95% identity to (SEQ ID NO: 78).

[0168] In some embodiments, an antibody that binds to EphA2 has a V_H comprising an amino acid sequence having at least 95% identity to any one of SEQ ID NO: 67-77 and 591-655; and a V_L comprising an amino sequence having at 95% identity to any one of SEQ ID NOs: 78-88 and 656-720.

[0169] In some embodiments, an antibody that binds to EphA2 comprises a heavy chain variable (V_H) region and a light chain variable (V_L) region. The V_H region has at least 70% amino acid sequence identity to SEQ ID NO:67; and comprises a CDR1 of SEQ ID NO:1, or the CDR1 of SEQ ID NO:1 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:83, or the CDR2 of SEQ ID NO:12 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:23 or the CDR3 of SEQ ID NO:23 in which 1, 2, 3, 4, or 5 are substituted. The V_L region has at least 70% amino acid sequence identity to SEQ ID NO: 78, and comprises a CDR1 of SEQ ID NO:34 or the CDR1 of SEQ ID NO:34 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:45, or the CDR2 of SEQ ID NO:45 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:56 or the CDR3 of SEQ ID NO:56 in which 1, 2, 3, 4, or 5 are substituted.

[0170] In some embodiments, an antibody that binds to EphA2 comprises: a V_H region comprising amino acid sequence SEQ ID NO:67 and a V_L region comprising amino acid sequence SEQ ID NO:78; a V_H region comprising amino acid sequence SEQ ID NO:68 and a V_L region comprising amino acid sequence SEQ ID NO:79; a V_H region comprising amino acid sequence SEQ ID NO:69 and a V_L region comprising amino acid sequence SEQ ID NO:80; a V_H region comprising amino acid sequence SEQ ID NO:70 and a V_L region comprising amino acid sequence SEQ ID NO:81; a V_H region comprising amino acid sequence SEQ ID NO:71 and a V_L region comprising amino acid sequence SEQ ID NO:82; a V_H region comprising amino acid sequence SEQ ID NO:72 and a V_L region comprising amino acid sequence SEQ ID NO:83; a V_H region comprising amino acid sequence SEQ ID NO:73 and a V_L region comprising amino acid sequence SEQ ID NO:84; a V_H region comprising amino acid sequence SEQ ID NO:74 and a V_L region comprising amino acid sequence SEQ ID NO:85; a V_H region comprising amino acid sequence SEQ ID NO:75 and a V_L region comprising amino acid sequence SEQ ID NO:86; a V_H region comprising amino acid sequence SEQ TD NO:76 and a V_L region comprising amino acid sequence SEQ ID NO:87; or a V_H

region comprising amino acid sequence SEQ ID NO:77 and a V_L region comprising amino acid sequence SEQ ID NO:88.

[0171] In some embodiments, an EphA2 antibody of the present disclosure has one, two, or three CDRs of a V_L sequence in Table 7. In some embodiments, the EphA2 antibody has at least one mutation and no more than 10, 20, 30, 40 or 50 mutations in the V_L amino acid sequences compared to a V_L sequence set forth in Table 8. In some embodiments, the V_L amino acid sequence may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid insertions or deletions compared to a V_L sequence set forth in Table 8. In some embodiments, the V_L amino acid sequence may comprise a deletion or insertion, e.g., a 1, 2, 3, 4, 5, 6, or 7 amino acid deletion or insertion, relative to a CDR sequence shown in Table 7. In some embodiments, the V_L region comprises a CDR1 having 1 or 2 substitutions in relative to a CDR1 sequence shown in Table 7. In some embodiments, a CDR1 has 3, 4, or 5 substitutions relative to a CDR1 sequence shown in Table 7. In some embodiments, the V_L region comprises a CDR2 with 1 or 2; or 1, 2, or 3; substitutions relative to the CDR2 sequence shown in Table 7. In some embodiments, the V_L region comprises a CDR3 with 1, 2, or 3; or 1, 2, 3, or 4; substitutions relative to a CDR3 sequence shown in Table 7. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3, each having at least 70% identity to a CDR1, CDR2, and CDR3 as shown in Table 7. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3, each having at least 80% identity to a CDR1, CDR2, and CDR3 as shown in Table 7. In some embodiments, an anti-tumor antibody of the present disclosure comprises a CDR1, CDR2, and CDR3 as shown in Table 7. In some embodiments, an EphA2 antibody of the present disclosure comprises HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702 in Table 7. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3 of the VL of an antibody designated as AB-010361 or AB-010699. Of these antibodies, AB-008873; AB-009805; AB-009806; AB-009807; AB-009808 are sibling antibodies as they are derived from the same lineage as AB-008873. See Example 1.

[0172] In some embodiments, the antibody comprises a heavy chain variable region and a light chain variable region of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699. AB-010700, AB-010701, or AB-010702, or a variant thereof, wherein the variant comprises a heavy chain variable region having a sequence that is at least 95% identical to that of the corresponding heavy chain variable region and a light chain variable region having a sequence that is at least 95% identical to the corresponding light chain variable region.

[0173] In some embodiments, an EphA2 antibody of the present disclosure has one, two, or three CDRs of a V_H sequence in Table 8. In some embodiments, the EphA2 antibody has at least one mutation and no more than 10, 20, 30, 40 or 50 mutations in the V_H amino acid sequences compared to a V_H sequence set forth in Table 8. In some embodiments, the V_H amino acid sequence may comprise 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid insertions or deletions compared to a V_H sequence set forth in Table 8. In some embodiments, the V_H amino acid sequence may comprise a deletion or insertion, e.g., a 1, 2, 3, 4, 5, 6, or 7 amino acid deletion or insertion, relative to a CDR sequence shown in Table 6. In some embodiments, the V_H region comprises a CDR1 having 1 or 2 substitutions in relative to a CDR1 sequence shown in Table 6. In some embodiments, a CDR1 has 3, 4, or 5 substitutions relative to a CDR1 sequence shown in Table 6. In some embodiments, the V_H region comprises a CDR2 with 1 or 2; or 1, 2, or 3; substitutions relative to the CDR2 sequence shown in Table 6. In some embodiments, the V_H region comprises a CDR3 with 1, 2, or 3; or 1, 2, 3, or 4; substitutions relative to a CDR3 sequence shown in Table 6. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3, each having at least 70% identity to a CDR1, CDR2, and CDR3 as shown in Table 6. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3, each having at least 80% identity to a CDR1, CDR2, and CDR3 as shown in Table 6. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3 as shown in Table 6. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3 of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359,

AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702 in Table 6. In some embodiments, an EphA2 antibody of the present disclosure comprises a CDR1, CDR2, and CDR3 of the VH of an antibody designated as AB-010361 and AB-010699.

[0174] Exemplary EphA2 antibodies include AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699. AB-010700, AB-010701, or AB-010702. In some embodiments, the EphA2 antibody is AB-010361 or AB-010699. These exemplary EphA2 antibodies have structures (HCDRs, LCDRs, V_H and/or V_L sequences) shown in Tables 6-8.

TABLE 6

Heavy chain CDRs			
Antibody ID	HCDR1	HCDR2	HCDR3
AB-008873	(SEQ ID NO: 1)	(SEQ ID NO: 12)	(SEQ ID NO: 23)
AB-009805	(SEQ ID NO: 2)	(SEQ ID NO: 13)	(SEQ ID NO: 24)
AB-009806	(SEQ ID NO: 3)	(SEQ ID NO: 14)	(SEQ ID NO: 25)
AB-009807	(SEQ ID NO: 4)	(SEQ ID NO: 15)	(SEQ ID NO: 26)
AB-009808	(SEQ ID NO: 5)	(SEQ ID NO: 16)	(SEQ ID NO: 27)
AB-009812	(SEQ ID NO: 6)	(SEQ ID NO: 17)	(SEQ ID NO: 28)
AB-009813	(SEQ ID NO: 7)	(SEQ ID NO: 18)	(SEQ ID NO: 29)
AB-009814	(SEQ ID NO: 8)	(SEQ ID NO: 19)	(SEQ ID NO: 30)
AB-009815	(SEQ ID NO: 9)	(SEQ ID NO: 20)	(SEQ ID NO: 31)
AB-009816	(SEQ ID NO: 10)	(SEQ ID NO: 21)	(SEQ ID NO: 32)
AB-009817	(SEQ ID NO: 11)	(SEQ ID NO: 22)	(SEQ ID NO: 33)
AB-010141	(SEQ ID NO: 201)	(SEQ ID NO: 266)	(SEQ ID NO: 331)
AB-010142	(SEQ ID NO: 202)	(SEQ ID NO: 267)	(SEQ ID NO: 332)
AB-010143	(SEQ ID NO: 203)	(SEQ ID NO: 268)	(SEQ ID NO: 333)
AB-010144	(SEQ ID NO: 204)	(SEQ ID NO: 269)	(SEQ ID NO: 334)
AB-010145	(SEQ ID NO: 205)	(SEQ ID NO: 270)	(SEQ ID NO: 335)
AB-010146	(SEQ ID NO: 206)	(SEQ ID NO: 271)	(SEQ ID NO: 336)
AB-010147	(SEQ ID NO: 207)	(SEQ ID NO: 272)	(SEQ ID NO: 337)
AB-010148	(SEQ ID NO: 208)	(SEQ ID NO: 273)	(SEQ ID NO: 338)
AB-010149	(SEQ ID NO: 209)	(SEQ ID NO: 274)	(SEQ ID NO: 339)
AB-010150	(SEQ ID NO: 210)	(SEQ ID NO: 275)	(SEQ ID NO: 340)
AB-010151	(SEQ ID NO: 211)	(SEQ ID NO: 276)	(SEQ ID NO: 341)
AB-010152	(SEQ ID NO: 212)	(SEQ ID NO: 277)	(SEQ ID NO: 342)
AB-010357	(SEQ ID NO: 213)	(SEQ ID NO: 278)	(SEQ ID NO: 343)
AB-010358	(SEQ ID NO: 214)	(SEQ ID NO: 279)	(SEQ ID NO: 344)
AB-010359	(SEQ ID NO: 215)	(SEQ ID NO: 280)	(SEQ ID NO: 345)
AB-010360	(SEQ ID NO: 216)	(SEQ ID NO: 281)	(SEQ ID NO: 346)
AB-010361	(SEQ ID NO: 217)	(SEQ ID NO: 282)	(SEQ ID NO: 347)
AB-010362	(SEQ ID NO: 218)	(SEQ ID NO: 283)	(SEQ ID NO: 348)
AB-010363	(SEQ ID NO: 219)	(SEQ ID NO: 284)	(SEQ ID NO: 349)
AB-010364	(SEQ ID NO: 220)	(SEQ ID NO: 285)	(SEQ ID NO: 350)
AB-010365	(SEQ ID NO: 221)	(SEQ ID NO: 286)	(SEQ ID NO: 351)
AB-010366	(SEQ ID NO: 222)	(SEQ ID NO: 287)	(SEQ ID NO: 352)
AB-010367	(SEQ ID NO: 223)	(SEQ ID NO: 288)	(SEQ ID NO: 353)
AB-010661	(SEQ ID NO: 224)	(SEQ ID NO: 289)	(SEQ ID NO: 354)
AB-010662	(SEQ ID NO: 225)	(SEQ ID NO: 290)	(SEQ ID NO: 355)
AB-010663	(SEQ ID NO: 226)	(SEQ ID NO: 291)	(SEQ ID NO: 356)
AB-010664	(SEQ ID NO: 227)	(SEQ ID NO: 292)	(SEQ ID NO: 357)
AB-010665	(SEQ ID NO: 228)	(SEQ ID NO: 293)	(SEQ ID NO: 358)
AB-010666	(SEQ ID NO: 229)	(SEQ ID NO: 294)	(SEQ ID NO: 359)
AB-010667	(SEQ ID NO: 230)	(SEQ ID NO: 295)	(SEQ ID NO: 360)
AB-010668	(SEQ ID NO: 231)	(SEQ ID NO: 296)	(SEQ ID NO: 361)
AB-010669	(SEQ ID NO: 232)	(SEQ ID NO: 297)	(SEQ ID NO: 362)
AB-010670	(SEQ ID NO: 233)	(SEQ ID NO: 298)	(SEQ ID NO: 363)
AB-010671	(SEQ ID NO: 234)	(SEQ ID NO: 299)	(SEQ ID NO: 364)
AB-010672	(SEQ ID NO: 235)	(SEQ ID NO: 300)	(SEQ ID NO: 365)
AB-010673	(SEQ ID NO: 236)	(SEQ ID NO: 301)	(SEQ ID NO: 366)

TABLE 6-continued

Heavy chain CDRs			
Antibody ID	HCDR1	HCDR2	HCDR3
AB-010674	(SEQ ID NO: 237)	(SEQ ID NO: 302)	(SEQ ID NO: 367)
AB-010675	(SEQ ID NO: 238)	(SEQ ID NO: 303)	(SEQ ID NO: 368)
AB-010676	(SEQ ID NO: 239)	(SEQ ID NO: 304)	(SEQ ID NO: 369)
AB-010677	(SEQ ID NO: 240)	(SEQ ID NO: 305)	(SEQ ID NO: 370)
AB-010678	(SEQ ID NO: 241)	(SEQ ID NO: 306)	(SEQ ID NO: 371)
AB-010679	(SEQ ID NO: 242)	(SEQ ID NO: 307)	(SEQ ID NO: 372)
AB-010680	(SEQ ID NO: 243)	(SEQ ID NO: 308)	(SEQ ID NO: 373)
AB-010681	(SEQ ID NO: 244)	(SEQ ID NO: 309)	(SEQ ID NO: 374)
AB-010682	(SEQ ID NO: 245)	(SEQ ID NO: 310)	(SEQ ID NO: 375)
AB-010683	(SEQ ID NO: 246)	(SEQ ID NO: 311)	(SEQ ID NO: 376)
AB-010684	(SEQ ID NO: 247)	(SEQ ID NO: 312)	(SEQ ID NO: 377)
AB-010685	(SEQ ID NO: 248)	(SEQ ID NO: 313)	(SEQ ID NO: 378)
AB-010686	(SEQ ID NO: 249)	(SEQ ID NO: 314)	(SEQ ID NO: 379)
AB-010687	(SEQ ID NO: 250)	(SEQ ID NO: 315)	(SEQ ID NO: 380)
AB-010688	(SEQ ID NO: 251)	(SEQ ID NO: 316)	(SEQ ID NO: 381)
AB-010689	(SEQ ID NO: 252)	(SEQ ID NO: 317)	(SEQ ID NO: 382)
AB-010690	(SEQ ID NO: 253)	(SEQ ID NO: 318)	(SEQ ID NO: 383)
AB-010691	(SEQ ID NO: 254)	(SEQ ID NO: 319)	(SEQ ID NO: 384)
AB-010692	(SEQ ID NO: 255)	(SEQ ID NO: 320)	(SEQ ID NO: 385)
AB-010693	(SEQ ID NO: 256)	(SEQ ID NO: 321)	(SEQ ID NO: 386)
AB-010694	(SEQ ID NO: 257)	(SEQ ID NO: 322)	(SEQ ID NO: 387)
AB-010695	(SEQ ID NO: 258)	(SEQ ID NO: 323)	(SEQ ID NO: 388)
AB-010696	(SEQ ID NO: 259)	(SEQ ID NO: 324)	(SEQ ID NO: 389)
AB-010697	(SEQ ID NO: 260)	(SEQ ID NO: 325)	(SEQ ID NO: 390)
AB-010698	(SEQ ID NO: 261)	(SEQ ID NO: 326)	(SEQ ID NO: 391)
AB-010699	(SEQ ID NO: 262)	(SEQ ID NO: 327)	(SEQ ID NO: 392)
AB-010700	(SEQ ID NO: 263)	(SEQ ID NO: 328)	(SEQ ID NO: 393)
AB-010701	(SEQ ID NO: 264)	(SEQ ID NO: 329)	(SEQ ID NO: 394)
AB-010702	(SEQ ID NO: 265)	(SEQ ID NO: 330)	(SEQ ID NO: 395)

TABLE 7

Light chain CDR sequences			
Antibody ID	LCDR1	LCDR2	LCDR3
AB-008873	(SEQ ID NO: 34)	(SEQ ID NO: 45)	(SEQ ID NO: 56)
AB-009805	(SEQ ID NO: 35)	(SEQ ID NO: 46)	(SEQ ID NO: 57)
AB-009806	(SEQ ID NO: 36)	(SEQ ID NO: 47)	(SEQ ID NO: 58)
AB-009807	(SEQ ID NO: 37)	(SEQ ID NO: 48)	(SEQ ID NO: 59)
AB-009808	(SEQ ID NO: 38)	(SEQ ID NO: 49)	(SEQ ID NO: 60)
AB-009812	(SEQ ID NO: 39)	(SEQ ID NO: 50)	(SEQ ID NO: 61)
AB-009813	(SEQ ID NO: 40)	(SEQ ID NO: 51)	(SEQ ID NO: 62)
AB-009814	(SEQ ID NO: 41)	(SEQ ID NO: 52)	(SEQ ID NO: 63)
AB-009815	(SEQ ID NO: 42)	(SEQ ID NO: 53)	(SEQ ID NO: 64)
AB-009816	(SEQ ID NO: 43)	(SEQ ID NO: 54)	(SEQ ID NO: 65)
AB-009817	(SEQ ID NO: 44)	(SEQ ID NO: 55)	(SEQ ID NO: 66)
AB-010141	(SEQ ID NO: 396)	(SEQ ID NO: 461)	(SEQ ID NO: 526)
AB-010142	(SEQ ID NO: 397)	(SEQ ID NO: 462)	(SEQ ID NO: 527)
AB-010143	(SEQ ID NO: 398)	(SEQ ID NO: 463)	(SEQ ID NO: 528)
AB-010144	(SEQ ID NO: 399)	(SEQ ID NO: 464)	(SEQ ID NO: 529)
AB-010145	(SEQ ID NO: 400)	(SEQ ID NO: 465)	(SEQ ID NO: 530)
AB-010146	(SEQ ID NO: 401)	(SEQ ID NO: 466)	(SEQ ID NO: 531)
AB-010147	(SEQ ID NO: 402)	(SEQ ID NO: 467)	(SEQ ID NO: 532)
AB-010148	(SEQ ID NO: 403)	(SEQ ID NO: 468)	(SEQ ID NO: 533)
AB-010149	(SEQ ID NO: 404)	(SEQ ID NO: 469)	(SEQ ID NO: 534)
AB-010150	(SEQ ID NO: 405)	(SEQ ID NO: 470)	(SEQ ID NO: 535)
AB-010151	(SEQ ID NO: 406)	(SEQ ID NO: 471)	(SEQ ID NO: 536)
AB-010152	(SEQ ID NO: 407)	(SEQ ID NO: 472)	(SEQ ID NO: 537)
AB-010357	(SEQ ID NO: 408)	(SEQ ID NO: 473)	(SEQ ID NO: 538)
AB-010358	(SEQ ID NO: 409)	(SEQ ID NO: 474)	(SEQ ID NO: 539)
AB-010359	(SEQ ID NO: 410)	(SEQ ID NO: 475)	(SEQ ID NO: 540)
AB-010360	(SEQ ID NO: 411)	(SEQ ID NO: 476)	(SEQ ID NO: 541)
AB-010361	(SEQ ID NO: 412)	(SEQ ID NO: 477)	(SEQ ID NO: 542)
AB-010362	(SEQ ID NO: 413)	(SEQ ID NO: 478)	(SEQ ID NO: 543)
AB-010363	(SEQ ID NO: 414)	(SEQ ID NO: 479)	(SEQ ID NO: 544)
AB-010364	(SEQ ID NO: 415)	(SEQ ID NO: 480)	(SEQ ID NO: 545)
AB-010365	(SEQ ID NO: 416)	(SEQ ID NO: 481)	(SEQ ID NO: 546)
AB-010366	(SEQ ID NO: 417)	(SEQ ID NO: 482)	(SEQ ID NO: 547)
AB-010367	(SEQ ID NO: 418)	(SEQ ID NO: 483)	(SEQ ID NO: 548)

TABLE 7-continued

Light chain CDR sequences			
Antibody ID	LCDR1	LCDR2	LCDR3
AB-010661	(SEQ ID NO: 419)	(SEQ ID NO: 484)	(SEQ ID NO: 549)
AB-010662	(SEQ ID NO: 420)	(SEQ ID NO: 485)	(SEQ ID NO: 550)
AB-010663	(SEQ ID NO: 421)	(SEQ ID NO: 486)	(SEQ ID NO: 551)
AB-010664	(SEQ ID NO: 422)	(SEQ ID NO: 487)	(SEQ ID NO: 552)
AB-010665	(SEQ ID NO: 423)	(SEQ ID NO: 488)	(SEQ ID NO: 553)
AB-010666	(SEQ ID NO: 424)	(SEQ ID NO: 489)	(SEQ ID NO: 554)
AB-010667	(SEQ ID NO: 425)	(SEQ ID NO: 490)	(SEQ ID NO: 555)
AB-010668	(SEQ ID NO: 426)	(SEQ ID NO: 491)	(SEQ ID NO: 556)
AB-010669	(SEQ ID NO: 427)	(SEQ ID NO: 492)	(SEQ ID NO: 557)
AB-010670	(SEQ ID NO: 428)	(SEQ ID NO: 493)	(SEQ ID NO: 558)
AB-010671	(SEQ ID NO: 429)	(SEQ ID NO: 494)	(SEQ ID NO: 559)
AB-010672	(SEQ ID NO: 430)	(SEQ ID NO: 495)	(SEQ ID NO: 560)
AB-010673	(SEQ ID NO: 431)	(SEQ ID NO: 496)	(SEQ ID NO: 561)
AB-010674	(SEQ ID NO: 432)	(SEQ ID NO: 497)	(SEQ ID NO: 562)
AB-010675	(SEQ ID NO: 433)	(SEQ ID NO: 498)	(SEQ ID NO: 563)
AB-010676	(SEQ ID NO: 434)	(SEQ ID NO: 499)	(SEQ ID NO: 564)
AB-010677	(SEQ ID NO: 435)	(SEQ ID NO: 500)	(SEQ ID NO: 565)
AB-010678	(SEQ ID NO: 436)	(SEQ ID NO: 501)	(SEQ ID NO: 566)
AB-010679	(SEQ ID NO: 437)	(SEQ ID NO: 502)	(SEQ ID NO: 567)
AB-010680	(SEQ ID NO: 438)	(SEQ ID NO: 503)	(SEQ ID NO: 568)
AB-010681	(SEQ ID NO: 439)	(SEQ ID NO: 504)	(SEQ ID NO: 569)
AB-010682	(SEQ ID NO: 440)	(SEQ ID NO: 505)	(SEQ ID NO: 570)
AB-010683	(SEQ ID NO: 441)	(SEQ ID NO: 506)	(SEQ ID NO: 571)
AB-010684	(SEQ ID NO: 442)	(SEQ ID NO: 507)	(SEQ ID NO: 572)
AB-010685	(SEQ ID NO: 443)	(SEQ ID NO: 508)	(SEQ ID NO: 573)
AB-010686	(SEQ ID NO: 444)	(SEQ ID NO: 509)	(SEQ ID NO: 574)
AB-010687	(SEQ ID NO: 445)	(SEQ ID NO: 510)	(SEQ ID NO: 575)
AB-010688	(SEQ ID NO: 446)	(SEQ ID NO: 511)	(SEQ ID NO: 576)
AB-010689	(SEQ ID NO: 447)	(SEQ ID NO: 512)	(SEQ ID NO: 577)
AB-010690	(SEQ ID NO: 448)	(SEQ ID NO: 513)	(SEQ ID NO: 578)
AB-010691	(SEQ ID NO: 449)	(SEQ ID NO: 514)	(SEQ ID NO: 579)
AB-010692	(SEQ ID NO: 450)	(SEQ ID NO: 515)	(SEQ ID NO: 580)
AB-010693	(SEQ ID NO: 451)	(SEQ ID NO: 516)	(SEQ ID NO: 581)
AB-010694	(SEQ ID NO: 452)	(SEQ ID NO: 517)	(SEQ ID NO: 582)
AB-010695	(SEQ ID NO: 453)	(SEQ ID NO: 518)	(SEQ ID NO: 583)
AB-010696	(SEQ ID NO: 454)	(SEQ ID NO: 519)	(SEQ ID NO: 584)
AB-010697	(SEQ ID NO: 455)	(SEQ ID NO: 520)	(SEQ ID NO: 585)
AB-010698	(SEQ ID NO: 456)	(SEQ ID NO: 521)	(SEQ ID NO: 586)
AB-010699	(SEQ ID NO: 457)	(SEQ ID NO: 522)	(SEQ ID NO: 587)
AB-010700	(SEQ ID NO: 458)	(SEQ ID NO: 523)	(SEQ ID NO: 588)
AB-010701	(SEQ ID NO: 459)	(SEQ ID NO: 524)	(SEQ ID NO: 589)
AB-010702	(SEQ ID NO: 460)	(SEQ ID NO: 525)	(SEQ ID NO: 590)

TABLE 8

Heavy and Light variable region sequences		
Antibody ID	V_H	V_L
AB-008873	(SEQ ID NO: 67)	(SEQ ID NO: 78)
AB-009805	(SEQ ID NO: 68)	(SEQ ID NO: 79)
AB-009806	(SEQ ID NO: 69)	(SEQ ID NO: 80)
AB-009807	(SEQ ID NO: 70)	(SEQ ID NO: 81)
AB-009808	(SEQ ID NO: 71)	(SEQ ID NO: 82)
AB-009812	(SEQ ID NO: 72)	(SEQ ID NO: 83)
AB-009813	(SEQ ID NO: 73)	(SEQ ID NO: 84)
AB-009814	(SEQ ID NO: 74)	(SEQ ID NO: 85)
AB-009815	(SEQ ID NO: 75)	(SEQ ID NO: 86)
AB-009816	(SEQ ID NO: 76)	(SEQ ID NO: 87)
AB-009817	(SEQ ID NO: 77)	(SEQ ID NO: 88)
AB-010141	(SEQ ID NO: 591)	(SEQ ID NO: 656)
AB-010142	(SEQ ID NO: 592)	(SEQ ID NO: 657)
AB-010143	(SEQ ID NO: 593)	(SEQ ID NO: 658)
AB-010144	(SEQ ID NO: 594)	(SEQ ID NO: 659)
AB-010145	(SEQ ID NO: 595)	(SEQ ID NO: 660)
AB-010146	(SEQ ID NO: 596)	(SEQ ID NO: 661)
AB-010147	(SEQ ID NO: 597)	(SEQ ID NO: 662)
AB-010148	(SEQ ID NO: 598)	(SEQ ID NO: 663)
AB-010149	(SEQ ID NO: 599)	(SEQ ID NO: 664)
AB-010150	(SEQ ID NO: 600)	(SEQ ID NO: 665)

TABLE 8-continued

Heavy and Light variable region sequences		
Antibody ID	V_H	V_L
AB-010151	(SEQ ID NO: 601)	(SEQ ID NO: 666)
AB-010152	(SEQ ID NO: 602)	(SEQ ID NO: 667)
AB-010357	(SEQ ID NO: 603)	(SEQ ID NO: 668)
AB-010358	(SEQ ID NO: 604)	(SEQ ID NO: 669)
AB-010359	(SEQ ID NO: 605)	(SEQ ID NO: 670)
AB-010360	(SEQ ID NO: 606)	(SEQ ID NO: 671)
AB-010361	(SEQ ID NO: 607)	(SEQ ID NO: 672)
AB-010362	(SEQ ID NO: 608)	(SEQ ID NO: 673)
AB-010363	(SEQ ID NO: 609)	(SEQ ID NO: 674)
AB-010364	(SEQ ID NO: 610)	(SEQ ID NO: 675)
AB-010365	(SEQ ID NO: 611)	(SEQ ID NO: 676)
AB-010366	(SEQ ID NO: 612)	(SEQ ID NO: 677)
AB-010367	(SEQ ID NO: 613)	(SEQ ID NO: 678)
AB-010661	(SEQ ID NO: 614)	(SEQ ID NO: 679)
AB-010662	(SEQ ID NO: 615)	(SEQ ID NO: 680)
AB-010663	(SEQ ID NO: 616)	(SEQ ID NO: 681)
AB-010664	(SEQ ID NO: 617)	(SEQ ID NO: 682)
AB-010665	(SEQ ID NO: 618)	(SEQ ID NO: 683)
AB-010666	(SEQ ID NO: 619)	(SEQ ID NO: 684)
AB-010667	(SEQ ID NO: 620)	(SEQ ID NO: 685)
AB-010668	(SEQ ID NO: 621)	(SEQ ID NO: 686)

TABLE 8-continued

Heavy and Light variable region sequences		
Antibody ID	V_H	V_L
AB-010669	(SEQ ID NO: 622)	(SEQ ID NO: 687)
AB-010670	(SEQ ID NO: 623)	(SEQ ID NO: 688)
AB-010671	(SEQ ID NO: 624)	(SEQ ID NO: 689)
AB-010672	(SEQ ID NO: 625)	(SEQ ID NO: 690)
AB-010673	(SEQ ID NO: 626)	(SEQ ID NO: 691)
AB-010674	(SEQ ID NO: 627)	(SEQ ID NO: 692)
AB-010675	(SEQ ID NO: 628)	(SEQ ID NO: 693)
AB-010676	(SEQ ID NO: 629)	(SEQ ID NO: 694)
AB-010677	(SEQ ID NO: 630)	(SEQ ID NO: 695)
AB-010678	(SEQ ID NO: 631)	(SEQ ID NO: 696)
AB-010679	(SEQ ID NO: 632)	(SEQ ID NO: 697)
AB-010680	(SEQ ID NO: 633)	(SEQ ID NO: 698)
AB-010681	(SEQ ID NO: 634)	(SEQ ID NO: 699)
AB-010682	(SEQ ID NO: 635)	(SEQ ID NO: 700)
AB-010683	(SEQ ID NO: 636)	(SEQ ID NO: 701)
AB-010684	(SEQ ID NO: 637)	(SEQ ID NO: 702)
AB-010685	(SEQ ID NO: 638)	(SEQ ID NO: 703)
AB-010686	(SEQ ID NO: 639)	(SEQ ID NO: 704)
AB-010687	(SEQ ID NO: 640)	(SEQ ID NO: 705)
AB-010688	(SEQ ID NO: 641)	(SEQ ID NO: 706)
AB-010689	(SEQ ID NO: 642)	(SEQ ID NO: 707)
AB-010690	(SEQ ID NO: 643)	(SEQ ID NO: 708)
AB-010691	(SEQ ID NO: 644)	(SEQ ID NO: 709)
AB-010692	(SEQ ID NO: 645)	(SEQ ID NO: 710)
AB-010693	(SEQ ID NO: 646)	(SEQ ID NO: 711)
AB-010694	(SEQ ID NO: 647)	(SEQ ID NO: 712)
AB-010695	(SEQ ID NO: 648)	(SEQ ID NO: 713)
AB-010696	(SEQ ID NO: 649)	(SEQ ID NO: 714)
AB-010697	(SEQ ID NO: 650)	(SEQ ID NO: 715)
AB-010698	(SEQ ID NO: 651)	(SEQ ID NO: 716)
AB-010699	(SEQ ID NO: 652)	(SEQ ID NO: 717)
AB-010700	(SEQ ID NO: 653)	(SEQ ID NO: 718)
AB-010701	(SEQ ID NO: 654)	(SEQ ID NO: 719)
AB-010702	(SEQ ID NO: 655)	(SEQ ID NO: 720)

[0175] In some embodiments, the EphA2 antibody disclosed herein comprises a heavy chain variable region and a light chain variable region of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697. AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702 or a variant thereof. The variant comprises a heavy chain variable region having a sequence that is at least 95% identical to that of the corresponding heavy chain variable region and a light chain variable region having a sequence that is at least 95% identical to the corresponding light chain variable region. For example, the variant may comprise a V_H that is at least 95% identical to that of AB-008873, AB-010148, AB-010363, or AB-010699 and/or a V_L that is at least 95% identical to that of AB-008873, AB-010148, AB-010363, or AB-010699.

[0176] In some embodiments, the EphA2 antibody disclosed herein comprises a heavy chain variable region and a light chain variable region of an antibody designated as AB-010361 or AB-010699.

[0177] In some embodiments, the EphA2 antibody disclosed herein comprises an HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702, or a variant thereof, and at least one, two, three, four, five, or all six of the CDRs of the variant contain 1 or 2 amino acid substitutions compared to the corresponding CDR. For example, at least one, two, three, four, five, or all six of the CDRs of the variant contain 1 or 2 amino acid substitutions compared to the CDRs of AB-008873, AB-010148, AB-010363, or AB-010699

[0178] In some embodiments, the EphA2 antibody disclosed herein comprises an HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of an antibody designated as AB-010361 or AB-010699.

Epitope

[0179] An EphA2 antibody disclosed herein binds to an epitope in the EphA2 protein (SEQ ID NO: 94). In some embodiments, an EphA2 antibody disclosed herein binds to an epitope in the FN2 domain of the EphA2 protein. In some embodiments, the binding can be detected using yeast display experiments illustrated in FIG. **6**. The FN2 domain consists of residues 437-534 of SEQ ID NO: 94 and has a sequence of SEQ ID NO: 95 (TEPPKVRLEGRSTT-SLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYN-VRRTEGFSVTLDDLA PDTTYLVQVQALTQEGQGAG-SKVHEFQTLSPEGSGN). In some embodiments, the epitope recognized by an EphA2 antibody disclosed herein overlaps with that of AB-010018.

[0180] Epitopes for the EphA2 antibodies disclosed herein can be determined by any method well known in the art, for example, by conventional immunoassays. In one example, the epitope to which the EphA2 antibody binds can be determined in a systematic screening by using overlapping peptides derived from the EphA2 sequence and determining binding by the antibody. In some cases, during the discovery process, the generation and characterization of antibodies may elucidate information about desirable epitopes. This information may allow screening for antibodies that compete for binding to the same epitope, e.g., an epitope in the FN2 domain of the EphA2 protein.

[0181] In some cases, an epitope binning method is performed, e.g., using Bio-layer interferometry (BLI) as described in Example 4. The underlying principle of epitope binning is that proteins with overlapping binding sites will sterically inhibit simultaneous binding. For each pair of antibodies, the antigen is pre-saturated with the 1^{st} monoclonal antibody (mAb), the 2^{nd} mAb is applied and signal from the binding of the 2^{nd} mAb to the antigen is detected and quantified. A significantly reduced signal from binding of 2^{nd} mAb to the antigen indicates that there are overlapping binding sites between the two antibodies. Pair-wise interactions of the antibodies are scored and used to bin the antibodies into sets defined by mutual competition. Epitope binning is useful for antibody characterization because antibodies in a bin may exhibit similar MOAs. Binning with antibodies with antibodies of known reactivity can inform epitope identification.

[0182] An EphA2 antibody disclosed herein binds to at least one, at least two, at least three, at least four, at least five residues selected from the group consisting of Pro439, Lys441, Arg443, Leu444, Arg447, Lys476, Gly477, Leu504, Gln506, Ser519, Lys520, Val521, His522, Glu523, Phe524, and Gln525, with the residue numbering referring to the EphA2 amino acid sequence of SEQ ID NO: 94. In one embodiment, the EphA2 antibody binds to at least one, at least two, at least three, at least four, at least five residues, at least six residues, at least seven residues, at least eight residues, or at least nine residues selected from the group consisting of Leu444, Arg447, Lys476, Gln506, Ser519, Lys520, Val521, Glu523, Phe524, and Gln525. In one embodiment, the EphA2 antibody binds to all the residues Leu444, Arg447, Lys476, Gln506, Ser519, Lys520, Val521, Glu523, Phe524, and Gln525.

[0183] In some embodiments, the EphA2 antibody bind to a conformational epitope that is in one, two, three and/or four regions in the EphA2 protein: a first region consisting of amino acid residues 439-447, a second region consisting of amino acid residues 476-477, a third region consisting of amino acid residues 504-506, and a fourth region consisting of amino acid residues 519-525. In some embodiments, the epitope that the EphA2 antibody binds is conserved across species of cyno, rat, mouse and human, all of which are commonly used in toxicology studies. FIG. **46**, FIGS. **44** and **45**. Seiradake et al., Nat. Struct. Mol. Biol 17, 398-402 (2010).

Tumor-Binding Activity

[0184] The activity of the EphA2 antibodies as described herein can be assessed for binding in binding assays. Non-limiting examples of suitable assays include surface plasmon resonance analysis using a biosensor system such as a Biacore® system or a flow cytometry assay, which are further described in the EXAMPLES section.

[0185] In some embodiments, binding to EphA2 protein is assessed in a competitive assay format with a reference antibody AB-008873 or a reference antibody having the variable regions of AB-008873. In some embodiments, a variant EphA2 antibody in accordance with the present disclosure may block binding of the reference antibody in a competition assay by about 50% or more.

[0186] In some embodiments, binding assays to assess variant activity are performed on tumor tissues or tumor cells ex vivo, e.g., on tumor cells that were grown as a tumor graft in a syngeneic (immune-matched) mouse in vivo then harvested and processed within 24-48 hrs. Binding can be assessed by any number of means including flow cytometry.

[0187] The antibodies disclosed herein bind specifically to tumor cells. In some embodiments the antibody is added to a cancer cell line and the binding is analyzed using a flow cytometry. AB-008873 showed a strong binding to 786-O, A375, A549, H522, LoVo, MDA-MB-231, PC3, RKO, SKOV3, SW1116, and CT26. Additionally, AB-008873 is also capable of binding to CT26 ex vivo cells. FIG. **1**, Tables 23 and 24. Its sibling antibodies AB-009805, AB-009806, AB-009807, and AB-009808 also showed binding activity to A549 cells, but less than that of AB-008873. FIG. **20**. Unlike AB-010018 (mouse monoclonal antibody precursor of humanized DS-8895a. See U.S. Pat. No. 9,150,657, Sequences 35 and 37 Daiichi Sankyo (IgG)), which showed binding and ADCC activity on A549 cells (human lung cancer cells) but did not on CT26 cells (murine colon cancer cells), AB-008873 demonstrated binding and ADCC activity on both A549 cells and CT26 cells. FIG. **4**.

[0188] In some embodiments, the binding of the antibodies to bind to tumor cells are assessed by immunofluorescence methods, as described in the EXAMPLES. AB-008873 and its variants preferentially bind to various tumors but not to normal human tissues (FIGS. **27**-**34**). As compared to AB-010018 and LS-C100255, AB-008873 advantageously showed less staining in normal tissues and a higher staining in tumor tissues. In one illustrative example, the AB-008873 showed preferential binding to ovarian cancer and soft tissue sarcoma (STS) than the respective tumor adjacent tissue (TATs); in contrast, all commercial anti-EphA2 antibodies tested, exception for LS-C 36249-100, did not show any detectable binding in these two tumors (FIG. **27**). In some embodiments, for certain cancer types, such as uterine cancer, head and neck cancer, and NSCLS, an EphA2 antibody disclosed herein demonstrate tumor selectivity (i.e., preferentially bind to tumor tissue than TATs) in different donors. That is to say, the antibody shows tumor selectivity in some donors but not other donors. In one exemplary assay, AB-008873 and its variant antibodies, AB-009812, AB-009813, AB-009815, and AB-009816 showed tumor selectivity in different donors in uterine cancer, head and neck cancer, and NSCLS (FIG. **28**-**30**). This suggests that there are possible differential epitope availability and/or processing between donors, or possibly a unique epitope recognition by the AB-008873 paratope.

[0189] In some embodiments, the antibody's binding activity of functional activity is assessed by determining EC_{50} values, and in some embodiments additionally determining delta activity, i.e., the difference in specific activity between lower and upper plateaus of the activation curve expressed as percent of activity of a selected antibody having known in vitro activity. In typical embodiments, EC_{50} values are compared to a reference antibody. For purposes of this disclosure, an antibody comprising the VH and VL regions of an EphA2 antibody disclosed herein and a mouse IgG2a Fc region when testing binding or functional activity using a tumor cell line, is employed as a reference antibody and included in an assay to assess variant activity relative to the reference antibody. The fold over EC_{50} is calculated by dividing the EC_{50} of the reference antibody by the EC_{50} of the test antibody. Based on the resulting values, the antibodies were assigned to groups and given a ranking from 0-4 as follows: 0=(>500 nM); 1=<0.5; 2=0.5 to 2; 3=2 to 4; 4=>4.

EphA2 Signaling Effect

[0190] In some embodiments, an EphA2 antibody of the present disclosure exhibits no measurable agonistic effect on the ephrinA1-EphA2 signaling axis. In some embodiments, an EphA2 antibody of the present disclosure exhibits less agonistic effect on the ephrinA1-EphA2 signaling axis than the ephrin-A1 ligand. In some embodiments, an EphA2 antibody of the present disclosure activates the ephrin A1-EphA2 signaling axis at least 99%, 98% 95%, 90%, 85%, 80%, 75%, 70%, 65%, 60%, 55%, or 50% less than ephrin-A1 activates the ephrin A1-EphA2 signaling axis. In some embodiments, an EphA2 antibody of the present disclosure activates the ephrin A1-EphA2 signaling axis at least 99%, 98%, 95%, 90%, 85%, or 80% less than ephrin-A1 activates the ephrin A1-EphA2 signaling axis. FIG. **43**A.

[0191] In some embodiments, an EphA2 antibody of the present disclosure shows weak agonistic effect on the ephrinA1-EphA2 signaling axis, with the effect occurring at apotency less than the potency of the ADCC of the antibody. In some embodiments, an EphA2 antibody of the present disclosure has an EC_{50} for activation of the ephrinA1-EphA2 signaling axis that is at least 10-, 20-, 50-, 100-, 200-, 500-fold less potent than the ADCC EC_{50} of the antibody.

[0192] In some embodiments, an EphA2 antibody of the present disclosure exhibits no antagonistic effect on the ephrinA1-EphA2 signaling axis. In some embodiments, an EphA2 antibody of the present disclosure exhibits less antagonistic effect on the ephrinA1-EphA2 signaling axis than AB-010018. FIG. **43**B. Methods for measuring ephrinA1-EphA2 signaling axis and the effect of the EphA2 antibodies described herein are well known. In some embodiments, it can be accomplished by detecting the phosphorylation of EphA2 protein for example, one or more of the residues Y588, Y772, S897 of the EphaA2 protein. The degree of phosphorylation can be quantified using methods well known in the art, for example, Western Blot analysis using antibodies specific to described phosphorylation sites. One exemplary method of measuring ephrinA1-EphA2 signaling axis is disclosed in Example 5.

[0193] The limited effect on EphaA2 signaling provides a therapeutic advantage over other EphA2 antibodies that act as agonists or antagonists of the ephrin A1-EphA2 signaling axis by increasing the therapeutic index. For example, an EphaA2 antibody, or an immunoconjugate or bispecific version thereof, with limited effect on EphA2 signaling may be dosed at a level that provides both efficacy and safety while an EphaA2 antibody, or an immunoconjugate or bispecific version thereof, that agonizes the ephrin A1-EphA2 signaling axis could show a smaller, and potentially non-viable, therapeutic index.

Fe Effector Function

[0194] In some embodiments, an EphA2 antibody of the present disclosure comprises an Fc region with effector function. Examples of effector functions include, but are not limited to, C1q binding and complement-dependent cytotoxicity (CDC), Fc receptor binding (e.g., FcγR binding), ADCC, antibody-dependent cell-mediated phagocytosis (ADCP), down-regulation of cell surface receptors (e.g., B cell receptor), and B-cell activation. Effector functions may vary with the antibody class. For example, native human IgG1 and IgG3 antibodies can elicit ADCC and CDC activities upon binding to an appropriate Fc receptor present on an immune system cell; and native human IgG1, IgG2, IgG3, and IgG4 can elicit ADCP functions upon binding to the appropriate Fc receptor present on an immune cell.

[0195] In some embodiments, the Fc region of the EphA2 antibodies disclosed herein may be an Fc region engineered to alter one or more functional properties of the antibody, such as serum half-life, complement fixation, Fc receptor binding, and/or ADCC. Accordingly, an Fc region can comprise additional mutations to increase or decrease effector functions, i.e., the ability to induce certain biological functions upon binding to an Fc receptor expressed on an immune cell. Immune cells include, but are not limited to, monocytes, macrophages, neutrophils, dendritic cells, eosinophils, mast cells, platelets, B cells, large granular lymphocytes, Langerhans' cells, natural killer (NK) cells, and cytotoxic T cells. In some embodiments, an antibody of the present disclosure has enhanced ADCC and/or serum stability compared to antibody AB-008873 when the antibodies are assayed in a human IgG1 isotype format.

[0196] The EphA2 antibodies of the present disclosure may be evaluated in various assays for their ability to mediate FcR-dependent activity. In one assay, the binding activity of an EphA2 antibody is evaluated in an Fc receptor engagement assay. For purposes of testing variants, "engagement" of an Fc receptor occurs when a variant antibody binds to both a target tumor cell via its Fv region and an FcγR present on an immune cell via the antibody Fc region in such as manner to transduce a signal. If the Fc region is kept constant among variants that differ in their Fv regions, then the assay allows an evaluation of tumor binding activity across such variants in the context of potential signal transduction through a particular Fc region binding a particular Fc receptor. In some embodiments, binding of the antibody Fc region can result in clustering and/or internalization of the FcR, resulting in a luminescence signal in cells harboring a NFAT-RE-Luciferase reporter construct.

ADCC Activity

[0197] In some embodiments, an EphA2 antibody of the present disclosure has ADCC when the antibodies are assayed in a mouse IgG2a isotype format. In some embodiments, an EphA2 antibody of the present disclosure has ADCC activity that is comparable to that of AB-008873. The term "comparable activity," refers to the ADCC activity of the EphA2 antibody in a range of 40% to 200% of the activity of a reference antibody that exhibits ADCC when evaluated under the same assay conditions. In one exemplary assay, AB-008873 showed dose dependent ADCC on A549, MDA-MB-231, CT26, and to a small extent EMT6. See FIG. **2**A-**2**B and Table 25. Its sibling antibodies AB-009805, AB-009806, AB-009807, and AB-009808 also showed ADCC activity on A549 cells, but less than that of AB-008873. Certain variants exhibited improved ADCC activity, including AB-010699, AB-010361 and AB-010685. FIG. **50**A-**50**B and FIG. **52**.

ADCP Activity

[0198] In some embodiments, ADCP activity of an EphA2 antibody (e.g., a variant of AB-008873) is assessed using fluorescently labeled, in vitro cultured tumor cells and Raw264.7 murine macrophages. In certain embodiments, opsonization of the tumor cell by the antibody leads to phagocytosis detected by flow cytometry. Variations of this assay have been described and can include co-labeling of tumor and effector cells or assessment of phagocytosis through FecyRIIa engagement (e.g., FcγRIIa-H ADCP Reporter Bioassay from Promega).

[0199] In one illustrative embodiment, The ADCP activity of an EphA2 antibody is evaluated using the method described above and in more detail in EXAMPLE 3. In one illustrative example, AB-008873 showed strong ADCP activity on A549 cells. FIG. **2**B and FIG. **20**. The sibling antibodies AB-009805, AB-009806, AB-009807, and AB-009808 also showed ADCP activity, but less than that of AB-008873. In one illustrative example, AB-008873 showed strong ADCP activity on A549 cells. FIG. **2**A-**2**B, Table 25, and FIG. **20**. The sibling antibodies AB-009805, AB-009806, AB-009807, and AB-009808 also showed ADCP activity, but less than that of AB-008873.

ADC Activity

[0200] An EphA2 antibody is deemed to have ADC activity if, when the antibody is conjugated to a drug molecule (toxin) to form an antibody drug conjugate (ADC), said ADC can kill target cells. In some embodiments, the antibody is deemed to have ADC activity if the EC50 of the assay measuring the cell killing activity of the ADC is less than 1E-08. In one exemplary assay, the ADC activity of the EphA2 antibody is evaluated using a drug-conjugated secondary antibody. The antibody-drug conjugate assay involves tumor target cells, primary antibodies of interest (the EphA2 antibody to be tested), and a secondary antibody that is conjugated to a drug molecule, where the secondary antibody recognizes the primary antibody. Briefly, primary antibody dilutions are incubated with target cells at room temperature for a first period (for example, 10-30 minutes). The drug-conjugated secondary antibody is then added to the incubation mixture containing the target cells and the primary antibody. The mixture is then incubated for second period before measuring the extent of target cell lysis. In some embodiments, the assay generates a 100% cell lysis value by adding cell lysis buffer directly to target cell sample, which are not treated by the primary or drug-conjugated second antibody, and cell killing data from samples treated the antibody mixtures as disclosed above can be normalized to the value of 100% cell lysis. The results of the assay can be used to predict whether an ADC produced by conjugating a particular antibody and the drug molecule can kill target cells.

In Vivo Activity

[0201] In some embodiments, activity of an EphA2 antibody variant is evaluated in vivo in a suitable animal tumor model. A reduction in tumor load reflects the anti-tumor function of an antibody.

[0202] In some embodiments, a variant of an antibody as described herein has at least 50%, or at least 60%, or 70%, or greater, of the anti-tumor activity of a reference antibody as shown in Tables 1-4 when evaluated under the same assay conditions to measure the anti-tumor activity in vivo. In some embodiments, an anti-tumor antibody exhibits improved activity, i.e., greater than 100% activity, compared to the reference antibody.

Antibody Formats

[0203] In a further aspect of the invention, an EphA2 antibody in accordance with the disclosure may be an antibody fragment, e.g., a Fv, Fab, Fab', scFv, diabody, or $F(ab')_2$ fragment. In another embodiment, the antibody is a substantially full-length antibody, e.g., an IgG antibody or other antibody class or isotype as defined herein. For a review of certain antibody fragments, see Hudson et al. Nat. Med. 9: 129-134 (2003). Antibody fragments can be made by various techniques, including but not limited to proteolytic digestion of an intact antibody as well as production by recombinant host cells.

[0204] In some embodiments, an EphA2 antibody according to the present disclosure that is administered to a patient is an IgG of the IgG1 subclass. In some embodiments, such an antibody is an IgG of the IgG2, IgG3, or IgG4 subclass. In some embodiments, such an antibody is an IgM. In some embodiments, such an antibody has a lambda light chain constant region. In some embodiments, such an antibody has a kappa light chain constant region.

[0205] In some embodiments an EphA2 antibody in accordance with the present disclosure is in a monovalent format. In some embodiments, the tumor-targeting antibody is in a fragment format, e.g., a Fv, Fab, Fab', scFv, diabody, or $F(ab')_2$ fragment.

[0206] In some embodiments, EphA2 antibodies disclosed herein, including antibody fragments, of the present disclosure comprises an Fc region with effector function, e.g., exhibits antibody-dependent cellular cytotoxicity (ADCC), antibody-dependent cellular phagocytosis (ADCP), and/or complement-dependent cytotoxicity (CDC). In some embodiments, the Fc region may be an Fc region engineered to alter one or more functional properties of the antibody, such as serum half-life, complement fixation, Fc receptor binding, and/or ADCC. Accordingly, an Fc region can comprise additional mutations to increase or decrease effector functions, i.e., the ability to induce certain biological functions upon binding to an Fc receptor expressed on an immune cell. Immune cells include, but are not limited to, monocytes, macrophages, neutrophils, dendritic cells, eosinophils, mast cells, platelets, B cells, large granular lymphocytes, Langerhans' cells, natural killer (NK) cells, and cytotoxic T cells.

[0207] In some embodiments, an Fc region described herein can include additional modifications that modulate effector function. Examples of Fc region amino acid mutations that modulate an effector function include, but are not limited to, one or more substitutions at positions 228, 233, 234, 235, 236, 237, 238, 239, 243, 265, 269, 270, 297, 298, 318, 326, 327, 329, 330, 331, 332, 333, and 334 (EU numbering scheme) of an Fc region.

[0208] Illustrative substitutions that decrease effector functions include the following: position 329 may have a mutation in which proline is substituted with a glycine or arginine or an amino acid residue large enough to destroy the Fc/Fcγ receptor interface that is formed between proline 329 of the Fc and tryptophan residues Trp 87 and Trp 110 of FcγRIII. Additional illustrative substitutions that decrease effector functions include S228P, E233P, L235E, N297A, N297D, and P331S. Multiple substitutions may also be present, e.g., L234A and L235A of a human IgG1 Fc region; L234A, L235A, and P329G of a human IgG1 Fc region; S228P and L235E of a human IgG4 Fc region; L234A and G237A of a human IgG1 Fc region; L234A, L235A, and G237A of a human IgG1 Fc region; V234A and G237A of a human IgG2 Fc region; L235A, G237A, and E318A of a human IgG4 Fc region; and S228P and L236E of a human IgG4 Fc region, to decrease effectors functions. Examples of substitutions that increase effector functions include, e.g., E333A, K326W/E333S, S239D/I332E/G236A, S239D/A330L/I332E, G236A/S239D/A330L/I332E, F243L, G236A, and S298A/E333A/K334A. In some embodiments, the Fc mutations include P329G, L234A, L235A, or a combination thereof. Descriptions of amino acid mutations in an Fc region that can increase or decrease effector functions can be found in, e.g., Wang et al., Protein Cell. 9(1): 63-73, 2018; Saunders, Front Immunol. June 7, eCollection, 2019; Kellner et al., Transfus Med Hemother. 44(5): 327-336, 2017; and Lo et al., J Biol Chem. 292(9):3900-3908, 2017.

[0209] In some embodiments, an Fc region may have one or more amino acid substitutions that modulate ADCC, e.g., substitutions at positions 298, 333, and/or 334 of the Fc region, according to the EU numbering scheme. Specifically, S298A, E333A, and K334A can be introduced to an Fc region to increase the affinity of the Fc region to FcγRIIIa and decrease the affinity of the Fc region to FcγRIIa and FcγRIIb.

[0210] An Fc region can also comprise additional mutations to increase serum half-life. Through enhanced binding to the neonatal Fc receptor (FcRn), such mutations in an Fc region can improve pharmacokinetics of the antibody. Examples of substitutions in an Fc region that increase the serum half-life of an antibody include, e.g., M252Y/S254T/T256E, T250Q/M428L, N434A, N434H, T307A/E380A/N434A, M428L/N434S. M252Y/M428L, D259I/V308F, N434S, V308W, V308Y, and V308F. Descriptions of amino acid mutations in an Fc region that can increase the serum half-life of an antibody can be found in, e.g., Dumet et al., MAbs. 26:1-10, 2019; Booth et al., MAbs. 10(7):1098-1110, 2018; and Dall'Acqua et al., J Biol Chem. 281(33):23514-24, 2006.

[0211] Furthermore, in some embodiments, an antibody of the disclosure may be chemically modified (e.g., one or more chemical moieties can be attached to the antibody) or be modified, e.g., produced in cell lines and/or in cell culture conditions to alter its glycosylation (e.g., hypofucosylation, afucosylation, or increased sialylation), to alter one or more functional properties of the antibody. For example, the antibody can be linked to one of a variety of polymers, for example, polyethylene glycol. In some embodiments, an antibody may comprise mutations to facilitate linkage to a chemical moiety and/or to alter residues that are subject to post-translational modifications, e.g., glycosylation.

[0212] In some embodiments, an EphA2 antibody described herein comprise an Fc region having altered glycosylation that increases the ability of the antibody to recruit NK cells and/or increase ADCC. In some embodiments, the Fc region comprises glycan containing no fucose (i.e., the Fc region is afucosylated). Afucosylated antibodies can be produced using cell lines that express a heterologous enzyme that depletes the fucose pool inside the cell (e.g., GlymaxX® by ProBioGen AG, Berlin, Germany). Non-fucosylated antibodies can also be produced using a host cell line in which the endogenous α-1,6-fucosyltransferase (FUT8) gene is deleted. See Satoh, M. et al., "Non-fucosylated therapeutic antibodies as next-generation therapeutic antibodies," Expert Opinion on Biological Therapy, 6:11, 1161-1173, DOI: 10.1517/14712598.6.11.1161.

[0213] In some embodiments, an EphA2 antibody is constructed as a multivalent antibody. In some embodiments, an EphA2 antibody is constructed as a tetravalent molecule, comprising four EphA2 binding arms per molecule. Such constructs exhibit increased ADCC activity, as well as increased binding to tumor cells a measured by flow cytometry.

[0214] In some embodiments, an EphA2 antibody of the present disclosure is employed in a bispecific or multi-specific format, e.g., a tri-specific format. For example, in some embodiments, the antibody may be incorporated into a bispecific or multi-specific antibody that comprises a further binding domain that binds to the same or a different antigen.

[0215] There are a variety of possible formats that can be used in bispecific or multi-specific antibodies. The formats can vary elements such as the number of binding arms, the format of each binding arm (e.g., Fab, scFv, scFab, or VH-only), the number of antigen binding domains present on the binding arms, the connectivity and geometry of each arm with respect to each other, the presence or absence of an Fc domain, the Ig class (e.g., IgG or IgM), the Fc subclass (e.g., hIgG1, hIgG2, or hIgG4), and any mutations to the Fc (e.g., mutations to reduce or increase effector function or extend serum half-life). Also see Speiss et al., Alternative Molecular Formats and Therapeutic Applications for Bispecific Antibodies, Mol Immunol, 67, 95-106 (2015), in particular FIG. **1**, for examples of bispecific and multispecific formats.

[0216] Illustrative antigens that can be targeted by a further binding domain in a bispecific or multi-specific antibody that comprises an antigen binding domain of an EphA2 antibody described herein, include, but are not limited to, antigens on T cells to enhance T cell engagement and/or activate T cells. Illustrative examples of such an antigen include, but are not limited to, CD3, CD2, CD4, CD5, CD6, CD8, CD28, CD40L, CD44, IL-15Rα, CD122, CD132, or CD25. In some embodiments, the antigen is CD3. In some embodiments, the antigen is in a T cell activating pathway, such as a 4-1BB/CD137, 4-1BBL/CD137L, OX40, OX40L, GITRL, GITR, CD27, CD70, CD28, ICOS, HVEM, or LIGHT antigen.

[0217] In some embodiments, an EphA2 antibody is incorporated into a bispecific or multi-specific antibody that comprises a binding domain that binds to a T-cell antigen. These bispecific antibodies or multi-specific antibodies can direct T cells to attach and lyse targeted tumor cells, i.e., the EphA2 expressing tumor cells. In some embodiments, the bispecific or multispecific antibody comprises a binding domain that binds to CD3. In some embodiments, the bispecific or multispecific antibody comprises a binding domain that binds to human CD3 comprising the anti-tumor antibodies described herein.

[0218] Table 9 provides specific examples of anti-CD3 binding arms that can be combined with any of the anti-EphA2 antibodies described herein. FIG. **53** provides examples of a variety of formats according to which anti-EphA2/anti-CD3 bispecific constructs that can be generated.

TABLE 9

CD3 heavy chain (VH) and light chain (VL) sequences.

AB ID	VH	VL
AB-008703	(SEQ ID NO: 133)	(SEQ ID NO: 134)
AB-008704	(SEQ ID NO: 135)	(SEQ ID NO: 136)
AB-008705	(SEQ ID NO: 137)	(SEQ ID NO: 138)
AB-008706	(SEQ ID NO: 139)	(SEQ ID NO: 140)
AB-008707	(SEQ ID NO: 141)	(SEQ ID NO: 142)
AB-008708	(SEQ ID NO: 143)	(SEQ ID NO: 144)
AB-008709	(SEQ ID NO: 145)	(SEQ ID NO: 146)
AB-008710	(SEQ ID NO: 147)	(SEQ ID NO: 148)
AB-008711	(SEQ ID NO: 149)	(SEQ ID NO: 150)
AB-008712	(SEQ ID NO: 151)	(SEQ ID NO: 152)
AB-008713	(SEQ ID NO: 153)	(SEQ ID NO: 154)
AB-008714	(SEQ ID NO: 155)	(SEQ ID NO: 156)
AB-008715	(SEQ ID NO: 157)	(SEQ ID NO: 158)
AB-008716	(SEQ ID NO: 159)	(SEQ ID NO: 160)
AB-008717	(SEQ ID NO: 161)	(SEQ ID NO: 162)
AB-008718	(SEQ ID NO: 163)	(SEQ ID NO: 164)
AB-008719	(SEQ ID NO: 165)	(SEQ ID NO: 166)
AB-008720	(SEQ ID NO: 167)	(SEQ ID NO: 168)
AB-008721	(SEQ ID NO: 169)	(SEQ ID NO: 170)
AB-008722	(SEQ ID NO: 171)	(SEQ ID NO: 172)
AB-008723	(SEQ ID NO: 173)	(SEQ ID NO: 174)
AB-008724	(SEQ ID NO: 175)	(SEQ ID NO: 176)
AB-008725	(SEQ ID NO: 177)	(SEQ ID NO: 178)
AB-008726	(SEQ ID NO: 179)	(SEQ ID NO: 180)
AB-008727	(SEQ ID NO: 181)	(SEQ ID NO: 182)
AB-008728	(SEQ ID NO: 183)	(SEQ ID NO: 184)
AB-008729	(SEQ ID NO: 185)	(SEQ ID NO: 186)
AB-008730	(SEQ ID NO: 187)	(SEQ ID NO: 188)
AB-008731	(SEQ ID NO: 189)	(SEQ ID NO: 190)
AB-008732	(SEQ ID NO: 191)	(SEQ ID NO: 192)
AB-008733	(SEQ ID NO: 193)	(SEQ ID NO: 194)
AB-008734	(SEQ ID NO: 195)	(SEQ ID NO: 196)
AB-008735	(SEQ ID NO: 197)	(SEQ ID NO: 198)
AB-008736	(SEQ ID NO: 199)	(SEQ ID NO: 200)

[0219] FIG. **54** shows treatment of mice carrying the tumors formed by PC3 tumor cells using an exemplary bispecific antibody led to tumor burden reduction and/or elimination of tumors in mice. Said bispecific antibody was generated using a 1+1 (Fab−)(scFv−)Fc format (shown in FIG. **53**) and the CD3 arm of which comprises the VH/VL sequence of AB-008707.

[0220] Additional antigens that can be targeted by a further binding domain in a bispecific or multi-specific antibody that comprises an antigen binding domain of an EphA2 antibody described herein, include, but are not limited to, antigens on NK cells to activate or inhibit NK cell pathways.

[0221] Illustrative examples of such an antigen include activating NK cell receptors such as activating human Killer Immunoglobulin-like Receptor (KIR) family members, activating Ly49 family members, CD94-NKG2C/E/H heterodimeric receptors, NKG2D, SLAM family receptors including 2B4/CD244, CRACC/SLAMF7, NTB-A/SLAMF6, Fc gamma RIIIA/CD16a, CD27, CD100/Semaphorin 4D, CD160, natural cytotoxicity receptors, including NKp30, NKp44, and NKp46, DNAM-1/CD226, IL-2 receptor subunit beta (IL-2RB), IL-2 receptor subunit gamma (IL-2RG), 4-1BB/CD137, and CRTAM. Illustrative examples of such an antigen include inhibiting NK cell receptors such as inhibiting human KIR family members, inhibiting Ly49 family members, CD94/NKG2A, TIGIT and CD96, sialic acid-binding Siglecs (Siglec-3, -7, and -9), ILT2/LILRB1, KLRG1, LAIR-1, CD161/NKR-P1A, and CEACAM-1.

[0222] In some embodiments, an EphA2 antibody is incorporated into a bispecific or multi-specific antibody that comprises a binding domain from an agonist antibody that binds to 4-1BB.

[0223] In one embodiment the 4-1 in agonist antibody is a bispecific antibody that is capable of binding to both EphA2 and 4-1BB. For purposes of this application, the term "4-1BB engager," refers to the portion of a molecule (e.g., a bispecific antibody capable of binding to both 4-1BB and EphA2) that binds to 4-1BB. In some embodiments, the 4-1BB engager is an antibody or an antibody fragment (e.g., scFv) that binds to 4-1BB. In some embodiments, the 4-1BB engager is a multimeric 4-1BB ligand ("4-1BBL"), for example, a 4-1BBL trimer. In some embodiments, as further described below, the bispecific antibody comprises one or more scFv fragments of an anti-4-1BB antibody and an EphA2 antibody disclosed herein. In one embodiment, the 4-1BB agonist antibody is a trispecific antibody. Examples of bispecific and trispecific antibody constructs are described in US 20190010248, FIG. 1; WO2020025659, FIG. 1; Berezhnoy A. et al., Converting PD-L1-induced T-lymphocyte Inhibition into CD137-mediated Costimulation via PD-L1×CD137 Bispecific DART® Molecules. Poster presented at 30th EORTC/AACR/NCI Symposium, Nov. 13-16, 2018, Dublin, Ireland, Compte, M. et al., A tumor-targeted trimeric 4-1BB-agonistic antibody induces potent tumor-targeting immunity without systemic toxicity. Nat Com 9, 4809 (2018), FIG. **1**; U.S. Ser. No. 10/239,949, FIG. 10; and WO2019/092452, Example 2.

[0224] Non-limiting examples of the scFv of the anti 4-1BB antibodies that can be used in the fusion proteins are shown in Table 10. Exemplary linkers that can be used to connect the heavy chain or light chain of the EphA2 antibody and the 4-1BB ligand domains in the fusion are shown in Table 14. Various fusion proteins comprising the 4-1BBL and the EphA2 antibodies are shown in Table 14.

TABLE 10

scFv4-1BB sequences used in the fusion construct

Part Name	Description	Part Sequence
SCFV-009036-01	Anti-4-1BB SCFV	(SEQ ID NO: 111)
SCFV-009036-02	Anti-4-1BB SCFV	(SEQ ID NO: 112)
SCFV-009365-01	Anti-4-1BB SCFV	(SEQ ID NO: 113)
SCFV-009365-02	Anti-4-1BB SCFV	(SEQ ID NO: 114)
SCFV-009366-01	Anti-4-1BB SCFV	(SEQ ID NO: 115)
SCFV-009366-02	Anti-4-1BB SCFV	(SEQ ID NO: 116)
SCFV-009037-01	Anti-4-1BB SCFV	(SEQ ID NO: 117)
SCFV-009037-02	Anti-4-1BB SCFV	(SEQ ID NO: 118)
SCFV-009809-01	Anti-4-1BB SCFV	(SEQ ID NO: 119)
SCFV-009809-02	Anti-4-1BB SCFV	(SEQ ID NO: 120)
SCFV-009810-01	Anti-4-1BB SCFV	(SEQ ID NO: 121)
SCFV-009810-02	Anti-4-1BB SCFV	(SEQ ID NO: 122)

TABLE 11

Heavy Chain Sequences Used in 4-1BB Bispecific antibody		
IgG Identifier	Sequence of CH123 fused to 4-1BB engager	Sequence of CH123 not fused (if applicable)
hIgG1	(SEQ ID NO: 123)	N/A
hIgG4_S228P_R409K	(SEQ ID NO: 124)	N/A
hIgG1_P329G_L234A_L235A	(SEQ ID NO: 125)	N/A
mIgG1	(SEQ ID NO: 126)	N/A
hIgG1_kih	(SEQ ID NO: 127)	(SEQ ID NO: 128)
hIgG4_S228P_R409K_kih	(SEQ ID NO: 129)	(SEQ ID NO: 130)
hIgG1-Xmab	(SEQ ID NO: 131)	(SEQ ID NO: 132)

Note: "CH123" refers to an IgG heavy chain constant region comprising CH1-CH2-hinge-CH3.

[0225] Additional 4-1BB antibody sequences suitable for use in generating bispecific constructs that are capable of binding to both EphA2 and 4-1BB are provided in Table 12.

TABLE 12

4-1BB antibody sequences suitable for generating bispecific constructs		
4-1BB antibody	V_H	V_L
hu106-1	(SEQ ID NO: 721)	(SEQ ID NO: 743)
TABBY106	(SEQ ID NO: 722)	(SEQ ID NO: 744)
TABBY107	(SEQ ID NO: 723)	(SEQ ID NO: 745)
BBK4	(SEQ ID NO: 724)	(SEQ ID NO: 746)
hu39E3G.4	(SEQ ID NO: 725)	(SEQ ID NO: 747)
Urelumab	(SEQ ID NO: 726)	(SEQ ID NO: 748)
CTX-471	(SEQ ID NO: 727)	(SEQ ID NO: 749)
CTX-471AF	(SEQ ID NO: 728)	(SEQ ID NO: 750)
MOR-6032	(SEQ ID NO: 729)	(SEQ ID NO: 751)
MOR-7361	(SEQ ID NO: 730)	(SEQ ID NO: 752)
MOR-7480	(SEQ ID NO: 731)	(SEQ ID NO: 753)
MOR-7480.1/Utomilumab	(SEQ ID NO: 732)	(SEQ ID NO: 754)
MOR-7483	(SEQ ID NO: 733)	(SEQ ID NO: 755)
Utomilumab	(SEQ ID NO: 734)	(SEQ ID NO: 756)
EU-101	(SEQ ID NO: 735)	(SEQ ID NO: 757)
STA-551	(SEQ ID NO: 736)	(SEQ ID NO: 758)
LVGN-6051	(SEQ ID NO: 737)	(SEQ ID NO: 759)
AGEN-2373	(SEQ ID NO: 738)	(SEQ ID NO: 760)
ADG-106/AG10131	(SEQ ID NO: 739)	(SEQ ID NO: 761)
ATOR-1017	(SEQ ID NO: 740)	(SEQ ID NO: 762)
Hz4B4-1	(SEQ ID NO: 741)	(SEQ ID NO: 763)
Hz4B4-2	(SEQ ID NO: 742)	(SEQ ID NO: 764)

[0226] As shown in FIG. **49**, one exemplary bispecific antibody that comprises two scFv fragments of anti-4-1BB antibody linked to the HC of the EphA2 antibody AB-010361 showed capability of reducing tumor growth in mice carrying tumors formed from CT26 cells.

[0227] In one embodiment, the fusion molecule comprises a silenced human IgG1 with three human 4-1BB ligand ectodomains attached via flexible linkers. See WO2019086499, FIGS. **1-3**. Other 4-1BBL fusion molecules can be utilized with the tumor-targeting antibodies described herein, see Zhang et al., Targeted and Untargeted CD137L Fusion Proteins for the Immunotherapy of Experimental Solid Tumors, Clin Cancer Res., 2758-2767 (2007), FIG. 1; (Kermer et al., Combining Antibody-Directed Presentation of IL-15 and 4-1BBL in a Trifunctional Fusion Protein for Cancer Immunotherapy, Mol Cancer Ther, 112-121 (2014), FIG. **1**.

[0228] In some embodiments, an EphA2 antibody is incorporated into a fusion molecule comprising one or more 4-1BB ligands (4-1BBL). In one embodiment, a trimer of 4-1BBL is C-terminally fused to either the light chain or heavy chain of an EphA2 antibody. In one embodiment, one or more individual 4-1BBL domains are connected via linkers, with one of the domains additionally fused to the EphA2 antibody via a linker. The 4-1BBL domains comprise entire ECD portion of the molecule or truncated forms that can still bind and activate 4-1BB. See WO2019086499, FIGS. 1-3. Non-limiting examples of the 4-1BBL domains that can be used in the fusion proteins are shown in Table 13. Exemplary linkers that can be used to connect the heavy chain or light chain of the EphA2 antibody and the 4-1BB ligand domains in the fusion are shown in Table 14. Various fusion proteins comprising the 4-1BBL and the EphA2 antibodies are shown in Table 14.

TABLE 13

4-1BBL domains used in the fusion constructs		
Part Name	Description	Part Sequence
h4-1BBL (THD)	Human 4-1BB ligand, residues 89-242 of ECD	(SEQ ID NO: 106)
h4-1BBL (THD)3-10	Human 4-1BB Ligand (89-242) Trimer with (GGSGG)2 linkers (SEQ ID NO: 781) between promoters	(SEQ ID NO: 107)
h41BBL (THD)3-2	Human 4-1BB Ligand (89-242) Trimer with GG linkers between promoters	(SEQ ID NO: 108)
h4-1BBL (71-248)	Human 4-1BB ligand, residues 71-248 of ECD	(SEQ ID NO: 109)
h4-1BBL (71-248)3-10	Human 4-1BB Ligand (71-248) Trimer with (GGSGG)2 linkers between promoters	(SEQ ID NO: 110)

Note:

Part names are comprised of the 4-1BBL domain, number of domains, and length of linker between domains.

TABLE 14

4-1BB fusion constructs

Molecule Name	IgG Identifier	Valency (tumor:4-1BB)	Fusion Location	Linker to 4-1BB-engaging Part	4-1BB Engager*
h1-2p2-LC-h41BBL(THD)3-2	hIgG1	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-2
h1-2p2-LC-h41BBL(THD)3-10	hIgG1	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-10
h1-2p2-LC-h41BBL(71-248)3-10	hIgG1	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(71-248)3-10
h4-2p2-LC-h41BBL(THD)3-2	hIgG4_S228P_R409K	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-2
h4-2p2-LC-h41BBL(THD)3-10	hIgG4_S228P_R409K	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-10
h4-2p2-LC-h41BBL(71-248)3-10	hIgG4_S228P_R409K	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(71-248)3-10
h1s-2p2-LC-h41BBL(THD)3-2	hIgG1_P329G_L234A_L235A	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-2
h1s-2p2-LC-h41BBL(THD)3-10	hIgG1_P329G_L234A_L235A	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(THD)3-10
h1s-2p2-LC-h41BBL(71-248)3-10	hIgG1_P329G_L234A_L235A	2:2	LC-Cter	(SEQ ID NO: 105)	h41BBL(71-248)3-10
h4-2p2-HC-SCFV9365-02	hIgG4_S228P_R409K	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9365-02
m1-2p2-HC-SCFV9365-02	mIgG1	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9365-02
h4-2p2-HC-SCFV9036-02	hIgG4_S228P_R409K	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9036-02
m1-2p2-HC-SCFV9036-02	mIgG1	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9036-02
h4-2p2-HC-SCFV9809-01	hIgG4_S228P_R409K	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9809-01
m1-2p2-HC-SCFV9809-01	mIgG1	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9809-01
h4-2p2-HC-SCFV9810-01	hIgG4_S228P_R409K	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9810-01
m1-2p2-HC-SCFV9810-01	mIgG1	2:2	HC-Cter	(SEQ ID NO: 91)	SCFV9810-01
h1-2p1-HC-h41BBL(THD)3-2	hIgG1_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-2
h1-2p1-HC-h41BBL(THD)3-10	hIgG1_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-10
h1-2p1-HC-h41BBL(71-248)3-10	hIgG1_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(71-248)3-10
h4-2p1-HC-h41BBL(THD)3-2	hIgG4_S228P_R409K_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-2
h4-2p1-HC-h41BBL(THD)3-10	hIgG4_S228P_R409K_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-10
h4-2p1-HC-h41BBL(71-248)3-10	hIgG4_S228P_R409K_kih	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(71-248)3-10
hl-Xmab-2p1-HC-h41BBL(THD)3-2	hIgG1-Xmab	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-2
h1-Xmab-2p1-HC-h41BBL(THD)3-10	hIgG1-Xmab	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(THD)3-10
hl-Xmab-2p1-HC-h41BBL(71-248)3-10	hIgG1-Xmab	2:1	HC-Cter	(SEQ ID NO: 91)	h41BBL(71-248)3-10

*Terminology is as follows, using "h42BBL(THD)3-2" as an illustrative example:

1. h stands for "human" and m stands for "mouse" (or m = mouse)

2. 41BBL stand for 4-1BB ligand (native sequence)

3. THD stand for "TNF homology domain" portion of 41BBL. In instances where a number range in parentheses after 41BBL, e.g., h41BBL(71-248)3-10, refers to that residue numbers 71-248 of the 4-1BB ligand were used.

4. "3" refers to three copies connected by polypeptide linker

5. "–2" typically refers to the length of the linker used to connect the ligands together. In one example, h41BBL(THD)3-2 represent h41BBL(THD) + GG + h41BBL(THD) + GG + h41BBL(THD); In another example, h41BBL(71-248)3-10 represents h41BBL(71-248) + (GGSGG)2 (SEQ ID NO: 781) + h41BBL(71-248) + (GGSGG)2 (SEQ ID NO: 781) + h41BBL(71-248)

[0229] In some embodiments, the tumor-targeting antibody is conjugated to one or more TLR agonists. In one embodiment, the tumor-targeting antibody conjugated to a TLR agonist is a bispecific or multispecific antibody. In some embodiments, the tumor-targeting antibody is a bispecific or multispecific antibody comprising an antigen binding domain of an antibody described herein that further comprises a TLR agonist.

[0230] In some embodiments, a bispecific or multispecific antibody comprising an antigen binding domain of an antibody described herein further comprises a binding domain that binds to a checkpoint antigen, PD1, PDL1, CTLA-4, ICOS, PDL2, IDO1, IDO2, B7-H3, B7-H4, BTLA, HVEM, TIM3, GAL9, GITR, HAVCR2, LAG3, KIR, LAIR1, LIGHT, MARCO, OX-40, SLAM, 2B4, CD2, CD27, CD28, CD30, CD40, CD70, CD80, CD86, CD137, CD160. CD39,

VISTA, TIGIT, CGEN-15049, 2B4, CHK 1, CHK2, A2aR, or a B-7 family ligand or its receptor.

[0231] In some embodiments, a bispecific or multispecific antibody comprising an antigen binding domain of an antibody described herein further comprises a binding domain that targets a tumor-associated antigen. Illustrative tumor-associate antigens include, but are not limited to, EpCAM, HER2/neu, HER3/neu, G250, CEA, MAGE, proteoglycans. VEGF, VEGFR, EGFR, ErbB2, ErbB3, MET, IGF-1R, PSA, PSMA, EphA2, EphA3, EphA4, folate binding protein $\alpha V\beta 3$-integrin, integrin $\alpha 5\beta 1$ HLA, HLA-DR, ASC, CD1, CD2, CD4, CD6, CD7, CD8, CD11, CD13, CD14, CD19, CD20, CD21, CD22, CD23, CD24, CD30, CD33, CD37, CD40, CD41, CD47, CD52, c-erb-2, CALLA, MHCII, CD44v3, CD44v6, p97, ganglioside GM1, GM2, GM3, GD1 a, GD1 b, GD2, GD3, GT1 b, GT3, GQ1, NY-ESO-1, NFX2, SSX2, SSX4 Trp2, gp100, tyrosinase, Muc-1, telomerase, survivin, SLAMF7 EphG250, p53, CA125 MUC, Wue antigen, Lewis Y antigen, HSP-27, HSP-70, HSP-72, HSP-90, Pgp, PMEL17, MCSP, and cell surface targets GC182, GT468 or GT512.

[0232] In some embodiments, a binding domain of an antibody described herein may be incorporated into a chimeric antigen receptor (CAR-T) construct, an engineered TCR-T cell construct, or a combined CAR-T/TCR-T construct comprising the complete TCR complex (see Hardy et al. "Implications of T cell receptor biology on the development of new T cell therapies for cancer, Immunotherapy Vol. 12, No. 1. Published Online: 6 Jan. 2020 https://doi.org/10.2217/imt-2019-0046). Such constructs can be used to generate a modified immune cell such as a T-cell, NK-cell, or monocyte/macrophage comprising the binding domain of an antibody described herein. For example, in some embodiments, a first-generation CAR joins a single-chain variable region from the antibody to a CD3zeta (ζ) intracellular signaling domain of a CD3 T-cell receptor through hinge and transmembrane domains. In some embodiments, the CAR may contain co-stimulating domains, e.g., a second or third generation CAR may include an additional one or two-co-stimulating domains, such as 4-1BB, CD28, or OX-40). In additional embodiments, a CAR-containing cell, e.g., a CAR-T cell, may additionally be engineered to with an inducible expression component such as a cytokine, e.g., IL-12 or IL-15 to increase activation of CAR-T cells and activate innate immune cells.

[0233] In one embodiment of any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises all six CDRs (HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3) sequences from the individual antibodies disclosed in Tables 6 and 7.

[0234] In one embodiment of any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises the V_H and V_L sequences from the individual antibodies disclosed in Table 8.

[0235] In one embodiment of any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises all six CDRs (HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3) sequences from any one of antibodies AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699. AB-010700, AB-010701, or AB-010702. In some embodiments, in any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises all six CDRs (HCDR1, HCDR2, HCDR3, LCDR1. LCDR2, and LCDR3) sequences from AB-010361 or AB-010699.

[0236] In one embodiment of any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises the V_H and V_L sequences any one of AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; and AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666. AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698. AB-010699, AB-010700, AB-010701, or AB-010702. In one embodiment of any of the above bispecific or multispecific antibody constructs, the tumor-targeting binding domain comprises the V_H and V_L sequences of AB-010361 or AB-010699.

Generation of Antibodies

[0237] EphA2 antibodies as disclosed herein are commonly produced using vectors and recombinant methodology well known in the art (see, e.g., Sambrook & Russell, Molecular Cloning: A Laboratory Manual, Cold Spring Harbor Laboratory Press; Ausubel, Current Protocols in Molecular Biology). Reagents, cloning vectors, and kits for genetic manipulation are available from commercial vendors. Accordingly, in a further aspect of the invention, provided herein are isolated nucleic acids encoding a V_H and/or V_L region, or fragment thereof, of any of the tumor-targeting antibodies as described herein; vectors comprising such nucleic acids and host cells into which the nucleic acids are introduced that are used to replicate the antibody-encoding nucleic acids and/or to express the antibodies. Such nucleic acids may encode an amino acid sequence containing the V_L and/or an amino acid sequence containing the V_H of the tumor-targeting antibody (e.g., the light and/or heavy chains of the antibody). In some embodiments, the host cell contains (1) a vector containing a polynucleotide that encodes the V_L amino acid sequence and a polynucleotide that encodes the V_H amino acid sequence, or (2) a first vector containing a polynucleotide that encodes the V_L amino acid sequence and a second vector containing a polynucleotide that encodes the V_H amino acid sequence.

[0238] In a further aspect, the invention provides a method of making an EphA2 antibody as described herein. In some embodiments, the method includes culturing a host cell as described in the preceding paragraph under conditions suitable for expression of the antibody. In some embodiments, the antibody is subsequently recovered from the host cell (or host cell culture medium).

[0239] Suitable vectors containing polynucleotides encoding antibodies of the present disclosure, or fragments thereof, include cloning vectors and expression vectors. While the cloning vector selected may vary according to the host cell intended to be used, useful cloning vectors generally can self-replicate, may possess a single target for a particular restriction endonuclease, and/or may carry genes for a marker that can be used in selecting clones containing the vector. Examples include plasmids and bacterial viruses, e.g., pUC18, pUC19, Bluescript (e.g., pBS SK+) and its derivatives, mpl8, mpl9, pBR322, pMB9, ColE1 plasmids, pCR1, RP4, phage DNAs, and shuttle vectors. These and many other cloning vectors are available from commercial vendors.

[0240] Expression vectors generally are replicable polynucleotide constructs that contain a nucleic acid of the present disclosure. The expression vector can be replicable in the host cells either as episomes or as an integral part of the chromosomal DNA. Suitable expression vectors include but are not limited to plasmids and viral vectors, including adenoviruses, adeno-associated viruses, retroviruses, and any other vector.

[0241] Suitable host cells for expressing an EphA2 antibody as described herein include both prokaryotic and eukaryotic cells. For example, an EphA2 antibody may be produced in bacteria, when glycosylation and Fc effector function are not needed. After expression, the antibody may be isolated from the bacterial cell paste in a soluble fraction and can be further purified. Alternatively, the host cell may be a eukaryotic host cell, including eukaryotic microorganisms, such as filamentous fungi or yeast, including fungi and yeast strains whose glycosylation pathways have been "humanized," resulting in the production of an antibody with a partially or fully human glycosylation pattern, vertebrate, invertebrate, and plant cells. Examples of invertebrate cells include insect cells. Numerous baculoviral strains have been identified which may be used in conjunction with insect cells. Plant cell cultures can also be utilized as host cells.

[0242] In some embodiments, vertebrate host cells are used for producing an EphA2 antibody of the present disclosure. For example, mammalian cell lines such as a monkey kidney CV1 line transformed by SV40 (COS-7); human embryonic kidney line (293 or 293 cells as described, e.g., in Graham et al., *J. Gen Virol.* 36:59, 1977; baby hamster kidney cells (BHK); mouse sertoli cells (TM4 cells as described, e.g., in Mather, *Biol. Reprod.* 23:243-251, 1980 monkey kidney cells (CVI); African green monkey kidney cells (VERO-76); human cervical carcinoma cells (HELA); canine kidney cells (MDCK; buffalo rat liver cells (BRL 3A); human lung cells (W138); human liver cells (Hep G2); mouse mammary tumor (MMT 060562); TRI cells, as described, e.g., in Mather et al., Annals N.Y. Acad. Sci. 383:44-68, 1982; MRC 5 cells; and FS4 cells may be used to express an tumor-targeting antibodies. Other useful mammalian host cell lines include Chinese hamster ovary (CHO) cells, including DHFR− CHO cells (Urlaub et al., *Proc. Natl. Acad. Sci. USA* 77:4216, 1980); and myeloma cell lines such as Y0, NS0 and Sp2/0. Host cells of the present disclosure also include, without limitation, isolated cells, in vitro cultured cells, and ex vivo cultured cells. For a review of certain mammalian host cell lines suitable for antibody production, see, e.g., Yazaki and Wu, Methods in Molecular Biology, Vol. 248 (B. K. C. Lo, ed., Humana Press, Totowa, NJ), pp. 255-268, 2003.

[0243] In some embodiments, an EphA2 antibody of the present disclosure is produced by a CHO cell line, e.g., the CHO-K1 cell line. One or more expression plasmids can be introduced that encode heavy and light chain sequences. For example, in one embodiment, an expression plasmid encoding a heavy chain disclosed herein, e.g., SEQ ID NO: 67, and an expression plasmid encoding a light chain disclosed herein, e.g., SEQ ID NO: 68 are transfected into host cells as linearized plasmids at a ratio of 1:1 in the CHO-K1 host cell line using reagents such as Freestyle Max reagent. Fluorescence-activated cell sorting (FACS) coupled with single cell imaging can be used as a cloning method to obtain a production cell line.

[0244] A host cell transfected with an expression vector encoding an EphA2 antibody of the present disclosure, or fragment thereof, can be cultured under appropriate conditions to allow expression of the polypeptide to occur. The polypeptides may be secreted and isolated from a mixture of cells and medium containing the polypeptides. Alternatively, the polypeptide may be retained in the cytoplasm or in a membrane fraction and the cells harvested, lysed, and the polypeptide isolated using a desired method.

[0245] In some embodiments, an EphA2 antibody of the present disclosure can be produced by in vitro synthesis (see, e.g., Sutro Biopharma biochemical protein synthesis platform).

[0246] In some embodiments, provided herein is a method of generating variants of an EphA2 antibody as disclosed herein. Thus, for example, a construct encoding a variant of a V_H CDR3 as described herein can be modified and the V_H region encoded by the modified construct can be tested for binding activity to CT26 cells and/or in vivo tumor-targeting activity in the context of a V_H region as described herein, that is paired with a V_L region or variant region as described herein. Similarly, a construct encoding a variant of a V_L CDR3 as described herein can be modified and the V_L region encoded by the modified construct can be tested for binding to CT26 cells, or other tumor cells, and/or in vivo tumor-targeting activity efficacy. Such an analysis can also be performed with other CDRs or framework regions and an antibody having the desired activity can then be selected.

Tumor-Targeting Antibody Conjugates/Co-Stimulatory Agents

[0247] In a further aspect, an EphA2 antibody of the present disclosure may be conjugated or linked to therapeutic, imaging/detectable moieties, or enzymes. For example, the tumor-targeting antibody may be conjugated to a detectable marker, a cytotoxic agent, an immunomodulating agent, an imaging agent, a therapeutic agent, an oligonucleotide, or an enzyme. Methods for conjugating or linking antibodies to a desired molecule are well known in the art. The desired molecule may be linked to the antibody covalently or by non-covalent linkages.

[0248] In some embodiments, the antibody is conjugated, either directly or via a cleavable or non-cleavable linker, to a cytotoxic moiety or other moiety that exerts their effects on critical cellular processes required for survival ("payload") to form an antibody-drug conjugate ("ADC").

[0249] In some embodiments, the antibody is conjugated to an auristatin. In some embodiments, the ADC is an EphA2 antibody disclosed herein conjugated to a ZymeLink™ Auristatin (ZLA) payload. Such ADC constructs are referred to as the ZymeLink™ ADC constructs, as further disclosed below.

[0250] In some embodiments, ZymeLink™ ADCs of the present disclosure comprise an EphA2 antibody conjugated to a ZLA having structure (1) (also referred to herein as "Compound 1") via a linker L.

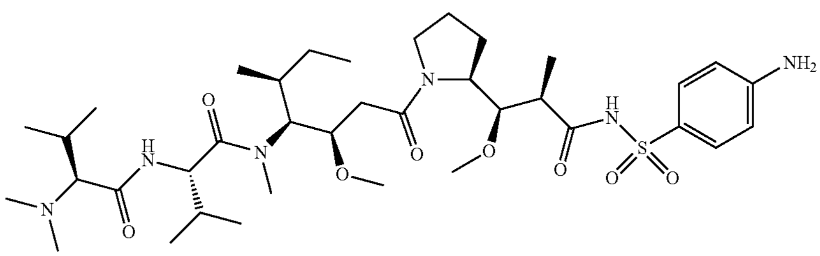
[0258] In some embodiments, in the ZymeLink™ ADCs of Formula (I), linker L is a peptide-containing linker. In some embodiments, in the ADCs of Formula (I), linker L comprises a dipeptide selected from Val-Lys. Ala-Lys, Phe-Lys. Val-Cit, Phe-Cit, Leu-Cit, Ile-Cit and Trp-Cit.

[0259] In some embodiments, in the ZymeLink™ ADCs of Formula (I), linker L comprises a dipeptide and a stretcher.

[0260] In certain embodiments, linker L in ZymeLink™ ADCs of Formula (I) is conjugated to the antibody via a cysteine residue or a lysine residue on the antibody. In some embodiments, linker L in ADCs of Formula (I) is conjugated to the antibody via a lysine residue on the antibody.

[0261] In some embodiments, in the ZymcLink™ ADCs of Formula (I), n is an integer from 1 to 8. In some embodiments, in the ADCs of Formula (I), n has a value

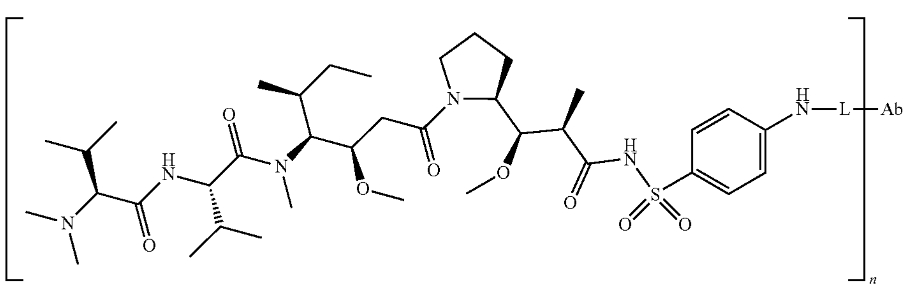
1



[0251] In certain embodiments, the ZymeLink™ ADCs have Formula (I):

from 2 to 8, e.g., from 2 to 6, from 3 to 7, or from 4 to 8. In terms of upper limits, n can be less than 12, e.g., less than

(I)



[0252] wherein:

[0253] L is a cleavable linker;

[0254] n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and

[0255] Ab is the antibody.

[0256] In some embodiments, the each of the n drugs in formula (I) is independently conjugated to the antibody Ab via one of n linkers L.

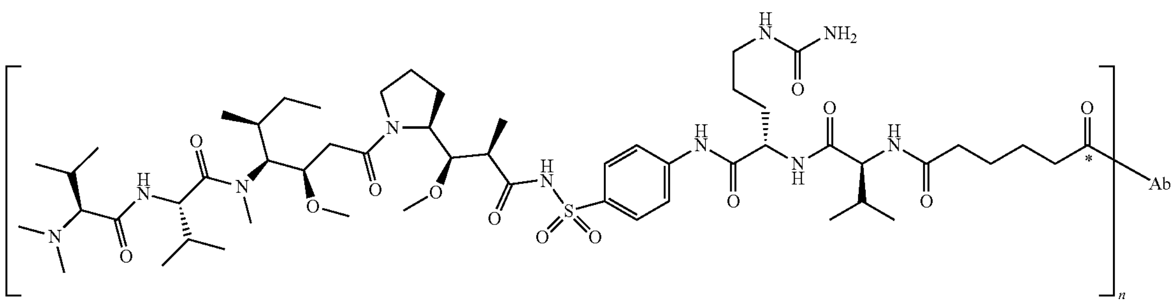
[0257] In some embodiments, in the ZymeLink™ ADCs of general Formula (I), linker L is a protease-cleavable linker.

11, less than 10, less than 9, less than 8, less than 7, less than 6, less than 5, less than 4, or less than 3. In terms of lower limits, n can be greater than 2, e.g., greater than 3, greater than 4, greater than 5, greater than 6, greater than 7, greater than 8, greater than 9, greater than 10, or greater than 11. Larger values of n, e.g., greater than 12, are also contemplated.

[0262] Combinations of any of the foregoing embodiments for ZymeLink™ ADCs of Formula (I) are also contemplated and each combination forms a separate embodiment for the purposes of the present disclosure.

[0263] In certain embodiments, the ZymeLink™ ADCs of the present disclosure have Formula (II):

(II)



[0264] wherein:

[0265] n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and

[0266] Ab is the antibody.

[0267] In some embodiments, the each of the n drugs in formula (II) is independently conjugated to the antibody Ab via one of n linkers L.

[0268] In certain embodiments, the carbonyl group of Formula (II) marked with an asterisk forms a peptide bond with the side chain amine group of a lysine residue on the antibody (Ab).

[0269] In some embodiments, in the ZymeLink™ ADCs of general Formula (II), n is an integer from 1 to 8. In some embodiments, in the ADCs of general Formula (II), n has a value from 2 to 8, e.g., from 2 to 6, from 3 to 7, or from 4 to 8. In terms of upper limits, n can be less than 12, e.g., less than 11, less than 10, less than 9, less than 8, less than 7, less than 6, less than 5, less than 4, or less than 3. In terms of lower limits, n can be greater than 2, e.g., greater than 3, greater than 4, greater than 5, greater than 6, greater than 7, greater than 8, greater than 9, greater than 10, or greater than 11. Larger values of n, e.g., greater than 12, are also contemplated.

Linker L

[0270] In the ADCs of the present disclosure, the antibody is linked to an auristatin analogue disclosed herein by a linker L. Linkers are bifunctional or multifunctional moieties capable of linking one or more drug molecules to an antibody. A linker may be bifunctional (or monovalent) such that it links a single drug to a single site on the antibody, or it may be multifunctional (or polyvalent) such that it links more than one drug molecule to a single site on the antibody. Linkers capable of linking one drug molecule to more than one site on the antibody may also be multifunctional.

[0271] Certain linkers useful in the present disclosure can be up to 30 carbon atoms in length. For example, the linkers can each independently be from 5 to 20 carbon atoms in length. The types of bonds used to link the linker to the cytotoxic agent and antibody of the present disclosure include, but are not limited to, amides, amines, esters, carbamates, ureas, thioethers, thiocarbamates, thiocarbonate and thioureas. One of skill in the art will appreciate that other types of bonds are useful in the provided ADCs.

[0272] Attachment of a linker to an antibody can be accomplished in a variety of ways, such as through surface lysines on the antibody, reductive coupling to oxidized carbohydrates on the antibody, or through cysteine residues on the antibody liberated by reducing interchain disulfide linkages. Alternatively, attachment of a linker to an antibody may be achieved by modification of the antibody to include additional cysteine residues (see, for example, U.S. Pat. Nos. 7,521,541; 8,455,622 and 9,000,130) or non-natural amino acids that provide reactive handles, such as selenomethionine, p-acetylphenylalanine, formylglycine or p-azidomethyl-L-phenylalanine (see, for example, Hofer et al., Biochemistry, 48:12047-12057 (2009); Axup et al., PNAS, 109:16101-16106 (2012); Wu et al., PNAS, 106:3000-3005 (2009); Zimmerman et al., Bioconj. Chem., 25:351-361 (2014)), to allow for site-specific conjugation.

[0273] Linkers include at least one functional group capable of reacting with the target group or groups on the antibody, and one or more functional groups capable of reacting with a target group on the drug. Suitable functional groups are known in the art and include those described, for example, in Bioconjugate Techniques (G. T. Hermanson, 2013, Academic Press).

[0274] Examples of target groups on the antibody to which a linker may be conjugated include the thiol groups of cysteine residues and the amine groups of lysine residues. Non-limiting examples of functional groups for reacting with free cysteines or thiols include maleimide, haloacetamide, haloacetyl, activated esters such as succinimide esters, 4-nitrophenyl esters, pentafluorophenyl esters, tetrafluorophenyl esters, anhydrides, acid chlorides, sulfonyl chlorides, isocyanates and isothiocyanates. Also useful in this context are "self-stabilizing" maleimides as described in Lyon et al., Nat. Biotechnol., 32:1059-1062 (2014). Non-limiting examples of functional groups for reacting with free amines include activated esters (such as N-hydroxysuccinamide (NHS) esters, sulfo-NHS esters, imido esters such as Traut's reagent, tetrafluorophenyl (TFP) esters and sulfodichlorophenyl esters), isothiocyanates, aldehydes and acid anhydrides (such as diethylenetriaminepentaacetic anhydride (DTPA)). Other examples include succinimido-1,1,3,3-tetra-methyluronium tetrafluoroborate (TSTU) and benzotriazol-1-yl-oxytripyrrolidinophosphonium hexafluorophosphate (PyBOP).

[0275] Other linkers include those having a functional group that allows for bridging of two interchain cysteines on the antibody, such as a ThioBridge® linker (Badescu et al., Bioconjug. Chem., 25:1124-1136 (2014)), a dithiomaleimide (DTM) linker (Behrens et al., Mol. Pharm., 12:3986-3998 (2015)), a dithioarylpyridazinedione-based linker (Lee et al., Chem. Sci., 7:799-802 (2016)), a dibromopyridazinedione-based linker (Maruani et al., Nat. Commun., 6:6645 (2015)) and others known in the art.

[0276] A linker may comprise one or more linker components. Typically, a linker will comprise two or more linker components. Exemplary linker components include functional groups for reaction with the antibody, functional groups for reaction with the drug, stretchers, peptide components, self-immolative groups, self-elimination groups, hydrophilic moieties, and the like. Various linker components are known in the art, some of which are described below.

[0277] Certain useful linker components can be obtained from various commercial sources, such as Pierce Biotechnology, Inc. (now Thermo Fisher Scientific Corporation, Waltham, MA) and Molecular Biosciences Inc. (Boulder, Colo.), or may be synthesized in accordance with procedures described in the art (see, for example, Toki et al., J. Org. Chem., 67:1866-1872 (2002); Dubowchik et al., Tetrahedron Letters, 38:5257-60 (1997); Walker, M. A., J. Org. Chem., 60:5352-5355 (1995); Frisch et al., Bioconjugate Chem., 7:180-186 (1996); U.S. Pat. Nos. 6,214,345 and 7,553,816, and International Patent Application Publication No. WO 02/088172).

[0278] Examples of linker components include, but are not limited to, N-(β-maleimidopropyloxy)-N-hydroxy succinimide ester (BMPS), N-(ε-maleimidocaproyloxy) succinimide ester (EMCS), N-[(R)-maleimidobutyryloxy]succinimide ester (GMBS), 1,6-hexane-bis-vinylsulfone (HBVS), succinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxy-(6-amidocaproate) (LC-SMCC), m-maleimidobenzoyl-N-hydroxysuccinimide ester (MBS), 4-(4-N-maleimidophenyl)butyric acid hydrazide (MPBH), succinimidyl 3-(bromoacetamido)propionate (SBAP), succinimidyl iodoacetate (SIA), succinimidyl (4-iodoacetyl)aminobenzoate (STAB), N-succinimidyl-3-(2-pyridyldithio) propionate (SPDP), N-succinimidyl-4-(2-pyridylthio)pentanoate (SPP), succinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxylate (SMCC), succinimidyl 4-(p-maleimidophenyl)butyrate (SMPB), succinimidyl 6-[(0-maleimidopropionamido)hexanoate](SMPH), iminothiolane (IT), sulfo-EMCS, sulfo-GMBS, sulfo-KMUS, sulfo-MBS, sulfo-SIAB, sulfo-SMCC, sulfo-SMPB and succinimidyl-(4-vinylsulfone)benzoate (SVSB).

[0279] Additional examples include bis-maleimide reagents such as dithiobismaleimidoethane (DTME), bismaleimido-trioxyethylene glycol (BMPEO), 1,4-bismaleimidobutane (BMB), 1,4 bismaleimidyl-2,3-dihydroxybutane (BMDB), bismaleimidohexane (BMH), bismaleimidoethane (BMOE), BM(PEG)2 and BM(PEG)3; bifunctional derivatives of imidoesters (such as dimethyl adipimidate HCl), active esters (such as disuccinimidyl suberate), aldehydes (such as glutaraldehyde), bis-azido compounds (such as bis (p-azidobenzoyl) hexanediamine), bis-diazonium derivatives (such as bis-(p-diazoniumbenzoyl)-ethylenediamine), diisocyanates (such as toluene 2,6-diisocyanate), and bis-active fluorine compounds (such as 1,5-difluoro-2,4-dinitrobenzene).

[0280] Suitable linkers typically are more chemically stable to conditions outside the cell than to conditions inside the cell, although less stable linkers may be contemplated in certain situations, such as when the drug is selective or targeted and has a low toxicity to normal cells. Linkers may be "cleavable linkers" or "non-cleavable linkers." A cleavable linker is typically susceptible to cleavage under intracellular conditions, for example, through lysosomal processes. Examples include linkers that are protease-sensitive, acid-sensitive, reduction-sensitive or photolabile. Non-cleavable linkers by contrast, rely on the degradation of the antibody in the cell, which typically results in the release of an amino acid-linker-toxin moiety.

[0281] In accordance with the present disclosure, linker L comprised by the ADCs of Formula (I) is a cleavable linker. Suitable cleavable linkers include, for example, linkers comprising a peptide component that includes two or more amino acids and is cleavable by an intracellular protease, such as lysosomal protease or an endosomal protease. A peptide component may comprise amino acid residues that occur naturally and/or minor amino acids and/or non-naturally occurring amino acid analogues, such as citrulline. Peptide components may be designed and optimized for enzymatic cleavage by a particular enzyme, for example, a tumor-associated protease, cathepsin B, C or D, or a plasmin protease.

[0282] In certain embodiments, linker L comprised by the ADCs of Formula (I) may be a peptide-containing linker. In some embodiments, linker L comprised by the ADCs may be a dipeptide-containing linker, such as a linker containing valine-citrulline (Val-Cit) or phenylalanine-lysine (Phe-Lys). Other examples of suitable dipeptides for inclusion in linker L include Val-Lys, Ala-Lys, Me-Val-Cit, Phe-homo-Lys, Phe-Cit, Leu-Cit, Ile-Cit, Trp-Cit, Phe-Arg, Ala-Phe, Val-Ala, Met-Lys, Asn-Lys, Ile-Pro, Ile-Val, Asp-Val, His-Val, Met-(D)Lys, Asn-(D)Lys, Val-(D)Asp, NorVal-(D)Asp, Ala-(D)Asp, Me3Lys-Pro, PhenylGly-(D)Lys, Met-(D)Lys, Asn-(D)Lys, Pro-(D)Lys and Met-(D)Lys. Cleavable linkers may also include longer peptide components such as tripeptides, tetrapeptides or pentapeptides. Examples include, but are not limited to, the tripeptides Met-Cit-Val, Gly-Cit-Val, (D)Phe-Phe-Lys and (D)Ala-Phe-Lys, and the tetrapeptides Gly-Phe-Leu-Gly, Gly-Gly-Phe-Gly and Ala-Leu-Ala-Leu. In some embodiments, linker L comprised by the ADCs may be a peptide-containing linker, where the peptide is between two and five amino acids in length, for example, between two and four amino acids in length.

[0283] Additional examples of cleavable linkers include disulfide-containing linkers, such as, N-succinimydyl-4-(2-pyridyldithio) butanoate (SPBD) and N-succinimydyl-4-(2-pyridyldithio)-2-sulfo butanoate (sulfo-SPBD). Disulfide-containing linkers may optionally include additional groups to provide steric hindrance adjacent to the disulfide bond to improve the extracellular stability of the linker, for example, inclusion of a geminal dimethyl group. Other suitable linkers include linkers hydrolyzable at a specific pH or within a pH range, such as hydrazone linkers. Linkers comprising combinations of these functionalities may also be useful, for example, linkers comprising both a hydrazone and a disulfide are known in the art.

[0284] A further example of a cleavable linker is a linker comprising a β-glucuronide, which is cleavable by β-glucuronidase, an enzyme present in lysosomes and tumor interstitium (see, for example, De Graaf et al., Curr. Pharm. Des., 8:1391-1403 (2002)).

[0285] Cleavable linkers may optionally further comprise one or more additional components such as self-immolative and self-elimination groups, stretchers or hydrophilic moieties.

[0286] Self-immolative and self-elimination groups that find use in linkers include, for example, p-aminobenzyloxycarbonyl (PABC) and p-aminobenzyl ether (PABE) groups, and methylated ethylene diamine (MED). Other examples of self-immolative groups include, but are not limited to, aromatic compounds that are electronically similar to the PABC or PABE group such as heterocyclic derivatives, for example 2-aminoimidazol-5-methanol derivatives as described in U.S. Pat. No. 7,375,078. Other examples include groups that undergo cyclization upon amide bond hydrolysis, such as substituted and unsubstituted 4-aminobutyric acid amides (Rodrigues et al., Chemistry Biology, 2:223-227 (1995)) and 2-aminophenylpropionic acid amides (Amsberry et al., J. Org. Chem., 55:5867-5877 (1990)).

[0287] Stretchers that find use in linkers for ADCs include, for example, alkylene groups and stretchers based on aliphatic acids, diacids, amines or diamines, such as diglycolate, malonate, caproate and caproamide. Other stretchers include, for example, glycine-based stretchers, polyethylene glycol (PEG) stretchers and monomethoxy polyethylene glycol (mPEG) stretchers. PEG and mPEG stretchers also function as hydrophilic moieties.

[0288] Examples of components commonly found in cleavable linkers include, but are not limited to, SPBD, sulfo-SPBD, hydrazone, Val-Cit, maleidocaproyl (MC), MC-Val-Cit, MC-Val-Cit-PABC, Phe-Lys, MC-Phe-Lys, MC-Phe-Lys-PABC, maleimido triethylene glycolate (MT), MT-Val-Cit, MT-Phe-Lys, TFP and adipate (AD).

[0289] Selection of an appropriate linker for a given ADC may be readily made by the skilled person having knowledge of the art and taking into account relevant factors, such as the site of attachment to the antibody, any structural constraints of the payload drug and the hydrophobicity of the payload drug (see, for example, review in Nolting, Chapter 5, Antibody-Drug Conjugates: Methods in Molecular Biology, 2013, Ducry (Ed.), Springer).

[0290] In certain embodiments, linker L included in the ADCs of the present disclosure is a peptide-based linker having Formula (V):

$$\#—Z—[Str]_s—AA_1—[AA_2]_m—[X]_o—\% \qquad (V)$$

[0291] wherein:

[0292] Z is a linking group that joins the linker to a target group on the antibody;

[0293] Str is a stretcher;

[0294] AA_1 and AA_2 are each independently an amino acid, wherein AA_1-$[AA_2]_m$ forms a protease cleavage site;

[0295] X is a self-immolative group;

[0296] s is 0 or 1;

[0297] m is 1, 2, 3 or 4;

[0298] o is 0, 1 or 2;

[0299] # is the point of attachment to the antibody, and

[0300] % is the point of attachment to the auristatin analogue.

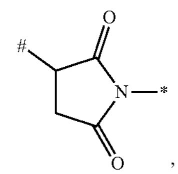
[0301] In some embodiments, in Formula (V), m is 1, 2 or 3.

[0302] In some embodiments, in Formula (V), s is 1.

[0303] In some embodiments, in Formula (V), o is 0 (i.e., X is absent).

[0304] In some embodiments, in Formula (V):

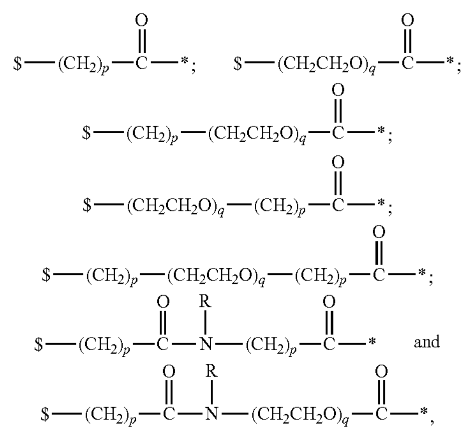
[0305] Z is



where # is the point of attachment to the antibody, and * is the point of attachment to the remainder of the linker.

[0306] In some embodiments, in Formula (V), Z is a carbonyl group (—C(O)—).

[0307] In some embodiments, in Formula (V), Str is selected from:



[0308] wherein:

[0309] each R is independently H or C_1-C_6 alkyl;

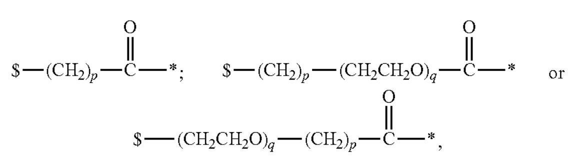
[0310] each p is independently an integer from 2 to 10;

[0311] each q is independently an integer from 1 to 10,

[0312] $ is the point of attachment to Z, and

[0313] * is the point of attachment to the remainder of the linker.

[0314] In some embodiments, in Formula (V), Str is:



[0315] where p, q, $ and * are as defined above.

[0316] In some embodiments, in Formula (V), Str is:

$$\$-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*;\quad \$-(CH_2)_p-(CH_2CH_2O)_q-\overset{\overset{O}{\|}}{C}-*\quad \text{or}$$

$$\$-(CH_2CH_2O)_q-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*,$$

[0317] where $^\$$ and * are as defined above, p is an integer from 2 to 6, and q is an integer from 2 to 8.

[0318] In some embodiments, in Formula (V), AA1-[AA2]m is selected from Val-Lys, Ala-Lys, Phc-Lys, Val-Cit, Phc-Cit, Leu-Cit, Ile-Cit, Trp-Cit, Phc-Arg, Ala-Phe, Val-Ala, Met-Lys, Asn-Lys, Ile-Pro, Ile-Val, Asp-Val, His-Val, Met-(D)Lys, Asn-(D)Lys, Val-(D)Asp, NorVal-(D)Asp, Ala-(D)Asp, Me3Lys-Pro, PhenylGly-(D)Lys, Met-(D)Lys, Asn-(D)Lys, Pro-(D)Lys, Met-(D)Lys, Met-Cit-Val, Gly-Cit-Val, (D)Phe-Phe-Lys, (D)Ala-Phe-Lys, Gly-Phe-Leu-Gly and Ala-Leu-Ala-Leu.

[0319] In some embodiments, in Formula (V), m is 1 (i.e., AA1-[AA2]m is a dipeptide).

[0320] In some embodiments, in Formula (V), AA1-[AA2]m is a dipeptide selected from Val-Lys, Ala-Lys, Phe-Lys, Val-Cit, Phe-Cit, Leu-Cit, Ile-Cit and Trp-Cit.

[0321] In some embodiments, in Formula (V):

[0322] Z is a carbonyl group (—C(O)—);

[0323] Str is

$$\$-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*\quad \text{or}\quad \$-(CH_2CH_2O)_q-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*,$$

where $^\$$ is the point of attachment to Z, * is the point of attachment to the remainder of the linker, p is an integer from 2 to 6, and q is an integer from 2 to 8;

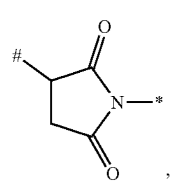
[0324] m is 1 and AA_1-$[AA_2]_m$ is a dipeptide selected from Val-Lys, Ala-Lys, Phe-Lys, Val-Cit, Phe-Cit, Leu-Cit, Ile-Cit and Trp-Cit;

[0325] s is 1; and

[0326] o is 0.

[0327] In some embodiments, in Formula (V):

[0328] Z is



where # is the point of attachment to the antibody, and * is the point of attachment to the remainder of the linker;

[0329] Str is

$$\$-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*\quad \text{or}\quad \$-(CH_2CH_2O)_q-(CH_2)_p-\overset{\overset{O}{\|}}{C}-*,$$

where $^\$$ is the point of attachment to Z, * is the point of attachment to the remainder of the linker, p is an integer from 2 to 6, and q is an integer from 2 to 8;

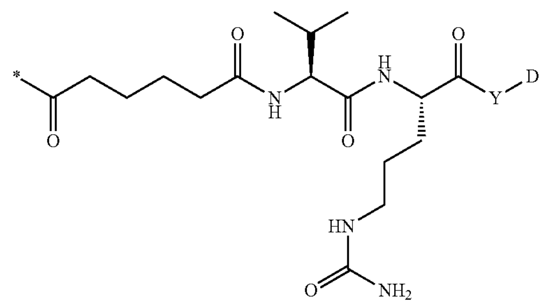
[0330] m is 1 and AA_1-$|AA_2|_m$ is a dipeptide selected from Val-Lys, Ala-Lys, Phe-Lys, Val-Cit, Phe-Cit, Leu-Cit, Ile-Cit and Trp-Cit;

[0331] s is 1; and

[0332] o is 0.

[0333] In certain embodiments, the linker included in the ZymeLink™ ADCs of the present disclosure has Formula (VI):

(VI)



[0334] where:

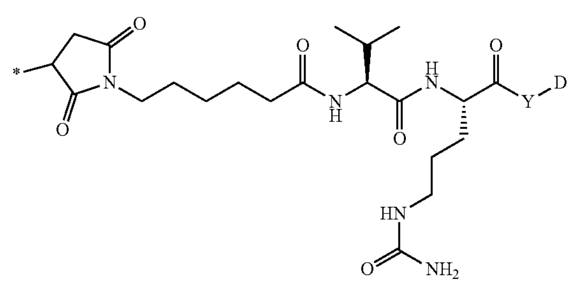
[0335] * is the point of attachment to the antibody;

[0336] Y is one or more additional linker components, or is absent; and

[0337] D is the point of attachment to the auristatin analogue.

[0338] In some embodiments, in the linker of Formula (VI), Y is absent.

[0339] In certain embodiments, the linker included in the ZymeLink™ ADCs of the present disclosure has Formula (VII):



[0340] where:

[0341] * is the point of attachment to the antibody;

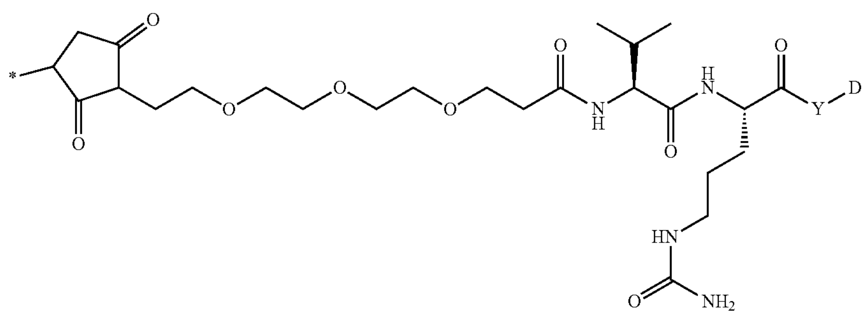
[0342] Y is one or more additional linker components, or is absent; and

[0343] D is the point of attachment to the auristatin analogue.

[0344] In some embodiments, in the linker of Formula (VII), Y is absent.

[0345] In certain embodiments, the linker included in the ZymeLink™ ADCs of the present disclosure has Formula (VIII):

(VIII)



[0346] where:

[0347] * is the point of attachment to the antibody;

[0348] Y is one or more additional linker components, or is absent, and

[0349] D is the point of attachment to the auristatin analogue.

[0350] In some embodiments, in the linker of Formula (VIII), Y is absent.

[0351] In some embodiments, the antibody is conjugated to a microtubule inhibitor that induces apoptosis in cells undergoing mitosis by, for example, causing cell cycle arrest at G2/M. Nonlimiting examples of microtubule inhibitors that can be used include maytansine derivatives (DM1/DM4), or auristatins (MMAE/MMAF) and variants thereof, such as monomethyl auristatin D, PF-06380101, duostatin5, AS269, Tap18Hr1, AGD-0182, HPA-Auristatin F. In some embodiments, the payload is a tubulin-targeting agent, for example, hemiasterlin, tubulysin, or eribulin. In some embodiments, the payloads are DNA-damaging payloads, which include enediynes (calicheamicin), duocarmycin derivatives, pyrrolobenzodiazepine dimers (PBD dimers), and indolinobenzodiazepine pseudo-dimers.

[0352] In some embodiments, the antibody is conjugated to a cytotoxic agent including, but not limited to, e.g., ricin A chain, doxorubicin, daunorubicin, a maytansinoid, taxol, ethidium bromide, mitomycin, etoposide, tenoposide, vincristine, vinblastine, colchicine, dihydroxy anthracin dione, methotrexact, actinomycin, a diphtheria toxin, extotoxin A from *Pseudomonas, Pseudomonas* exotoxin40, abrin, abrin A chain, modeccin A chain, alpha sarcin, gelonin, mitogellin, restrictocin, cobran venom factor, a ribonuclease, engineered Shiga toxin, phenomycin, enomycin, curicin, crotin, calicheamicin, *Saponaria officinalis* inhibitor, glucocorticoid, auristatin, auromycin, yttrium, bismuth, combrestatin, duocarmycins, dolastatin, cc1065, or a cisplatin. In some embodiments, the antibody may be linked to an agent such as an enzyme inhibitor, a proliferation inhibitor, a lytic agent, a DNA or RNA synthesis inhibitors, a membrane permeability modifier, a DNA metabolite, a dichloroethylsulfide derivative, a protein production inhibitor, a ribosome inhibitor, or an inducer of apoptosis. In some embodiments, the antibody is conjugated to a drug such as a topoisomeriase inhibitor, e.g., a topoisomeraise I inhibitor. Topoisomeraise I inhibitors include but are not limited to quinoline alkaloids (SN-38, DXd).

[0353] In some embodiments, the antibody is conjugated to a cancer chemotherapeutic agent, as described below.

[0354] In some embodiments, the antibody is conjugated to one or more of the cytotoxic and/or anti-mitotic compounds as disclosed in WO2014144871A1 and WO2016041082A1, the entire content of both applications are herein incorporated by reference.

[0355] In some embodiments, an EphA2 antibody as described herein is joined to a molecule that facilitates transport of the antibody across a biological membrane, e.g., by enhancing penetration of the membrane, facilitating protein translocation across membranes. Thus, for example, the antibody may be linked to a cell penetration agent, such as a cell-penetrating peptide. Examples of cell penetrating peptides include TAT, penetrating, polyarginine molecules, Kunitz domain-derived peptides, e.g., angiopep-2, SynB, buforin, transportan, amphiphathic peptides and others. In some embodiments, the antibody may be conjugated with a cationic molecule such as a polyamine. In some embodiments, the antibody may be conjugated to an agent that facilitates transport across the blood brain barrier, e.g., transcytosis. Thus, for example, the antibody may be conjugated to an agent that binds to endothelial cell receptors that are internalized, e.g., transferrin receptor, insulin receptor, insulin-like growth factor receptor, or a low-density lipoprotein receptor, and the like. In some embodiments, the antibody may be conjugated to a toxin facilitating entry of the antibody into the cytoplasm, e.g., Shiga toxin. In some embodiments, an EphA2 antibody as described herein can be conjugated to an engineered toxin body (ETBs) to facilitate internalization of the antibody into a cell.

[0356] In some embodiments, an EphA2 antibody described herein is conjugated or administered with a polypeptide immunomodulating agent, e.g., an adjuvant. Examples of immunomodulating agents include, but are not limited to, cytokines (e.g., transforming growth factor-β (TGFβ)), growth factors, lymphotoxins, tumor necrosis factor (TNF), hematopoietic factors, interleukins (e.g., interleukin-1 (IL-1), IL-2, IL-3, IL-6, IL-10, IL-12, IL-15, an IL-15/IL-15Rα, e.g., sushi domain, complex, IL-18, and IL-21), colony stimulating factors (e.g., granulocyte-colony stimulating factor (G-CSF) and granulocyte macrophage-colony stimulating factor (GM-CSF), interferons (e.g., interferon-α, -β or -γ, erythropoietin and thrombopoietin, or a combination thereof. In some embodiments, the antibody is linked or administered with a compound that stimulates the innate immune system, such as an adjuvant, a Toll-like receptor (TLR) agonist, a C-type lectin receptor (CLR) agonist, a retinoic acid-inducible gene I-like receptor (RLR) agonist, a saponin, a polysaccharide such as chitin, chitosan, β-glucan, an ISCOM, QS-21, a stimulator of interferon genes (STING) agonist, or another immunopotentiating agent.

[0357] In some embodiments, an EphA2 antibody described herein is conjugated to or administered with an IL-15 receptor agonist, such as an IL-15 fusion construct, an IL-15:IL-15Rα fusion construct or a single-chain IL-15:IL-15Rα (sushi) fusion construct. In one embodiment, the tumor-targeting antibody conjugated to an IL-15 receptor agonist is a bispecific or multispecific antibody. In some embodiments, the antibody is a bispecific or multispecific antibody comprising an antigen binding domain described herein that further comprises an IL-15 receptor agonist.

[0358] In one embodiment, an EphA2 antibody described herein is administered with a single-chain IL-15:IL-15Rα (sushi) fusion construct. In some embodiments, an EphA2 antibody is administered with a polymer-conjugated IL-15 construct, such as NKTR-255.

[0359] The IL-15:IL-15Rα single chain constructs can be administered to a subject comprising a therapeutically effective dose, for example in the range of less than 0.01 mg/kg body weight to about 25 mg/kg body weight, or 0.1-10 mg/kg, or in the range 1 mg-2 g per patient, or approximately 50 mg-1000 mg/patient.

[0360] In one embodiment, the single-chain IL-15 fusion construct comprises IL-15 joined to IL-15Rα (sushi) with a polypeptide linker. In one embodiment, the single-chain IL-15 fusion construct is joined via a polypeptide linker to another protein, such as an Fc for long half-life. See, for example, FIG. 9B in WO2018071919A1 (corresponding to U.S. patent Ser. No. 10/550,185). In one embodiment, the TL-15 is joined or fused to the N-terminus of the heavy chain of an Fc, and IL-15Rα(sushi) is joined or fused to the other Fc heavy chain N-terminus, using a heavy chain heterodimerization technology to form the desired hybrid Fc. See, for example, FIG. 9A in WO2018071919A1.

[0361] In one embodiment, the IL-15:IL-15Rα (sushi) single chain constructs are fused to the C-terminus of an antibody light chain, or the C-terminus of an antibody heavy chain, in both cases producing a molecule with two tumor-targeting binding sites (the Fab arms), and two IL15:IL15Ra units. Illustrative configurations of the fusion construct is shown in FIG. **36**. In another embodiment, one copy of an IL15:IL15Rα fusion construct is fused to an EphA2 antibody, thereby producing an antibody molecule comprising two tumor-targeting binding sites (the Fab anns) and only one IL15:IL15Rα unit, for example using a knob-in-holes approach to heavy chain heterodimerization, or other heterodimerization technology.

[0362] In some embodiments, the IL-15:IL-15Rα (sushi) fusion constructs or the antibodies comprising the fusion constructs comprise a low affinity IL-15 variant having improved pharmacokinetics (PK). In some embodiments, the IL-15:IL-15Rα (sushi) fusion constructs comprise a high affinity IL-15 variant having increased agonist activity. In some embodiments, the high affinity IL-15 variant has an N72D mutation. In some embodiments, the high affinity variant is fused to a dimeric IL-15Rα sushi domain-IgG1 Fc fusion protein. In some embodiments, the IL-15:IL-15Rα (sushi) fusion construct is ALT-803. See Liu B, et al. (November 2016). "A Novel Fusion of ALT-803 (Interleukin (IL)-15 Superagonist) with an Antibody Demonstrates Antigen-specific Antitumor Responses". The Journal of Biological Chemistry. 291 (46): 23869-23881. doi:10.1074/jbc.M116.733600. Sequences of the IL-15:IL-15Rα fusion constructs and linkers are provided in the Examples.

[0363] In some embodiments, antibodies comprising the IL15:IL15Rα fusion construct comprise one or more mutations in the Fc region described herein, for example E333A, K326W/E333S, S239D/I332E/G236A, S239D/A330L/I332E, G236A/S239D/A330L/I332E, F243L, G236A, and S298A/E333A/K334A. In some embodiments, antibodies comprising the IL15:IL15Rα fusion comprise one or more mutations in the Fc region that increase binding of the antibody to tumor cells, for example the mutations P329G, L234A, L235A, or a combination thereof.

[0364] In some embodiments, antibodies comprising the IL15:IL15Rα fusion construct comprises one or more sequences shown in Table 15. In some embodiments, the fusion protein comprises or consists of SEQ ID NO: 89, SEQ ID NO: 90, SEQ ID NO: 91, and SEQ ID NO: 92. In some embodiments, the fusion protein comprises or consists of SEQ ID NO: 89, SEQ ID NO: 90, SEQ ID NO: 91, and SEQ ID NO: 93.

TABLE 15

Sequences in antibodies comprising the IL15:IL15Rα fusion construct

Description	Sequence
AB-008873 mIgG2a Heavy Chain	(SEQ ID NO: 89)
AB-008873 mIgG2a Light Chain	(SEQ ID NO: 90)
Linker (G4S)3	(SEQ ID NO: 91)
Linker TS(G4S)3	(SEQ ID NO: 105)
IL15(wt/RLI)	(SEQ ID NO: 92)
IL15(low/RLI)	(SEQ ID NO: 93)

[0365] In some embodiments, an EphA2 antibody described herein is conjugated to or administered with an IL-2 receptor agonist. In one embodiment, the tumor-targeting antibody that is conjugated to an IL-2 receptor agonist is a bispecific or multispecific antibody. In some embodiments, the antibody is a bispecific or multispecific antibody comprising an antigen binding domain of an antibody described herein and an IL-2 receptor agonist. In some embodiments, the IL-2 receptor agonist is pegylated IL-2.

[0366] In some embodiments, an EphA2 antibody described herein is conjugated to or administered with a construct that can act as a trap for transforming growth factor-β (TGFβ). In one embodiment, the TGFβ trap comprises the extracellular domain (ECD) of TGFβ. In one embodiment, the TGFβ trap comprises the extracellular domain (ECD) of TGFβRII. In one embodiment, the TGFβ trap is in the form of a bispecific antibody (see US2018/0118832A1, FIG. **1**). The TGFβRII ECD can preferably trap TGFβ1, and its low affinity to TGFβ2 may mitigate potential cardiac toxicity.

[0367] Thus, in another aspect, an EphA2 antibody described herein comprises an extracellular domain (ECD) of the TGFβ Receptor fused to the C-terminus of the heavy chain or to the C-terminus of the light chain. In some embodiments, the TGFβ trap is a single trap construct. In some embodiments, the single TGFβ trap is a bispecific tumor-targeting TGFβ trap comprising a TGFβ RII ECD fused to any one of the antibodies disclosed herein via a flexible linker to the C-terminus of the heavy chain or to the C-terminus of the light chain.

[0368] In some embodiments, the TGFβ trap is a tandem trap construct. In some embodiments, the tandem TGFβ trap comprises an IgG fused to two TGFβRII ECDs. In some embodiments, the tandem TGFβ trap comprises two TGFβ2RII ECDs. In some embodiments, the two TGFβ2RII ECDs are fused in series and are linked by a short linker (for example L10 or L25). In some embodiments, the two TGFβ2RII ECDs are fused directly in series without a linker (L0). In some embodiments, the tandem TGFβRII ECDs are fused to the C-terminus of the heavy chain (HC-Cter), and the heavy chains were designed as an asymmetric pair such that the tandem trap is on only one heavy chain. In some embodiments, the asymmetric pair of heavy chains comprise knob-in-hole mutation that promote pairing of the heavy chains. For example, in some embodiments, one heavy chain comprises the amino acid substitutions T366S+L368A+Y407V (and optionally Y349C), and the other heavy chain comprises the amino acid substitution T336W (and optionally S354C). In some embodiments, the asymmetric single heavy chain C-ter fusion improves steric access of the Fc region to Fc gamma receptors and thereby improve function.

[0369] In some embodiments, the tandem TGFβ trap is fused to the C-terminus of the light chain (LC-Cter), such that both light chains comprise two TGFβRII ECDs. In these embodiments, the net molecule exhibits twice the TGFβ trapping capacity per molecule, and therefore may exhibit improved function.

[0370] In some embodiments, the bispecific TGFβ trap construct comprises human variable regions. In some embodiments, the bispecific TGFβ trap construct comprises a IgG1 or IgG2 constant region. In some embodiments, the bispecific TGFβ trap construct comprises a human IgG1 constant region. In some embodiments, the bispecific TGFβ trap construct comprises a mouse IgG2a constant region. In some embodiments, the variable regions of the TGFβ trap construct are fused in frame to the IgG constant regions.

[0371] Binding of the TGFβ trap construct can be determined using, for example, an ELISA assay, as described in the Examples. The ability of TGFβ trap constructs to bind to target tumor cells can be determined, for example, using flow-cytometry, as described in the Examples. The ability of TGFβ trap constructs to engage and stimulate Fc-gamma receptor in the presence of target tumor cells can be determined using a reporter bioassay, as described in the Examples. The ability of TGFβ trap constructs to inhibit tumor growth can be determined, for example, in a syngeneic mouse model, as described in the Examples.

[0372] In some embodiments, the antibody may be linked to a radionuclide, an iron-related compound, a dye, a fluorescent agent, or an imaging agent. In some embodiments, an antibody may be linked to agents, such as, but not limited to, metals; metal chelators; lanthanides; lanthanide chelators; radiometals; radiometal chelators; positron-emitting nuclei; microbubbles (for ultrasound); liposomes; molecules microencapsulated in liposomes or nanosphere; monocrystalline iron oxide nanocompounds; magnetic resonance imaging contrast agents; light absorbing, reflecting and/or scattering agents; colloidal particles; fluorophores, such as near-infrared fluorophores.

[0373] In one embodiment of any of the above constructs, the EphA2 antibody is any one of AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665. AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702. In one embodiment of any of the above constructs, the tumor-targeting binding domain comprises the V_H and V_L sequences of AB-010361 or AB-010699.

Methods of Inducing an Immune Response

[0374] In a further aspect, provided herein are methods of inducing an immune response by administering an EphA2 antibody as described herein to a subject that has a tumor. In some embodiments, the EphA2 antibody is an antibody set forth in Tables 6-8, or a variant thereof as described above. In some embodiments, the antibody or variant thereof comprises a modified Fc region comprising mutations described herein. For example, in some embodiments, the antibody comprises an Fc mutation that increases effect function selected from E333A, K326W/E333S, S239D/I332E/G236A, S239D/A330L/I332E, G236A/S239D/A330L/I332E, F243L, G236A, S298A/E333A/K334A, and P329G/L234A/L235A, or a combination thereof. In some embodiments, the antibody comprises a modified Fc region that is a-fucosylated. In some embodiments, the antibody is conjugated to or administered with an IL-15 receptor agonist, a TGFβ trap, a TLR agonist, or an agonist anti-4-1BB antibody. In some embodiments, the antibody is a bispecific or multispecific antibody described herein.

[0375] An immune response induced by administration of an antibody as described herein can be either an innate or adaptive immune response. In some embodiments, the antibody activates an immune response directly, e.g., via binding of the antibody to a target tumor cell and engagement with an Fc receptor on an effector cell such that the effector cell is activated. In some embodiments, the antibody indirectly activates an immune response by inducing immune responses that are initiated by antibody binding to the target cell and an effector cell with subsequent induction of downstream immune responses. In some embodiments, the antibody activates monocytes, myeloid cells, and/or NK cells, e.g., macrophages, neutrophils, dendritic cells, mast cells, basophils, eosinophile, and/or NK cells. In some embodiments, the antibody activates T lymphocytes and/or B cells.

Treatment of Cancer

[0376] In a further aspect, an EphA2 antibody as provided herein, or a variant thereof as described herein, can be used and a therapeutic agent to treat cancer. In some embodiments, the antibody or variant thereof comprises a modified Fc region comprising mutations described herein. For example, in some embodiments, the antibody comprises an

Fc mutation that increases effector function selected from E333A, K326W/E333S, S239D/I332E/G236A, S239D/A330L/I332E, G236A/S239D/A330L/I332E, F243L, G236A, S298A/E333A/K334A, and P329G/L234A/L235A, or a combination thereof. In some embodiments, the antibody comprises a modified Fc region that is afucosylated. In some embodiments, the antibody is conjugated to or administered with an IL-15 receptor agonist, a TGFβ trap, a TLR agonist, or an agonist anti-4-1BB antibody. In some embodiments, the antibody is a bispecific or multispecific antibody described herein.

[0377] In some aspects, the disclosure additionally provides methods of identifying subjects who are candidates for treatment with an EphA2 antibody having tumor-targeting effects. Thus, in one embodiment, the invention provides a method of identifying a patient who can benefit from treatment with an EphA2 antibody of the present disclosure. In one embodiment, the patient has tumor that expresses EphA2. In one embodiment, the patient has tumor that overexpresses EphA2. In some embodiments, the tumor sample is from a primary tumor. In alternative embodiments, the tumor sample is a metastatic lesion. Binding of antibody to tumor cells through a binding interaction with the EphA2 can be measured using any assay, such as immunohistochemistry or flow cytometry. In some embodiments, binding of antibody to at least 0.2%, 0.5%, or 1%, or at least 5% or 10%, or at least 20%, 30%, or 50%, of the tumor cells in a sample may be used as a selection criterion for determining a patient to be treated with an EphA2 antibody as described herein. In other embodiments, analysis of components of the blood, e.g., circulating exosomes and/or extracellular RNA-protein complex and/or extracellular protein, is used to identify a patient whose tumor cells are overexpressing EphA2.

[0378] An EphA2 antibody disclosed herein can be used to treat several different cancers. In some embodiments, a cancer patient who can benefit from the treatment of the EphA2 antibody has a cancer that expresses EphA2. In some embodiments, a cancer patient who can benefit from the treatment of the EphA2 antibody has a cancer overexpressing EphA2. In some embodiments, the cancer is a carcinoma or a sarcoma. In some embodiments, the cancer is a hematological cancer. In some embodiments, the cancer is breast cancer, prostate cancer, testicular cancer, renal cell cancer, bladder cancer, ovarian cancer, cervical cancer, endometrial cancer, lung cancer, colorectal cancer, anal cancer, pancreatic cancer, gastric cancer, esophageal cancer, hepatocellular cancer, head and neck cancer, a brain cancer, e.g., glioblastoma, melanoma, or a bone or soft tissue sarcoma. In one embodiment, the cancer is acral melanoma. In some embodiments, the cancer is acute lymphoblastic leukemia, acute myeloid leukemia, adrenocortical carcinoma, astrocytoma, basal-cell carcinoma, bile duct cancer, bone tumor, brainstem glioma, cerebellar astrocytoma, cerebral astrocytoma, ependymoma, medulloblastoma, supratentorial primitive neuroectodermal tumors, visual pathway and hypothalamic glioma, bronchial adenomas, Burkitt's lymphoma, central nervous system lymphoma, cerebellar astrocytoma, chondrosarcoma, chronic lymphocytic leukemia, chronic myelogenous leukemia, chronic myeloproliferative disorders, colon cancer, cutaneous T-cell lymphoma, desmoplastic small round cell tumor, endometrial cancer, ependymoma, epithelioid hemangioendothelioma (EHE), esophageal cancer, Ewing's sarcoma, extracranial germ cell tumor, extragonadal germ cell tumor, extrahepatic bile duct cancer, a cancer of the eye, intraocular melanoma, retinoblastoma, gallbladder cancer, gastrointestinal carcinoid tumor, gastrointestinal stromal tumor (GIST), germ cell tumor, gestational trophoblastic tumor, gastric carcinoma, hairy cell leukemia, hepatocellular carcinoma, Hodgkin lymphoma, hypopharyngeal cancer, hypothalamic and visual pathway glioma, childhood, intraocular melanoma, islet cell carcinoma, Kaposi sarcoma, kidney cancer, laryngeal cancer, leukemias, lip and oral cavity cancer, liposarcoma, liver cancer, non-small cell lung cancer, small-cell lung cancer, lymphomas, macroglobulinemia, male breast cancer, malignant fibrous histiocytoma of bone, medulloblastoma, Merkel cell cancer, mesothelioma, metastatic squamous neck cancer, mouth cancer, multiple endocrine neoplasia syndrome, multiple myeloma, mycosis fungoides, myelodysplastic syndromes, myelogenous leukemia, myeloid leukemia, adult acute, myeloproliferative disorders, chronic, myxoma, nasal cavity and paranasal sinus cancer, nasopharyngeal carcinoma, neuroblastoma, non-Hodgkin lymphoma, oligodendroglioma, oral cancer, oropharyngeal cancer, osteosarcoma, ovarian epithelial cancer, ovarian germ cell tumor, ovarian low malignant potential tumor, paranasal sinus and nasal cavity cancer, parathyroid cancer, penile cancer, pharyngeal cancer, pheochromocytoma, pineal astrocytoma, pineal germinoma, pineoblastoma, supratentorial primitive neuroectodermal tumors, pituitary adenoma, plasma cell neoplasia, pleuropulmonary blastoma, primary central nervous system lymphoma, rectal cancer, renal cell carcinoma, retinoblastoma, rhabdomyosarcoma, salivary gland cancer, Ewing sarcoma, Kaposi sarcoma, soft tissue sarcoma, uterine sarcoma, Sezary syndrome, non-melanoma skin cancer, melanoma, small intestine cancer, squamous cell carcinoma, squamous neck cancer, stomach cancer, cutaneous T-Cell lymphoma, throat cancer, thymoma, thyroid cancer, transitional cell cancer of the renal pelvis and ureter, trophoblastic tumor, gestational, urethral cancer, uterine cancer, vaginal cancer, vulvar cancer, Waldenström macroglobulinemia, or Wilms tumor.

[0379] In some embodiments, the cancer is lung cancer, e.g., non-small cell lung adenocarcinoma or squamous cell carcinoma; breast cancer, e.g., Triple$^-$, ER/PR$^+$Her2$^-$, ER/PR$^-$Her2$^+$, or Triple$^-$; colorectal cancer, e.g., adenocarcinoma, mucinous adenocarcinoma, or papillary adenocarcinoma; esophageal cancer; stomach cancer; kidney cancer, e.g., kidney clear cell cancer; ovarian cancer, e.g., ovarian endometrioid carcinoma, ovarian mucinous cystadenocarcinoma, or ovarian serous cystadenocarcinoma; melanoma, e.g., acral melanoma, cutaneous melanoma, or mucosal melanoma; uterine or cervical cancer; liver cancer, e.g., hepatocellular carcinoma or bile duct carcinoma; bladder cancer, e.g., transitional or urothelial bladder cancer; or testicular cancer.

[0380] In some embodiments, an EphA2 antibody disclosed herein can be used to treat a gastric cancer, ovarian cancer, or soft tissue sarcoma. As discussed above, the EphA2 antibody, e.g., AB-008873, showed better tumor selectivity than several EphA2 antibodies derived from clinical candidates and commercial EphA2 antibodies as referenced in Table 34.

[0381] In one aspect, methods of the disclosure comprise administering an EphA2 antibody disclosed herein, or a variant thereof, as a pharmaceutical composition to a cancer patient in a therapeutically effective amount using a dosing regimen suitable for treatment of the cancer. The composition can be formulated for use in a variety of drug delivery systems. One or more physiologically acceptable excipients or carriers can also be included in the compositions for proper formulation. Suitable formulations for use in the present disclosure are found, e.g., in Remington: The Science and Practice of Pharmacy, 21st Edition, Philadelphia, PA. Lippincott Williams & Wilkins, 2005.

[0382] The tumor-targeting antibody is provided in a solution suitable for administration to the patient, such as a sterile isotonic aqueous solution for injection. The antibody is dissolved or suspended at a suitable concentration in an acceptable carrier. In some embodiments the carrier is aqueous, e.g., water, saline, phosphate buffered saline, and the like. The compositions may contain auxiliary pharmaceutical substances as required to approximate physiological conditions, such as pH adjusting and buffering agents, tonicity adjusting agents, and the like.

Administration

[0383] The pharmaceutical compositions are administered to a patient in an amount sufficient to cure or at least partially arrest the disease or symptoms of the disease and its complications. An amount adequate to accomplish this is defined as a "therapeutically effective dose." A therapeutically effective dose is determined by monitoring a patient's response to therapy. Typical benchmarks indicative of a therapeutically effective dose include the amelioration of symptoms of the disease in the patient. Amounts effective for this use will depend upon the severity of the disease and the general state of the patient's health, including other factors such as age, weight, gender, administration route, and the like Single or multiple administrations of the antibody may be administered depending on the dosage and frequency as required and tolerated by the patient. In any event, the methods provide a sufficient quantity of tumor-targeting antibody to effectively treat the patient.

[0384] An EphA2 antibody can be administered by any suitable means, including, for example, parenteral, intrapulmonary, and intranasal, administration, as well as local administration, such as intratumor administration. Parenteral infusions include intramuscular, intravenous, intraarterial, intraperitoneal, or subcutaneous administration. In some embodiments, the antibody may be administered by insufflation. In an illustrative embodiment, the antibody may be stored at 10 mg/ml in sterile isotonic aqueous saline solution for injection at 4° C. and is diluted in either 100 ml or 200 ml 0.9% sodium chloride for injection prior to administration to the patient. In some embodiments, the antibody is administered by intravenous infusion over the course of 1 hour at a dose of between 0.01 and 25 mg/kg. In other embodiments, the antibody is administered by intravenous infusion over a period of between 15 minutes and 2 hours. In still other embodiments, the administration procedure is via subcutaneous bolus injection.

[0385] The dose of antibody is chosen to provide effective therapy for the patient and is in the range of less than 0.01 mg/kg body weight to about 25 mg/kg body weight or in the range 1 mg-2 g per patient. Preferably the dose is in the range 0.1-10 mg/kg or approximately 50 mg-1000 mg/patient. The dose may be repeated at an appropriate frequency which may be in the range once per day to once every three months, or every six months, depending on the pharmacokinetics of the antibody (e.g., half-life of the antibody in the circulation) and the pharmacodynamic response (e.g., the duration of the therapeutic effect of the antibody). In some embodiments, the in vivo half-life of between about 7 and about 25 days and antibody dosing is repeated between once per week and once every 3 months or once every 6 months. In other embodiments, the antibody is administered approximately once per month.

[0386] In an illustrative embodiment, the antibody may be stored at 10 mg/ml or 20 mg/ml in a sterile isotonic aqueous solution. The solution can comprise agents such as buffering agents and stabilizing agents. For example, in some embodiments, a buffering agent such as histidine is included to maintain a formulation pH of about 5.5. Additional reagents such as sucrose or alternatives can be added to prevent aggregation and fragmentation in solution and during freezing and thawing. Agents such as polysorbate 80 or an alternative can be included to lower surface tension and stabilizes the antibody against agitation-induced denaturation and air-liquid and ice-liquid surface denaturation. In some embodiments, the solution for injection is stored at 4° C. and is diluted in either 100 ml or 200 ml 0.9% sodium chloride for injection prior to administration to the patient.

[0387] In some embodiments, antibody for IV administration is formulated at a target concentration of 20 mg/mL in 20 mM histidine buffer, 8% (w/v) sucrose and 0.02% (w/v) polysorbate 80, pH 5.5.

Combination Therapy

[0388] An EphA2 antibody disclosed herein may be administered with one or more additional therapeutic agents, e.g., radiation therapy, chemotherapeutic agents and/or immunotherapeutic agents.

[0389] In some embodiments, an EphA2 antibody can be administered in conjunction with an agent that targets an immune checkpoint antigen. In one aspect, the agent is a biologic therapeutic or a small molecule. In another aspect, the agent is a monoclonal antibody, a humanized antibody, a human antibody, a fusion protein or a combination thereof. In certain embodiments, the agents inhibit, e.g., by blocking ligand binding to receptor, a checkpoint antigen that may be PD1, PDL1, CTLA-4, ICOS, PDL2, IDO1, IDO2, B7-H3, B7-H4, BTLA, HVEM, TIM3, GAL9, GITR, HAVCR2, LAG3, KIR, LAIR1, LIGHT, MARCO, OX-40, SLAM, 2B4, CD2, CD27, CD28, CD30, CD40, CD70, CD80, CD86, CD137 (4-1BB), CD160, CD39, VISTA, TIGIT, a SIGLEC, CGEN-15049, 2B4, CHK1, CHK2, A2aR, B-7 family ligands or their receptors, or a combination thereof. In some embodiments, the agent targets PD-1, e.g., an antibody that blocks PD-L1 binding to PD-1 or otherwise inhibits PD-1. In some embodiments, the agent targets CTLA-4. In some embodiments, the agent targets LAG3. In some embodiments, the agent targets TIM3. In some embodiments, the agents target ICOS.

[0390] In some embodiments, an EphA2 antibody can be administered in conjunction with a therapeutic antibody, such as an antibody that targets a tumor cell antigen. Examples of therapeutic antibodies include as rituximab, trastuzumab, tositumomab, ibritumomab, alemtuzumab, atezolizumab, avelumab, durvalumab, pidilizumab, AMP-224, AMP-514, PDR001, cemiplimab, BMS-936559, CK-301, epratuzumab, bevacizumab, elotuzumab, necitumumab, blinatumomab, brentuximab, cetuximab, daratumumab, denosumab, dinutuximab, gemtuzumab ibritumomab ipilimumab, nivolumab, obinutuzumab, ofatumumab, ado-trastuzumab, panitumumab, pembrolizumab, pertuzumab, ramucirumab, and ranibizumab. In some embodiments, an EphA2 antibody can be administered in conjunction with a therapeutic antibody that binds an extracellular RNA-protein complex comprising polyadenylated RNA, such as the antibody designated ATRC-101, see WO2020168231 incorporated herein in its entirety.

[0391] In some embodiments, an EphA2 antibody as described herein is administered with an agonist anti-4-1BB antibody such as urelumab (Bristol Meyers Squibb, BMS-663513, U.S. Pat. No. 7,288,638), utomilumab, (Pfizer, PF-05082566, WO2012145183), 1D8 and 5B9 (US20100279932), hu106-1, TABBY 101-TABBY 110 (WO2017205745, FIGS. **2**A-**2**F), Hz4B4-1 (U.S. Pat. No. 6,458,934), BBK-1 and BBK-4 (U.S. Pat. No. 6,559,997), hu39E3.G4 (U.S. Pat. No. 6,887,673), CTX-471 and CTX-471AF (U.S. Ser. No. 10/279,038), MOR-6032, MOR-7361, MOR-7480 and MOR-7483 (U.S. Pat. No. 8,337,850), LVGN-6051 (US20210246218), AGEN-2373 (US 2021/0106693), ADG-106/AG10131 (US20200369776), ATOR-1017 (US20190352414).

[0392] In some embodiments, an EphA2 antibody is administered with a chemotherapeutic agent. Examples of cancer chemotherapeutic agents include alkylating agents such as thiotepa and cyclophosphamide; alkyl sulfonates such as busulfan, improsulfan and piposulfan; aziridines such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and methylamelamines including altretamine, triethylenemelamine, trietylenephosphoramide, triethylenethiophosphaoramide and trimethylolomelamine; nitrogen mustards such as chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard; nitrosureas such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, ranimustine; antibiotics such as aclacinomysins, actinomycin, authramycin, azaserine, bleomycins, cactinomycin, calicheamicin, carabicin, caminomycin, carzinophilin, chromomycins, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, doxorubicin, epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins, mycophenolic acid, nogalamycin, olivomycins, peplomycin, potfiromycin, puromycin, quelamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin; antimetabolites such as methotrexate and 5-fluorouracil; folic acid analogues such as denopterin, methotrexate, pteropterin, trimetrexate; purine analogs such as fludarabine, 6-mercaptopurine, thiamiprine, thioguanine; pyrimidine analogs such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, floxuridine, androgens such as calusterone, dromostanolone propionate, epitiostanol, mepitiostane, testolactone; anti-adrenals such as aminoglutethimide, mitotane, trilostane; folic acid replenisher such as frolinic acid; aceglatone; aldophosphamide glycoside; aminolevulinic acid; amsacrine; bestrabucil; bisantrene; edatraxate; defofamine; demecolcine; diaziquone; elformithine; elliptinium acetate; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidamine; mitoguazone; mitoxantrone; mopidamol; nitracrine; pentostatin; phenamet; pirarubicin; podophyllinic acid; 2-ethylhydrazide; procarbazine; razoxane; sizofiran; spirogermanium; tenuazonic acid; triaziquone; 2,2',2"-trichlorotriethylamine; urethan; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside; cyclophosphamide; thiotepa; taxoids, e.g. paclitaxel and doxetaxel; chlorambucil; gemcitabine; 6-thioguanine; mercaptopurine; methotrexate; platinum analogs such as cisplatin and carboplatin; vinblastine; docetaxel, platinum; etoposide (VP-16); ifosfamide; mitomycin C; mitoxantrone; vincristine; vinorelbine; navelbine; novantrone; teniposide; daunomycin; aminopterin; xeloda; ibandronate; CPT-1 1; topoisomerase inhibitor RFS 2000; difluoromethylomithine (DMFO); retinoic acid derivatives such as bexarotene, alitretinoin; denilcukin diftitox; esperamicins; capecitabinc; and pharmaceutically acceptable salts, acids or derivatives of any of the above. Also included in this definition are anti-hormonal agents that act to regulate or inhibit honnone action on tumors such as anti-estrogens including for example tamoxifen, raloxifene, mifepristone, aromatase inhibiting 4(5)-imidazoles, 4-hydroxytamoxifen, trioxifene, keoxifene, LY 1 17018, onapristone, and toremifene (Fareston); and anti-androgens such as flutamide, nilutamide, bicalutamide, leuprolide, and goserelin; and pharmaceutically acceptable salts, acids or derivatives of any of the above. Further cancer therapeutic agents include sorafenib and other protein kinase inhibitors such as afatinib, axitinib, crizotinib, dasatinib, erlotinib, fostamatinib, gefitinib, imatinib, lapatinib, lenvatinib, mubritinib, nilotinib, pazopanib, pegaptanib, ruxolitinib, vandetanib, vemurafenib, and sunitinib; sirolimus (rapamycin), everolimus and other mTOR inhibitors. Examples of additional chemotherapeutic agents include topoisomerase I inhibitors (e.g., irinotecan, topotecan, camptothecin and analogs or metabolites thereof, and doxorubicin); topoisomerase II inhibitors (e.g., etoposide, teniposide, and daunorubicin); alkylating agents (e.g., melphalan, chlorambucil, busulfan, thiotepa, ifosfamide, carmustine, lomustine, semustine, streptozocin, decarbazine, methotrexate, mitoinycin C, and cyclophosphamide); DNA intercalators (e.g., cisplatin, oxaliplatin, and carboplatin); DNA intercalators and free radical generators such as bleomycin; and nucleoside mimetics (e.g., 5-fluorouracil, capecitibine, gemcitabine, fludarabine, cytarabine, mercaptopurine, thioguanine, pentostatin, and hydroxyurea). Illustrative chemotherapeutic agents additionally include paclitaxel, docetaxel, and related analogs; vincristine, vinblastin, and related analogs; thalidomide, lenalidomide, and related analogs (e.g., CC-5013 and CC-4047); protein tyrosine kinase inhibitors (e.g., imatinib mesylate and gefitinib); proteasome inhibitors (e.g., bortezomib); NF-κB inhibitors, including inhibitors of IκB kinase1 and other inhibitors of proteins or enzymes known to be upregulated, over-expressed or activated in cancers, the inhibition of which down regulates cell replication. Additional agents include asparaginase and a *Bacillus* Calmete-Guerin preparation.

[0393] In some embodiments, an EphA2 antibody as described herein is administered after, or at the same time, as a therapeutic agent, e.g., a chemotherapeutic agent, such as doxorubicin, that induces stress granules ("SG-inducing agent"). Increasing the amount of stress granules in cancer cells can promote targeting the tumor cells by the tumor-targeting antibody. Other exemplary therapeutic agents that can induce stress granules include pyrimidine analogs (e.g., 5-FU, under trade names of Adrucil®, Carac®, Efudex®, Efudix®); protease inhibitors (e.g., Bortezomib, under the trade name of Velcade®); kinase inhibitors (e.g, Sorafenib and Imatinib, under the trade names of Nexavar® and Gleevec®, respectively); Arsenic compounds (e.g., Arsenic trioxide, under the trade name of Trisenox®); Platinum-based compounds that induce DNA damage (e.g., Cisplatin and Oxaliplatin®, under the trade names of Platinol® and Eloxatin®, respectively); agents that disrupt microtubules (e.g., Vinblastin, under the trade name of Velban® or alkabban-AQ®; vincristin, under the trade name of Vincasar®, Margibo®, or Oncovin®; Vinorelbin, under the trade name of Navelbin®); topoisomerase II inhibitor (e.g., Etoposide, under the trade name of Etopophos, Toposar®, VePesid®); and agents that induce DNA damage, e.g., irradiation. Several exemplary therapeutic agents that can induce stress granules formation are disclosed in Mahboubi et al., *Biochimica et Biophysica Acta* 1863 (2017) 884-895.

[0394] Various combinations with the tumor-targeting antibody and the SG-inducing agent (or a combination of such agents) described herein may be employed to treat a cancer patient. By "combination therapy" or "in combination with", it is not intended to imply that the therapeutic agents must be administered at the same time and/or formulated for delivery together, although these methods of delivery are within the scope described herein. The tumor-targeting antibody and the SG-inducing agent can be administered following the same or different dosing regimen. In some embodiments, the tumor-targeting antibody and the SG-inducing agent is administered sequentially in any order during the entire or portions of the treatment period. In some embodiments, the tumor-targeting antibody and the SG-inducing agent is administered simultaneously or approximately simultaneously (e.g., within about 1, 5, 10, 15, 20, or 30 minutes of each other). In still other embodiments, the SG-inducing agent may be administered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more days before administration of the tumor-targeting antibody. In some embodiments, the SG-inducing agent is administered from 1 to 4 weeks, or longer, before the tumor-targeting antibody is administered.

[0395] An EphA2 antibody may also be administered to a cancer patient in conjunction with a cell-based therapy, such as natural killer (NK) cell therapy or a cancer vaccine. In some instances, a cancer vaccine is a peptide-based vaccine, a nucleic acid-based vaccine, a cell-based vaccine, a virus-based or viral fragment-based vaccine or an antigen presenting cell (APC) based vaccine (e.g., dendritic cell based vaccine). Cancer vaccines include Gardasil®, Cervarix®, sipuleucel-T (Provenge®), NeuVax™ HER-2 ICD peptide-based vaccine, HER-2/neu peptide vaccine, AdHER2/neu dendritic cell vaccine, HER-2 pulsed DC1 vaccine, Ad-sig-hMUC-1/ecdCD40L fusion protein vaccine, MVX-ONCO-1, hTERT/survivin/CMV multipeptide vaccine, E39, J65, P10s-PADRE, rV-CEA-Tricom, GVAX®, Lucanix®, HER2 VRP, AVX901, ONT-10, ISAlOl, ADXSl 1-001, VGX-3100, INO-9012, GSK1437173A, BPX-501, AGS-003, IDC-G305, HyperAcute®-Renal (HAR) immunotherapy, Prevenarl3, MAGER-3.A1, NA17.A2, DCVax-Direct, latent membrane protein-2 (LMP2)-loaded dendritic cell vaccine (NCT02115126), HS410-101 (NCT02010203, Heat Biologies), EAU RF 2010-01 (NCT01435356, GSK), 140036 (NCT02015104, Rutgers Cancer Institute of New Jersey), 130016 (NCTO 1730118, National Cancer Institute), MVX-201101 (NCT02193503, Maxivax SA), ITL-007-ATCR-MBC (NCT01741038, Immunovative Therapies, Limited), CDR0000644921 (NCT00923143, Abramson cancer center of the University of Pennsylvania), SuMo-Sec-01 (NCT00108875, Julius Maximilians Universitaet Hospital), or MCC-15651 (NCT01176474, Medarex, Inc, BMS).

[0396] In some embodiments, an EphA2 antibody of the present disclosure may be administered with an agent, e.g., a corticosteroid, that mitigates side-effects resulting from stimulation of the immune system.

[0397] In the context of the present disclosure a therapeutic agent that is administered in conjunction with an EphA2 antibody of the present disclosure can be administered prior to administrations of the tumor-targeting antibody or after administration of the tumor-targeting antibody. In some embodiments, an EphA2 antibody may be administered at the same time as the additional therapeutic agent. In some embodiments, an EphA2 antibody and an additional therapeutic agent described above can be administered following the same or different dosing regimens. In some embodiments, the tumor-targeting antibody and the therapeutic agent are administered sequentially in any order during the entire treatment period or portions thereof. In some embodiments, the tumor-targeting antibody and the therapeutic agent are administered simultaneously or approximately simultaneously (e.g., within about 1, 5, 10, 15, 20, or 30 minutes of each other). In still other embodiments, the therapeutic agent may be administered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more days before the administration of the tumor-targeting antibody. In still other embodiments, the therapeutic agent may be administered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more days after the administration of the tumor-targeting antibody.

Functional Assays

[0398] Also described herein are functional assays that can be used to determine the ability of the antibodies described herein to mediate FcR-dependent activity. In some embodiments, the assay measures antibody dependent cellular cytotoxicity (ADCC), antibody dependent cellular phagocytosis (ADCP), or complement-dependent cytoxicity (CDC).

In Vivo Assays

[0399] In some embodiments, the activity of the antibodies is evaluated in vivo in an animal model that is known for specific human tumors. One exemplary model is the CT26 mouse model, as described in the EXAMPLES. Tumor-targeting activity of these antibodies in vivo may be assessed by using several assays, including but not limited to using flow cytometry to analyze the immune profiling of the blood and tumor, monitoring tumor growth, and performing immunofluorescence to semi-quantitative estimate tumor infiltration. In some embodiments, the effect of the antibody can be assessed using Survival, a normalized area above the curve metric (NAAC), and a normalized growth rate metric (NGRM), where NAAC and NGRM were both developed at Atreca. An "in vivo active" determination can be based on the in vivo activity was assessed by a p-value≤0.05 in at least one of the analyses of survival, NAAC, and NGRM, i.e., if an antibody exhibited a p-value of less than or equal to 0.05 for survival, NAAC, and/or NGRM (any one alone being sufficient), the antibody is considered "in vivo active".

[0400] In one aspect, provided herein are antibodies that exhibit inhibitory effects on tumors, including decreasing rate of tumor growth, size, tumor invasion and/or metastasis. Such antibodies exhibit tumor-targeting effects in vivo, e.g., when administered to subjects that has a tumor expressing or overexpressing EphA2.

Engineering Variants

[0401] In some embodiments, an antibody or variant thereof described herein is modified to have improved developability (i.e., reduced development liabilities), including but not limited to, decreased heterogeneity, increased yield, increased stability, improved net charges to improve pharmacokinetics, and or/reduced immunogenicity. In some embodiments, antibodies having improved developability can be obtained by introducing mutations to reduce or eliminate potential development liabilities. In some embodiments, antibodies having improved developability possess modifications as compared to a reference or control antibody in their amino acid sequence.

[0402] In some embodiments, the antibodies or variants thereof disclosed herein have improved developability while maintaining comparable or improved binding affinity to the target antigen as compared to a reference or control (unmodified) antibody. In some embodiments, the antibodies or variants thereof disclosed herein have improved developability while maintaining activities similar to a reference or control (unmodified) antibody.

[0403] In some embodiments, the antibodies or variants thereof have improved developability, e.g., as identified through various in vitro assays, such as aggregation assessment by HPLC or UPLC, hydrophobic interaction chromatography (HIC), polyspecificity assays (e.g., baculovirus particle binding), self-interaction nanoparticle spectroscopy (SINS), or mass spec analysis after incubation in an accelerated degradation condition such as high temperature, low pH, high pH, or oxidative H_2O_2. Mutations are successful if activity is maintained (or enhanced) while removing or reducing the severity of the liability.

[0404] Improved properties of antibodies or variants thereof as described herein include: (1) fits a standard platform (expression, purification, formulation); (2) high yield; (3) low heterogeneity (glycosylation, chemical modification, and the like); (4) consistent manufacturability (batch-to-batch, and small-to-large scale); (5) high stability (years in liquid formulation), e.g., minimal chemical degradation, fragmentation, and aggregation; and (6) long PK (in vivo half-life), e.g., no off-target binding, no impairment of FcRn recycling, and stable. Antibody liabilities are further described in Table 16.

TABLE 16

Description of potential development liabilities

Free cysteine[6]	Yield, heterogeneity, stability, activity	Sequence comprises an odd number of cysteines	
N-linked glycosylation	Yield, heterogeneity, activity	N(~P)(S, T)[1]	High
Abnormal net charge	Platform fit, PK	Sharma 2014[2]	High
Patches of hydrophobicity	Stability, PK	Sharma 2014	High
Patches of same charge	Stability, PK	N/A (based on structure)	Medium
Proteolysis	Stability, PK	(K, R)(K, R)[3]	Medium
Proteolysis	Stability, PK	DP	Medium
Asparagine deamidation	Heterogeneity, stability, activity	NG; N(A, N, S, T)[4]	Medium; Low
Aspartate isomerization	Heterogeneity, stability, activity	DG; D(A, D, S, T)[5]	Medium; Low
Lysine glycation	Heterogeneity, stability, activity	K	Low
Methionine oxidation	Heterogeneity, stability, activity	M	Low
Tryptophan oxidation	Heterogeneity, stability, activity	W	Low

Note:

[1]The N-linked glycosylation site is N-X-S/T, where X is any residue other than proline.

[2]Sharma et al., *Proc. Natl. Acad. Sci. USA* 111: 18601-18606, 2014

[3]This motif consists of a K or R, followed by a K or R. Stated differently, the motif can be KK, KR, RK, or RR.

[4]The dipeptide NG poses a medium risk of development liability. The dipeptides NA, NN, NS, and NT pose a low risk of development liability. N may also exhibit low risk of liability for other successor residues, e.g., D, H, or P. Stated differently, dipeptide ND, NH, or NP poses a low risk of development liability.

[5]Similarly to the above, the dipeptide DG poses a medium risk of development liability. The dipeptides DA, DD, DS, and DT pose a low risk of development liability. D may also exhibit low risk of development liability for other successor residues, e.g., N, H, or P.

[6]"Free cysteine" refers to a cysteine that does not form a disulfide bond with another cysteine and thus is left "free" as thiols. The presence of free cysteines in the antibody can be a potential development liability. Typically, an odd net number of cysteines in the protein shows a likelihood there is a free cysteine.

As one illustrative example, a variant was generated by introducing H76PT mutation to the heavy chain variable region of AB-009815 (SEQ ID NO: 75). This mutation removes a medium risk DP proteolytic cleavage site that could cause AB degradation during production, while in storage, or in the serum which decreases half-life after injection in mice/humans. The variant retains the ADC activity of AB-009815 in tumor cells, e.g., H522, LoVo, A375, SKOV3, A549, and MDAMB231 cells (FIG. 35)

[0405] Another goal for engineering variants is to reduce the risk of clinical immunogenicity: the generation of anti-drug antibodies against the therapeutic antibody. To reduce the risk, the antibody sequences are evaluated to identify residues that can be engineered to increase similarity to the intended population's native immunoglobulin variable region sequences.

[0406] The factors that drive clinical immunogenicity can be classified into two groups. First are factors that are intrinsic to the drug, such as: sequence; post-translational modifications; aggregates; degradation products; and contaminants. Second are factors related to how the drug is used, such as: dose level; dose frequency; route of administration; patient immune status; and patient HLA type.

[0407] One approach to engineering a variant to be as much like self as possible is to identify a close germline sequence and mutate as many mismatched positions (also known as "germline deviations") to the germline residue type as possible. This approach applies for germline genes IGHV, IGHJ, IGKV, IGKJ, IGLV, and IGLJ, and accounts for all the variable heavy (V_H) and variable light (V_L) regions except for part of H-CDR3. Germline gene IGHD codes for part of the H-CDR3 region but typically exhibits too much variation in how it is recombined with IGHV and IGHJ (e.g., forward or reverse orientation, any of three translation frames, and 5' and 3' modifications and non-templated additions) to present a "self" sequence template from a population perspective.

[0408] Each germline gene can present as different alleles in the population. The least immunogenic drug candidate, in terms of minimizing the percent of patients with an immunogenic response, would likely be one which matches an allele commonly found in the patient population. Single nucleotide polymorphism (SNP) data from the human genome can be used to approximate the frequency of alleles in the population.

[0409] Another approach to engineering a lead for reduced immunogenicity risk is to use in silico predictions of immunogenicity, such as the prediction of T cell epitopes, or use in vitro assays of immunogenicity, such as ex vivo human T cell activation. For example, services such as those offered by Lonza, United Kingdom, are available that employ platforms for prediction of HLA binding and in vitro assessment to further identify potential epitopes.

[0410] Antibody variants can be designed to enhance the efficacy of the antibody. In some embodiments, design parameters can focus on CDRs, e.g., CDR3. Positions to be mutated can be identified based on structural analysis of antibody-antigen co-crystals (Oyen et al., *Proc. Natl. Acad Sci. USA* 114: E10438-E10445, 2017; Epub Nov. 14, 2017) and based on sequence information of other antibodies from the same lineage.

Approaches to Mutation Design

[0411] Development liabilities can be removed or reduced by one or more mutations. Mutations are designed to preserve antibody structure and function while removing or reducing development liabilities and to improve function. In some embodiments, mutations to chemically similar residues can be identified that maintain size, shape, charge, and/or polarity. Illustrative mutations are described in Table 17.

TABLE 17

Preferred mutations to remove development liabilities

Free cysteine	Odd # C	High	C→(A, S)
N-linked glycosylation	N(~P)(S, T)	High	N→(Q, D, S, A); (S, T)→(A, N)
Proteolytic cleavage	(K, R)(K, R)	Medium	K, R→(Q, S, A)
Proteolytic cleavage	DP	Medium	D→(E, S, A)
Asparagine deamidation	NG; N(A, N, S, T)*	Medium; Low	N→(Q, S, A); G→(A, S)
Aspartate isomerization	DG; D(A, D, S, T)*	Medium; Low	D→(E, S, A); G→(A, S)
Lysine glycation	K	Low	K→(R, Q, S, A)
Methionine oxidation	M	Low	M→(Q, L, S, A)
Tryptophan oxidation	W	Low	W→(Y, F)

Note:
the last column of Table 17 shows preferred mutations. For example, C→(A, S) refers to C can be mutated to either A or S to remove development liabilities.

Methods for Altering the Glycosylation of an Antibody

[0412] In another aspect, the antibodies described herein comprise an Fc region having altered glycosylation that increase the ability of the antibody to recruit NK cells and/or increase ADCC. In some embodiments, the Fc region comprises glycan containing no fucose (i.e., the Fc region is afucosylated). Fucosylated antibodies can be produced using cell lines that express a heterologous enzyme that depletes the fucose pool inside the cell (e.g., GlymaxX® by ProBioGen AG, Berlin, Germany). Non-fucosylated antibodies can also be produced using a host cell line in which the endogenous α-1,6-fucosyltransferase (FUT8) gene is deleted. See Satoh. M. et al., "Non-fucosylated therapeutic antibodies as next-generation therapeutic antibodies," Expert Opinion on Biological Therapy, 6:11, 1161-1173, DOI: 10.1517/14712598.6.11.1161.

Diagnosis of Cancer

[0413] The EphA2 antibodies disclosed herein can also be used for diagnosing a patient having a cancer suitable for treatment. In one aspect, the method comprises contacting a tumor sample from the patient with an antibody disclosed above and detecting binding of the antibody to the tumor sample. In some cases, the antibodies are conjugated to a detectable label that produces fluorescent, luminescent or colorimetric signals, and detecting the signal from the label indicates that tumor is suitable for treatment with an EphA2 antibody disclosed herein. In some cases, after the antibodies are contacted with the tumor sample, a labeled secondary antibody is added to the tumor sample that have been contacted with the antibody disclosed herein and detecting the signal from the secondary antibody indicates that the tumor expressing or overexpressing EphA2.

[0414] As discussed above, in some tumor types, e.g., uterine cancer, head and neck cancer, and NSCLC, an EphA2 antibody may display donor-specific tumor selectivity, i.e., the antibody may show tumor selectivity in some donors but not other donors. In these cases, it is desirable to determine the binding of the tumor to the EphA2 antibody before treating the patient.

Screening Antibodies

[0415] Also disclosed herein is a method for selecting a tumor-targeting antibody. The method comprises contacting an antibody disclosed above with a polypeptide comprising SEQ ID NO: 94 or SEQ ID NO: 95 and contacting the antibody with a tumor cell and selecting the antibody if the antibody binds to the polypeptide and binds preferentially to the tumor cell as compared to normal cell.

[0416] The following examples are offered for illustrative purposes and are not intended to limit the invention. Those of skill in the art will readily recognize a variety of non-critical parameters that can be changed or modified to yield essentially the same results.

EXAMPLES

Example 1. AB-008873

[0417] AB-008873 was discovered in antibody repertoires generated by Immune Repertoire Capture® (IRC™) technology from plasmablast B cells isolated a non-small cell lung cancer patient with an active anti-tumor immune response after treatment with the anti-PD-1 antibody OPDIVO® (nivolumab) (Bristol Myers Squibb). AB-008873 was tested for cell surface binding to 6,000 human membrane proteins expressed on the surface of HEK293 cells by flow cytometry. Two membrane proteins showed binding over the background: EphA2 and FcGR1. No other ephrin type-A or type-B receptors had a binding signal above the background.

[0418] AB-008873 was further shown to bind the extracellular domain of EphA2 via an indirect and reverse indirect ELISA. The ELISAs also confirmed binding to mouse EphA2.

[0419] Next, AB-008873 was tested to determine if the previously observed the in vitro surface binding to A549 of the antibody was mediated by interaction with EphA2. 3 EphA2 guide RNAs were introduced into Cas9 overexpressing A549 cells. The EphA2 guide RNAs were designed by Synthego and Chop CRISPR tool. Exact sequences were selected based on low off-target score. The designed sgRNAs were synthesized and modified by Synthego. Electroporations of guide RNA was performed using the Neon™ Transfection System 10 µL Kit #MPK1096 according to the manufacturer's protocol. Final concentrations of the sgRNA were 100 µM (100 pmol/l). This resulted in a knockout of EphA2 gene in >90% of cells. The polyclonal EphA2 KO A549 cell population was then cultured for ~4 days and assayed for cell surface binding by AB-008873. The fraction of cells in the AB-008873 positive gate was reduced from ~98% (irrelevant guide RNA) to ~10%, thus confirming EphA2 is the primary driver of AB-008873 A549 binding.

Example 2. AB-008873 Variants

[0420] The sequence of AB-008873 was analyzed for potential liabilities. There are no high-risk liabilities. There is a medium-risk proteolytic cleavage site at H75D-H76P (i.e., heavy chain position 75 aspartate followed by heavy chain position 76 proline) and a medium-risk deamidation site at H108N-H109G. Low-risk liabilities include tryptophan oxidation at H34W and L90W; asparagine deamidation at H60N-H61N and L25N-L26N; lysine glycation at H67K, H100K, and L30K; and aspartate isomerization at H115D-H116A, L49D-L50D, L50D-L51S, and L91D-L92S.

[0421] AB-008873 was aligned to its closest human germline genes (IGHV4-34*02, IGHD2-8*01, IGHJ3*02, IGLV3-21*02, and IGLJ2*01), and to four of its known siblings (AB-009805, AB-009806, AB-009807, and AB-009808). Three variants were designed to explore differences between AB-008873 and its siblings. AB-009812 was designed with mutation H54RS to AB-008873 (heavy chain position 54 mutated from arginine to serine), to match H54S as found in the siblings and germline, and to create the N-linked glycosylation motif found in the siblings and germline (H52N-H53H-H54S). AB-009813 was designed with mutation H54RA to AB-008873, to remove the H54 arginine from AB-008873, but not create an N-linked glycosylation motif (ie, H52N-H53H-H54A is not an N-linked glycosylation motif). AB-009814 was designed with deletion of three residues, H61N-H62Y-H63N, to match the siblings and germline in that region. Note that AB-009814 could equivalently be described as a deletion of H60N-H61N-H62Y, H59Y-H60N-H61N, or H58N-H59Y-H60N and the resulting sequence is the same.

[0422] AB-009815 and AB-009816 were designed to remove the medium-risk proteolytic cleavage liability in AB-008873 at H75D-H76P. The variants make the mutation H76PT or H76PA, respectively. AB-009815 with the H76PT mutation also changes the sequence o germline.

[0423] AB-009817 was designed to remove the medium-risk deamidation site at H108N-H109G. The variant makes the mutation H109GA relative to AB-008873, to reduce the liability to a low risk deamidation site (i.e., NG is medium risk, but NA is low risk). H106C and H111C are predicted to form an intra-H3 disulfide bond, with H107-H110 in a 4-residue turn. Alternative approaches to removing the medium risk deamidation site include H108NS, H108NA, H108NQ, H108NL, or H108NY.

[0424] AB-008873, siblings AB-009805, AB-009806, AB-009807, AB-009808, and variants AB-009812, AB-009813, AB-009814, AB-009815, AB-009816, and AB-009817 were synthesized on a mouse IgG2a framework. All siblings show less binding, less ADCC, and less ADCP activity than AB-008873. See, FIG. **20**. Variants AB-009815 and AB-009816, which remove the same medium-risk cleavage site, show similar levels of binding and activity as AB-008873. AB-009817 completely loses binding (tested up to 1 µM) and functional activity (tested up to 100 nM). Variants AB-009812, AB-009813, and AB-009814 show some binding and ADCC activity, though less than AB-008873. These results show that the N-linked glycosylation motif and the 3-aa insertion in H2 region affect binding, but they are not essential. The ADC (antibody-drug conjugate) activity of AB-009815 is the same as that of AB-008873. See, FIG. **21**A-C and Table 18.

TABLE 18

Functional activity of engineered AB-008873 variants						
		Flow		ADCC		
Reg ID	Mutation	EC_{50} (nM)	Delta	EC_{50} (nM)	Delta	Rationale
AB-008873		3.6	680293	8.9	46.8	
AB-009812	H54RS	9.6	567759	51.0	41.1	Re-create NHS glycosylation motif as found in sibs and germline
AB-009813	H54RA	6.8	578049	26.0	39.9	Keep glycosylation motif out bt via alanine instead of arginine
AB-009814	H61NYNxxx	11.5	407293	62.8	36	Undo 3-residue insertion, back to as found in sibs and germline
AB-009815	H76PT	5.5	745456	10.4	59.8	Remove medium-risk DP proteolytic cleavage site
AB-009816	H76PA	5.5	670105	12.5	58	Remove medium-risk DP proteolytic cleavage site
AB-009817	H109GA	inactive	0	inactive	0	Remove medium-risk NG asparagine deamidation site (leave as low-risk NA)

[0425] The 12 variants AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, and AB-010152 were designed to mutate AB-009815 closer to germline. Each variant exhibits 1 to 4 mutations at positions that differ from the closest germline. Some positions are framework; some are CDR. AB-010141 makes H31DG. AB-010142 makes L31NS. AB-010143 makes L97LV. AB-010144 makes H51VI. AB-010145 makes H86TS. AB-010146 makes H129AS. AB-010147 makes L60QR. AB-010148 combines H51VI and H86TS. AB-010149 combines H51VI and H129AS. AB-010150 combines H86TS and H129AS. AB-010151 combines H51VI, H86TS, and H129AS. AB-010152 combines H51VI, H86TS, H129AS, and L60QR. Further combinations of mutations to germline are of interest.

[0426] Anti-EphA2 Variants AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, and AB-010152 were synthesized on a mouse IgG2a framework and assayed for ADCC, both by EC_{50} and delta-activity as compared AB-008873 and melting temperature (Tm) measured using UNcle instrument. The results of these assays are shown in Table 19.

TABLE 19

Properties of anti-EphA2 antibodies					
				ADCC	
AB regid	Parent	Mutations	Tm	EC_{50} (nM)	dAct (%)
AB-008873	N/A	N/A	64.1	29.0	18.6
AB-010141	AB-009815	H31DG	63.1	4.6	24.2
AB-010142	AB-009815	L31NS	64.2		
AB-010143	AB-009815	L97LV	65.4	25.1	21.6
AB-010144	AB-009815	H51VI	64.5	35.8	17.1
AB-010145	AB-009815	H86TS	64.1	48.9	19.2
AB-010146	AB-009815	H129AS	63.9	66.8	17.3
AB-010147	AB-009815	L60QR	66.5	79.4	15.8
AB-010148	AB-009815	H51VI_H86TS	64.4	45	17.4
AB-010149	AB-009815	H51VI_H129AS	64.6		
AB-010150	AB-009815	H86TS_H129AS	64.4	65.9	17.1
AB-010151	AB-009815	H51_H86_H129	64.6	33.3	17.5
AB-010152	AB-009815	H51_H86_H129_L60	67.0		

[0427] AB-010141 with the H31DG mutation exhibits improved ADCC EC_{50} but decreases Tm by approximately 1° C. AB-010143 with L97LV exhibits improved Tm by approximately 1° C. AB-010147 with L60QR exhibits improved Tm by approximately 2° C. but might exhibit somewhat decreased ADCC activity. AB-010148 with H51VI and H86TS exhibits similar Tm and ADCC activity as compared to AB-008873.

[0428] Additional variants were generated based on AB-010148 and designated as AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, and AB-010367.

[0429] AB-010148 was selected based on Tm and ADCC activity similar to that of AB-008873, while incorporating H76PT to remove the medium-risk DP proteolytic cleavage site and incorporating H51VI and H86TS to remove germline deviations to reduce potential immunogenicity. Combinations of L31DG, L97LV, and/or L60QR were added to AB-010148 to improve Tm, improve ADCC activity, and/or remove germline deviations. H31DG is predicted to remove an intramolecular salt bridge between H31D and H54R. To potentially compensate for this removal, H31DG was combined with either H54RA or H54RQ, along with L97LV and optionally L60QR, in variants AB-010364, AB-010365, AB-010366, and AB-010367.

[0430] Antibody variants AB-010357-AB-010367 were synthesized on a mouse IgG2a framework and assayed for ADCC, both by EC_{50} and delta-activity (i.e., change in activity) as compared AB-008873 and melting temperature (Tm) measured using UNcle instrument. The results of these assays are shown in Table 20.

TABLE 20

Properties of Anti-EphA2 antibodies

AB regid	Parent	Mutations	Tm	EC_{50} (nM)	Fold-improved	dAct (%)
AB-008873	N/A	N/A		34.8	1.00	19.3
AB-010357	AB-010148	L97LV	65.3	11.6	3.00	23.1
AB-010358	AB-010148	L60QR	66.9	73.3	0.47	15.7
AB-010359	AB-010148	L97LV__L60QR	67.6	25.3	1.38	16.9
AB-010360	AB-010148	H31DG	64.0	4.3	8.09	26.1
AB-010361	AB-010148	H31DG__L97LV	65.0	3.1	11.23	24.1
AB-010362	AB-010148	H31DG__L60QR	65.9	12.8	2.72	19.6
AB-010363	AB-010148	H31DG__L60QR__L97LV	66.5	9.5	3.66	27.8
AB-010364	AB-010148	H31DG__H54RA__L97LV	64.5	14.5	2.40	26.3
AB-010365	AB-010148	H31DG__H54RA__L97LV__L60QR	66.7	34.6	1.01	21.3
AB-010366	AB-010148	H31DG__H54RQ__L97LV	64.7	28.9	1.20	25.9
AB-010367	AB-010148	H31DG__H54RQ__L97LV__L60QR	67.6	42.7	0.81	21.3

[0431] The results show that multiple variants exhibit improvements in both Tm and ADCC activity relative to AB-008873.

[0432] Antibodies AB-010357, AB-010361, and AB-010363 were then used as the basis for generation of further variants. AB-010357 was mutated to generate AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671. AB-010672, AB-010673, and AB-010674.

[0433] AB-010361 was mutated to generate AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, and AB-010688.

[0434] AB-010363 was mutated to generate AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, and AB-010702.

Example 3. In Vitro Studies

Immunophenotyping

[0435] Peripheral blood was blocked with 1:100 TruStain FcX PLUS Antibody (BioLegend 156604) for 10 min at 4° C. and stained for 30 min at 4° C. with a mastermix of antibodies for each panel. Next, 1.6 ml of 1× fix/lyse solution prepared according to manufacturer's protocol (BD 558049) was added to each sample and incubated for 10 min at RT. Samples were centrifuged at 500×g for 5 min, supernatant removed and transferred to a 96-well plate. Samples were centrifuged at 300×g for 5 min, washed with 200 μl of PBS, and resuspended in 100 μl of 1% FBS/5 mM EDTA/PBS for analysis on Beckman Coulter's Cytoflex flow cytometer. The results were analyzed using FlowJo_v10.7.1. Reagents for immunophenotyping are shown in Table 21.

TABLE 21

Reagents for immunophenotyping

Lymphoid Panel	BioLegend	Myeloid Panel	BioLegend
CD45-PerCP	103129	CD45-PerCP	103129
CD3-APC	152306	Ly6C-APC	128016
CD8-AF700	100730	F4/80-AF700	123130
CD4-APCCy7	100414	B220-APCCy7	103224
CD25-PE	101904	Tim3-PE	119704
Tim3-PEDazzle594	134014	CD11c-PEDazzle594	117348
CD49b-PECy7	108922	CD103-PECy7	121426
CD62L-BV421	104436	MHCII-BV421	107632
PD1-BV510	135241	CD24-BV510	101831
CD44-BV605	103047	Ly6G-BV605	127639
CD19-BV650	115541	CD11b-BV650	101259

[0436] Tumors were digested using Miltenyi's gentleMACS Octo Dissociator with heaters and Tumor Dissociation Kit, mouse (130-096-730), according to the manufacturer's instructions. The dissociated cells were cryopreserved in 10% FBS/DMSO. Dissociated cells were thawed using pre-warmed 2% FBS/RPMI dropwise, centrifuged at 300×g for 10 min, with supernatant decanted. Cells are blocked TStain FcX PLUS Antibody (BioLegend 156604) for 10 min at 4° C. and stained with a mastermix of antibodies for 30 min at 4° C. Next, 100 μl of PBS was added, cells were centrifuged at 300×g for 5 min supernatant decanted, and washed with another 200 μl PBS. Cells were resuspended in 100 μl of 1% FBS/5 mM EDTA/PBS with 1:500 Membrane integrity Dye (Intellicyt 90365) for analysis on Beckman Coulter's Cytoflex flow cytometer. Resulting data was analyzed using FlowJo_v10.7.1. Reagents for immunophenotyping in tumor cells are provided in Table 22.

TABLE 22

Reagents for immunophenotyping in tumor cells			
Lymphoid Panel	BioLegend	Myeloid Panel	BioLegend
CD3-APC	152306	Ly6C-APC	128016
CD8-AF700	100730	F4/80-AF700	123130
CD4-APCCy7	100414	B220-APCCy7	103224
CD25-PE	101904	CD45-PE	103106
NKG2D-PEDazzle594	130214	CD11c-PEDazzle594	117348
CD49b-PECy7	108922	CD103-PECy7	121426
PD1-BV421	135221	MHCII-BV421	107632
Gr1-BV510	108457	CD24-BV510	101831
CD45-BV605	103155	Ly6G-BV605	127639
CD19-BV650	115541	CD11b-BV650	101259

Immunophenotyping Results Summary:

[0437] Early increases in total T cells, cytotoxic T cells, and helper T cells observed in the blood 2 days after the first dose. At the same time, decreases in overall myeloid, monocytes, or M-MDSCs were detected after treatment with 40 mg/kg AB-008873. These changes were no longer detected 9 days after the first dose and instead, an increase in CD11b+Ly6Chi monocytes or M-MDSCs was observed. FIG. **8**

[0438] Tumors treated with AB-008873 showed no significant changes in the percentage of tumor infiltrating leukocytes (TILs) by flow cytometer at days 10 and 17 post inoculation compared to PBS treated controls. FIG. **10**. Similarly, no significant changes in T cells, NK cells, monocytes/M-MDSCs, granulocytes/G-MDSCs were detected in the tumor at days 10 and 17 (FIG. **11** and FIG. **18**) while total macrophages were increased at day 10 in mice treated with 40 mg/kg AB-008873. Tumors collected at endpoint and treated with 40 mg/kg AB-008873 showed evidence of decreased TILs, M-MDSCs, and slight increases in cDC1. FIG. **12** and FIG. **17**.

Binding Assays

Flow Screen (In Vitro) Method:

[0439] Surface binding of the antibodies to in vitro grown cell lines was assessed by flow cytometry. Tumor cells were detached from their culture plate using Versene and counted. Cells were staining in BSA-containing buffer with primary antibody for 30 minutes at 4° C. with shaking. Following, cells were washed and stained with secondary PE-labeled antibody for 30 minutes at 4° C. with shaking. Before analysis on an Intellicyt iQue3 scanner, cells were counter-stained with DAPI. MedFI values from live, single cells was expressed as fold over isotype control.

Flow Screen (Ex Vivo) Method:

[0440] AB-008873 was conjugated using Thermo's Site-Click™ R-PE Antibody Labeling Kit for testing on dissociated CT26 ex vivo cells. Dissociated cells were thawed and cells were blocked with TruStain FcX™ (anti-mouse CD16/32) Antibody and stained with PE-conjugated antibodies and CD45-BV605 for 30 min at 4° C. Cells were washed 3 times with 200 1 1% FBS/1 mM EDTA/PBS and resuspended in assay buffer containing DAPI. Cells were analyzed using the Cytoflex and FlowJo_v10.7.1.

[0441] Surface binding of AB-008873 on several in vitro mouse and human tumor cell lines was assessed using flow cytometry. Among the cell lines screened, AB-008873 show binding on 786-0, A375, A549, H522, LoVo, MDA-MB-231, PC3, RKO, SKOV3, SW1116, and CT26. Additionally, AB-008873 bound CT26 ex vivo cells. FIG. **1**, Tables 23 and 24.

TABLE 23

Surface binding profile of AB-008873 in human cell lines Secondary Screens: In Vitro Cell Lines Flow-Based Binding Assay															
293T	786O	A375	A549	BT474	COLO205	H522	LoVo	MDAMB231	PC3	RKO	SKBR3	SKMEL28	SKOV3	SW1116	T84
2	3	2	3	0	0	3	3	3	4	4	0	0	3	2	1

TABLE 24

Surface binding profile of AB-008873 in mouse cell lines Secondary Screens: In Vitro Cell Lines Flow-Based Binding Assay			
CT26	EMT6	MC38	RENCA
2.66667	0.2	0	0

ADCC

[0442] To detect Antibody-dependent cellular cytotoxicity (ADCC), KILR™-transfected target cells were opsonized with different concentrations of the antibody at room temperature for 20 minutes. Following, NK92 cells carrying a mouse extracellular-human intracellular CD16 chimera were added at a 5:1 ratio of effector to target cells. After a 4 h co-culture at 37° C. and 5% CO_2, pre-mixed KILR™ detection reagent was added to each well at equal volume and allowed to incubate for 30 min at room temperature in the dark before reading luminescence on a pate reader. Cytotoxicity was calculated after background (spontaneous release) subtraction and expressed as percent of a maximum lysis control.

[0443] ADCC activity of AB-008873 was assessed on several in vitro human and mouse tumor cell lines. Among the cell lines screened, AB-008873 showed dose-dependent ADCC on A549, MDA-MB-231, CT26, and to a small extent, on EMT6 cells. AB-008873 did not show any ADCC when the antibody was used up to 100 nM on SKBR3. FIG. **2**A-**2**B and Table 25.

TABLE 25

Results of assessing the ADCC activity (FIG. 2A) and ADCP activity (FIG. 2B) of AB-008873 in vitro functional assays.

	n	EC_{50} (nM)	Delta Toxicity (%)	NAUC (%)
A549	15	15.3 ± 11.8	37.8 ± 19.2	30.7 ± 13.1
MDAMB231	2	18.7 ± 1.4	53.2 ± 5.7	17.1 ± 1.7
SKBR3	2	>100	0	0
CT26	4	10.2 ± 9.5	37.5 ± 5.9	52.6 ± 15.4
EMT6	2	>100	15.1 ± 10.6	7.9 ± 4.9

Note:
Table 25 summarizes the results in FIG. 2A. The values are shown as mean + SD. N = number of biological replicates; delta toxicity (%) = difference between lower and upper plateau of the sigmoidal dose response curve; NAUC (%) = Normalized Area Under the Curve, internal metric quantifying area under the curve of the sigmoidal dose response normalized to a known positive control.

[0444] Antibodies derived from the same lineage as AB-008873 (sibling antibodies) were tested for binding and for ADCC and ADCP activity on tumor cell line A549. Compared with AB-008873, all sibling antibodies showed less potent and less maximum binding and less ADCC and ADCP activity. Out of the sibling antibodies, AB-009806 showed the best binding and functional activity, followed by AB-009805. AB-009807 and AB-009808 showed little binding and ADCC activity but some ADCP activity. FIG. **20**

[0445] Engineered variant antibodies of AB-0008873 were tested for binding and for ADCC activity on A549 tumor cells. Compared with AB-008873, AB-009815 and AB-009816 showed similar levels of binding and ADCC activity as AB-008873. AB-009812 and AB-009813 showed weaker binding and ADCC activity, followed by AB-009814. AB-009817 did not show any substantial binding or ADCC activity on A549 up to 100 nM. FIG. **21**.

[0446] AB-010357, AB-010361, and AB-010363 and variants (Table 27) based on those antibodies exhibited ADCC activity greater than AB-008873 on the A549 tumor cells. All the variants tested exhibited ADCC activity at least as good as their parental antibody. Several variants showed a 10-to-100-fold increase in potency as compared to their parent, with some variants showed a greater than 700-fold increase. AB-010685. AB-010671, and AB-010699 had EC_{50} values under these assay conditions of approximately 6 pM. AB-010681 and AB-010695 had EC_{50} values of approximately 8 pM and 10 pM, respectively. FIG. **40** and Table 26. It is notable that both AB-010685 and AB-010699 have the same double mutations L29SY and L92SH. See Table 28.

TABLE 26

ADCC of anti-EphA2 antibodies

		ADCC - EC_{50} (nM)		ADCC - dAct rel. parent (%)	
AB ID	Mutations	Trial 1	Trial 2	Trial 1	Trial 2
AB-008873	N/A	24			
AB-010357		12.3	9.6	100.0	100.0
AB-010661	10357_H107TI_L23GR	5.7		160.1	
AB-010662	10357_L23GR_L29SY	0.57		166.4	
AB-010663	10357_L23GR_L30KM	4.7		163.1	
AB-010664	10357_L23GR_L92SH	0.54		167.7	
AB-010665	10357_H107TI_L29SY	~0.23	0.14	177.0	120.5
AB-010666	10357_L29SY_L30KM	~0.19	0.16	193.7	132.8
AB-010667	10357_L29SY_L92SH	<0.03	0.012	154.8	108.9
AB-010668	10357_H107TI_L30KM	3.7		162.9	
AB-010669	10357_L30KM_L92SH	0.44		177.7	
AB-010670	10357_H107TI_L92SH	0.36		178.2	
AB-010671	10357_H107TI_L29SY_L92SH	<0.03	0.0064	157.1	106.7
AB-010672	10357_L93SR	12.4		162.9	
AB-010673	10357_L93SE	3.9		166.4	
AB-010674	10357_L29SY_L93SE	~0.13	0.10	171.2	97.2
AB-010361		4.0	2.2	100.0	100.0
AB-010675	10361_H107TI_L23GR	1.5		122.5	
AB-010676	10361_L23GR_L29SY	~0.1	0.039	121.6	88.7
AB-010677	10361_L23GR_L30KM	1.6		165.5	
AB-010678	10361_L23GR_L92SH	~0.14	0.047	154.2	93.9
AB-010679	10361_H107TI_L29SY	~0.15	0.040	171.5	102.9
AB-010680	10361_L29SY_L30KM	~0.081	0.031	161.1	106.1
AB-010681	10361_L29SY_L92SH	<0.03	0.0079	152.1	104.6
AB-010682	10361_H107TI_L30KM	3.2		147.9	
AB-010683	10361_L30KM_L92SH	~0.14	0.046	174.8	100.0
AB-010684	10361_H107TI_L92SH	~0.14	0.048	166.6	115.4
AB-010685	10361_H107TI_L29SY_L92SH	<0.03	0.0057	161.1	106.4
AB-010686	10361_L93SR	6.4		132.3	
AB-010687	10361_L93SE	2.8		166.8	
AB-010688	10361_L29SY_L93S	~0.052	0.032	129.9	108.0
AB-010363		5.1	4.2	100.0	100.0
AB-010689	10363_H107TI_L23R	2.5		88.0	
AB-010690	10363_L23GR_L29S	0.19		112.4	

TABLE 26-continued

ADCC of anti-EphA2 antibodies

AB ID	Mutations	ADCC - EC_{50} (nM) Trial 1	ADCC - EC_{50} (nM) Trial 2	ADCC - dAct rel. parent (%) Trial 1	ADCC - dAct rel. parent (%) Trial 2
AB-010691	10363__L23GR__L30K	2.2		85.3	
AB-010692	10363__L23GR__L92S\	0.21		105.0	
AB-010693	10363__H107TI__L29	0.15		129.0	
AB-010694	10363__L29SY__L30K	0.14		131.3	
AB-010695	10363__L29SY__L92SH	~0.021	0.0096	148.3	118.8
AB-010696	10363__H107TI__L30KM	1.9		114.7	
AB-010697	10363__L30KM__L92SH	0.30		147.1	
AB-010698	10363__H107TI__L92SH	0.23		142.1	
AB-010699	10363__H107TI__L29SY__L92SH	~0.014	0.0058	152.5	114.6
AB-010700	10363__L93SR	9.5		75.7	
AB-010701	10363__L93SE	3.0		99.2	
AB-010702	10363__L29SY__L93SE	0.065		124.7	

TABLE 27

Sequence features of selected anti-EphA2 antibody variants

	Variant based on AB-010357	Variant based on AB-010361	Variant based on AB-010363
H107TI__L23GR	AB-010661	AB-010675	AB-010689
L23GR__L29SY	AB-010662	AB-010676	AB-010690
L23GR__L30KM	AB-010663	AB-010677	AB-010691
L23GR__L92SH	AB-010664	AB-010678	AB-010692
H107TI__L29SY	AB-010665	AB-010679	AB-010693
L29SY__L30KM	AB-010666	AB-010680	AB-010694
L29SY__L92SH	AB-010667	AB-010681	AB-010695
H107TI__L30KM	AB-010668	AB-010682	AB-010696
L30KM__L92SH	AB-010669	AB-010683	AB-010697
H107TI__L92SH	AB-010670	AB-010684	AB-010698
H107TI__L29SY__L92SH	AB-010671	AB-010685	AB-010699
L93SR	AB-010672	AB-010686	AB-010700
L93SE	AB-010673	AB-010687	AB-010701
L29SY__L93SE	AB-010674	AB-010688	AB-010702

TABLE 28

Activities of anti-EphA2 antibodies

Mutations	Variant based on AB-010357 EC_{50} (nM)	Variant based on AB-010361 EC_{50} (nM)	Variant based on AB-010363 EC_{50} (nM)
Parent (AB-010357, AB-010361, or AB-010363)	10.95	3.1	4.65
H107TI__L23GR	5.7	1.5	2.5
L23GR__L29SY	0.57	0.039	0.19
L23GR__L30KM	4.7	1.6	2.2
L23GR__L92SH	0.54	0.047	0.21
H107TI__L29SY	0.14	0.04	0.15
L29SY__L30KM	0.16	0.031	0.14
L29SY__L92SH	0.012	0.0079	0.0096
H107TI__L30KM	3.7	3.2	1.9
L30KM__L92SH	0.44	0.046	0.30
H107TI__L92SH	0.36	0.048	0.23
H107TI__L29SY__L92SH	0.0064	0.0057	0.0058
L93SR	12.4	6.4	9.5
L93SE	3.9	2.8	3.0
L29SY__L93SE	0.10	0.032	0.065

[0447] AB-010699 and AB-010361 exhibited greater ADCC activity on A549 cells than AB-010018, while both AB-010699 and AB-010695 were more toxic than cetuximab on A549 cells. FIGS. **50**A and **50**B. AB-010699 showed several thousand-fold higher ADCC activity on A549 cells and higher thermostability (as reflected in a 2.0° C. higher Tm1) as compared to AB-008873. AB-010685 was also highly potent on PC3-KILR cells and exhibited a 10-fold improvement in ADCC as compared to AB-010018. FIG. **52**.

ADCP

[0448] For antibody-dependent cellular phagocytosis (ADCP), the cell membranes of target cells were labeled with green fluorescent membrane dye preceding opsonization with different concentrations of antibody at room temperature for 30 minutes. RAW264.7 mouse macrophage (effector) cells were then added at a 1:1 ratio with the target cells and co-cultured at 37° C. and 5% CO_2 for 2 hours. After incubation, the cell mixture was dissociated with 2 mM ethylenediaminetetraacetic acid (EDTA, pH 7.4). The cell suspension was then washed several times in an isotonic salt solution and probed with a commercially available anti-CD11b antibody, conjugated with Allophycocyanin (APC). After a 30-minute incubation at 4° C. with this antibody, the cell suspension was washed again before being analyzed by flow cytometry. Cytotoxicity was calculated as the fraction of effector cells recorded that also displayed signal for green fluorescent dye and was expressed as a percentage of the total effector cell population.

[0449] The activity of AB-008873 was tested on A549 and CT26 cells. The results, as shown in FIG. **2**B, demonstrated dose-dependent activity in both models.

[0450] AB-008873 was tested for ADCP activity on A549 cells alongside its siblings. All siblings showed reduced activity compared to AB-008873. FIG. **20**.

ADC

[0451] AB-008873 was tested for its antibody-drug conjugate activity using a secondary, toxin-conjugated antibody. In brief, target cells were detached from the culture plate and cell concentration was adjusted to 31,250 cells/mL in assay media. 2,500 cells were added to each well of a 96 well plate and incubated with different concentrations of primary antibody for 15 min at room temperature. Following, secondary Fab anti-mouse IgG Fc conjugated to Duocannycin with a cleavable linker (Moradec, #AM-202-DD) was added at a final concentration of 250 ng/mL. Cells were incubated for 72 h at 37° C. and 5% CO_2. At the end of the incubation period, 100 µl CellTiter-Glo® was added to each well and allowed to incubate for 5-10 min at room temperature before reading luminescence in a BMG ClarioSTAR plate reader. Data was then normalized to a maximum lysis control and plotted using graph pad prism.

[0452] The ADC activity of AB-008873 was assessed on several in vitro human and mouse tumor cell lines. Among the cell lines screened, AB-008873 showed dose dependent ADC on A549, SKOV3, MDA-MB-231, A375, LoVo, H522, RKO, and PC3. As shown in FIG. **3**, AB-008873 demonstrated ADC activity on these tumor cells.

[0453] The ADC activity of AB-008873 was compared to the engineered variant AB-009815 (H76PT mutation) on H522, LoVo, A375, SKOV3, A549, and MDA-MB-231 cells. AB-009815 showed similar activity compared to AB-008873 in all cell lines tested. As shown in FIG. **35** show that the ADC activity of AB-009815 is substantially identical to AB-008873 on these tumor cells.

[0454] The ADC activity of AB-010361 and AB-010699 was assayed as described above and was compared to that of AB-008873, with the results shown in FIG. **51**. Both AB-010361 and AB-010699 had similar ADC activity as AB-008873 in cells expressing high levels of EphA2 and exhibited enhanced ADC activity in cells expressing low levels of EphA2.

Thermostability

[0455] AB-010357, AB-010361, and AB-010363 and variants based on those antibodies were assayed for thermostability using an Unchained Labs UNcle instrument. The concentration of purified antibodies was adjusted as needed to between 0.2 mg/ml and 0.5 mg/ml in PBS immediately prior to analysis. To determine melting temperature (Tm) intrinsic protein fluorescence was measured at 473 nm every 1.1° C. as temperature was increased linearly from 25° C. to 95° C. at a rate of 0.3° C./min. The UNcle Analysis software (version 4.01) was used to find the Tm as the first derivative of the barycentric mean (BCM). All antibodies tested were at least as stable as AB-008873, with AB-010363 and variants thereof exhibiting improved thermostability as indicated by the increased Tm1. Table 29 provides thermostability of anti-EphA2 antibodies.

TABLE 29

Thermostability of anti-EphA2 antibodies

Mutations	Variant based on AB-010357 Average Tm1 (° C.)	Variant based on AB-010361 Average Tm1 (° C.)	Variant based on AB-010363 Average Tm1 (° C.)
Parent (AB-010357, AB-010361, or AB-010363)	64.8	64.7	68.0
H107TI_L23GR	66.7	65.5	66.5
L23GR_L29SY	64.7	65.5	66.4
L23GR_L30KM	64.5	64.7	67.0
L23GR_L92SH	64.4	64.1	65.8
H107TI_L29SY	64.3	63.6	65.8
L29SY_L30KM	64.5	63.9	65.7

TABLE 29-continued

Thermostability of anti-EphA2 antibodies			
Mutations	Variant based on AB-010357 Average Tm1 (° C.)	Variant based on AB-010361 Average Tm1 (° C.)	Variant based on AB-010363 Average Tm1 (° C.)
L29SY_L92SH	64.0	64.4	65.9
H107TI_L30KM	64.5	64.4	66.8
L30KM_L92SH	64.5	64.2	66.5
H107TI_L92SH	65.0	64.4	66.9
H107TI_L29SY_L92SH	64.9	64.2	66.4
L93SR	64.7	65.2	66.8
L93SE	65.3	64.9	67.7
L29SY_L93SE	64.9	64.5	66.7

Example 4. In Vivo Studies

Tumor Cell Implantation/Inoculation

[0456] Mouse CT26 tumor cells were propagated in culture by passaging cells every 2 to 3 days (1:10 subcultures) for 6 passages. On the day of inoculation, cells were collected, counted, and diluted to 5×10^6 cells/mL in RPMI medium without supplements. Cell viability was recorded as 86.8% at lime of inoculation.

[0457] Female, 6-week-old BALB/c mice were inoculated in the right hind flank by subcutaneous injection with 1×10^6 CT26 cells/mouse in 0.2 mL Waymouth's media without supplements. The day of cell inoculation was designated as Study Day 0.

Randomization

[0458] Mice were inoculated with CT26 tumor cells on Study Day 0. To achieve study groups with consistent and homogenous tumor volumes an overage of approximately 50% was included. Mouse tumors routinely became visible and palpable five days after cell inoculation. Tumor volumes were measured twice prior to randomization. On Study Day 9, mice were randomized to ensure homogenous tumor volumes using the 'matched distribution' randomization function of the StudyLog lab animal management software. Mice with pre-ulcerated tumors, irregular shaped tumors, or multiple tumors were excluded from randomization. Animal IDs were assigned randomly within each treatment group on the day of randomization.

Test Article Dosing

[0459] Test article AB-008873 was administered starting on Day 8 by intraperitoneal (IP) injection twice weekly at 20 or 40 mg/kg based on group mean body weight. Mice received either 1 or 3 doses before they were removed from study for analysis two days after dosing (Day 10 or Day 17, respectively).

[0460] Mice in the vehicle control groups were dosed at 10 mL/kg DPBS based on group mean body weight using the same dosing schedule as the test article.

Termination

[0461] Mice were euthanized by (1) carbon dioxide asphyxiation followed by cervical dislocation or (2) isoflurane inhalation and cardiac puncture followed by cervical dislocation at predetermined time points (i.e., Day 10 or Day 17).

[0462] Several assays were performed to analyze the effect of the antibody. Flow cytometry was used to analyze the immune profiling of the blood and tumor. Tumor volumes were measured. Semi-quantitative estimates of tumor infiltrates were generated using immunofluorescence staining.

Tissue Harvest for Immunofluorescence

[0463] Mice were euthanized by carbon dioxide asphyxiation followed by cervical dislocation. Tumors were excised from the mice, weighed, and placed in pre-filled tubes containing 10% neutral buffered formalin (Caplugs). Tumors were fixed for 24 hours at room temperature and then transferred to 80% ethanol prior to slicing and embedding in paraffin.

Immunofluorescence Analysis

Tumor Processing for Immunofluorescence Analysis

[0464] Each fixed tumor was dissected using a slicer matrix with 2.0 or 3.0 mm section slide intervals (Zivic 5526 or 5527), to allow for assessment of tumor microenvironment at 3 to 6 evenly spaced locations to account for tumor heterogeneity. The resultant tissue slices from each tumor were placed in tissue cassettes (up to three tumors in one cassette; including a piece of mouse spleen as a landmark for orientation) and subsequently processed in graded alcohols and xylene and embedded in paraffin per manufacturer's instructions using a Tissue Tek VIP® 6 A1 Vacuum Infiltration Processor (Sakura) and Tissue-Tek® TEC™ 5 embedding console system (Sakura). Blocks containing tumors were sectioned at 5 μm using a Accu-Cut® SRM™ 200 Rotary Microtome (Sakura). Sections were mounted to slide and dried. In one exemplary procedure, samples (e.g., tumor samples) were sectioned by longitudinal slicing into 2 or 3 mm thick slices using a mold. Sections are then embedded face down for examination. This approach is ideal for assessing infiltrates across the tumor and fitting multiple tumors per block. FIG. **7** and Table 30 show the procedures and reagents used in an in vivo study to assess the effect of the AB-008873 treatment to mice.

TABLE 30

Reagents and dosing information							
n■	Test Art.	Lot#	dose level (mg/kg)	Dose Freq.	# of Doses	Route	MODEL
15	AB-008873	2	40	2/Wk	7	i.p.	CT26
15	PBS	12620008	0	2/Wk	7	i.p.	CT26
5	AB-008873	2	20	2/Wk	1	i.p.	CT26
5	AB-008873	2	40	2/Wk	1	i.p.	CT26
5	PBS	12020008	0	2/Wk	1	i.p.	CT26
5	AB-008873	2	20	2/Wk	3	i.p.	CT26
5	AB-008873	2	40	2/Wk	3	i.p.	CT26
5	PBS	12620008	0	2/Wk	3	i.p.	CT26

Immunofluorescence Staining

[0465] Immunofluorescence staining was performed on 5 μm formalin-fixed paraffin-embedded tissue sections of mouse CT26 tumors following standard immunostaining protocols. The primary antibodies were utilized with species-specific secondary antibodies and detected using standard methodology.

[0466] The tissue sections were baked at 65° C. for 30 minutes, dewaxed in xylene and rehydrated. Antigen retrieval was performed under high pressure either at 95° C. for 20 minutes or 110° C. for 15 minutes using Target Retrieval Solution (Dako) in Decloaking Chamber NxGen (Biocare Medical). The sections were blocked for 15 minutes at room temperature with Bloxall (Vector Labs).

T Cell Marker Protocol

[0467] Dual labeling for cytotoxic T cells ("T_{cyt} cells", which are CD3+CD8+) and regulatory T cells (T_{reg} cells, i.e., CD4+/forkhead-box protein P3; FoxP3+) populations was performed as follows. Following antigen retrieval and blocking with Bloxall, tissue sections were blocked in blocking buffer (3% bovine serum albumin with 3% normal donkey serum in 1× phosphate-buffered saline) for 30 minutes or 1 hour at room temperature. After blocking, the slides were incubated with primary antibodies against CD8 for 1 hour at room temperature, FoxP3 overnight at 4° C., or with species-appropriate isotype controls at corresponding times and temperatures. The slides were subsequently washed with Wash Buffer (Thermo-Fisher Scientific), incubated with species-specific secondary PowerVision Poly-HRP (Leica) for 30 minutes at room temperature followed by incubation with CF Tyrainide 647 (for CD8) or 488 (for FoxP3) Working Solution (Biotium) for 3 minutes at room temperature, respectively.

[0468] To provide signal in distinct channels for the second primary antibody, elution immunofluorescence was utilized through pressure and heat application. After washing and block steps, the second primary antibody against CD3 and CD4 or species-appropriate isotype control was applied to sections as describe in the previous paragraph. The slides were subsequently washed with Wash Buffer (Thermo-Fisher Scientific), incubated with species-specific secondary PowerVision Poly-HRP (Leica) for 30 minutes at room temperature followed by incubation with CF Tyramide 488 (for CD3) or 647 (for CD4) Working Solution (Biotium) for 3 minutes at room temperature, respectively.

[0469] All tissue slides were counterstained with Hoechst dye, mounted in PermaFluor Aqueous Mounting Medium (ThermoFisher Scientific), and coverslipped. Images were acquired using an AxioScan whole slide scanner (Carl Zeiss Microscopy).

Dendritic Cell Marker Protocol

[0470] Dual labeling for dendritic cells (DC cells, i.e., CD3− CD103+) populations was performed as follows. Following antigen retrieval and blocking with Bloxall, tissue sections were blocked in blocking buffer (3% bovine serum albumin with 3% normal donkey serum in 1× phosphate-buffered saline) for 30 minutes or 1 hour at room temperature. After blocking, the slides were incubated with the primary antibody against CD103 for 1 hour at room temperature or with the species-appropriate isotype control at corresponding time and temperature. The slides were subsequently washed with Wash Buffer (Thermo-Fisher Scientific), incubated with species-specific secondary PowerVision Poly-HRP (Leica) for 30 minutes at room temperature followed by incubation with CF Tyramide 488 (for CD103) Working Solution (Biotium) for 3 minutes at room temperature.

[0471] To provide signal in distinct channels for the second primary antibody, elution immunofluorescence was utilized through pressure and heat application. After washing and block steps, the second primary antibody against CD3 or species-appropriate isotype control was applied to sections and incubated overnight at 4° C. The slides were subsequently washed with Wash Buffer (Thermo-Fisher Scientific), incubated with species-specific secondary PowerVision Poly-HRP (Leica) for 30 minutes at room temperature followed by incubation with CF Tyramide 647 (for CD3) Working Solution (Biotium) for 3 minutes at room temperature.

[0472] All tissue slides were counterstained with Hoechst dye, mounted in PermaFluor Aqueous Mounting Medium (ThermoFisher Scientific), and coverslipped. Images were acquired using an AxioScan whole slide scanner (Carl Zeiss Microscopy).

Macrophage Marker Protocol

[0473] Following antigen retrieval and blocking with Bloxall, tissue sections were successively blocked for 15 minutes with Avidin block (Biocare), Biotin block (Biocare), and blocking buffer (3% bovine serum albumin with 3% normal donkey serum in 1×PBS) for 1 hour at room temperature. After blocking, the slides were incubated with primary antibodies against F4/80 and inducible nitric oxide synthase (iNOS), F4/80 and Arginase-1 (Arg-1), or species-appropriate isotype controls overnight at 4° C. The next day, slides were washed with Wash Buffer (Thermo-Fisher Scientific), incubated for 30 minutes at room temperature with species-specific (for F4/80 and IgG) secondary antibody conjugated to biotin and then 5 minutes at room temperature with Streptavidin Conjugate CF® 647 Working Solution (Biotium). The slides were subsequently washed twice with Wash Buffer, incubated with species-specific (for iNOS/Arg-1 and IgG) secondary PowerVision Poly-HRP for 30 minutes at room temperature followed by incubation with CF Tyramide 488 Working Solution (Biotium) for 3 minutes at room temperature.

[0474] All tissue slides were counterstained with Hoechst dye, mounted in PermaFluor Aqueous Mounting Medium (ThermoFisher Scientific), and coverslipped. Images were acquired using an AxioScan whole slide scanner (Carl Zeiss Microscopy).

Algorithm-Based Digital Image Analysis

[0475] Positive immunoreactivity was assessed by analyzing whole slide images (Zeiss AxioScan) using Indica Labs HALO software through application of the HighPlex FL v3.2.1 algorithm to identify the double positive cell populations (Table 31). For each of the two to six tumor tissue sections per tumor, regions of interest were manually annotated, while peripheral tissues such as skin, adipose, and connective tissues as well as tissue folds and other artifacts were excluded. Individual cells were detected based on HOECHST nuclear stain. Detection thresholds for fluorescent signals were adjusted based on antibody-specific signal strength and nonspecific background signal for each experiment. Cell count estimates were determined for each tumor and normalized to cell counts observed in the corresponding IgG antibody control-stained tumor.

TABLE 31

Marker Phenotype for Immune Cell Subsets

Cell Type	Marker Phenotype
T_{cyt} cells	$CD3^+$ $CD8^+$
T_{reg} cells	$CD4^+$ $FoxP3^+$
Dendritic cells	CD3− CD103+
M1 macrophages	$F4/80^+$ $iNOS^+$
M2 macrophages	$F4/80^+$ $Arg\text{-}1^+$

Arg-1 = arginase-1;
CD = cluster of differentiation;
FoxP3 = forkhead-box protein P3;
iNOS = inducible nitric oxide synthase;
M1 = M1-type (classically activated) macrophage;
M2 = M2-type (alternatively activated) macrophage;
T_{cyt} cell = cytotoxic T cell;
T_{reg} cell = regulatory T cell

Results

[0476] A significant increase in CD3+ and CD4+ cell populations with a concomitant decrease in Treg cells was observed in CT26 tumors from mice treated with 40 mg/kg AB-008873 vs PBS by 10 days after administration of the antibody. FIGS. **13-15**. By Day 17, a significant increase in cytotoxic T cells and M1 macrophages (i.e., pro-inflammatory macrophages) were noted in CT26 tumors from mice treated with AB-008873 vs. PBS. The results are summarized in Tables 32 and 33. AB-008873 did not demonstrate significant anti-tumor activity vs PBS at either 20 mg/kg or 40 mg/kg. The results also show that the AB-008873 treatment showed early transient changes in blood and that the AB-008873 treatment induced TME changes associated with anti-tumorigenic, pro-inflammatory changes vs PBS in the CT26 model. FIG. **9** and FIG. **16**.

TABLE 32

Changes in % of CD45 by Flow cytometry with AB-008873 treatment

40 mg/kg	Day	TILs	T_{cyt}	T_{help}	T_{reg}	NK	M-MDSCs	G-MDSCs	MΦ	pDC	cDC1
Blood	10	—	↑*	↑*	ns	—	↓*	ns	—	—	—
	17	—	ns	ns	ns	—	↑*	ns	—	—	—
Tumor	10	ns	ns	ns	ns	ns	ns	ns	↑*	ns	ns
	17	ns	ns	ns	ns	ns	ns	ns	ns	ns	ns
	end	↓*	ns	ns	ns	ns	↓*	ns	ns	ns	↑*

TABLE 33

Changes by IF with AB-008873 treatment

	Day	CD3+	T_{cyt}	CD4+	T_{reg}	DC	MΦ	M1	M2
AB-008873	10	↑*	ns	↑*	↓*	ns	ns	ns	ns
40 mg/kg	17	ns	↑*	ns	ns	ns	ns	↑*	ns
AB-008873	10	ns	ns	ns	ns	ns	ns	ns	ns
20 mg/kg	17	ns	ns	ns	ns	ns	ns	↑*	ns

Tissue Reactivity

[0477] AB-008873 and its variants derived against distinct EphA2 epitopes, including AB-010016, AB-010017, and AB-010018 and MAB 3035, LS-C36249-100, LS-C100255-400, LS-B1794-50, and AB-0073254 were tested on human tumor and tumor adjacent tissues (TAT) (Table 34). FIG. **23** depicts binding sites of commercial anti-EphA2 antibodies. Typically, commercial anti-EphA2 antibodies bind to the extracellular domain or intracellular domain of Eph2A. For instance, MAB3035 (a.a. Gln25-Asn534), LS-C36249-100 (extracellular domain), LS-C100255-400 (a.a. 30-60), LS-B1794-50 (a.a. 450-500), and Ab73254 (extracellular domain) bind to sites in the extracellular domain, while 6997S (around a.a. Arg907) binds to sites in the intracellular domain. Briefly, fixed tissues were washed with 2.5% donkey serum to block non-specific binding. Sections were incubated with a primary antibody in 2.5% donkey serum diluent. Sections were incubated with Cy5 conjugated secondary antibody and counterstained with Hoechst, mounted in anti-fade mountant and stained using H&E. Tumor types and their related TAT (n=2-6 per indication) used include non-small cell lung cancer (NSCLC)-squamous cell carcinoma, NSCLC-adenocarcinoma, renal cell carcinoma (i.e., clear cell and papillary subtypes), melanoma, esophageal cancer, breast cancer (i.e., triple negative, hormone receptor-positive, and HER2-positive molecular phenotypes), colorectal cancer, soft tissue sarcoma, and pancreatic, prostate, and urothelial cancers.

TABLE 34

Antibodies tested on human tumor and tumor adjacent tissues in immunofluorescence studies

Category	Name	
8873 and variants	AB-008873	
	AB-009812	
	AB-009813	
	AB-009815	
	AB-009816	
EphA2 antibodies derived from clinical candidates	AB-010016	
	AB-010017	
	AB-010018	
Commercial anti-EphA2 Abs	Extracellular	MAB3035 (Gln25-Asn534)
		LS-C36249-100 (extracellular domain)
		LS-C100255-400 (AA 30-60)
		LS-B1794-50 (AA 450-500)
		Ab73254 (extracellular domain)
	Cytoplasmic	6997S (around Arg907)

[0478] In brief, commercially obtained frozen resected tumor samples and tumor adjacent tissue from non-autologous patients were sectioned to slide and lightly fixed using 4% paraformaldehyde. Following a buffer wash, the slides were incubated in a blocking reagent containing 2.5% normal donkey serum in phosphate-buffered saline (PBS), pH 7.0, at room temperature and then incubated overnight at 4° C. in primary antibody diluted in 2.5% donkey serum in PBS (concentration range of 0.1-30 μg/mL). The primary antibody is a chimeric sequence comprised of a human Fv derived from the Atreca library and a mouse IgG2a Fc sequence. After the primary antibody and subsequent wash in buffer solution, the slide is incubated in donkey anti-mouse secondary antibody conjugated to AlexaFluor 647 at room temperature. Following a wash step, the tissue is counterstained with Hoescht dye which ubiquitously labels cell nuclei, washed again, and coverslipped.

[0479] AB-008873 immunoreactivity was detected in NSCLC adenocarcinoma, melanoma, esophageal cancer, ovarian cancer, HER2+ breast cancer, soft tissue sarcoma, anal cancer, gastric cancer, uterine cancer, and head and neck cancer. FIGS. **25-30**. Nearly identical profiles were detected for its engineered variant antibodies. FIGS. **25-27**. AB-008873 showed approximately 80% overlap of immunoreactive tumor cores with the AB-010018 and 90% with the commercial antibody derived against the EphA2 extracellular domain. Table 35. Interestingly, in these experiments, AB-008873 signal, and that of all engineered variants, was detectable in gastric cancer whereas no reactivity was noted with anti-EphA2 compounds (i.e., antibodies AB-010018 (mouse monoclonal antibody precursor of clinical candidate humanized DS-8895a. See U.S. Pat. No. 9,150,657, SEQ ID NO: 35 and 37), AB-010016 (derived from MedImmune's clinical candidate Medi-547), and AB-010017 (derived from Merrimack's clinical candidate MM-310)) nor the commercial anti-EphA2s. FIG. **25**. Furthermore, AB-008873 and its engineered variants demonstrated a punctate binding pattern across multiple cancer types including ovarian and uterine cancers and soft tissue sarcoma while any labeling by antibodies AB-010018, AB-010016, and AB-010017 or commercial antibodies was diffused with cytoplasmic signal predominant. FIGS. **26-28**.

TABLE 35

IHC score of AB-008873 and its variants on 21 types of fresh frozen human tumor and TAT

		IHC score										
	Ab Info	NSCLC-SQ	NSCLC-Adeno	RCC-clear cell	RCC-papillary	MELA	ESO	OVA	TNBC	ER+ BRC	Her2+ BRC	CRC
8873 and variants	AB-008873	1	2	1	1	2	2	2	1	1	2	1
	AB-009812	1	2	1	1	2	2	2	1	1	2	1
	AB-009813	1	2	1	1	2	2	2	1	1	1	1
	AB-009815	1	2	1	1	2	2	2	1	1	2	1
	AB-009816	1	2	1	1	2	2	2	1	1	2	1

TABLE 35-continued

IHC score of AB-008873 and its variants on 21 types of fresh frozen human tumor and TAT

EphA2 antibodies derived from clinical candidates		AB-010018	2	2	1	1	3	2	2	2	2	2	1
		AB-010016	1	1	1	1	1.5	1	1	1	1	1	1
		AB-010017	1	2	1	1	2	1	1	1	1	1	2
Commercial anti-EphA2 Abs	Extra-cellular	MAB3035 (Gln25-Asn534)	2	2	1	1	3	2	2	1	1	1	1
		LS-C36249-100 (extra-cellular)	2	2	1	1	3	2	2	1	1	2	1
		LS-C100255-400 (AA 30-60)	1	1	1	1	2	1	1	1	1	1	1
		LS-B1794-50 (AA 450-500)	1	2	1	2	2	1	1	1	1	1	1
		Ab73254 (extra-cellular)	2	2	1	1	2	1	1	1	1	1	1
	Cytoplasmic	6997S (around Arg907)	1	1	1	1	2	2	1	1	1	1	1

Ab Info			IHC score											
			PANC	PROC	STS	URO	ANC	HNC	GAC	UCEC-endometrioid	UCEC-clear cell	UCEC-papil-lary	*	**
8873 and variants		AB-008873	1	1	2	1	2	2	2	1	2	2		69%
		AB-009812	1	1	2	1	2	2	2	1	1	2	91%	62%
		AB-009813	1	1	2	1	2	2	2	1	2	2	91%	62%
		AB-009815	1	1	2	1	2	2	2	1	2	2	100%	69%
		AB-009816	1	1	2	1	2	2	2	1	2	2	100%	69%
EphA2 antibodies derived from clinical candidates		AB-010018	1	1	2	1	2	1	1	2	2	2	82%	
		AB-010016	1	1	1	1	2	2	1	1	2	2	45%	31%
		AB-010017	1	1	1	1	2	2	1	2	2	3	55%	46%
Commercial anti-EphA2 Abs	Extra-cellular	MAB3035 (Gln25-Asn534)	2	1	1	1	2	2	1	2	2	2	73%	69%
		LS-C36249-100 (extra-cellular)	1	1	2	1	2	2	1	2	2	3	91%	85%
		LS-C100255-400 (AA 30-60)	1	1	1	1	1	1	1	1	2	2	27%	23%
		LS-B1794-50 (AA 450-500)	1	1	1	1	1	1	1	1	2	2	36%	31%
		Ab73254 (extra-cellular)	1	1	1	1	1	1	1	1	2	1	27%	31%
	Cytoplasmic	6997S (around Arg907)	1	1	1	1	2	1	1	1	2	2	45%	38%

*refers to % overlap of positive cancer types between Ab-008873 and other antibodies.

**refers to % overlap of positive cancer types between DS-8895 and other antibodies.

[0480] AB-010361 and AB-010363 were assayed as above. Both variants retained similar reactivity as AB-008873 to most of the tested cancer samples. However, there were some notable differences observed for reactivity to melanoma, soft tissue sarcoma, endometrioid uterine cancer, esophageal cancer, triple negative breast cancer, ovarian cancer, and urothelial cancer. AB-010361 and AB-010363 show higher binding affinity than AB-008873 on melanoma, soft tissue sarcoma, endometrioid uterine cancer. AB-010361 shows higher binding affinity than AB-008873 on esophageal cancer and triple negative breast cancer. AB-010363 shows higher binding affinity than AB-008873 on ovarian cancer, and urothelial cancer. Furthermore, AB-010361 shows enhanced reactivity in triple negative breast cancer as compared to AB-010363, but diminished reactivity as compared to AB-010363 in ovarian and urologic cancer. AB-010363 shows enhanced reactivity in urothelial cancer compared to AB-010361, AB-010018, AB-010016, and AB-010017, but diminished tumor-selective binding in esophageal cancer as compared to AB-008873. Table 36 illustrates reactivity of anti-EphA2 antibodies to cancers.

TABLE 36

Reactivity of anti-EphA2 antibodies to cancers

	AB-008873					AB-010361					AB-010363				
Indication	30 μg/ml	10 μg/ml	3 μg/?	1 μg/?	0.3 μg/ml	30 μg/ml	10 μg/ml	3 μg/?	1 μg/?	0.3 μg/ml	30 μg/ml	10 μg/ml	3 μg/?	1 μg/?	0.3 μg/ml
MELA	2	2	1	1	1	2	2	2	1	1	2	2	2	1	1
STS	2	2	2	2	1	2	2	2	2	2	2	2	2	2	2
UCEC-?	2	2	1	1	1	2	2	2	1	1	2	2	2	1	1
ESO	2	2	1	1	1	2	2	2	2	1	1	1	1	1	1
OVA	2	1	1	1	1	1	1	1	1	1	2	2	2	2	1
TNBC	1	1	1	1	1	2	2	1	1	1	1	1	1	1	1
URO	1	1	1	1	1	1	1	1	1	1	2	2	2	1	1
UCEC-clear ?	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
UCEC-?	2	2	2	2	1	2	2	2	2	1	2	2	2	2	1
NSCLC-SQ	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
NSCLC-?	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RCC-clear ?	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RCC-?	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
ER+ BRC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Her2+ BRC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
CRC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PANC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PROC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

	AB-010016					AB-010018				
Indication	30 μg/ml	10 μg/ml	3 μg/ml	1 μg/ml	0.3 μg/ml	30 μg/ml	10 μg/ml	3 μg/ml	1 μg/ml	0.3 μg/ml
MELA	1	1	1	1	1	2	2	2	2	2
STS	2	1	1	1	1	2	2	2	2	2
UCEC-?	1	1	1	1	1	2	2	2	2	2
ESO	1	1	1	1	1	2	2	2	2	2
OVA	1	1	1	1	1	1	1	1	1	1
TNBC	1	1	1	1	1	1	1	1	1	1
URO	2	1	1	1	1	1	1	1	1	1
UCEC-clear ?	2	1	1	1	1	2	2	2	2	2
UCEC-?	1	1	1	1	1	2	2	2	2	2
NSCLC-SQ	1	1	1	1	1	1	1	1	1	1
NSCLC-?	1	1	1	1	1	1	1	1	1	1
RCC-clear ?	1	1	1	1	1	1	1	1	1	1
RCC-?	1	1	1	1	1	1	1	1	1	1
ER+ BRC	1	1	1	1	1	1	1	1	1	1
Her2+ BRC	1	1	1	1	1	1	1	1	1	1

TABLE 36-continued

Reactivity of anti-EphA2 antibodies to cancers										
CRC	1	1	1	1	1	1	1	1	1	1
PANC	1	1	1	1	1	1	1	1	1	1
PROC	1	1	1	1	1	1	1	1	1	1

Score of 1 indicates no signal above TAT or isotype control.

? indicates text missing or illegible when filed

TABLE 36

(Continued) Indication	AB-010016 30 ug/ml	10 ug/ml	3 ug/ml	1 ug/ml	0.3 ug/ml	AB-010018 30 ug/ml	10 ug/ml	3 ug/ml	1 ug/ml	0.3 ug/ml
MELA	1	1	1	1	1	2	2	2	2	2
STS	2	1	1	1	1	2	2	2	2	2
UCEC-	1	1	1	1	1	2	2	2	2	2
ESO	1	1	1	1	1	2	2	2	2	2
OVA	1	1	1	1	1	1	1	1	1	1
TNBC	1	1	1	1	1	1	1	1	1	1
URO	2	1	1	1	1	1	1	1	1	1
UCEC-clear cell	2	1	1	1	1	2	2	2	2	2
UCEC-papillary	1	1	1	1	1	2	2	2	2	2
NSCLC-SQ	1	1	1	1	1	1	1	1	1	1
NSCLC-Adeno	1	1	1	1	1	1	1	1	1	1
RCC-clear cell	1	1	1	1	1	1	1	1	1	1
RCC-papillary	1	1	1	1	1	1	1	1	1	1
ER+ BRC	1	1	1	1	1	1	1	1	1	1
Her2+ BRC	1	1	1	1	1	1	1	1	1	1
CRC	1	1	1	1	1	1	1	1	1	1
PANC	1	1	1	1	1	1	1	1	1	1
PROC	1	1	1	1	1	1	1	1	1	1

Score of 1 indicates no signal above TAT or isotype control.

[0481] AB-010699 showed increased potency in many tumor types as compared to AB-010018 and AB-00873 and AB-010361. However, AB-010699 is less potent than its parent AB-010361 against soft tissue carcinoma in this assay. FIG. **42**; Table 37, 38, and 39.

TABLE 37

Binding to EphA2 epitopes across cancer types by various EphA2 antibodies

	AB-008873 (μg/ml)						AB-010361 (μg/ml)						AB-010699 (μg/ml)						AB-010018 (μg/ml)					
Indication	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1
NSCLC-SQ	2	2	1	1	1	1	2	1	1	1	1	1	2	2	2	2	2	2	2	1	1	1	1	1
NSCLC-Adeno	2	1	1	1	1	1	2	2	1	1	1	1	2	2	1	1	1	1	1	1	1	1	1	1
RCC-papillary	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	1	1	1
MELA	2	2	1	1	1	1	2	2	2	1	1	1	2	2	2	2	2	1	2	2	2	2	2	1
ESO	2	1	1	1	1	1	2	1	1	1	1	1	2	2	1	1	1	1	2	1	1	1	1	1
PANC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2
STS	2	2	2	1	1	1	2	2	2	2	1	1	1	1	1	1	1	1	2	1	1	1	1	1
URO	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2
Uterine	2	1	1	1	1	1	2	2	1	1	1	1	2	2	2	2	1	1	2	2	2	2	2	1

Score of 1 indicates no signal above TAT or isotype control.

TABLE 38

Binding to EphA2 epitopes across cancer types by various EphA2 antibodies

	AB-008873 (μg/ml)						AB-010361 (μg/ml)						stage AB-010699 (μg/ml)						AB-010018 (μg/ml)					
Indication	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1	30	10	3	1	0.3	0.1
NSCLC-SQ	2	2	1	1	1	1	2	1	1	1	1	1	2	2	2	2	2	2	2	1	1	1	1	1
NSCLC-Adeno	2	1	1	1	1	1	2	2	1	1	1	1	2	2	1	1	1	1	1	1	1	1	1	1
RCC-papillary	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	1	1	1
MELA	2	2	1	1	1	1	2	2	2	1	1	1	2	2	2	2	2	1	2	2	2	2	2	1
ESO	2	1	1	1	1	1	2	1	1	1	1	1	2	2	1	1	1	1	2	1	1	1	1	1
PANC	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2
STS	2	2	2	1	1	1	2	2	2	2	1	1	1	1	1	1	1	1	2	1	1	1	1	1
URO	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2	2	2	2
Uterine	2	1	1	1	1	1	2	2	1	1	1	1	2	2	2	2	1	1	2	2	2	2	2	1

Score of 1 indicates no signal above TAT or isotype control.

TABLE 39

Binding to EphA2 epitopes across cancer types by various EphA2 antibodies

	AB-008873 (parental)			AB-010361			AB-010699		
Cancer Indication	Reactive #	Total #	Reactivity %	Reactive #	Total #	Reactivity %	Reactive #	Total #	Reactivity %
Esophageal (Squamous)	0	10	0%	3	10	30%	8	10	80%
Lung (Squamous)	5	20	25%	8	20	40%			
Breast	2	32	6%	10	32	31%			
Renal	1	34	3%	10	34	29%			
Ovarian	2	36	6%	6	36	17%			

[0482] The reactivity of AB-010699 at 3 μg/ml in human cancers was tested across a panel of human cancer samples. As shown in FIG. **22**, AB-010699 showed greater than 20% reactivity across all cancers tested, with results ranging from 100% reactivity in stomach adenocarcinoma (number of samples tested=8) to 23% reactivity in soft tissue sarcoma (number of samples tested –35). AB-010016 and AB-010017 (that bind to the ligand binding domain of EphA2) and AB-010018 were tested for reactivity over the same set of tissue samples as AB-010699, all at a concentration of 3 ug/ml. As shown in FIG. **24**, the signal of these three comparator antibodies was very low for many of the samples tested. The assay was repeated with increased concentrations, with AB-010016 and AB-010017 being tested at 30 μg/ml and AB-010018 being tested at 10 μg/ml while AB-010699 was maintained at 3 μg/ml. AB-010699 showed a differentiated reactivity over all three comparator antibodies tested. FIG. **31**.

[0483] Immunohistochemistry staining was performed on fresh frozen TMAs of select normal human tissues (i.e., liver, kidney, stomach, heart, pancreas, lung; n=3 per tissue type) with human uterine cancer as the positive control, following the manufacturer's instructions of Vector Laboratory's standard MOM protocol. In short, the frozen slides were dried at room temperature for 15 minutes and fixed with 4% PFA at room temperature for 15 minutes. After fixation, the slides were washed with wash buffer and incubated in MOM blocking reagent. After blocking the slides were incubated with AB-008873, its engineered variants, antibody AB-010018, or mouse IgG2a, the isotype control, at 3 μg/mL, overnight at 4° C. Commercial antibodies were also tested at 10 μg/mL. Next, the slides were washed with Wash Buffer and incubated with a secondary for 40 minutes at room temperature. The slides were washed and then the chromogenic reaction was developed by incubating them in Betazoid DAB for 2 minutes. After the color development the slides were counter stained with Hematoxylin QS for 15 seconds, dehydrated, mounted and then imaged in AxioScan whole slide scanner. Antibodies being tested include AB-008873 and engineered variants include AB-009812, AB-009813, AB-009815, AB-009816. AB-010018, and commercial anti-Eph2A antibody (LC-C100255-400, LS-B174-50).

[0484] The results show that AB-008873 and its engineered variants demonstrated no detectable to faint cytoplasmic signal in all normal human tissues tested. Table 40. Unlike AB-008873, the antibody AB-010018 revealed enhanced labeling of epithelium in stomach as well as moderate signal with renal tubules. Similar normal tissue staining profiles were noted with the commercial antibodies tested. FIG. **33-34**.

TABLE 40

Results of the binding of anti-EphA2 antibodies to normal human tissue

Ab Info		Normal Human Tissue IHC Screen					
		Liver	Kidney	Stomach	Heart	Pancreas	Lung
8873 and variants	AB-008873	Neg to faint	Faint	Faint	Faint	Faint	Neg to faint
	AB-009812	Neg to faint	Faint	Faint	Faint	Faint	Faint
	AB-009813	Neg to faint	Faint	Faint	Faint	Faint	Neg to faint
	AB-009815	Neg to faint	Faint	Faint	Faint	Faint	Faint
	AB-009816	Neg to faint	Faint	Faint	Faint	Faint	Neg to faint
EphA2 Abs derived from clinical candidates	AB-010018	Neg to faint	Moderate	Faint to mod	Neg to faint	Neg to faint	Neg to faint
Commercial EphA2 Ab	LS-C100255-400	Faint	Moderate	Faint to mod	Faint	Faint	Faint
	LS-B1794-50	Faint	Moderate	Faint to mod	Faint	Faint	Faint

Note all antibodies were used at 3 μg/ml, except for LS-C100255-400 and LS-B1794-50, which were used at 10 μg/ml.

Example 5. EPHA2 Signaling

[0485] The effect of AB-008873 and variants AB-010361 and AB-010699 on EphA2 signaling was tested along with antibodies AB-010016 and AB-010018. To assess EphA2 signaling, antibodies specific to described phosphorylation sites in the intracellular domain were used. In brief, MDA-MB-231 cells were seeded in 6 well plates and allowed to attach overnight. Next day, cells were serum-starved for 24 h. Following, antibodies and ephrin-A1 Fc were added at the respective concentrations and allowed to incubate for 15 min at 37° C. and 5% CO_2. Cells were immediately lysed and processed for Western Blot analysis. Samples were loaded on a gel and run for 70 min at 200V. After transfer to PVDF membrane, phospho-specific antibodies were used quantify receptor activation. In parallel, total EphA2 and GAPDH were detected and quantified and used for normalization post image analysis. To assess agonistic effects of the antibodies on the ephrinA1-EphA2 signaling axis, MDA-MB-231 cells were incubated with different concentrations of antibody for 15 min followed by ephrinA1-FC at 1 μg/mL. Cells were subsequently processed for Western Blot analysis as described above.

[0486] For all phosphorylation sites tested, weak or no agonistic activation and no interference in ephrinA1-EphA2 signaling axis were detected for AB-008873 and AB-010361. AB-010699 exhibited agonistic activity but only at 20% the activation induced by the natural ligand ephrin A1 (in Fc format). FIG. A.

[0487] In contrast AB-010016 and AB-010018 significantly interfere with EphA2 signaling, with AB-010016 acting as a strong agonist and AB-010018 acting as an antagonist of the signaling activity. FIGS. **43** A and B. AB-010016 activated signaling at comparable levels to the ephrin A1 ligand. FIG. **43**A

Example 6. Epitope Binning by Bio-Layer Interferometry

[0488] Epitope binning was performed using an Octet Red 96E instrument (Sartorius) using the in-tandem format. In this format, antigen is captured on a biosensor tip, a first antibody (or other binding partner) is bound to saturation, and subsequently the ability of a second antibody (or binding partner) known to recognize the antigen is assessed. If substantial binding by the second species is no longer observed, it is supportive that simultaneous binding is sterically hindered (i.e., the second species is "blocked" by pre-saturation of the 1^{st}) and that the two may bind overlapping epitopes. If, however, substantial binding of the second species is observed, despite saturation of the first, it is supportive that the two bind non-overlapping epitopes. Groups of antibodies that block one another can be considered to populate the same epitope bin.

[0489] In this experiment four antibodies and one natural ligand that recognize EphA2 were tested. The Fv regions for three of the antibodies were derived from therapeutic candidates that entered human clinical trials: AB-010016 derived from MedImmune's MEDI-547, disclosed in Peng et al., J. Mol. Biol. 2011 Oct. 21; 413(2): 390-405. doi: 10.1016/j.jmb.2011.08.018; AB-010017 derived from Merrimack's MM-310, disclosed in Geddie et al., MAbs. 2017 January; 9(1): 58-67. Doi: 10.1080/19420862.2016.125904; Kamnoun et al., Nat Biomed Eng. 2019 April; 3(4):264-280. doi: 10.1038/s41551-019-0385-4. Epub 2019 Apr. 5; and AB-010018 mouse monoclonal antibody precursor (U.S. Pat. No. 9,150,657, Sequences 35 and 37) of Daiichi-Sankyo's DS-8895a, disclosed in Hasegawa J. et al., Cancer Biol Ther. 2016 November; 17(11):1158-1167. doi: 10.1080/15384047.2016.1235663. Epub 2016 Sep. 21; Shitara K. et al., J Immunother Cancer, 2019 Aug. 14; 7(1):219. doi: 10.1186/s40425-019-0679-9, US844988; U.S. Pat. No. 9,150,657.

[0490] The fourth antibody Fv region (AB-008873) was discovered via Atreca's Immune Repertoire Capture (IRC™). Antibodies were produced with mouse IgG2a constant regions. The natural ligand tested was EphrinA1 fused to Fc (Acro, EF1-H5251). Antigen immobilized on the biosensor was the extracellular domain of human EphA2 (25-534) with a C-terminal HIS tag (R&D Systems, 3035-A2) and captured using Ni-NTA tips (Sartorius, 18-5102). Protein solutions were made in 1× Kinetics buffer (Satorius, 18-1092) supplemented with 2.5 mM Imidazole to reduce non-specific binding. Each pairwise interaction was assessed in both orders of addition, and controls for binding in the absence of antigen or with antigen but without a first binding species included. Binning indicated the five molecules populated only two Bins. The first bin contains molecules AB-008873 and AB-010018 and was interpreted to represent epitopes on the membrane proximal fibronectin domain of EphA2 based on the reported epitope location of DS-8895a (Hasegawa, J. et al., Novel anti-EPHA2 antibody, DS-8895a for cancer treatment. *Cancer Biology and Therapy,* 17(11), 1158-1167 (2016)). The second bin contains molecules AB-010017. AB-010016, and EphrinA1-Fc and was interpreted to represent epitopes on the ligand binding domain (Peng, L. et al., Structural and functional characterization of an agonistic anti-human EphA2 monoclonal antibody. *Journal of Molecular Biology,* 413(2), 390-405 (2011); Geddie,

M. L. et al., Improving the developability of an anti-EphA2 single-chain variable fragment for nanoparticle targeting. *MAbs,* 9(1), 58-67 (2017)). FIG. **5**, Tables 41-42. Referring to FIG. **5**, the FN2 location in the first bin is inferred from domain deletion data reported in Hasegawa et al., 2016, FIG. **1**, noting that the reported MMP cleavage sites are (385/395/432/435).

TABLE 41

Results of Octet in tandem binning for epitope mapping

Analyte 1	Analyte 2	Notes/Scoring	TIP	Ligand
8873-02	8873-02	Blocks (self) despite faster dissoc.	NiNTA	R&D ECD-06
10016-01	8873-02	Binding observed (curve behind light green)	NiNTA	R&D ECD-06
10017-01	8873-02	Binding observed	NiNTA	R&D ECD-06
10018-01	8873-02	Blocks	NiNTA	R&D ECD-06
Ephrin-A1	8873-02	Binding observed	NiNTA	R&D ECD-06
Buffer	8873-02	Positive control OK - EphA2 + competitor	NiNTA	R&D ECD-06
Buffer	8873-02	Background control OK - competitor alone	NiNTA	Buffer

TABLE 42

Epitope recognized by AB-008873 overlaps with that of AB-010018

	Bin #1 - FN2		Bin #2 - LBD		
	8873-2	10018-01	10017-01	10016-01	Ephrin-A1
8873-2	blocked	blocked	binds	binds	binds
100018-01	blocked	blocked	binds	binds	binds
10017-01	binds	binds	blocked	blocked	blocked
10016-01	binds	binds	blocked	blocked	blocked
Ephrin-A1	binds	binds	blocked	blocked	blocked

Epitope Domain Investigation by Yeast Surface Display

[0491] To further validate AB-008873 epitope domain mapping, subdomains of human EphA2 and chimeras of human EphA2 and human EphA4 were displayed on the surface of yeast. AB-008873 had shown binding to EphA2 but not homologous receptor EphA4 in other assays (screening at Integral Molecular). 5 constructs (shown in Table 43) were selected for display on the surface of yeast and staining via AB-008873:

TABLE 43

EphA2 subdomains

EphA2 (aa 25-534)	Full ECD	QGKEVVLLDFAAAGGELGWLTHPYGKGWDLMQNIMNDMPI YMYSVCNVMSGDQDNWLRTNWVYRGEAERIFIELKFTVRDC NSFPGGASSCKETFNLYYAESDLDYGTNFQKRLFTKIDTIAPD EITVSSDFEARHVKLNVEERSVGPLTRKGFYLAFQDIGACVAL LSVRVYYKKCPELLQGLAHFPETIAGSDAPSLATVAGTCVDH AVVPPGGEEPRMHCAVDGEWLVPIGQCLCQAGYEKVEDACQ ACSPGFFKFEASESPCLECPEHTLPSPEGATSCECEEGFFRAPQ DPASMPCTRPPSAPHYLTAVGMGAKVELRWTPPQDSGGREDI VYSVTCEQCWPESGECGPCEASVRYSEPPHGLTRTSVTVSDLE PHMNYTFTVEARNGVSGLVTSRSFRTASVSINQTEPPKVRLEG RSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVRRTE GFSVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTLSPE GSGN (SEQ ID NO: 101)
EphA2 (aa 437-534)	FN2 only	TEPPKVRLEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDS NSYNVRRTEGFSVTLDDLAPDTTYLVQVQALTQEGQGAGSK VHEFQTLSPEGSGN (SEQ ID NO: 95)
EphA4_ECD (aa 23-546)	Full ECD	SRVYPANEVTLLDSRSVQGELGWIASPLEGGWEEVSIMDEKN TPIRTYQVCNVMEPSQNNWLRTDWITREGAQRVYIEIKFTLRD CNSLPGVMGTCKETFNLYYYESDNDKERFIRENQFVKIDTIAA DESFTQVDIGDRIMKLNTEIRDVGPLSKKGFYLAFQDVGACIA LVSVRVFYKKCPLTVRNLAQFPDTITGADTSSLVEVRGSCVNN SEEKDVPKMYCGADGEWLVPIGNCLCNAGHEERSGECQACKI GYYKALSTDATCAKCPPHSYSVWEGATSCTCDRGFFRADND AASMPCTRPPSAPLNLISNVNETSVNLEWSSPQNTGGRQDISY NVVCKKCGAGDPSKCRPCGSGVHYTPQQNGLKTTKVSITDLL AHTNYTFEIWAVNGVSKYNPNPDQSVSVTVTTNQAAPSSIAL

TABLE 43-continued

		EphA2 subdomains
		VQAKEVTRYSVALAWLEPDRPNGVILEYEVKYYEKDQNERS YRIVRTAARNTDIKGLNPLTSYVFHVRARTAAGYGDFSEPLEV TTNTVPSRIIGDGANS (SEQ ID NO: 102)
EphA2_A4in FN1	EphA2 with EphA4 FN1 domain	QGKEVVLLDFAAAGGELGWLTHPYGKGWDLMQNIMNDMPI YMYSVCNVMSGDQDNWLRTNWVYRGEAERIFIELKFTVRDC NSFPGGASSCKETFNLYYAESDLDYGTNFQKRLFTKIDTIAPD EITVSSDFEARHVKLNVEERSVGPLTRKGFYLAFQDIGACVAL LSVRVYYKKCPELLQGLAHFPETIAGSDAPSLATVAGTCVDH AVVPPGGEEPRMHCAVDGEWLVPIGQCLCQAGYEKVEDACQ ACSPGFFKFEASESPCLECPEHTLPSPEGATSCECEEGFFRAPQ DPASMPCTRPPSAPLNLISNVNETSVNLEWSSPQNTGGRQDIS YNVVCKKCGAGDPSKCRPCGSGVHYTPQQNGLKTTKVSITDL LAHTNYTFEIWAVNGVSKYNPNPDQSVSVTVTTNQTEPPKVR LEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVR RTEGFSVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTL SPEGSGN (SEQ ID NO: 103)
EphA2_A4in FN2	EphA2 with EphA4 FN2 domain	QGKEVVLLDFAAAGGELGWLTHPYGKGWDLMQNIMNDMPI YMYSVCNVMSGDQDNWLRTNWVYRGEAERIFIELKFTVRDC NSFPGGASSCKETFNLYYAESDLDYGTNFQKRLFTKIDTIAPD EITVSSDFEARHVKLNVEERSVGPLTRKGFYLAFQDIGACVAL LSVRVYYKKCPELLQGLAHFPETIAGSDAPSLATVAGTCVDH AVVPPGGEEPRMHCAVDGEWLVPIGQCLCQAGYEKVEDACQ ACSPGFFKFEASESPCLECPEHTLPSPEGATSCECEEGFFRAPQ DPASMPCTRPPSAPHYLTAVGMGAKVELRWTPPQDSGGREDI VYSVTCEQCWPESGECGPCEASVRYSEPPHGLTRTSVTVSDLE PHMNYTFTVEARNGVSGLVTSRSFRTASVSINQAAPSSIALVQ AKEVTRYSVALAWLEPDRPNGVILEYEVKYYEKDQNERSYRI VRTAARNTDIKGLNPLTSYVFHVRARTAAGYGDFSEPLEVTT NTVPSRIIGDGANS (SEQ ID NO: 104)

[0492] Constructs were expressed as Aga2 fusions with an internal HA tag between Aga2 and the displayed protein as well as a C-terminal Myc tag (Boder, E. T., & Wittrup, K. D., Yeast surface display for screening combinatorial polypeptide libraries. Nature Biotechnology, 15(6), 553-557 (1997)). Yeast cells were cultured in SDCAA (Teknova, S0543) and induced with SGCAA (Teknova, S0542). Cells were stained with pairs of primary and secondary antibodies. AB-008873 Lot 2 (Atum, mIgG2a) and Donkey anti-Mouse IgG AF647 (JacksonImmunoResearch 715-605-150) were used for 10-point titrations to assess 8873 binding. Chicken IgY anti-cMyc (Life technologies A21281) and Goat anti Chicken IgY FITC (Life technologies A11039) were used to assess expression levels. The analysis was run on the iQue3 flow cytometer. Staining demonstrates binding of AB-008873 to the full extracellular domains EphA2, but not EphA4. The FN2 domain of EphA2 alone is sufficient to mediate binding and expresses at high levels. When the FN2 domain of EphA4 is swapped into EphA2, the binding to AB-008873 was lost. When the FN1 domain of EphA4 was swapped into in EphA2, the binding to AB-008873 was still observed. FIG. **6**. As shown in FIG. **6**, EphA4 chimeras offer stability advantages and can inform the finer mapping at the domain-level by expressing EphA2 truncations. Referring to FIG. **6** lower panel, FN2 domain (EphA2-FN2) alone shows strong binding signal (triangle). FN2 domain alone also has high relative expression. EphA2 full ECD shows a binding signal (circle) relatively lower than the binding signal driven by the FN2 domain alone. EphA4 full ECD shows no binding signal (square). Swapping the FN1 domain from EphA4 into EphA2 enables binding of the epitope (diamond). Binding is lost when the FN2 domain from EphA4 is swapped into EphA2. These results indicate that the epitope of AB-008873 is located at the membrane-proximal FN2 domain of EphA2.

[0493] Finer, residue-level resolution for AB-008873's epitope was determined by generating a library of FN2 mutants and testing their binding to AB-008873 via yeast display. The results of the testing indicated that AB-008873 binds a conformational epitope, spanning four stretches of primary sequence of EphA2, comprising the following residues: Pro439, Lys441, Arg443, Leu444, Arg447, Lys476, Gly477, Leu504, Gln506, Ser519, Lys520, Val521, His522, Glu523, Phe524, and Gln525. FIG. **44**. The residues contained in the epitope are not conserved within the EphA2 family. FIG. **45**. However, the epitope residues are conserved across species commonly used in toxicology studies. FIG. **46**. The epitope of AB-008873 is located on the opposite side of the FN2 domain from the reported LBD-FN2 site. See Nikolov. D. B et. al. (2014) Cell Adh Migr 8, 360-365.

[0494] Co-crystallization studies showed that the disulfide loop in the HCDR3 of AB-008873 demonstrated substantial rotation as compared to the structure without target interaction. FIG. **47**. Furthermore, the co-crystal and yeast display data aligned and provided additional details on the epitope. Combining these data indicated epitope residues that are not in direct contact with the CDRs of AB-008873 (as defined by a 4 Angstrom distance constraint) but can substantially reduce or abrogate binding when mutated in yeast display: Pro439, Lys441, Arg443, Gly477, Leu504, and His522.

[0495] Additional yeast display assays indicated that the epitopes of AB-008873 and AB-010018 overlap but are distinct. The epitope of AB-010018 is a protrusion on an adjacent face of FN2 relative to the AB-008873 epitope, with two residues shared between epitopes. AB-010018 was shown to bind to Thr472, Arg474, Asp478, Ser479, Asn480, Gly477, and Leu504 in this assay.

Example 7. Binding Affinity

Steady State KD

[0496] Binding of antibodies to the FN2 domain of hEphA2 was measured using the Biacore® T200 SPR system at 25° C. using HBS-EP+ buffer (0.01 M HEPES pH 7.4, 0.15 M NaCl, 3 mM EDTA, 0.005% v/v Surfactant P20). Antibodies in mIgG2a format were captured at levels between 475-715 RU via an anti-mouse Fc antibody immobilized to a CM5 chip. Binding was assessed using a titration of C-terminally His-tagged hEphA2-FN2 (Thr437-Asn534+ His tag) associated for 30 s or 75 s at 30 μL/min. Data was double-reference subtracted using both a surface without captured antibody and from injections of HBS-EP+ buffer. The equilibrium response was measured at 4 seconds before the end of association with a window of 5 seconds and fit using the Biacore T200 Evaluation Software 3.1 with the Steady State Affinity Model. The results are shown in Table 44.

TABLE 44

The binding kinetics of the EphA2 antibodies to the FN2 domain of hEphA2

Antibody	Steady-State Monovalent KD (M)	Standard Error of KD	Chi2 (RU2)	n
AB-008873	2.26E−06	5.9E−08	0.134	2
AB-010361	1.11E−06	5.5E−08	0.363	3
AB-010363	1.31E−06	2.6E−08	0.068	3
AB-010699	1.89E−08	2.9E−10	0.286	3

[0497] Monovalent affinity to hEphA2-FN2 increased from ~2 M from parental antibody AB-008873 to ~20 nM for the engineered variant AB-010699.

Binding of EphA2 Antibodies to the FN2 Domain of hEphA2

[0498] Binding of EphA2 antibodies to the FN2 domain of hEphA2 was measured using the Biacore® T200 SPR system. Biacore® assays were conducted at 25° C. using HBS-EP+ buffer (0.01 M HEPES pH 7.4, 0.15 M NaCl, 3 mM EDTA, 0.005% v/v Surfactant P20) using the FN2 domain as the analyte with the antibodies in mIgG2a format as the immobilized ligand. Data was analyzed using 1:1 Langmuir binding model and steady-state model in Biacore Evaluation software.

[0499] C-terminally His-tagged human EphA2 FN2 domain (Thr437-Asn534+His tag) was used as the analyte. The analyte was associated for 120 seconds and dissociated for 100 seconds or 300 seconds at 30 μL/min. Titration of 0, 11.1 nM, 33.3 nM, 100 nM, 300 nM, and 900 nM run for AB-010699 and AB-010018 as mIgG2a. The results are shown in Table 45.

TABLE 45

Measurement results of binding of EphA2 antibodies to the FN2 domain of hEphA2

Antibody	ka (1/Ms)	kd (1/s)	KD (M)	Rmax (RU)
AB-010699	3.9E+6	0.075	1.9E−08	110.2
AB-010018	4.9E+6	0.0021	4.3E−09	74.9

Binding of EphA2 Antibodies to the FN2 Domain, and Fragments of the FN1-FN2 Domains of the EphA2

[0500] Additional Biacore® assays were conducted to determine binding of AB-010699 human Fab fragment and AB-010018 human Fab fragment to the ECD, FN1-FN2 domains, FN2 domain, and fragments of the FN1-FN2 domains of the EphA2 protein. These assays were conducted using the EphA2 proteins fused to the N-terminal side of a human Fe (hEphA2-Fc fusion) as the immobilized ligand and AB-010699 and AB-010018 in Fab format as the analyte. The hEphA2-Fc fusions were captured with an anti-human Fe immobilized to a CM5 chip.

[0501] Biacore® assays were conducted at 25° C. using HBS-EP+ buffer (0.01 M HEPES pH 7.4, 0.15 M NaCl, 3 mM EDTA, 0.005% v/v Surfactant P20). Data was double-reference subtracted and data was analyzed using 1:1 Langmuir binding model and steady-state model in Biacore Evaluation software.

[0502] The analyte was associated for 120 seconds and dissociated for 100 seconds for AB-010699 or 300 seconds for AB-010018 at 30 L/min. Binding to hEphA2-ECD-Fc tested using 100, 300, and 900 nM analyte. Binding to all other ligands tested using 33.3, 100, and 300 nM analyte. The results are shown in Tables 46-47.

TABLE 46

Binding kinetics of AB-010699

Ligand	ka (1/Ms)	kd (1/s)	KD (M)	Rmax (RU)
hEphA2-FN12-(327-534) - Fc	5.0E+6	4.3E−2	8.6E−9	149.9
hEphA2-FN2-(437-534) - Fc	1.0E+7	8.5E−2	8.6E−9	59.2
hEphA2-clv385-534-Fc	6.05E+6	7.2E−2	1.2E−8	136.4
hEphA2-clv395-534-Fc	5.72E+6	7.0E−2	1.2E−8	155.5
hEphA2-clv432-534-Fc	7.78E+6	7.5E−2	9.6E−9	145.9
hEphA2-clv435-534-Fc	8.42E+6	7.8E−2	9.3E−9	150.1

TABLE 47

Binding kinetics of for AB-010018

Ligand	ka (1/Ms)	kd (1/s)	KD (M)	Rmax (RU)
hEphA2-ECD(25-534)	3.2E+5	2.0E−3	6.3E−9	120.2
hEphA2-FN12-(327-534) - Fc	1.8E+5	1.9E−3	1.1E−8	115.7
hEphA2-FN2-(437-534) - Fc	4.3E+5	2.7E−3	6.1E−9	52.4
hEphA2-clv385-534-Fc	2.6E+5	2.7E−3	1.0E−8	111.9
hEphA2-clv395-534-Fc	2.6E+5	2.6E−3	1.0E−8	138.4
hEphA2-clv432-534-Fc	4.0E+5	3.0E−3	7.5E−9	112.3
hEphA2-clv435-534-Fc	4.3E+5	3.1E−3	7.3E−9	114.5

[0503] The calculated monovalent KD of AB-010699 is similar across all hEphA2 variants (ranging from 9 to 12 nM) and there appears to be no differentiated binding to different hEphA2 recombinant fragment representing cleavage forms (ranging from 9 to 12 nM).

[0504] The calculated monovalent KD of AB-010018 is also similar across all hEphA2 variants (ranging from 6 to 11 nM) and there also appears to be no differentiated binding to different hEphA2 cleavage forms (ranging from 7 to 10 nM).

Binding of EphA2 Antibodies as SCFV

[0505] Single-chain variable fragments (SCFVs) of 10699 were expressed with C-terminal His-tags in mammalian cells and purified by Nickel Sepharose affinity chromatography. SCFVs were assayed in triplicate for thermostability using an Unchained Labs UNcle instrument. The concentration of purified antibodies was adjusted to 0.5 mg/ml in PBS immediately prior to analysis. To determine melting temperature (Tm) intrinsic protein fluorescence was measured at 473 nm every 1.1° C. as temperature was increased linearly from 25° C. to 95° C. at a rate of 0.3° C./min. The UNcle Analysis software (version 4.01) was used to find the Tm as the first derivative of the barycentric mean (BCM). Binding of the SCFVs to the FN2 domain of hEphA2 was measured in triplicate using the Biacore T200 SPR system at 25° C. using HBS-EP+ buffer (0.01 M HEPES pH 7.4, 0.15 M NaCl, 3 mM EDTA, 0.005% v/v Surfactant P20). SCFVs were captured at levels between 200-300 RU via an anti-His antibody immobilized to a CM5 chip. Binding was assessed using a titration of FLAG-tagged hEphA2-FN2 (300 nM, 100 nM, 33 nM, 11 nM, 3.7 nM) associated for 120 s and dissociated for 30 s at 30 μL/min. Data was double-reference subtracted using both a surface without captured SCFV and from injections of HBS-EP+ buffer. The 1:1 Langmuir binding model was used to fit each titration and the average of three replicates is reported. Table 48 shows amino acid sequences in SCFV constructs, and the results are shown in Table 49.

TABLE 48

SCFV Constructs			
Construct	Orientation	Linker	Sequence
SCFV-010699-01	VH-VL	(G4S)4 (SEQ ID NO: 782)	(SEQ ID NO: 765)
SCFV-010699-02	VL-VH	(G4S)4 (SEQ ID NO: 782)	(SEQ ID NO: 766)
SCFV-010699-05	VH-VL	(G4S)3 (SEQ ID NO: 91)	(SEQ ID NO: 767)
SCFV-010699-06	VL-VH	(G4S)3 (SEQ ID NO: 91)	(SEQ ID NO: 768)

TABLE 49

Binding kinetics of EphA2 scFvs scFV construct					
	mg/L yield	Average Tm (° C.) [Standard Deviation]	Average KD (M) to FN2_FLAG [% CV]	Average ka (1/Ms) [% CV]	Average kd (1/s) [% CV]
SCFV-010699-01	160	66.3 [0.6]	2.4E−08 [0.6%]	4.2E+06 [1.4%]	0.10 [1.0%]
SCFV-010699-02	82	65.0 [0.3]	3.0E−08 [0.9%]	5.2E+06 [0.7%]	0.16 [1.6%]
SCFV-010699-05	97	65.0 [0.6]	2.7E−08 [1.1%]	3.9E+06 [3.2%]	0.11 [2.4%]
SCFV-010699-06	179	63.7 [0.7]	3.1E−08 [1.2%]	4.4E+06 [1.4%]	0.13 [0.7%]

All the SCFV constructs exhibited an average KD (24-31 nM) that is comparable to that of the native AB-010669 monovalent KD.

Example 8. Multivalent Formats

[0506] AB-008873 was used to construct tetravalent molecules that had four EphA2 binding arms per molecule. FIG. **32** shows schematics of the formats, including the format of the Fc regions. The constructs were tested for ADCC, and cell binding activity as described above. Certain of the tetravalent constructs tested exhibited improved ADCC and cell binding over the bivalent format. FIG. **32** (RR) and Table 50. Those constructs that included components fused to the C-terminus of Fc exhibited substantial reduction of ADCC activity.

TABLE 50

Tetravalent constructs								
					Flow Binding - EC_{50}		Flow Binding - dAct	
Molecule	ADCC - EC_{50} M	Fold rel AB-8873 IgG	ADCC - dAct %	Fold rel AB-8873 IgG	M	Fold rel AB-8873 IgG	MedFl	Fold rel AB-8873 IgG
A B-008873-hlgG1	1.6E−08	1.0	30.0	1.0	6.1E−09	1.0	609705	1.0
SCFV-8873-01_hFc1_F2	1.8E−08	0.9	28.3	0.9	1.4E−08	0.4	223880	0.4
Fab8873-(GGSGG)3-HC-AB-008873-h1-4p0_F26	1.5E−09	10.5	32.8	1.1	1.3E−09	4.6	2230775	3.7
Fab8873-(GGSPG)3-HC-AB-008873-h1-4p0_F26	1.2E−09	13.3	31.0	1.0	4.8E−10	17.6	1556556	3.4

TABLE 50-continued

Tetravalent constructs								
					Flow Binding - EC_{50}		Flow Binding - dAct	
Molecule	ADCC - EC_{50} M	Fold rel AB-8873 IgG	ADCC - dAct %	Fold rel AB-8873 IgG	M	Fold rel AB-8873 IgG	MedFl	Fold rel AB-8873 IgG
SCFV8873-01-(GGSGG)3-HC-AB-008873-h1-2p2__F13	1.9E−09	8.3	27.6	0.9	4.4E−10	13.5	1835412	3.1
SCFV8873-01-(GGSPG)3-HC-AB-008873-h1-2p2__F13	1.4E−09	11.9	34.4	1.1	1.8E−09	3.8	2658851	4.5
SCFV8873-01-(GGSGG)3-LC-AB-008873-h1-2p2__F14	1.1E−09	15.0	29.8	1.0	2.1E−09	3.0	2558140	4.4
SCFV8873-01-(GGSPG)3-LC-AB-008873-h1-2p2__F14	3.1E−10	52.3	36.9	1.2	4.4E−10	13.6	2873733	5.0
AB-8873-h1-2p2-LC-SCFV-008873-01__F4	1.2E−08	1.3	30.9	1.0	7.5E−10	7.8	1307497	2.2
Ab8873-h1-2p2-HC-(GGSGG)3-SCFV8873-01__F3	1.0E−07	0.2	3.3	0.1	6.7E−10	10.2	737080	1.2
Ab8873-h1-2p2-HC-(GGSPG)3-SCFV8873-01__F3	1.0E−07	0.2	3.6	0.1	3.0E−09	1.9	913380	1.6
Ab8873-h1-4p0-HC-(GGSGG)3-Fab8873__F27	1.0E−07	0.2	1.6	0.1	1.3E−09	4.5	1042829	1.8
Ab8873-h1-4p0-HC-(GGSPG)3-Fab8873__F27	1.0E−07	0.2	8.9	0.3	1.7E−09	5.0	615030	1.3
Fab8873-SCFV8873-01__h1-2p2__F28	2.9E−09	5.5	33.8	1.1	9.9E−10	5.9	1442867	2.4
SCFV8873-01-h1-2p2-HC-(GGSGG)3-SCFV8873-01__F29	1.0E−07	0.2	3.5	0.1	1.3E−09	4.6	1109331	2.0
SCFV8873-01-h1-2p2-HC-(GGSPG)3-SCFV8873-01__F29	1.7E−09	9.7	7.6	0.3	3.4E−09	2.2	1393777	2.3

Example 9. AB-008873-IL15 Fusions

[0507] Receptor-linked interleukin-15 (RLI) is made by covalently joining the IL-15 receptor alpha sushi domain to IL-15 via a glycine-serine linker. RLI enables trans presentation of IL-15 to the IL-15 receptor complex (IL-15 receptor beta and IL-15 common gamma chain) on T and NK immune cells, and this trans presentation encourages immune responses such as proliferation and survival of naïve CD8+ T cells, development of memory CD8+ T cells, improved survival and upregulation of lytic molecules and CD16a on NK cells, etc.

[0508] Bispecific antibodies, referred to as IL-15 MultiMabs, that couple RLI to the anti-tumor antibody AB-008873 were developed. In one example of an IL-15 MultiMab, a 2-plus-2 molecule (2p2) that links an RLI domain to the C-terminus of each light chain of an IgG molecule via a glycine-serine linker was created. These RLI domains can be either wildtype ("wt/RLI") or a mutated variant expected to have lower affinity for the IL-15 receptor complex ("low/RLI"). The two molecules described are shown here as FIG. **36**, and sequences of the component parts are provided in Table 15 above.

[0509] These IL-15 MultiMabs have been assessed for their ability to induce proliferation of primary NK cells. Primary human NK cells were labeled with CellTrace CFSE cell proliferation dye (Invitrogen Cat. C4554) and mixed with AB-008873 IL-15 MultiMabs. After 3 days of treatment, cells were analyzed with a flow cytometer and gated for cells that lost signal from proliferation dye relative to untreated cells. See FIG. **37**.

[0510] The AB-008873 IL-15 MultiMabs showed dose-dependent increases in human NK cell proliferation. Antibodies fused with wild-type RLI ("wt/RLI") were more potent than those fused with low affinity RLI ("low/RLI"), and unmodified antibodies ("AB-008873-mIgG2a") did not show a measurable response up to 100 nM.

[0511] AB-008873 IL15 MultiMabs were assessed for their ability to bind to target tumor cells using flow cytometry. Parental antibody or IL-15 MultiMabs were added to CT26 ex vivo cells at 100 nM and incubated on ice, followed by washing. Secondary Alexa Fluor 647-conjugated anti-mouse antibody and cell membrane integrity dye was added and incubated on ice, followed by washing. Stained cells were analyzed by flow cytometry, gated for viable cells, and quantified for Alexa Fluor 647 detection. The data is shown in FIG. **38**. The results indicate that addition of the RLI domain to AB-008873 did not negatively impact binding of the antibody to tumor target cells.

[0512] As shown in FIG. **48**, AB-008873-m2p2-LC-IL15 (wt/RLI) was active in tumor growth inhibition and regression (TGI/R) studies in syngeneic mouse models using CT26 tumor cells.

Example 10. 4-1BB and 4-1BBL Constructs

[0513] 4-1BB fusion proteins were tested for their tumor targeting activities. These fusion proteins comprise (i) at least the antigen binding domains of the EphA2 antibodies and 4-1BB ligand domains (Table 13) or (ii) at least the antigen binding domains of the EphA2 antibodies and the ScFv portions of the anti-4-1BB antibodies (Table 10).

[0514] Bispecific antibodies that comprise two scFv fragments of anti-4-1BB antibody linked to the HC of the EphA2 antibodies AB-008873 or AB-010361 (EphA2-4-1BB bispecific antibodies) were constructed. Anti-hen egg lysozyme targeting anti-4-1BB antibody (5554-41BB) was also constructed and used as a negative control.

[0515] After confirming that they retained the ability to bind to PC3 cells, the EphA2-4-1BB bispecific antibodies were assayed for their ability to activate human 4-1BB using an inducible reporter cell system (Promega cat JA2351). 4-1n reporter cells were either cultured alone with the EphA2-4-1BB bispecific antibodies or co-cultured with A549 tumor cells that had been treated with the EphA2-4-1BB bispecific antibodies. The 4-1BB antibody Urelumab (AB-009811) was included as a non-tumor targeting control. After 5 hours of culture, 4-1BB activation was determined from the reporter cells by measuring bioluminescence output.

[0516] Results from this assay indicated that the EphA2-4-1BB bispecific antibodies only exhibited 4-1BB activation in the presence of tumor cells while the Urelumab control activated 4-1BB with us without the presence of tumor cells.

[0517] The activation assay was repeated using MDA-MB-231, CT26, or PC3 tumor cells. The EphA2-4-1BB bispecific antibodies showed activation of 4-1BB on all tumor cell lines tested. FIG. **19**A-D.

[0518] The bispecific antibody that comprises two scFv fragments of anti-4-1BB antibody linked to the HC of the EphA2 antibody AB-010361 was tested in BALB/c mice that developed tumors after they have been inoculated with CT26 cells. Mice were dosed intraperitoneally with test article once a week for four weeks (7-, 14-, 21-, and 28-days post CT26 implantation) at a dose level of 10 mg/kg. Vehicle control, PBS, was dosed at 10 mL/kg. As compared to the group that were treated with PBS and the group of mice that were treated with an anti-hen egg lysozyme targeting anti-4-1BB antibody ("5554-41BB"), mice treated with the EphA2-4-1BB bispecific antibody showed reduced tumor growth during the treatment period. FIG. **49**.

[0519] The mice treated with the EphA2-4-1BB bispecific antibodies as above were assayed for the presence of liver toxicity, a known side effect of certain 4-1BB agonist-based therapeutics, including anti-4-1BB antibody Urelumab. Liver toxicity was determined by measuring ALT (alanine aminotransferase) and AST (aspartate aminotransferase) which are released into the bloodstream when liver damage occurs. The ALT was measured using the MAK052 assay kit (Sigma-Aldrich) which measures the amount of pyruvate generated and AST was measured using the MAK055 assay kits (Sigma-Aldrich) which measures the amount of glutamate generated.

[0520] Results of these assays showed that mice treated with the AB-01361-4-1BB bispecific antibody and the AB-5554-41BB negative control do not show increase over vehicle of ALT & AST levels while the antibody 3H3 at 10 mg/kg (mouse surrogate for Urelumab) shows increased serum ALT & AST levels (ALT ~8-fold increase, $p<0.0001$).

[0521] Furthermore, livers from mice treated with the AB-01361-4-1BB bispecific antibody showed no visible signs of liver inflammation as determined by tissue staining using hematoxylin and eosin.

[0522] Portal vein infiltration and expansion to parenchyma was detected with 10 mpk 3H3 but not AB-010361-41BB.

[0523] Constructs that comprise AB-008873 fused with h41BBL (THD)3-2 were generated and assayed for the ability to activate 4-1BB for tumor cell lines A549, MDA-MB231, SKOV3, and PC3, as described above for the EphA2-4-1BB bispecific antibodies. AB-009811 was included as a positive control and 5554-41BBL (THD) 3-10 or 5554-41BBL (THD) 3-2 was used as a negative control. The AB-008873-h41BBL construct activated 4-1BB in all cell lines tested. FIG. **19**E.

Example 11. CD3 Bispecific Constructs

[0524] This example describes CD3 bispecific antibodies comprising the anti-tumor antibodies described herein. Table 9 provides specific examples of anti-CD3 binding arms that can be combined with any of the anti-EphA2 antibodies described herein. FIG. **53** provides examples of anti-EphA2/anti-CD3 bispecific constructs that can be generated.

[0525] Bispecific constructs generated using a 1+1 format with the CD3 arm comprising the VH/VL sequence of AB-008707 were assayed for in vivo activity in a mouse model as follows. NSG-DKO mice were inoculated subcutaneously in the flank with 5e6 PC3 tumor cells in 50% matrigel on Day 0. Human PBMCs from three individual donors were engrafted via IV tail vein injection on day I following inoculation (10e6 cells/mouse). Mice were randomized into 5 mice per group per donor based on tumor volume and treated with either vehicle, 5 mg/kg anti-hen egg lysozyme non-targeting control 5554/CD3, 5 mg/kg AB-010361/CD3, or 1 mg/kg cetuximab/CD3 positive control intraperitoneally 1×/week for 3 weeks (indicated by the dotted vertical lines). Tumor volumes were measured twice per week until mice were euthanized. The results show that treatment with AB-010361/CD3 ("AB-010361/CD3") and cetuximab/CD3 positive control ("Positive CD3 control"), but not AB-05554/CD3 negative control ("Non-targeting CD3 control"), led to a decrease in tumor burden, effectively eliminating tumors. FIG. **54**. Averaging across the 3 hPBMC donors, comparing 10361-CD3 vs 5554-CD3 is significant with p=3e-7 for NAAC. Positive control 129-CD3 is significant vs 5554-CD3 with p=2e-7 for NAAC.

Example 12. ADC Constructs

ADC In Vitro

[0526] This experiment tested the cytotoxicity of antibody-drug conjugates ("conjugates") each comprising an EphA2 antibody directly conjugated via a lysine to a ZymeLink™ Auristatin (ZLA) payload having structure (1) as disclosed above on SKOV3, MDAMB231, and PC3 tumor cells. Each of the ADCs has Formula (I). The average drug to antibody ratio (DAR) for each molecule ranges from 2.9 and 3.6 as determined by Mass Spectrometry. The EphA2 antibodies assessed include AB-010361 (IgG1), AB-10699 (IgG1), AB-010361 (IgG4), and AB-010671 (IgG1) and their conjugates are shown as "AB-0010361-ZLA," "AB-

010699-ZLA," "AB-010361-ZLA-IgG4," and "AB-010671-ZLA" in FIG. **55**A-D. Target tumor cells were detached from the culture plate and cell concentration was adjusted to 31,250 cells/mL in assay media. 2,500 cells were added to each well of a 96 well plate and incubated with different concentrations of directly conjugated primary antibody for 15 min at room temperature. Cells were then incubated for 72 h at 37° C. and 5% CO2. At the end of the incubation period, 100 μl CellTiter-Glo® was added to each well and allowed to incubate for 5-10 min at room temperature before reading luminescence in a BMG ClarioSTAR plate reader. Data were then normalized to a maximum lysis control and plotted using graph pad prism software.

[0527] All ZLA-conjugated EphA2 antibodies tested showed dose-dependent cytotoxic activity over isotype-ZLA control in vitro on PC3, MDA-MB-231, and SKOV3 cells. FIG. **55**A-C. EC_{50} values are listed in FIG. **55**D. The results show that AB-010699-ZLA and AB-010671 had higher potency but similar maximum cytotoxicity compared to AB-010361-ZLA and AB-010361-IgG4-ZLA. AB-010361-ZLA and AB-010361-IgG4-ZLA showed similar activity.

[0528] The assay was repeated in PC3 cells with AB-010699 and AB-010671 in both IgG1 and IgG4 formats. FIG. **59**A. ADC constructs comprising AB-010699 Fvs exhibited slightly higher potency as compared to constructs comprising AB-010671 Fvs. No appreciable differences in potency was observed between the IgG1 and IgG4 of the same Fv. FIG. **59**B.

In Vivo PC3 Model

[0529] This experiment assessed the activity of ZLA-Conjugated EphA2 antibodies AB-010671-ZLA and AB-010699-ZLA in vivo. 2×10^6 PC3 cells were inoculated in 0.2 mL of 1:1 DPBS with Matrigel subcutaneously into each of the male NPG mice. Tumors were allowed to establish and then randomized at an average size of 150 mm^3 with 10 mice per treatment cohort. Treatment started at the day of randomization (Day 0) with weekly intravenous injections of the conjugates at the indicated doses. Body weight as well as tumor sizes were monitored twice weekly. Study was ended at day 35 after 5 doses. At endpoint, mice were euthanized, and tumors excised and weighted.

[0530] The results showed that AB-010671-ZLA (FIG. **56**C and AB-010699-ZLA (FIG. **56**D) exhibited dose dependent anti-tumor activity with significant tumor reduction over isotype-ZLA observed at 3 and 6 mg/kg. AB-010361-ZLA and AB-010361-IgG4-ZLA showed weak activity at the highest dose tested (FIGS. **56**A and **56**B) with AB-010361-IgG4-ZLA being more efficacious as AB-010361-ZLA. This tumor size observation is confirmed by endpoint tumor weight analysis (FIG. **57**). AB-016699-ZLA and AB-010671-ZLA showed significantly smaller tumors in mice dosed with the conjugates at 3 mg/kg and 6 mg/kg whereas AB-010361-IgG4-ZLA showed significantly smaller tumors at 6 mg/kg.

[0531] Body weights were measured throughout the study and inversely correlated with tumor burden (FIG. **58**). Weight loss occurred in mice towards the end of the study starting from day 28 and was most prominent for vehicle and isotype control groups as well as tumors who didn't respond to therapy as outlined above. Mice in 3 and 6 mg/kg cohorts of AB-010699-ZLA and AB-010671 ZLA showed no loss in body weight as did mice in the 6 mg/kg cohort of AB-010361-IgG4-ZLA.

[0532] The assay was repeated with AB-010699 and AB-010671 in both IgG1 and IgG4 formats at 1, 3, and 6 mg/kg, AB-010016, and vehicle and isotype controls in mice carrying PC3 tumor cells measuring ~150 mm^3 or ~400 mm^3. AB-010699 and AB-010671 in both IgG1 and IgG4 formats showed anti-tumor activity at 3 and 6 mg/kg in both the ~150 mm^3 (FIG. **60**A) and the larger more established tumor models (FIG. **61**A), with AB-010699-IgG4 exhibiting slightly better efficacy compared to AB-010671-IgG4. FIG. **60**B and FIG. **61**B show the body weight of the mice over the course of the studies.

INCORPORATION BY REFERENCE

[0533] Each and every publication and patent document referred to in this disclosure is incorporated herein by reference in its entirety for all purposes to the same extent as if each such publication or document was specifically and individually indicated to be incorporated herein by reference.

[0534] While the invention has been described with reference to the specific examples and illustrations, changes can be made and equivalents can be substituted to adapt to a particular context or intended use as a matter of routine development and optimization and within the purview of one of ordinary skill in the art, thereby achieving benefits of the invention without departing from the scope of what is claimed and their equivalents.

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SEQ ID NO: 40           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
```

-continued

```
                      organism = synthetic construct
SEQUENCE: 40
GGNNIGSKNV H                                                      11

SEQ ID NO: 41         moltype = AA  length = 11
FEATURE               Location/Qualifiers
REGION                1..11
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..11
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 41
GGNNIGSKNV H                                                      11

SEQ ID NO: 42         moltype = AA  length = 11
FEATURE               Location/Qualifiers
REGION                1..11
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..11
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 42
GGNNIGSKNV H                                                      11

SEQ ID NO: 43         moltype = AA  length = 11
FEATURE               Location/Qualifiers
REGION                1..11
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..11
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 43
GGNNIGSKNV H                                                      11

SEQ ID NO: 44         moltype = AA  length = 11
FEATURE               Location/Qualifiers
REGION                1..11
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..11
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 44
GGNNIGSKNV H                                                      11

SEQ ID NO: 45         moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 45
DDSDRPS                                                           7

SEQ ID NO: 46         moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 46
DDSDRPS                                                           7

SEQ ID NO: 47         moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 47
DDSDRPS                                                           7

SEQ ID NO: 48         moltype = AA  length = 7
FEATURE               Location/Qualifiers
```

-continued

```
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 48
DDSDRPS                                                                  7

SEQ ID NO: 49           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 49
DDSDRPS                                                                  7

SEQ ID NO: 50           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 50
DDSDRPS                                                                  7

SEQ ID NO: 51           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 51
DDSDRPS                                                                  7

SEQ ID NO: 52           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 52
DDSDRPS                                                                  7

SEQ ID NO: 53           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 53
DDSDRPS                                                                  7

SEQ ID NO: 54           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 54
DDSDRPS                                                                  7

SEQ ID NO: 55           moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 55
```

-continued

```
DDSDRPS                                                      7

SEQ ID NO: 56           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 56
QVWDSSSDHL V                                                 11

SEQ ID NO: 57           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 57
QVWDSSSDHL V                                                 11

SEQ ID NO: 58           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 58
QVWDSSSDHL V                                                 11

SEQ ID NO: 59           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 59
QVWDSSSDHL V                                                 11

SEQ ID NO: 60           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 60
QVWDSSSDHL V                                                 11

SEQ ID NO: 61           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 61
QVWDSSSDHL V                                                 11

SEQ ID NO: 62           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 62
QVWDSSSDHL V                                                 11

SEQ ID NO: 63           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
```

-continued

```
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 63
QVWDSSSDHL V                                                  11

SEQ ID NO: 64           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 64
QVWDSSSDHL V                                                  11

SEQ ID NO: 65           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 65
QVWDSSSDHL V                                                  11

SEQ ID NO: 66           moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 66
QVWDSSSDHL V                                                  11

SEQ ID NO: 67           moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 67
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDPSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                    130

SEQ ID NO: 68           moltype = AA  length = 127
FEATURE                 Location/Qualifiers
REGION                  1..127
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..127
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 68
QVQLQQWGAG LLKPSETLSL TCAVYGGSLS DYHWSWIRQP PGKGLEWIGE INHSGSTNYN  60
PSLKSRVTIS VDTSKNQFSL KLSSVSAADT AVYYCAKPFR PHCTNGVCHS GDAFDIWGQG  120
TMATVSS                                                       127

SEQ ID NO: 69           moltype = AA  length = 127
FEATURE                 Location/Qualifiers
REGION                  1..127
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..127
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 69
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHSGSTNYN  60
PSLKSRVTIS VDPSKNQFSL KLTSVTAADT AVYYCAKPFR PHCTNGVCYS GDAFDIWGQG  120
TMVTVSS                                                       127

SEQ ID NO: 70           moltype = AA  length = 127
```

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```
FEATURE                Location/Qualifiers
REGION                 1..127
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..127
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 70
QVQLQQWGAG LLKPSETLSL TCAVYGGSFN DYYWSWIRQP PGKGLEWIGE VNHSGSTSYN  60
PSLKSRVTIS VDTSKNQFSL KLSSVTAADT AVYYCAKPFR PHCTNGVCYS GDAFDIWGQG  120
TMVTVSS                                                            127

SEQ ID NO: 71          moltype = AA  length = 127
FEATURE                Location/Qualifiers
REGION                 1..127
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..127
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 71
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHSGSTNYN  60
PSLKSRVTIS VDPSKNQFSL KLTSVTAADT AVYYCAKPFR PHCTNGVCYS GDAFDIWGQG  120
TMVTVSS                                                            127

SEQ ID NO: 72          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 72
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHSGSINYN  60
NYNPSLKSRV TISVDPSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 73          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 73
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHAGSINYN  60
NYNPSLKSRV TISVDPSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 74          moltype = AA  length = 127
FEATURE                Location/Qualifiers
REGION                 1..127
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..127
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 74
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
PSLKSRVTIS VDPSKNQFSL KLTSVTAADT AVYYCAKPLR PHCTNGVCYS GDAFDIWGQG  120
TMVTVAS                                                            127

SEQ ID NO: 75          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 75
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130
```

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```
SEQ ID NO: 76          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 76
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDASKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 77          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 77
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDPSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNAV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 78          moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 78
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 79          moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 79
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 80          moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 80
SYVLTQPPSV SVAPGQTARI TCGGNNIGTK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 81          moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 81
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPALVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 82          moltype = AA  length = 108
```

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```
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 82
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 83         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 83
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 84         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 84
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 85         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 85
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 86         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 86
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 87         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
source                1..108
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 87
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 88         moltype = AA  length = 108
FEATURE               Location/Qualifiers
REGION                1..108
                      note = Description of Artificial Sequence: Synthetic
                       polypeptide
```

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```
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 88
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 89           moltype = AA  length = 459
FEATURE                 Location/Qualifiers
REGION                  1..459
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..459
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 89
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDPSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS AKTTAPSVYP LAPVCGDTTG SSVTLGCLVK GYFPEPVTLT WNSGSLSSGV  180
HTFPAVLQSD LYTLSSSVTV TSSTWPSQSI TCNVAHPASS TKVDKKIEPR GPTIKPCPPC  240
KCPAPNLLGG PSVFIFPPKI KDVLMISLSP IVTCVVVDVS EDDPDVQISW FVNNVEVHTA  300
QTQTHREDYN STLRVVSALP IQHQDWMSGK EFKCKVNNKD LPAPIERTIS KPKGSVRAPQ  360
VYVLPPPEEE MTKKQVTLTC MVTDFMPEDI YVEWTNNGKT ELNYKNTEPV LDSDGSYFMY  420
SKLRVEKKNW VERNSYSCSV VHEGLHNHHT TKSFSRTPG                        459

SEQ ID NO: 90           moltype = AA  length = 214
FEATURE                 Location/Qualifiers
REGION                  1..214
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..214
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 90
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVLGQ PKSSPSVTLF  120
PPSSEELETN KATLVCTITD FYPGVVTVDW KVDGTPVTQG METTQPSKQS NNKYMASSYL  180
TLTARAWERH SSYSCQVTHE GHTVEKSLSR ADCS                             214

SEQ ID NO: 91           moltype = AA  length = 15
FEATURE                 Location/Qualifiers
REGION                  1..15
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..15
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 91
GGGGSGGGGS GGGGS                                                  15

SEQ ID NO: 92           moltype = AA  length = 212
FEATURE                 Location/Qualifiers
REGION                  1..212
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..212
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 92
ITCPPPMSVE HADIWVKSYS LYSRERYICN SGFKRKAGTS SLTECVLNKA TNVAHWTTPS  60
LKCIRDPALV HQRPAPPGGG GSGGGGSGGG SGGGGSLQNW VNVISDLKKI EDLIQSMHID  120
ATLYTESDVH PSCKVTAMKC FLLELQVISL ESGDASIHDT VENLIILANN SLSSNGNVTE  180
SGCKECEELE EKNIKEFLQS FVHIVQMFIN TS                               212

SEQ ID NO: 93           moltype = AA  length = 212
FEATURE                 Location/Qualifiers
REGION                  1..212
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..212
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 93
ITCPPPMSVE HADIWVKSYS LYSRERYICN SGFKRKAGTS SLTECVLNKA TNVAHWTTPS  60
LKCIRDPALV HQRPAPPGGG GSGGGGSGGG SGGGGSLQNW VNVISDLKKI EDLIQSMHID  120
ATLYTESNVH PSCKVTAMKC FLLELQVISL ESGDASIHDT VQDLIILANN SLSSNGNVTE  180
SGCKECEELE EKNIKEFLQS FVHIVQMFIN TS                               212
```

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```
SEQ ID NO: 94          moltype = AA  length = 976
FEATURE                Location/Qualifiers
source                 1..976
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 94
MELQAARACF ALLWGCALAA AAAAQGKEVV LLDFAAAGGE LGWLTHPYGK GWDLMQNIMN  60
DMPIYMYSVC NVMSGDQDNW LRTNWVYRGE AERIFIELKF TVRDCNSFPG GASSCKETFN  120
LYYAESDLDY GTNFQKRLFT KIDTIAPDEI TVSSDFEARH VKLNVEERSV GPLTRKGFYL  180
AFQDIGACVA LLSVRVYYKK CPELLQGLAH FPETIAGSDA PSLATVAGTC VDHAVVPPGG  240
EEPRMHCAVD GEWLVPIGQC LCQAGYEKVE DACQACSPGF FKFEASESPC LECPEHTLPS  300
PEGATSCECE EGFFRAPQDP ASMPCTRPPS APHYLTAVGM GAKVELRWTP PQDSGGREDI  360
VYSVTCEQCW PESGECGPCE ASVRYSEPPH GLTRTSVTVS DLEPHMNYTF TVEARNGVSG  420
LVTSRSFRTA SVSINQTEPP KVRLEGRSTT SLSVSWSIPP PQQSRVWKYE VTYRKKGDSN  480
SYNVRRTEGF SVTLDDLAPD TTYLVQVQAL TQEGQGAGSK VHEFQTLSPE GSGNLAVIGG  540
VAVGVVLLLV LAGVGFFIHR RRKNQRARQS PEDVYFSKSE QLKPLKTYVD PHTYEDPNQA  600
VLKFTTEIHP SCVTRQKVIG AGEFGEVYKG MLKTSSGKKE VPVAIKTLKA GYTEKQRVDF  660
LGEAGIMGQF SHHNIIRLEG VISKYKPMMI ITEYMENGAL DKFLREKDGE FSVLQLVGML  720
RGIAAGMKYL ANMNYVHRDL AARNILVNSN LVCKVSDFGL SRVLEDDPEA TYTTSGGKIP  780
IRWTAPEAIS YRKFTSASDV WSFGIVMWEV MTYGERPYWE LSNHEVMKAI NDGFRLPTPM  840
DCPSAIYQLM MQCWQQERAR RPKFADIVSI LDKLIRAPDS LKTLADFDPR VSIRLPSTSG  900
SEGVPFRTVS EWLESIKMQQ YTEHFMAAGY TAIEKVVQMT NDDIKRIGVR LPGHQKRIAY  960
SLLGLKDQVN TVGIPI                                                  976

SEQ ID NO: 95          moltype = AA  length = 98
FEATURE                Location/Qualifiers
source                 1..98
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 95
TEPPKVRLEG RSTTSLSVSW SIPPPQQSRV WKYEVTYRKK GDSNSYNVRR TEGFSVTLDD  60
LAPDTTYLVQ VQALTQEGQG AGSKVHEFQT LSPEGSGN                          98

SEQ ID NO: 96          moltype = AA  length = 24
FEATURE                Location/Qualifiers
source                 1..24
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 96
MELQAARACF ALLWGCALAA AAAA                                         24

SEQ ID NO: 97          moltype = AA  length = 175
FEATURE                Location/Qualifiers
source                 1..175
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 97
QGKEVVLLDF AAAGGELGWL THPYGKGWDL MQNIMNDMPI YMYSVCNVMS GDQDNWLRTN  60
WVYRGEAERI FIELKFTVRD CNSFPGGASS CKETFNLYYA ESDLDYGTNF QKRLFTKIDT  120
IAPDEITVSS DFEARHVKLN VEERSVGPLT RKGFYLAFQD IGACVALLSV RVYYK       175

SEQ ID NO: 98          moltype = AA  length = 62
FEATURE                Location/Qualifiers
source                 1..62
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 98
KCPELLQGLA HFPETIAGSD APSLATVAGT CVDHAVVPPG GEEPRMHCAV DGEWLVPIGQ  60
CL                                                                 62

SEQ ID NO: 99          moltype = AA  length = 65
FEATURE                Location/Qualifiers
source                 1..65
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 99
CQAGYEKVED ACQACSPGFF KFEASESPCL ECPEHTLPSP EGATSCECEE GFFRAPQDPA  60
SMPCT                                                              65

SEQ ID NO: 100         moltype = AA  length = 110
FEATURE                Location/Qualifiers
source                 1..110
                       mol_type = protein
                       organism = Homo sapiens
SEQUENCE: 100
```

-continued

```
RPPSAPHYLT AVGMGAKVEL RWTPPQDSGG REDIVYSVTC EQCWPESGEC GPCEASVRYS  60
EPPHGLTRTS VTVSDLEPHM NYTFTVEARN GVSGLVTSRS FRTASVSINQ            110

SEQ ID NO: 101          moltype = AA  length = 510
FEATURE                 Location/Qualifiers
source                  1..510
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 101
QGKEVVLLDF AAAGGELGWL THPYGKGWDL MQNIMNDMPI YMYSVCNVMS GDQDNWLRTN  60
WVYRGEAERI FIELKFTVRD CNSFPGGASS CKETFNLYYA ESDLDYGTNF QKRLFTKIDT  120
IAPDEITVSS DFEARHVKLN VEERSVGPLT RKGFYLAFQD IGACVALLSV RVYYKKCPEL  180
LQGLAHFPET IAGSDAPSLA TVAGTCVDHA VVPPGGEEPR MHCAVDGEWL VPIGQCLCQA  240
GYEKVEDACQ ACSPGFFKFE ASESPCLECP EHTLPSPEGA TSCECEEGFF RAPQDPASMP  300
CTRPPSAPHY LTAVGMGAKV ELRWTPPQDS GGREDIVYSV TCEQCWPESG ECGPCEASVR  360
YSEPPHGLTR TSVTVSDLEP HMNYTFTVEA RNGVSGLVTS RSFRTASVSI NQTEPPKVRL  420
EGRSTTSLSV SWSIPPPQQS RVWKYEVTYR KKGDSNSYNV RRTEGFSVTL DDLAPDTTYL  480
VQVQALTQEG QGAGSKVHEF QTLSPEGSGN                                  510

SEQ ID NO: 102          moltype = AA  length = 524
FEATURE                 Location/Qualifiers
source                  1..524
                        mol_type = protein
                        organism = Homo sapiens
SEQUENCE: 102
SRVYPANEVT LLDSRSVQGE LGWIASPLEG GWEEVSIMDE KNTPIRTYQV CNVMEPSQNN  60
WLRTDWITRE GAQRVYIEIK FTLRDCNSLP GVMGTCKETF NLYYYESDND KERFIRENQF  120
VKIDTIAADE SFTQVDIGDR IMKLNTEIRD VGPLSKKGFY LAFQDVGACI ALVSVRVFYK  180
KCPLTVRNLA QFPDTITGAD TSSLVEVRGS CVNNSEEKDV PKMYCGADGE WLVPIGNCLC  240
NAGHEERSGE CQACKIGYYK ALSTDATCAK CPPHSYSVWE GATSCTCDRG FFRADNDAAS  300
MPCTRPPSAP LNLISNVNET SVNLEWSSPQ NTGGRQDISY NVVCKKCGAG DPSKCRPCGS  360
GVHYTPQQNG LKTTKVSITD LLAHTNYTFE IWAVNGVSKY NPNPDQSVSV TVTTNQAAPS  420
SIALVQAKEV TRYSVALAWL EPDRPNGVIL EYEVKYYEKD QNERSYRIVR TAARNTDIKG  480
LNPLTSYVFH VRARTAAGYG DFSEPLEVTT NTVPSRIIGD GANS                  524

SEQ ID NO: 103          moltype = AA  length = 512
FEATURE                 Location/Qualifiers
REGION                  1..512
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..512
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 103
QGKEVVLLDF AAAGGELGWL THPYGKGWDL MQNIMNDMPI YMYSVCNVMS GDQDNWLRTN  60
WVYRGEAERI FIELKFTVRD CNSFPGGASS CKETFNLYYA ESDLDYGTNF QKRLFTKIDT  120
IAPDEITVSS DFEARHVKLN VEERSVGPLT RKGFYLAFQD IGACVALLSV RVYYKKCPEL  180
LQGLAHFPET IAGSDAPSLA TVAGTCVDHA VVPPGGEEPR MHCAVDGEWL VPIGQCLCQA  240
GYEKVEDACQ ACSPGFFKFE ASESPCLECP EHTLPSPEGA TSCECEEGFF RAPQDPASMP  300
CTRPPSAPLN LISNVNETSV NLEWSSPQNT GGRQDISYNV VCKKCGAGDP SKCRPCGSGV  360
HYTPQQNGLK TTKVSITDLL AHTNYTFEIW AVNGVSKYNP NPDQSVSVTV TTNQTEPPKV  420
RLEGRSTTSL SVSWSIPPPQ QSRVWKYEVT YRKKGDSNSY NVRRTEGFSV TLDDLAPDTT  480
YLVQVQALTQ EGQGAGSKVH EFQTLSPEGS GN                               512

SEQ ID NO: 104          moltype = AA  length = 520
FEATURE                 Location/Qualifiers
REGION                  1..520
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..520
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 104
QGKEVVLLDF AAAGGELGWL THPYGKGWDL MQNIMNDMPI YMYSVCNVMS GDQDNWLRTN  60
WVYRGEAERI FIELKFTVRD CNSFPGGASS CKETFNLYYA ESDLDYGTNF QKRLFTKIDT  120
IAPDEITVSS DFEARHVKLN VEERSVGPLT RKGFYLAFQD IGACVALLSV RVYYKKCPEL  180
LQGLAHFPET IAGSDAPSLA TVAGTCVDHA VVPPGGEEPR MHCAVDGEWL VPIGQCLCQA  240
GYEKVEDACQ ACSPGFFKFE ASESPCLECP EHTLPSPEGA TSCECEEGFF RAPQDPASMP  300
CTRPPSAPHY LTAVGMGAKV ELRWTPPQDS GGREDIVYSV TCEQCWPESG ECGPCEASVR  360
YSEPPHGLTR TSVTVSDLEP HMNYTFTVEA RNGVSGLVTS RSFRTASVSI NQAAPSSIAL  420
VQAKEVTRYS VALAWLEPDR PNGVILEYEV KYYEKDQNER SYRIVRTAAR NTDIKGLNPL  480
TSYVFHVRAR TAAGYGDFSE PLEVTTNTVP SRIIGDGANS                       520

SEQ ID NO: 105          moltype = AA  length = 17
FEATURE                 Location/Qualifiers
REGION                  1..17
```

-continued

```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..17
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 105
TSGGGGSGGG GSGGGGS                                                 17

SEQ ID NO: 106          moltype = AA  length = 154
FEATURE                 Location/Qualifiers
REGION                  1..154
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..154
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 106
QGMFAQLVAQ NVLLIDGPLS WYSDPGLAGV SLTGGLSYKE DTKELVVAKA GVYYVFFQLE  60
LRRVVAGEGS GSVSLALHLQ PLRSAAGAAA LALTVDLPPA SSEARNSAFG FQGRLLHLSA  120
GQRLGVHLHT EARARHAWQL TQGATVLGLF RVTP                              154

SEQ ID NO: 107          moltype = AA  length = 482
FEATURE                 Location/Qualifiers
REGION                  1..482
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..482
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 107
QGMFAQLVAQ NVLLIDGPLS WYSDPGLAGV SLTGGLSYKE DTKELVVAKA GVYYVFFQLE  60
LRRVVAGEGS GSVSLALHLQ PLRSAAGAAA LALTVDLPPA SSEARNSAFG FQGRLLHLSA  120
GQRLGVHLHT EARARHAWQL TQGATVLGLF RVTPGGSGGG GSGGQGMFAQ LVAQNVLLID  180
GPLSWYSDPG LAGVSLTGGL SYKEDTKELV VAKAGVYYVF FQLELRRVVA GEGSGSVSLA  240
LHLQPLRSAA GAAALALTVD LPPASSEARN SAFGFQGRLL HLSAGQRLGV HLHTEARARH  300
AWQLTQGATV LGLFRVTPGG SGGGGSGGQG MFAQLVAQNV LLIDGPLSWY SDPGLAGVSL  360
TGGLSYKEDT KELVVAKAGV YYVFFQLELR RVVAGEGSGS VSLALHLQPL RSAAGAAALA  420
LTVDLPPASS EARNSAFGFQ GRLLHLSAGQ RLGVHLHTEA RARHAWQLTQ GATVLGLFRV  480
TP                                                                 482

SEQ ID NO: 108          moltype = AA  length = 466
FEATURE                 Location/Qualifiers
REGION                  1..466
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..466
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 108
QGMFAQLVAQ NVLLIDGPLS WYSDPGLAGV SLTGGLSYKE DTKELVVAKA GVYYVFFQLE  60
LRRVVAGEGS GSVSLALHLQ PLRSAAGAAA LALTVDLPPA SSEARNSAFG FQGRLLHLSA  120
GQRLGVHLHT EARARHAWQL TQGATVLGLF RVTPGGQGMF AQLVAQNVLL IDGPLSWYSD  180
PGLAGVSLTG GLSYKEDTKE LVVAKAGVYY VFFQLELRRV VAGEGSGSVS LALHLQPLRS  240
AAGAAALALT VDLPPASSEA RNSAFGFQGR LLHLSAGQRL GVHLHTEARA RHAWQLTQGA  300
TVLGLFRVTP GGQGMFAQLV AQNVLLIDGP LSWYSDPGLA GVSLTGGLSY KEDTKELVVA  360
KAGVYYVFFQ LELRRVVAGE GSGSVSLALH LQPLRSAAGA AALALTVDLP PASSEARNSA  420
FGFQGRLLHL SAGQRLGVHL HTEARARHAW QLTQGATVLG LFRVTP                 466

SEQ ID NO: 109          moltype = AA  length = 178
FEATURE                 Location/Qualifiers
REGION                  1..178
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..178
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 109
REGPELSPDD PAGLLDLRQG MFAQLVAQNV LLIDGPLSWY SDPGLAGVSL TGGLSYKEDT  60
KELVVAKAGV YYVFFQLELR RVVAGEGSGS VSLALHLQPL RSAAGAAALA LTVDLPPASS  120
EARNSAFGFQ GRLLHLSAGQ RLGVHLHTEA RARHAWQLTQ GATVLGLFRV TPEIPAGL    178

SEQ ID NO: 110          moltype = AA  length = 554
FEATURE                 Location/Qualifiers
REGION                  1..554
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..554
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 110
REGPELSPDD PAGLLDLRQG MFAQLVAQNV LLIDGPLSWY SDPGLAGVSL TGGLSYKEDT  60
KELVVAKAGV YYVFFQLELR RVVAGEGSGS VSLALHLQPL RSAAGAAALA LTVDLPPASS  120
EARNSAFGFQ GRLLHLSAGQ RLGVHLHTEA RARHAWQLTQ GATVLGLFRV TPEIPAGLGG  180
SGGGGSGGRE GPELSPDDPA GLLDLRQGMF AQLVAQNVLL IDGPLSWYSD PGLAGVSLTG  240
GLSYKEDTKE LVVAKAGVYY VFFQLELRRV VAGEGSGSVS LALHLQPLRS AAGAAALALT  300
VDLPPASSEA RNSAFGFQGR LLHLSAGQRL GVHLHTEARA RHAWQLTQGA TVLGLFRVTP  360
EIPAGLGGSG GGGSGGREGP ELSPDDPAGL LDLRQGMFAQ LVAQNVLLID GPLSWYSDPG  420
LAGVSLTGGL SYKEDTKELV VAKAGVYYVF FQLELRRVVA GEGSGSVSLA LHLQPLRSAA  480
GAAALALTVD LPPASSEARN SAFGFQGRLL HLSAGQRLGV HLHTEARARH AWQLTQGATV  540
LGLFRVTPEI PAGL                                                  554

SEQ ID NO: 111          moltype = AA  length = 252
FEATURE                 Location/Qualifiers
REGION                  1..252
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..252
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 111
QVQLQQSGAE LAKPGTSVKL SCKASGYTFT SYYIYWVKQR PGQGLEWIGN IWPGNGGTFY  60
GEKFMGKATF TADTSSSTAY MLLGSLTPED SAYYFCARRP DYSGDDYFDY WGQGVLVTVS  120
SGGGGSGGGG SGGGGSGGGG SQVVLTQPKS VSTSLKSTVK LSCKLNSGNI GSYYVHWYQQ  180
HAGRSPTTMI YRDDKRPDGV PDRFSGSIDS SSNSAFLTIN NVQTEDDAIY FCHSYDSTIT  240
PVFGGGTKLT VL                                                    252

SEQ ID NO: 112          moltype = AA  length = 252
FEATURE                 Location/Qualifiers
REGION                  1..252
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..252
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 112
QVVLTQPKSV STSLKSTVKL SCKLNSGNIG SYYVHWYQQH AGRSPTTMIY RDDKRPDGVP  60
DRFSGSIDSS SNSAFLTINN VQTEDDAIYF CHSYDSTITP VFGGGTKLTV LGGGGSGGGG  120
SGGGGSGGGG SQVQLQQSGA ELAKPGTSVK LSCKASGYTF TSYYIYWVKQ RPGQGLEWIG  180
NIWPGNGGTF YGEKFMGKAT FTADTSSSTA YMLLGSLTPE DSAYYFCARR PDYSGDDYFD  240
YWGQGVLVTV SS                                                    252

SEQ ID NO: 113          moltype = AA  length = 253
FEATURE                 Location/Qualifiers
REGION                  1..253
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..253
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 113
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA ISGSGGSTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDS PFLLDDYYYY YYMDVWGKGT  120
TVTVSSGGGG SGGGGSGGGG SGGGGSDIQM TQSPSSVSAS VGDRVTITCR ASQGISSWLA  180
WYQQKPGKAP KLLIYAASSL QSGVPSRFSG SGSGTDFTLT ISSLQPEDFA TYYCQQGHLF  240
PITFGGGTKV EIK                                                   253

SEQ ID NO: 114          moltype = AA  length = 253
FEATURE                 Location/Qualifiers
REGION                  1..253
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..253
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 114
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ GHLFPITFGG GTKVEIKGGG GSGGGGSGGG  120
GSGGGGSEVQ LLESGGGLVQ PGGSLRLSCA ASGFTFSSYA MSWVRQAPGK GLEWVSAISG  180
SGGSTYYADS VKGRFTISRD NSKNTLYLQM NSLRAEDTAV YYCAKDSPFL LDDYYYYYYM  240
DVWGKGTTVT VSS                                                   253

SEQ ID NO: 115          moltype = AA  length = 253
FEATURE                 Location/Qualifiers
```

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```
REGION                  1..253
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..253
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 115
EVQLLESGGG LVQPGGSLRL SCAASGFTFR NYAMSWVRQA PGKGLEWVSA ISGSGDTTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDS PFLLDDYYYY YYMDVWGKGT  120
TVTVSSGGGG SGGGGSGGGG SGGGGSDIQM TQSPSSVSAS VGDRVTITCR ASQGISSWLA  180
WYQQKPGKAP KLLIYAASSL QSGVPSRFSG SGSGTDFTLT ISSLQPEDFA TYYCQQGHLF  240
PITFGGGTKV EIK                                                     253

SEQ ID NO: 116          moltype = AA  length = 253
FEATURE                 Location/Qualifiers
REGION                  1..253
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..253
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 116
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ GHLFPITFGG GTKVEIKGGG GSGGGGSGGG  120
GSGGGGSEVQ LLESGGGLVQ PGGSLRLSCA ASGFTFRNYA MSWVRQAPGK GLEWVSAISG  180
SGDTTYYADS VKGRFTISRD NSKNTLYLQM NSLRAEDTAV YYCAKDSPFL LDDYYYYYYM  240
DVWGKGTTVT VSS                                                     253

SEQ ID NO: 117          moltype = AA  length = 250
FEATURE                 Location/Qualifiers
REGION                  1..250
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..250
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 117
QVKLVQSGAA LVKPGASVKM SCKASDYTFN DYWVSWVKQR HGESLEWIGE IYPNSGATNF  60
NGKFRGKATL TVDNPTSTAY MELSRLTSED SAIYYCTREY TRDWFAYWGQ GTLVTVSSGG  120
GGSGGGGSGG GGSGGGGSDV VLTQTPSILS ATIGQSVSIS CRSSQSLLDS DGNTYLYWFL  180
QRPGQSPQRL IYLVSNLGSG VPNRFSGSGS GTDFTLKISG VEAEDLGIYY CMQPTHAPYT  240
FGAGTKLELK                                                         250

SEQ ID NO: 118          moltype = AA  length = 250
FEATURE                 Location/Qualifiers
REGION                  1..250
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..250
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 118
DVVLTQTPSI LSATIGQSVS ISCRSSQSLL DSDGNTYLYW FLQRPGQSPQ RLIYLVSNLG  60
SGVPNRFSGS GSGTDFTLKI SGVEAEDLGI YYCMQPTHAP YTFGAGTKLE LKGGGGSGGG  120
GSGGGGSGGG GSQVKLVQSG AALVKPGASV KMSCKASDYT FNDYWVSWVK QRHGESLEWI  180
GEIYPNSGAT NFNGKFRGKA TLTVDNPTST AYMELSRLTS EDSAIYYCTR EYTRDWFAYW  240
GQGTLVTVSS                                                         250

SEQ ID NO: 119          moltype = AA  length = 246
FEATURE                 Location/Qualifiers
REGION                  1..246
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..246
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 119
QVQLQQSGAE VIKPGASVKL SCKASGYTFS SYWMHWVRQA PGQGLEWIGE INPGNGHTNY  60
NEKFKSRATL TGDTSTSTVY MELSSLRSED TAVYYCARSF TTARAFAYWG QGTLVTVSSG  120
GGGSGGGGSG GGGSGGGGSD IVMTQSPAFL SVTPGEKVTI TCRASQTISD YLHWYQQKPD  180
QAPKLLIKYA SQSISGIPSR FSGSGSGTDF TFTISSLEAE DAATYYCQDG HSFPPTFGQG  240
TKLEIK                                                             246

SEQ ID NO: 120          moltype = AA  length = 246
FEATURE                 Location/Qualifiers
REGION                  1..246
```

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```
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..246
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 120
DIVMTQSPAF LSVTPGEKVT ITCRASQTIS DYLHWYQQKP DQAPKLLIKY ASQSISGIPS  60
RFSGSGSGTD FTFTISSLEA EDAATYYCQD GHSFPPTFGQ GTKLEIKGGG GSGGGGSGGG  120
GSGGGGSQVQ LQQSGAEVIK PGASVKLSCK ASGYTFSSYW MHWVRQAPGQ GLEWIGEINP  180
GNGHTNYNEK FKSRATLTGD TSTSTVYMEL SSLRSEDTAV YYCARSFTTA RAFAYWGQGT  240
LVTVSS                                                             246

SEQ ID NO: 121          moltype = AA  length = 247
FEATURE                 Location/Qualifiers
REGION                  1..247
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..247
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 121
EVQLVESGGG LVQPGGSLRL SCAASGFTFS DYWMSWVRQA PGKGLEWVAD IKNDGSYTNY  60
APSLTNRFTI SRDNAKNSLY LQMNSLRAED TAVYYCAREL TGTWGQGTMV TVSSGGGGSG  120
GGGSGGGGSG GGGSDIVMTQ SPDSLAVSLG ERATINCKSS QSLLSSGNQK NYLAWYQQKP  180
GQPPKLLIYY ASTRQSGVPD RFSGSGSGTD FTLTISSLQA EDVAVYYCLQ YDRYPFTFGQ  240
GTKLEIK                                                            247

SEQ ID NO: 122          moltype = AA  length = 247
FEATURE                 Location/Qualifiers
REGION                  1..247
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..247
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 122
DIVMTQSPDS LAVSLGERAT INCKSSQSLL SSGNQKNYLA WYQQKPGQPP KLLIYYASTR  60
QSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCLQYDRY PFTFGQGTKL EIKGGGGSGG  120
GGSGGGGSGG GGSEVQLVES GGGLVQPGGS LRLSCAASGF TFSDYWMSWV RQAPGKGLEW  180
VADIKNDGSY TNYAPSLTNR FTISRDNAKN SLYLQMNSLR AEDTAVYYCA RELTGTWGQG  240
TMVTVSS                                                            247

SEQ ID NO: 123          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
REGION                  1..328
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..328
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 123
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPELLGGP  120
SVFLFPPKPK DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS  180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM  240
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ  300
QGNVFSCSVM HEALHNHYTQ KSLSLSPG                                     328

SEQ ID NO: 124          moltype = AA  length = 325
FEATURE                 Location/Qualifiers
REGION                  1..325
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..325
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 124
STKGPSVFPL APCSRSTSES TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTKTY TCNVDHKPSN TKVDKRVESK YGPPCPCPA PEFLGGPSVF  120
LFPPKPKDTL MISRTPEVTC VVVDVSQEDP EVQFNWYVDG VEVHNAKTKP REEQFNSTYR  180
VVSVLTVLHQ DWLNGKEYKC KVSNKGLPSS IEKTISKAKG QPREPQVYTL PPSQEEMTKN  240
QVSLTCLVKG FYPSDIAVEW ESNGQPENNY KTTPPVLDSD GSFFLYSKLT VDKSRWQEGN  300
VFSCSVMHEA LHNHYTQKSL SLSLG                                        325

SEQ ID NO: 125          moltype = AA  length = 328
FEATURE                 Location/Qualifiers
```

-continued

```
REGION                 1..328
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 125
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPEAAGGP  120
SVFLFPPKPK DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS  180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL GAPIEKTISK AKGQPREPQV YTLPPSREEM  240
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ  300
QGNVFSCSVM HEALHNHYTQ KSLSLSPG                                   328

SEQ ID NO: 126         moltype = AA  length = 322
FEATURE                Location/Qualifiers
REGION                 1..322
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..322
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 126
KTTPPSVYPL APGSAAQTNS MVTLGCLVKG YFPEPVTVTW NSGSLSSGVH TFPAVLQSDL  60
YTLSSSVTVP SSTWPSETVT CNVAHPASST KVDKKIVPRD CGCKPCICTV PEVSSVFIFP  120
PKPKDVLTIT LTPKVTCVVV DISKDDPEVQ FSWFVDDVEV HTAQTQPREE QFNSTFRSVS  180
ELPIMHQDWL NGKEFKCRVN SAAFPAPIEK TISKTKGRPK APQVYTIPPP KEQMAKDKVS  240
LTCMITDFFP EDITVEWQWN GQPAENYKNT QPIMDTDGSY FVYSKLNVQK SNWEAGNTFT  300
CSVLHEGLHN HHTEKSLSHS PG                                         322

SEQ ID NO: 127         moltype = AA  length = 328
FEATURE                Location/Qualifiers
REGION                 1..328
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 127
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPELLGGP  120
SVFLFPPKPK DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS  180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM  240
TKNQVSLSCA VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLVS KLTVDKSRWQ  300
QGNVFSCSVM HEALHNHYTQ KSLSLSPG                                   328

SEQ ID NO: 128         moltype = AA  length = 328
FEATURE                Location/Qualifiers
REGION                 1..328
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 128
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSN TKVDKRVEPK SCDKTHTCPP CPAPELLGGP  120
SVFLFPPKPK DTLMISRTPE VTCVVVDVSH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS  180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM  240
TKNQVSLWCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ  300
QGNVFSCSVM HEALHNHYTQ KSLSLSPG                                   328

SEQ ID NO: 129         moltype = AA  length = 325
FEATURE                Location/Qualifiers
REGION                 1..325
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..325
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 129
STKGPSVFPL APCSRSTSES TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTKTY TCNVDHKPSN TKVDKRVESK YGPPCPPCPA PEFLGGPSVF  120
LFPPKPKDTL MISRTPEVTC VVVDVSQEDP EVQFNWYVDG VEVHNAKTKP REEQFNSTYR  180
VVSVLTVLHQ DWLNGKEYKC KVSNKGLPSS IEKTISKAKG QPREPQVYTL PPSQEEMTKN  240
QVSLSCAVKG FYPSDIAVEW ESNGQPENNY KTTPPVLDSD GSFFLVSKLT VDKSRWQEGN  300
```

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VFSCSVMHEA LHNHYTQKSL SLSLG                                   325

SEQ ID NO: 130         moltype = AA  length = 325
FEATURE                Location/Qualifiers
REGION                 1..325
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..325
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 130
STKGPSVFPL APCSRSTSES TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTKTY TCNVDHKPSN TKVDKRVESK YGPPCPPCPA PEFLGGPSVF  120
LFPPKPKDTL MISRTPEVTC VVVDVSQEDP EVQFNWYVDG VEVHNAKTKP REEQFNSTYR  180
VVSVLTVLHQ DWLNGKEYKC KVSNKGLPSS IEKTISKAKG QPREPQVYTL PPSQEEMTKN  240
QVSLWCLVKG FYPSDIAVEW ESNGQPENNY KTTPPVLDSD GSFFLYSKLT VDKSRWQEGN  300
VFSCSVMHEA LHNHYTQKSL SLSLG                                   325

SEQ ID NO: 131         moltype = AA  length = 328
FEATURE                Location/Qualifiers
REGION                 1..328
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 131
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSD TKVDKKVEPK SCDKTHTCPP CPAPPVAGPS  120
VFLFPPKPKD TLMISRTPEV TCVVVDVKHE DPEVKFNWYV DGVEVHNAKT KPREEQYNST  180
YRVVSVLTVL HQDWLNGKEY KCKVSNKALP APIEKTISKA KGQPREPQVY TLPPSREQMT  240
KNQVKLTCLV KGFYPSDIAV EWESNGQPEN NYKTTPPVLD SDGSFFLYSK LTVDKSRWQQ  300
GNVFSCSVMH EALHNHYTQK SLSLSPGK                                328

SEQ ID NO: 132         moltype = AA  length = 328
FEATURE                Location/Qualifiers
REGION                 1..328
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..328
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 132
STKGPSVFPL APSSKSTSGG TAALGCLVKD YFPEPVTVSW NSGALTSGVH TFPAVLQSSG  60
LYSLSSVVTV PSSSLGTQTY ICNVNHKPSD TKVDKKVEPK SCDKTHTCPP CPAPPVAGPS  120
VFLFPPKPKD TLMISRTPEV TCVVVDVKHE DPEVKFNWYV DGVEVHNAKT KPREEEYNST  180
YRVVSVLTVL HQDWLNGKEY KCKVSNKALP APIEKTISKA KGQPREPQVY TLPPSREEMT  240
KNQVSLTCDV SGFYPSDIAV EWESDGQPEN NYKTTPPVLD SDGSFFLYSK LTVDKSRWEQ  300
GDVFSCSVMH EALHNHYTQK SLSLSPGK                                328

SEQ ID NO: 133         moltype = AA  length = 125
FEATURE                Location/Qualifiers
REGION                 1..125
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..125
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 133
EVQLVESGGG LVQPKGSLKL SCAASGFTFN TYAMNWVRQA PGKGLEWVAR IRSKYNNYAT  60
YYADSVKDRF TISRDDSQSI LYLQMNNLKT EDTAMYYCVR HGNFGNSYVS WFAYWGQGTL  120
VTVSA                                                         125

SEQ ID NO: 134         moltype = AA  length = 109
FEATURE                Location/Qualifiers
REGION                 1..109
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..109
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 134
QAVVTQESAL TTSPGETVTL TCRSSTGAVT TSNYANWVQE KPDHLFTGLI GGTNKRAPGV  60
PARFSGSLIG DKAALTITGA QTEDEAIYFC ALWYSNLWVF GGGTKLTVL         109

SEQ ID NO: 135         moltype = AA  length = 125
```

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```
FEATURE                Location/Qualifiers
REGION                 1..125
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..125
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 135
EVQLVESGGG LVQPGGSLKL SCAASGFTFN KYAMNWVRQA PGKGLEWVAR IRSKYNNYAT  60
YYADSVKDRF TISRDDSKNT AYLQMNNLKT EDTAVYYCVR HGNFGNSYIS YWAYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 136         moltype = AA  length = 109
FEATURE                Location/Qualifiers
REGION                 1..109
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..109
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 136
QTVVTQEPSL TVSPGGTVTL TCGSSTGAVT SGNYPNWVQQ KPGQAPRGLI GGTKFLAPGT  60
PARFSGSLLG GKAALTLSGV QPEDEAEYYC VLWYSNRWVF GGGTKLTVL             109

SEQ ID NO: 137         moltype = AA  length = 125
FEATURE                Location/Qualifiers
REGION                 1..125
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..125
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 137
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNS LYLQMNSLKT EDTAVYYCVR HGNFGNSYVS WFAYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 138         moltype = AA  length = 109
FEATURE                Location/Qualifiers
REGION                 1..109
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..109
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 138
QAVVTQEPSL TVSPGGTVTL TCRSSTGAVT TSNYANWVQQ KPGQAPRGLI GGTNKRAPWT  60
PARFSGSLLG GKAALTITGA QAEDEADYYC ALWYSNLWVF GGGTKLTVL             109

SEQ ID NO: 139         moltype = AA  length = 125
FEATURE                Location/Qualifiers
REGION                 1..125
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..125
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 139
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT  60
YYADSVKDRF TISRDDSKNS LYLQMNSLKT EDTAVYYCVR HGNFGNSYVS WFAYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 140         moltype = AA  length = 109
FEATURE                Location/Qualifiers
REGION                 1..109
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..109
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 140
QAVVTQEPSL TVSPGGTVTL TCRSSTGAVT TSNYANWVQQ KPGQAPRGLI GGTNKRAPWT  60
PARFSGSLLG GKAALTITGA QAEDEADYYC ALWYSNLWVF GGGTKLTVL             109

SEQ ID NO: 141         moltype = AA  length = 125
FEATURE                Location/Qualifiers
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```
REGION                  1..125
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..125
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 141
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL  120
VTVSS                                                             125

SEQ ID NO: 142          moltype = AA  length = 109
FEATURE                 Location/Qualifiers
REGION                  1..109
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..109
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 142
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV  60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL             109

SEQ ID NO: 143          moltype = AA  length = 125
FEATURE                 Location/Qualifiers
REGION                  1..125
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..125
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 143
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL  120
VTVSS                                                             125

SEQ ID NO: 144          moltype = AA  length = 109
FEATURE                 Location/Qualifiers
REGION                  1..109
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..109
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 144
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV  60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL             109

SEQ ID NO: 145          moltype = AA  length = 125
FEATURE                 Location/Qualifiers
REGION                  1..125
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..125
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 145
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKANNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL  120
VTVSS                                                             125

SEQ ID NO: 146          moltype = AA  length = 109
FEATURE                 Location/Qualifiers
REGION                  1..109
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..109
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 146
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV  60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL             109

SEQ ID NO: 147          moltype = AA  length = 125
FEATURE                 Location/Qualifiers
REGION                  1..125
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```
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..125
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 147
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFDYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 148           moltype = AA  length = 109
FEATURE                  Location/Qualifiers
REGION                   1..109
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..109
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 148
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV  60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL              109

SEQ ID NO: 149           moltype = AA  length = 125
FEATURE                  Location/Qualifiers
REGION                   1..125
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..125
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 149
EVQLVESGGG LVQPGGSLRL SCAASGFTFN TYAMNWVRQA PGKGLEWVAR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGNSYVS YFAYWGQGTT  120
VTVSS                                                              125

SEQ ID NO: 150           moltype = AA  length = 110
FEATURE                  Location/Qualifiers
REGION                   1..110
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..110
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 150
EIVVTQSPAT LSVSPGERAT LSCRSSTGAV TESNYANWVQ EKPGQAFRGL IGGANKRAPG  60
VPARFSGSLS GDEATLTISS LQSEDFAVYY CALFYSNTWV FGQGTKLEIK             110

SEQ ID NO: 151           moltype = AA  length = 125
FEATURE                  Location/Qualifiers
REGION                   1..125
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..125
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 151
EVQLLESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVSR IRSKYNNYAT  60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGNSYVS WFAYWGQGTL  120
VTVSS                                                              125

SEQ ID NO: 152           moltype = AA  length = 109
FEATURE                  Location/Qualifiers
REGION                   1..109
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..109
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 152
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQE KPGQAFRGLI GGTNKRAPGT  60
PARFSGSLLG GKAALTLSGA QPEDEAEYYC ALWYSNLWVF GGGTKLTVL              109

SEQ ID NO: 153           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
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```
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 153
EVQLVQSGAE VKKPGASVKV SCKASGYTFT NYYIHWVRQA PGQGLEWIGW IYPGDGNTKY  60
NEKFKGRATL TADTSTSTAY LELSSLRSED TAVYYCARDS YSNYYFDYWG QGTLVTVSS   119

SEQ ID NO: 154           moltype = AA  length = 112
FEATURE                  Location/Qualifiers
REGION                   1..112
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..112
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 154
DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSRTRKNYLA WYQQKPGQPP KLLIYWASTR  60
ESGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCTQSFIL RTFGQGTKVE IK          112

SEQ ID NO: 155           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 155
EVQLVQSGAE VKKPGASVKV SCKASGFTFT SYYIHWVRQA PGQGLEWIGW IYPENDNTKY  60
NEKFKDRVTI TADTSTSTAY LELSSLRSED TAVYYCARDG YSRYYFDYWG QGTLVTVSS   119

SEQ ID NO: 156           moltype = AA  length = 112
FEATURE                  Location/Qualifiers
REGION                   1..112
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..112
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 156
DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSRTRKNYLA WYQQKPGQSP KLLIYWTSTR  60
KSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCKQSFIL RTFGQGTKVE IK          112

SEQ ID NO: 157           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 157
EVQLVQSGAE VKKPGASVKV SCKASGFTFT SYYIHWVRQA PGQGLEWIGW IYPENDNTKY  60
NEKFKDRVTI TADTSTSTAY LELSSLRSED TAVYYCARDG YSRYYFDYWG QGTLVTVSS   119

SEQ ID NO: 158           moltype = AA  length = 112
FEATURE                  Location/Qualifiers
REGION                   1..112
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..112
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 158
DIVMTQSPDS LAVSLGERAT INCKSSQSLL NSRTRKNYLA WYQQKPGQSP KLLIYWTSTR  60
KSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCTQSFIL RTFGQGTKVE IK          112

SEQ ID NO: 159           moltype = AA  length = 121
FEATURE                  Location/Qualifiers
REGION                   1..121
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..121
                         mol_type = protein
                         organism = synthetic construct
```

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SEQUENCE: 159
QVQLVQSGPE VKKPGSSVKV SCKASGYTFS RSTMHWVRQA PGQGLEWIGY INPSSAYTNY 60
NQKFKDRVTI TADKSTSTAY MELSSLRSED TAVYYCARPQ VHYDYNGFPY WGQGTLVTVS 120
S 121

SEQ ID NO: 160 moltype = AA length = 106
FEATURE Location/Qualifiers
REGION 1..106
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..106
mol_type = protein
organism = synthetic construct
SEQUENCE: 160
DIQMTQSPST LSASVGDRVT MTCSASSSVS YMNWYQQKPG KAPKRWIYDS SKLASGVPSR 60
FSGSGSGTDY TLTISSLQPD DFATYYCQQW SRNPPTFGGG TKVEIK 106

SEQ ID NO: 161 moltype = AA length = 119
FEATURE Location/Qualifiers
REGION 1..119
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..119
mol_type = protein
organism = synthetic construct
SEQUENCE: 161
QVQLQQSGAE LARPGASVKM SCKASGYTFT RYTMHWVKQR PGQGLEWIGY INPSRGYTNY 60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYCLDYWG QGTTLTVSS 119

SEQ ID NO: 162 moltype = AA length = 106
FEATURE Location/Qualifiers
REGION 1..106
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..106
mol_type = protein
organism = synthetic construct
SEQUENCE: 162
QIVLTQSPAI MSASPGEKVT MTCSASSSVS YMNWYQQKSG TSPKRWIYDT SKLASGVPAH 60
FRGSGSGTSY SLTISGMEAE DAATYYCQQW SSNPFTFGSG TKLEIN 106

SEQ ID NO: 163 moltype = AA length = 119
FEATURE Location/Qualifiers
REGION 1..119
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..119
mol_type = protein
organism = synthetic construct
SEQUENCE: 163
QVQLQQSGAE LARPGASVKM SCKASGYTFT RYTMHWVKQR PGQGLEWIGY INPSRGYTNY 60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYSLDYWG QGTTLTVSS 119

SEQ ID NO: 164 moltype = AA length = 106
FEATURE Location/Qualifiers
REGION 1..106
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..106
mol_type = protein
organism = synthetic construct
SEQUENCE: 164
QIVLTQSPAI MSASPGEKVT MTCSASSSVS YMNWYQQKSG TSPKRWIYDT SKLASGVPAH 60
FRGSGSGTSY SLTISGMEAE DAATYYCQQW SSNPFTFGSG TKLEIN 106

SEQ ID NO: 165 moltype = AA length = 119
FEATURE Location/Qualifiers
REGION 1..119
note = Description of Artificial Sequence: Synthetic polypeptide
source 1..119
mol_type = protein
organism = synthetic construct
SEQUENCE: 165
QVQLQQSGAE LARPGASVKM SCKASGYTFT RYTMHWVKQR PGQCLEWIGY INPSRGYTNY 60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYCLDYWG QGTTLTVSS 119

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SEQ ID NO: 166          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 166
QIVLTQSPAI MSASPGEKVT MTCSASSSVS YMNWYQQKSG TSPKRWIYDT SKLASGVPAH  60
FRGSGSGTSY SLTISGMEAE DAATYYCQQW SSNPFTFGCG TKLEIN                 106

SEQ ID NO: 167          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 167
QVQLQQSGAE LARPGASVKM SCKASGYTFT RYTMHWVKQR PGQCLEWIGY INPSRGYTNY  60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYSLDYWG QGTTLTVSS   119

SEQ ID NO: 168          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 168
QIVLTQSPAI MSASPGEKVT MTCSASSSVS YMNWYQQKSG TSPKRWIYDT SKLASGVPAH  60
FRGSGSGTSY SLTISGMEAE DAATYYCQQW SSNPFTFGCG TKLEIN                 106

SEQ ID NO: 169          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 169
QVQLQQSGAE LARPGASVKM SCKASGYTFT RYTMHWVKQR PGQGLEWIGY INPSRGYTNY  60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYCLDYWG QGTTLAVSS   119

SEQ ID NO: 170          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 170
QIVLTCSPAI MSASPGEKVT MTCRASSSVS YMNWYQQKSG TSPKRWIYDT SKVASGVPYR  60
FSGSGSGTSY SLTISSMEAE DAATYYCQQW SSNPLTFGSG TKLEIN                 106

SEQ ID NO: 171          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 171
DIKLQQSGAE LARPGASVKM SCKTSGYTFT RYTMHWVKQR PGQGLEWIGY INPSRGYTNY  60
NQKFKDKATL TTDKSSSTAY MQLSSLTSED SAVYYCARYY DDHYCLDYWG QGTTLTVSS   119

SEQ ID NO: 172          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
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                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..106
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 172
DIQLTQSPAI MSASPGEKVT MTCRASSSVS YMNWYQQKSG TSPKRWIYDT SKVASGVPYR  60
FSGSGSGTSY SLTISSMEAE DAATYYCQQW SSNPLTFGAG TKLELK                 106

SEQ ID NO: 173           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 173
DVQLVQSGAE VKKPGASVKV SCKASGYTFT RYTMHWVRQA PGQGLEWIGY INPSRGYTNY  60
ADSVKGRFTI TTDKSTSTAY MELSSLRSED TATYYCARYY DDHYCLDYWG QGTTVTVSS   119

SEQ ID NO: 174           moltype = AA  length = 106
FEATURE                  Location/Qualifiers
REGION                   1..106
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..106
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 174
DIVLTQSPAT LSLSPGERAT LSCRASQSVS YMNWYQQKPG KAPKRWIYDT SKVASGVPAR  60
FSGSGSGTDY SLTINSLEAE DAATYYCQQW SSNPLTFGGG TKVEIK                 106

SEQ ID NO: 175           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 175
QVQLVQSGGG VVQPGRSLRL SCKASGYTFT RYTMHWVRQA PGKGLEWIGY INPSRGYTNY  60
NQKVKDRFTI SRDNSKNTAF LQMDSLRPED TGVYFCARYY DDHYCLDYWG QGTPVTVSS   119

SEQ ID NO: 176           moltype = AA  length = 106
FEATURE                  Location/Qualifiers
REGION                   1..106
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..106
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 176
DIQMTQSPSS LSASVGDRVT ITCSASSSVS YMNWYQQTPG KAPKRWIYDT SKLASGVPSR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW SSNPFTFGQG TKLQIT                 106

SEQ ID NO: 177           moltype = AA  length = 119
FEATURE                  Location/Qualifiers
REGION                   1..119
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..119
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 177
EVQLVESGGG LVQPGGSLRL SCAASGYTFT RYTMHWVRQA PGKGLEWIGY INPSRGYTYY  60
ADSVKGRFTL STDKSKNTAY LQMSSLRAED TAVYYCARYY DDHYCLDYWG QGTLVTVSS   119

SEQ ID NO: 178           moltype = AA  length = 106
FEATURE                  Location/Qualifiers
REGION                   1..106
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..106
                         mol_type = protein
```

-continued

```
                        organism = synthetic construct
SEQUENCE: 178
DIQLTQSPSS LSASVGDRVT ITCRASSSVS YVAWYQQKPG KAPKRWIYDT SKLYSGVPSR  60
FSGSGSGTDY TLTISSLQPE DFATYYCQQW SSNPPTFGQG TKVEIK                106

SEQ ID NO: 179          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 179
QVQLVQSGGG VVQPGRSLRL SCKASGYTFT RYTMHWVRQA PGKGLEWIGY INPSRGYTNY  60
NQKFKDRFTI SRDNSKNTAF LQMDSLRPED TGVYFCARYY DDHYSLDYWG QGTPVTVSS   119

SEQ ID NO: 180          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 180
DIQMTQSPSS LSASVGDRVT ITCSASSSVS YMNWYQQTPG KAPKRWIYDT SKLASGVPSR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW SSNPFTFGQG TKLQIT                106

SEQ ID NO: 181          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 181
QVQLVQSGGG VVQPGRSLRL SCKASGYTFT RYTMHWVRQA PGKCLEWIGY INPSRGYTNY  60
NQKFKDRFTI SRDNSKNTAF LQMDSLRPED TGVYFCARYY DDHYSLDYWG QGTPVTVSS   119

SEQ ID NO: 182          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 182
DIQMTQSPSS LSASVGDRVT ITCSASSSVS YMNWYQQTPG KAPKRWIYDT SKLASGVPSR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW SSNPFTFGCG TKLQIT                106

SEQ ID NO: 183          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 183
QVQLVQSGGG VVQPGRSLRL SCKASGYTFT RYTMHWVRQA PGKCLEWIGY INPSRGYTNY  60
NQKFKDRFTI SRDNSKNTAF LQMDSLRPED TGVYFCARYY DDHYCLDYWG QGTPVTVSS   119

SEQ ID NO: 184          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 184
DIQMTQSPSS LSASVGDRVT ITCSASSSVS YMNWYQQTPG KAPKRWIYDT SKLASGVPSR  60
FSGSGSGTDY TFTISSLQPE DIATYYCQQW SSNPFTFGCG TKLQIT                106
```

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```
SEQ ID NO: 185          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
REGION                  1..122
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..122
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 185
EVQLQQSGPE LVKPGASMKI SCKASGYSFT GYTMNWVKQS HGKNLEWMGL INPYKGVSTY  60
NQKFKDKATL TVDKSSSTAY MELLSLTSED SAVYYCARSG YYGDSDWYFD VWGQGTTLTV  120
FS                                                                122

SEQ ID NO: 186          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 186
DIQMTQTTSS LSASLGDRVT ISCRASQDIR NYLNWYQQKP DGTVKLLIYY TSRLHSGVPS  60
KFSGSGSGTD YSLTISNLEQ EDIATYFCQQ GNTLPWTFAG GTKLEIK               107

SEQ ID NO: 187          moltype = AA  length = 122
FEATURE                 Location/Qualifiers
REGION                  1..122
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..122
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 187
EVQLVESGGG LVQPGGSLRL SCAASGYSFT GYTMNWVRQA PGKGLEWVAL INPYKGVSTY  60
NQKFKDRFTI SVDKSKNTAY LQMNSLRAED TAVYYCARSG YYGDSDWYFD VWGQGTLVTV  120
SS                                                                122

SEQ ID NO: 188          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 188
AIQMTQSPSS LSASVGDRVT ITCRASQDIR NYLNWYQQKP GKAPKLLIYY TSRLESGVPS  60
RFSGSGSGTD YTLTISSLQP EDFATYYCQQ GNTLPWTFGQ GTKVEIK               107

SEQ ID NO: 189          moltype = AA  length = 128
FEATURE                 Location/Qualifiers
REGION                  1..128
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..128
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 189
QLQLQESGPG LVKPSQTLSL TCSVSGGSIR SGGHYWSWIR QHPGKGLEWI GYIHHSGSTY  60
YNPSLKSRVI ISVDTSKNQF SLKLRSVTAA DTAIYYCARW RHDIFTTYPY YYYGMDVWGQ  120
GTLVTVSS                                                          128

SEQ ID NO: 190          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 190
EIVMTQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKP GQAPRLLIYG ASTRATGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNNWPWTFGQ GTKVEIK               107
```

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```
SEQ ID NO: 191          moltype = AA  length = 128
FEATURE                 Location/Qualifiers
REGION                  1..128
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..128
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 191
QLQLQESGPG LVKPSQTLSL TCSVSGGSIR SGGHYWSWIR QHPGKGLEWI GYISYSGSTY  60
YNPSLKSRVT ISVDTSKNQF SLKLSSVTAA DTAVYYCARW RHDILTAYPY YYYGMDVWGQ  120
GTLVTVSS                                                           128

SEQ ID NO: 192          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 192
EIVMTQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKP GQAPRLLIYG ASTRATGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNNWPWTFGQ GTKVEIK                107

SEQ ID NO: 193          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
REGION                  1..123
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..123
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 193
EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYAMHWVRQA PGKGLEWVSG ISWNSGSIGY  60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TALYYCAKDS RGYGDYRLGG AYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 194          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 194
EIVMTQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKP GQAPRLLIYG ASTRATGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNNWPWTFGQ GTKVEIK                107

SEQ ID NO: 195          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
REGION                  1..123
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..123
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 195
EVQLVESGGG LVQPGRSLRL SCAASGFTFD NYAMHWVRQA PGKGLEWVSG ISWNSGSIGY  60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TALYYCAKDS RGYGDYSRGG AYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 196          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 196
EIVMTQSPAT LSVSPGERAT LSCRASQSVS SNLAWYQQKP GQAPRLLIYG ASTRATGIPA  60
RFSGSGSGTE FTLTISSLQS EDFAVYYCQQ YNNWPWTFGQ GTKVEIK                107

SEQ ID NO: 197          moltype = AA  length = 124
```

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```
FEATURE                Location/Qualifiers
REGION                 1..124
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..124
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 197
EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYSMHWVRQA PGKGLEWVSG ISWNSGSKGY  60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TALYYCAKYG SGYGKFYHYG LDVWGQGTTV  120
TVSS                                                               124

SEQ ID NO: 198         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 198
DIQMTQSPSS LSASVGDRVT ITCRASQSIS SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SYSTPPITFG QGTRLEIK               108

SEQ ID NO: 199         moltype = AA  length = 124
FEATURE                Location/Qualifiers
REGION                 1..124
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..124
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 199
EVQLVESGGG LVQPGRSLRL SCAASGFTFD DYSMHWVRQA PGKGLEWVSG ISWNSGSIGY  60
ADSVKGRFTI SRDNAKNSLY LQMNSLRAED TALYYCAKDG SGYGKFYYYG MDVWGQGTTV  120
TVSS                                                               124

SEQ ID NO: 200         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 200
DIQMTQSPSS LSASVGDRVT ITCRASQSIS SYLNWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ SYSTPPITFG QGTRLEIK               108

SEQ ID NO: 201         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 201
GGSFSGYYWS                                                         10

SEQ ID NO: 202         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 202
GGSFSDYYWS                                                         10

SEQ ID NO: 203         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 203
GGSFSDYYWS                                                    10

SEQ ID NO: 204          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 204
GGSFSDYYWS                                                    10

SEQ ID NO: 205          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 205
GGSFSDYYWS                                                    10

SEQ ID NO: 206          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 206
GGSFSDYYWS                                                    10

SEQ ID NO: 207          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 207
GGSFSDYYWS                                                    10

SEQ ID NO: 208          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 208
GGSFSDYYWS                                                    10

SEQ ID NO: 209          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 209
GGSFSDYYWS                                                    10

SEQ ID NO: 210          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 210
GGSFSDYYWS                                                    10

SEQ ID NO: 211          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
```

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```
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 211
GGSFSDYYWS                                                             10

SEQ ID NO: 212           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 212
GGSFSDYYWS                                                             10

SEQ ID NO: 213           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 213
GGSFSDYYWS                                                             10

SEQ ID NO: 214           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 214
GGSFSDYYWS                                                             10

SEQ ID NO: 215           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 215
GGSFSDYYWS                                                             10

SEQ ID NO: 216           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 216
GGSFSGYYWS                                                             10

SEQ ID NO: 217           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 217
GGSFSGYYWS                                                             10

SEQ ID NO: 218           moltype = AA  length = 10
FEATURE                  Location/Qualifiers
REGION                   1..10
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..10
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 218
GGSFSGYYWS                                                             10
```

-continued

```
SEQ ID NO: 219          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 219
GGSFSGYYWS                                                  10

SEQ ID NO: 220          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 220
GGSFSGYYWS                                                  10

SEQ ID NO: 221          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 221
GGSFSGYYWS                                                  10

SEQ ID NO: 222          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 222
GGSFSGYYWS                                                  10

SEQ ID NO: 223          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 223
GGSFSGYYWS                                                  10

SEQ ID NO: 224          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 224
GGSFSDYYWS                                                  10

SEQ ID NO: 225          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 225
GGSFSDYYWS                                                  10

SEQ ID NO: 226          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
```

-continued

```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 226
GGSFSDYYWS                                                10

SEQ ID NO: 227          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 227
GGSFSDYYWS                                                10

SEQ ID NO: 228          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 228
GGSFSDYYWS                                                10

SEQ ID NO: 229          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 229
GGSFSDYYWS                                                10

SEQ ID NO: 230          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 230
GGSFSDYYWS                                                10

SEQ ID NO: 231          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 231
GGSFSDYYWS                                                10

SEQ ID NO: 232          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 232
GGSFSDYYWS                                                10

SEQ ID NO: 233          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 233
GGSFSDYYWS                                                10

SEQ ID NO: 234          moltype = AA  length = 10
```

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```
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 234
GGSFSDYYWS                                                      10

SEQ ID NO: 235         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 235
GGSFSDYYWS                                                      10

SEQ ID NO: 236         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 236
GGSFSDYYWS                                                      10

SEQ ID NO: 237         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 237
GGSFSDYYWS                                                      10

SEQ ID NO: 238         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 238
GGSFSGYYWS                                                      10

SEQ ID NO: 239         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 239
GGSFSGYYWS                                                      10

SEQ ID NO: 240         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 240
GGSFSGYYWS                                                      10

SEQ ID NO: 241         moltype = AA  length = 10
FEATURE                Location/Qualifiers
REGION                 1..10
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..10
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 241
GGSFSGYYWS                                                  10

SEQ ID NO: 242          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 242
GGSFSGYYWS                                                  10

SEQ ID NO: 243          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 243
GGSFSGYYWS                                                  10

SEQ ID NO: 244          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 244
GGSFSGYYWS                                                  10

SEQ ID NO: 245          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 245
GGSFSGYYWS                                                  10

SEQ ID NO: 246          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 246
GGSFSGYYWS                                                  10

SEQ ID NO: 247          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 247
GGSFSGYYWS                                                  10

SEQ ID NO: 248          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 248
GGSFSGYYWS                                                  10

SEQ ID NO: 249          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
```

-continued

```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 249
GGSFSGYYWS                                                        10

SEQ ID NO: 250          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 250
GGSFSGYYWS                                                        10

SEQ ID NO: 251          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 251
GGSFSGYYWS                                                        10

SEQ ID NO: 252          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 252
GGSFSGYYWS                                                        10

SEQ ID NO: 253          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 253
GGSFSGYYWS                                                        10

SEQ ID NO: 254          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 254
GGSFSGYYWS                                                        10

SEQ ID NO: 255          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 255
GGSFSGYYWS                                                        10

SEQ ID NO: 256          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 256
GGSFSGYYWS                                                        10
```

-continued

```
SEQ ID NO: 257          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 257
GGSFSGYYWS                                                          10

SEQ ID NO: 258          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 258
GGSFSGYYWS                                                          10

SEQ ID NO: 259          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 259
GGSFSGYYWS                                                          10

SEQ ID NO: 260          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 260
GGSFSGYYWS                                                          10

SEQ ID NO: 261          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 261
GGSFSGYYWS                                                          10

SEQ ID NO: 262          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 262
GGSFSGYYWS                                                          10

SEQ ID NO: 263          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 263
GGSFSGYYWS                                                          10

SEQ ID NO: 264          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
```

-continued

```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 264
GGSFSGYYWS                                                     10

SEQ ID NO: 265          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 265
GGSFSGYYWS                                                     10

SEQ ID NO: 266          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 266
EVNHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 267          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 267
EVNHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 268          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 268
EVNHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 269          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 269
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 270          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 270
EVNHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 271          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 271
EVNHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 272          moltype = AA  length = 19
```

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```
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 272
EVNHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 273          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 273
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 274          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 274
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 275          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 275
EVNHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 276          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 276
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 277          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 277
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 278          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 278
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 279          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
```

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```
SEQUENCE: 279
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 280          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 280
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 281          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 281
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 282          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 282
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 283          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 283
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 284          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 284
EINHRGSINY NNYNPSLKS                                                    19

SEQ ID NO: 285          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 285
EINHAGSINY NNYNPSLKS                                                    19

SEQ ID NO: 286          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 286
EINHAGSINY NNYNPSLKS                                                    19

SEQ ID NO: 287          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 287
EINHQGSINY NNYNPSLKS                                              19

SEQ ID NO: 288          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 288
EINHQGSINY NNYNPSLKS                                              19

SEQ ID NO: 289          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 289
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 290          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 290
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 291          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 291
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 292          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 292
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 293          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 293
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 294          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 294
EINHRGSINY NNYNPSLKS                                              19
```

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```
SEQ ID NO: 295          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 295
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 296          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 296
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 297          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 297
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 298          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 298
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 299          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 299
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 300          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 300
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 301          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 301
EINHRGSINY NNYNPSLKS                                              19

SEQ ID NO: 302          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
```

-continued

```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 302
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 303          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 303
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 304          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 304
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 305          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 305
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 306          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 306
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 307          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 307
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 308          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 308
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 309          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 309
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 310          moltype = AA  length = 19
```

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```
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 310
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 311         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 311
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 312         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 312
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 313         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 313
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 314         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 314
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 315         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 315
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 316         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 316
EINHRGSINY NNYNPSLKS                                                19

SEQ ID NO: 317         moltype = AA  length = 19
FEATURE                Location/Qualifiers
REGION                 1..19
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..19
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 317
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 318          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 318
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 319          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 319
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 320          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 320
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 321          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 321
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 322          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 322
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 323          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 323
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 324          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 324
EINHRGSINY NNYNPSLKS                                           19

SEQ ID NO: 325          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 325
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 326          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 326
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 327          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 327
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 328          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 328
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 329          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 329
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 330          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 330
EINHRGSINY NNYNPSLKS                                               19

SEQ ID NO: 331          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 331
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 332          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 332
AKPLRPHCTN GVCYSGDAFD I                                            21
```

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```
SEQ ID NO: 333          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 333
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 334          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 334
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 335          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 335
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 336          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 336
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 337          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 337
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 338          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 338
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 339          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 339
AKPLRPHCTN GVCYSGDAFD I                                               21

SEQ ID NO: 340          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 340
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 341          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 341
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 342          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 342
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 343          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 343
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 344          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 344
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 345          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 345
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 346          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 346
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 347          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 347
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 348          moltype = AA  length = 21
```

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```
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 348
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 349         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 349
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 350         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 350
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 351         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 351
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 352         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 352
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 353         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 353
AKPLRPHCTN GVCYSGDAFD I                                                  21

SEQ ID NO: 354         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 354
AKPLRPHCIN GVCYSGDAFD I                                                  21

SEQ ID NO: 355         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 355
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 356          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 356
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 357          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 357
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 358          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 358
AKPLRPHCIN GVCYSGDAFD I                                            21

SEQ ID NO: 359          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 359
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 360          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 360
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 361          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 361
AKPLRPHCIN GVCYSGDAFD I                                            21

SEQ ID NO: 362          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 362
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 363          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 363
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 364          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 364
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 365          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 365
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 366          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 366
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 367          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 367
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 368          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 368
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 369          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 369
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 370          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 370
AKPLRPHCTN GVCYSGDAFD I                                              21
```

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SEQ ID NO: 371          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 371
AKPLRPHCTN GVCYSGDAFD I                                             21

SEQ ID NO: 372          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 372
AKPLRPHCIN GVCYSGDAFD I                                             21

SEQ ID NO: 373          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 373
AKPLRPHCTN GVCYSGDAFD I                                             21

SEQ ID NO: 374          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 374
AKPLRPHCTN GVCYSGDAFD I                                             21

SEQ ID NO: 375          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 375
AKPLRPHCIN GVCYSGDAFD I                                             21

SEQ ID NO: 376          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 376
AKPLRPHCTN GVCYSGDAFD I                                             21

SEQ ID NO: 377          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 377
AKPLRPHCIN GVCYSGDAFD I                                             21

SEQ ID NO: 378          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 378
AKPLRPHCIN GVCYSGDAFD I                                       21

SEQ ID NO: 379          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 379
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 380          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 380
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 381          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 381
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 382          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 382
AKPLRPHCIN GVCYSGDAFD I                                       21

SEQ ID NO: 383          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 383
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 384          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 384
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 385          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 385
AKPLRPHCTN GVCYSGDAFD I                                       21

SEQ ID NO: 386          moltype = AA  length = 21
```

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```
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 386
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 387          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 387
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 388          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 388
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 389          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 389
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 390          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 390
AKPLRPHCTN GVCYSGDAFD I                                              21

SEQ ID NO: 391          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 391
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 392          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 392
AKPLRPHCIN GVCYSGDAFD I                                              21

SEQ ID NO: 393          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
```

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```
SEQUENCE: 393
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 394         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 394
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 395         moltype = AA  length = 21
FEATURE                Location/Qualifiers
REGION                 1..21
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..21
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 395
AKPLRPHCTN GVCYSGDAFD I                                            21

SEQ ID NO: 396         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 396
GGNNIGSKNV H                                                       11

SEQ ID NO: 397         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 397
GGNNIGSKSV H                                                       11

SEQ ID NO: 398         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 398
GGNNIGSKNV H                                                       11

SEQ ID NO: 399         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 399
GGNNIGSKNV H                                                       11

SEQ ID NO: 400         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 400
GGNNIGSKNV H                                                       11

SEQ ID NO: 401         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
```

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```
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 401
GGNNIGSKNV H                                                                 11

SEQ ID NO: 402           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 402
GGNNIGSKNV H                                                                 11

SEQ ID NO: 403           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 403
GGNNIGSKNV H                                                                 11

SEQ ID NO: 404           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 404
GGNNIGSKNV H                                                                 11

SEQ ID NO: 405           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 405
GGNNIGSKNV H                                                                 11

SEQ ID NO: 406           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 406
GGNNIGSKNV H                                                                 11

SEQ ID NO: 407           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 407
GGNNIGSKNV H                                                                 11

SEQ ID NO: 408           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 408
GGNNIGSKNV H                                                                 11
```

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```
SEQ ID NO: 409          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 409
GGNNIGSKNV H                                                              11

SEQ ID NO: 410          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 410
GGNNIGSKNV H                                                              11

SEQ ID NO: 411          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 411
GGNNIGSKNV H                                                              11

SEQ ID NO: 412          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 412
GGNNIGSKNV H                                                              11

SEQ ID NO: 413          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 413
GGNNIGSKNV H                                                              11

SEQ ID NO: 414          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 414
GGNNIGSKNV H                                                              11

SEQ ID NO: 415          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 415
GGNNIGSKNV H                                                              11

SEQ ID NO: 416          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 416
GGNNIGSKNV H                                          11

SEQ ID NO: 417          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 417
GGNNIGSKNV H                                          11

SEQ ID NO: 418          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 418
GGNNIGSKNV H                                          11

SEQ ID NO: 419          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 419
RGNNIGSKNV H                                          11

SEQ ID NO: 420          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 420
RGNNIGYKNV H                                          11

SEQ ID NO: 421          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 421
RGNNIGSMNV H                                          11

SEQ ID NO: 422          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 422
RGNNIGSKNV H                                          11

SEQ ID NO: 423          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 423
GGNNIGYKNV H                                          11

SEQ ID NO: 424          moltype = AA  length = 11
```

-continued

```
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 424
GGNNIGYMNV H                                                         11

SEQ ID NO: 425          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 425
GGNNIGYKNV H                                                         11

SEQ ID NO: 426          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 426
GGNNIGSMNV H                                                         11

SEQ ID NO: 427          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 427
GGNNIGSMNV H                                                         11

SEQ ID NO: 428          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 428
GGNNIGSKNV H                                                         11

SEQ ID NO: 429          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 429
GGNNIGYKNV H                                                         11

SEQ ID NO: 430          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 430
GGNNIGSKNV H                                                         11

SEQ ID NO: 431          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
```

-continued

```
SEQUENCE: 431
GGNNIGSKNV H                                             11

SEQ ID NO: 432          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 432
GGNNIGYKNV H                                             11

SEQ ID NO: 433          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 433
RGNNIGSKNV H                                             11

SEQ ID NO: 434          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 434
RGNNIGYKNV H                                             11

SEQ ID NO: 435          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 435
RGNNIGSMNV H                                             11

SEQ ID NO: 436          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 436
RGNNIGSKNV H                                             11

SEQ ID NO: 437          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 437
GGNNIGYKNV H                                             11

SEQ ID NO: 438          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 438
GGNNIGYMNV H                                             11

SEQ ID NO: 439          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
```

-continued

```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 439
GGNNIGYKNV H                                                    11

SEQ ID NO: 440          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 440
GGNNIGSMNV H                                                    11

SEQ ID NO: 441          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 441
GGNNIGSMNV H                                                    11

SEQ ID NO: 442          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 442
GGNNIGSKNV H                                                    11

SEQ ID NO: 443          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 443
GGNNIGYKNV H                                                    11

SEQ ID NO: 444          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 444
GGNNIGSKNV H                                                    11

SEQ ID NO: 445          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 445
GGNNIGSKNV H                                                    11

SEQ ID NO: 446          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 446
GGNNIGYKNV H                                                    11
```

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```
SEQ ID NO: 447          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 447
RGNNIGSKNV H                                                               11

SEQ ID NO: 448          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 448
RGNNIGYKNV H                                                               11

SEQ ID NO: 449          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 449
RGNNIGSMNV H                                                               11

SEQ ID NO: 450          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 450
RGNNIGSKNV H                                                               11

SEQ ID NO: 451          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 451
GGNNIGYKNV H                                                               11

SEQ ID NO: 452          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 452
GGNNIGYMNV H                                                               11

SEQ ID NO: 453          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 453
GGNNIGYKNV H                                                               11

SEQ ID NO: 454          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
```

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```
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 454
GGNNIGSMNV H                                                11

SEQ ID NO: 455           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 455
GGNNIGSMNV H                                                11

SEQ ID NO: 456           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 456
GGNNIGSKNV H                                                11

SEQ ID NO: 457           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 457
GGNNIGYKNV H                                                11

SEQ ID NO: 458           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 458
GGNNIGSKNV H                                                11

SEQ ID NO: 459           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 459
GGNNIGSKNV H                                                11

SEQ ID NO: 460           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..11
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 460
GGNNIGYKNV H                                                11

SEQ ID NO: 461           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
REGION                   1..7
                         note = Description of Artificial Sequence: Synthetic peptide
source                   1..7
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 461
DDSDRPS                                                     7

SEQ ID NO: 462           moltype = AA  length = 7
```

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```
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 462
DDSDRPS                                                       7

SEQ ID NO: 463         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 463
DDSDRPS                                                       7

SEQ ID NO: 464         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 464
DDSDRPS                                                       7

SEQ ID NO: 465         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 465
DDSDRPS                                                       7

SEQ ID NO: 466         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 466
DDSDRPS                                                       7

SEQ ID NO: 467         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 467
DDSDRPS                                                       7

SEQ ID NO: 468         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 468
DDSDRPS                                                       7

SEQ ID NO: 469         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 469
DDSDRPS                                                                7

SEQ ID NO: 470          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 470
DDSDRPS                                                                7

SEQ ID NO: 471          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 471
DDSDRPS                                                                7

SEQ ID NO: 472          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 472
DDSDRPS                                                                7

SEQ ID NO: 473          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 473
DDSDRPS                                                                7

SEQ ID NO: 474          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 474
DDSDRPS                                                                7

SEQ ID NO: 475          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 475
DDSDRPS                                                                7

SEQ ID NO: 476          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 476
DDSDRPS                                                                7

SEQ ID NO: 477          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 477
DDSDRPS                                                                    7

SEQ ID NO: 478          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 478
DDSDRPS                                                                    7

SEQ ID NO: 479          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 479
DDSDRPS                                                                    7

SEQ ID NO: 480          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 480
DDSDRPS                                                                    7

SEQ ID NO: 481          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 481
DDSDRPS                                                                    7

SEQ ID NO: 482          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 482
DDSDRPS                                                                    7

SEQ ID NO: 483          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 483
DDSDRPS                                                                    7

SEQ ID NO: 484          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 484
DDSDRPS                                                                    7
```

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```
SEQ ID NO: 485          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 485
DDSDRPS                                                           7

SEQ ID NO: 486          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 486
DDSDRPS                                                           7

SEQ ID NO: 487          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 487
DDSDRPS                                                           7

SEQ ID NO: 488          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 488
DDSDRPS                                                           7

SEQ ID NO: 489          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 489
DDSDRPS                                                           7

SEQ ID NO: 490          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 490
DDSDRPS                                                           7

SEQ ID NO: 491          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 491
DDSDRPS                                                           7

SEQ ID NO: 492          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 492
DDSDRPS                                                                7

SEQ ID NO: 493          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 493
DDSDRPS                                                                7

SEQ ID NO: 494          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 494
DDSDRPS                                                                7

SEQ ID NO: 495          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 495
DDSDRPS                                                                7

SEQ ID NO: 496          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 496
DDSDRPS                                                                7

SEQ ID NO: 497          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 497
DDSDRPS                                                                7

SEQ ID NO: 498          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 498
DDSDRPS                                                                7

SEQ ID NO: 499          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 499
DDSDRPS                                                                7

SEQ ID NO: 500          moltype = AA  length = 7
```

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```
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 500
DDSDRPS                                                                    7

SEQ ID NO: 501         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 501
DDSDRPS                                                                    7

SEQ ID NO: 502         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 502
DDSDRPS                                                                    7

SEQ ID NO: 503         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 503
DDSDRPS                                                                    7

SEQ ID NO: 504         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 504
DDSDRPS                                                                    7

SEQ ID NO: 505         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 505
DDSDRPS                                                                    7

SEQ ID NO: 506         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 506
DDSDRPS                                                                    7

SEQ ID NO: 507         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..7
                       mol_type = protein
                       organism = synthetic construct
```

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```
SEQUENCE: 507
DDSDRPS                                                        7

SEQ ID NO: 508        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 508
DDSDRPS                                                        7

SEQ ID NO: 509        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 509
DDSDRPS                                                        7

SEQ ID NO: 510        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 510
DDSDRPS                                                        7

SEQ ID NO: 511        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 511
DDSDRPS                                                        7

SEQ ID NO: 512        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 512
DDSDRPS                                                        7

SEQ ID NO: 513        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 513
DDSDRPS                                                        7

SEQ ID NO: 514        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
                      note = Description of Artificial Sequence: Synthetic peptide
source                1..7
                      mol_type = protein
                      organism = synthetic construct
SEQUENCE: 514
DDSDRPS                                                        7

SEQ ID NO: 515        moltype = AA  length = 7
FEATURE               Location/Qualifiers
REGION                1..7
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 515
DDSDRPS                                                                    7

SEQ ID NO: 516          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 516
DDSDRPS                                                                    7

SEQ ID NO: 517          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 517
DDSDRPS                                                                    7

SEQ ID NO: 518          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 518
DDSDRPS                                                                    7

SEQ ID NO: 519          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 519
DDSDRPS                                                                    7

SEQ ID NO: 520          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 520
DDSDRPS                                                                    7

SEQ ID NO: 521          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 521
DDSDRPS                                                                    7

SEQ ID NO: 522          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 522
DDSDRPS                                                                    7
```

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```
SEQ ID NO: 523          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 523
DDSDRPS                                                                7

SEQ ID NO: 524          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 524
DDSDRPS                                                                7

SEQ ID NO: 525          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..7
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 525
DDSDRPS                                                                7

SEQ ID NO: 526          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 526
QVWDSSSDHL V                                                           11

SEQ ID NO: 527          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 527
QVWDSSSDHL V                                                           11

SEQ ID NO: 528          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 528
QVWDSSSDHV V                                                           11

SEQ ID NO: 529          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 529
QVWDSSSDHL V                                                           11

SEQ ID NO: 530          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
```

-continued

```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 530
QVWDSSSDHL V                                              11

SEQ ID NO: 531          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 531
QVWDSSSDHL V                                              11

SEQ ID NO: 532          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 532
QVWDSSSDHL V                                              11

SEQ ID NO: 533          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 533
QVWDSSSDHL V                                              11

SEQ ID NO: 534          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 534
QVWDSSSDHL V                                              11

SEQ ID NO: 535          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 535
QVWDSSSDHL V                                              11

SEQ ID NO: 536          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 536
QVWDSSSDHL V                                              11

SEQ ID NO: 537          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 537
QVWDSSSDHL V                                              11

SEQ ID NO: 538          moltype = AA  length = 11
```

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```
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 538
QVWDSSSDHV V                                                         11

SEQ ID NO: 539          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 539
QVWDSSSDHL V                                                         11

SEQ ID NO: 540          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 540
QVWDSSSDHV V                                                         11

SEQ ID NO: 541          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 541
QVWDSSSDHL V                                                         11

SEQ ID NO: 542          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 542
QVWDSSSDHV V                                                         11

SEQ ID NO: 543          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 543
QVWDSSSDHL V                                                         11

SEQ ID NO: 544          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 544
QVWDSSSDHV V                                                         11

SEQ ID NO: 545          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
```

-continued

```
SEQUENCE: 545
QVWDSSSDHV V                                                    11

SEQ ID NO: 546          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 546
QVWDSSSDHV V                                                    11

SEQ ID NO: 547          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 547
QVWDSSSDHV V                                                    11

SEQ ID NO: 548          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 548
QVWDSSSDHV V                                                    11

SEQ ID NO: 549          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 549
QVWDSSSDHV V                                                    11

SEQ ID NO: 550          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 550
QVWDSSSDHV V                                                    11

SEQ ID NO: 551          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 551
QVWDSSSDHV V                                                    11

SEQ ID NO: 552          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 552
QVWDHSSDHV V                                                    11

SEQ ID NO: 553          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 553
QVWDSSSDHV V                                                  11

SEQ ID NO: 554          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 554
QVWDSSSDHV V                                                  11

SEQ ID NO: 555          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 555
QVWDHSSDHV V                                                  11

SEQ ID NO: 556          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 556
QVWDSSSDHV V                                                  11

SEQ ID NO: 557          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 557
QVWDHSSDHV V                                                  11

SEQ ID NO: 558          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 558
QVWDHSSDHV V                                                  11

SEQ ID NO: 559          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 559
QVWDHSSDHV V                                                  11

SEQ ID NO: 560          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 560
QVWDSRSDHV V                                                  11
```

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```
SEQ ID NO: 561          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 561
QVWDSESDHV V                                                    11

SEQ ID NO: 562          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 562
QVWDSESDHV V                                                    11

SEQ ID NO: 563          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 563
QVWDSSSDHV V                                                    11

SEQ ID NO: 564          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 564
QVWDSSSDHV V                                                    11

SEQ ID NO: 565          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 565
QVWDSSSDHV V                                                    11

SEQ ID NO: 566          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 566
QVWDHSSDHV V                                                    11

SEQ ID NO: 567          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 567
QVWDSSSDHV V                                                    11

SEQ ID NO: 568          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
```

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```
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 568
QVWDSSSDHV V                                                11

SEQ ID NO: 569          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 569
QVWDHSSDHV V                                                11

SEQ ID NO: 570          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 570
QVWDSSSDHV V                                                11

SEQ ID NO: 571          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 571
QVWDHSSDHV V                                                11

SEQ ID NO: 572          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 572
QVWDHSSDHV V                                                11

SEQ ID NO: 573          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 573
QVWDHSSDHV V                                                11

SEQ ID NO: 574          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 574
QVWDSRSDHV V                                                11

SEQ ID NO: 575          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 575
QVWDSESDHV V                                                11

SEQ ID NO: 576          moltype = AA  length = 11
```

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```
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 576
QVWDSESDHV V                                                    11

SEQ ID NO: 577          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 577
QVWDSSSDHV V                                                    11

SEQ ID NO: 578          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 578
QVWDSSSDHV V                                                    11

SEQ ID NO: 579          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 579
QVWDSSSDHV V                                                    11

SEQ ID NO: 580          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 580
QVWDHSSDHV V                                                    11

SEQ ID NO: 581          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 581
QVWDSSSDHV V                                                    11

SEQ ID NO: 582          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 582
QVWDSSSDHV V                                                    11

SEQ ID NO: 583          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
```

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```
SEQUENCE: 583
QVWDHSSDHV V                                                   11

SEQ ID NO: 584         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 584
QVWDSSSDHV V                                                   11

SEQ ID NO: 585         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 585
QVWDHSSDHV V                                                   11

SEQ ID NO: 586         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 586
QVWDHSSDHV V                                                   11

SEQ ID NO: 587         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 587
QVWDHSSDHV V                                                   11

SEQ ID NO: 588         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 588
QVWDSRSDHV V                                                   11

SEQ ID NO: 589         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 589
QVWDSESDHV V                                                   11

SEQ ID NO: 590         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Artificial Sequence: Synthetic peptide
source                 1..11
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 590
QVWDSESDHV V                                                   11

SEQ ID NO: 591         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
```

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```
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 591
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 592          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 592
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 593          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 593
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 594          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 594
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 595          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 595
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 596          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 596
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVSS                                                         130

SEQ ID NO: 597          moltype = AA  length = 130
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FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 597
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 598         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 598
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 599         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 599
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLTSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVSS                                                        130

SEQ ID NO: 600         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 600
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE VNHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVSS                                                        130

SEQ ID NO: 601         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 601
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVSS                                                        130

SEQ ID NO: 602         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 602
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVSS                                                        130
```

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SEQ ID NO: 603          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 603
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 604          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 604
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 605          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 605
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 606          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 606
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 607          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 607
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 608          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 608
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
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NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 609          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 609
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 610          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 610
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHAGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 611          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 611
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHAGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 612          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 612
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHQGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 613          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 613
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHQGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 614          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
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SEQUENCE: 614
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 615         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 615
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 616         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 616
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 617         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 617
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 618         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 618
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 619         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 619
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 620         moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
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                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 620
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 621          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 621
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 622          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 622
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 623          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 623
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 624          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 624
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 625          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 625
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 626          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
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                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 626
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 627          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 627
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS DYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 628          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 628
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 629          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 629
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 630          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 630
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 631          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 631
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 632          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
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REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 632
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 633          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 633
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 634          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 634
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 635          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 635
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 636          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 636
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 637          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 637
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130
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SEQ ID NO: 638          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 638
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 639          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 639
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 640          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 640
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 641          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 641
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 642          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 642
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 643          moltype = AA  length = 130
FEATURE                Location/Qualifiers
REGION                 1..130
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..130
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 643
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
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GQGTMVTVAS                                                        130

SEQ ID NO: 644          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 644
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 645          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 645
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 646          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 646
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 647          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 647
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 648          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 648
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 649          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 649
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QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 650          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 650
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 651          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 651
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 652          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 652
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 653          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 653
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 654          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 654
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                        130

SEQ ID NO: 655          moltype = AA  length = 130
FEATURE                 Location/Qualifiers
REGION                  1..130
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..130
                        mol_type = protein
```

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                         organism = synthetic construct
SEQUENCE: 655
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCTNGV CYSGDAFDIW  120
GQGTMVTVAS                                                         130

SEQ ID NO: 656           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 656
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 657           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 657
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK SVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 658           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 658
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 659           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 659
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 660           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 660
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL               108

SEQ ID NO: 661           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 661
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
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FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 662          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 662
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 663          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 663
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 664          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 664
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 665          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 665
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 666          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 666
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 667          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 667
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 668          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
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REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 668
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 669          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 669
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 670          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 670
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 671          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 671
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 672          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 672
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 673          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 673
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHLVFG GGTKLTVL              108

SEQ ID NO: 674          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
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                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 674
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 675           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 675
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 676           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 676
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 677           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 677
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 678           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 678
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 679           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 679
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL               108

SEQ ID NO: 680           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 680
SYVLTQPPSV SVAPGQTARI TCRGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
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FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 681          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 681
SYVLTQPPSV SVAPGQTARI TCRGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 682          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 682
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 683          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 683
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 684          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 684
SYVLTQPPSV SVAPGQTARI TCGGNNIGYM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 685          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 685
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 686          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 686
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 687          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
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REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 687
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL               108

SEQ ID NO: 688         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 688
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL               108

SEQ ID NO: 689         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 689
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL               108

SEQ ID NO: 690         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 690
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSRSDHVVFG GGTKLTVL               108

SEQ ID NO: 691         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 691
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL               108

SEQ ID NO: 692         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 692
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL               108

SEQ ID NO: 693         moltype = AA  length = 108
FEATURE                Location/Qualifiers
REGION                 1..108
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..108
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```
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 693
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL                108

SEQ ID NO: 694           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 694
SYVLTQPPSV SVAPGQTARI TCRGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL                108

SEQ ID NO: 695           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 695
SYVLTQPPSV SVAPGQTARI TCRGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL                108

SEQ ID NO: 696           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 696
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL                108

SEQ ID NO: 697           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 697
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL                108

SEQ ID NO: 698           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 698
SYVLTQPPSV SVAPGQTARI TCGGNNIGYM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL                108

SEQ ID NO: 699           moltype = AA  length = 108
FEATURE                  Location/Qualifiers
REGION                   1..108
                         note = Description of Artificial Sequence: Synthetic
                          polypeptide
source                   1..108
                         mol_type = protein
                         organism = synthetic construct
SEQUENCE: 699
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
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FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 700          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 700
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 701          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 701
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 702          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 702
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 703          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 703
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 704          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 704
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSRSDHVVFG GGTKLTVL              108

SEQ ID NO: 705          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 705
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL              108

SEQ ID NO: 706          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
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REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 706
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPEQ  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL              108

SEQ ID NO: 707          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 707
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 708          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 708
SYVLTQPPSV SVAPGQTARI TCRGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 709          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 709
SYVLTQPPSV SVAPGQTARI TCRGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 710          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 710
SYVLTQPPSV SVAPGQTARI TCRGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 711          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 711
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 712          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
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                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 712
SYVLTQPPSV SVAPGQTARI TCGGNNIGYM NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 713          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 713
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 714          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 714
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSSSDHVVFG GGTKLTVL              108

SEQ ID NO: 715          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 715
SYVLTQPPSV SVAPGQTARI TCGGNNIGSM NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 716          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 716
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 717          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 717
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVL              108

SEQ ID NO: 718          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 718
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
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FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSRSDHVVFG GGTKLTVL              108

SEQ ID NO: 719          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 719
SYVLTQPPSV SVAPGQTARI TCGGNNIGSK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL              108

SEQ ID NO: 720          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 720
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DSESDHVVFG GGTKLTVL              108

SEQ ID NO: 721          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
REGION                  1..121
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..121
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 721
EVQLVQSGAE VKKPGSSVKV SCKASGYTFT SYYIYWVRQA PGQGLEWIGN IWPGNGGTFY  60
GEKFMGRATF TADTSTSTAY MELSSLRSED TAVYYCARRP DYSGDDYFDY WGQGTLVTVS  120
S                                                                 121

SEQ ID NO: 722          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
REGION                  1..121
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..121
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 722
QVQLQQSGAE LAKPGTSVKL SCKASGYTFT SYYIYWVKQR PGQGLEWIGN IWPGNGGTFY  60
GEKFMGKATF TADTSSSTAY MLLGSLTPED SAYYFCARRP DYSGDDYFDY WGQGVLVTVS  120
S                                                                 121

SEQ ID NO: 723          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
REGION                  1..118
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..118
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 723
QVKLVQSGAA LVKPGASVKM SCKASDYTFN DYWVSWVKQR HGESLEWIGE IYPNSGATNF  60
NGKFRGKATL TVDNPTSTAY MELSRLTSED SAIYYCTREY TRDWFAYWGQ GTLVTVSS    118

SEQ ID NO: 724          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 724
QVQLQQSGAE VIKPGASVKL SCKASGYTFS SYWMHWVRQA PGQGLEWIGE INPGNGHTNY  60
NEKFKSRATL TGDTSTSTVY MELSSLRSED TAVYYCARSF TTARAFAYWG QGTLVTVSS   119
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SEQ ID NO: 725          moltype = AA  length = 114
FEATURE                 Location/Qualifiers
REGION                  1..114
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..114
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 725
EVQLVESGGG LVQPGGSLRL SCAASGFTFS DYWMSWVRQA PGKGLEWVAD IKNDGSYTNY  60
APSLTNRFTI SRDNAKNSLY LQMNSLRAED TAVYYCAREL TGTWGQGTMV TVSS        114

SEQ ID NO: 726          moltype = AA  length = 121
FEATURE                 Location/Qualifiers
REGION                  1..121
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..121
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 726
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQS PEKGLEWIGE INHGGYVTYN  60
PSLESRVTIS VDTSKNQFSL KLSSVTAADT AVYYCARDYG PGNYDWYFDL WGRGTLVTVS  120
S                                                                  121

SEQ ID NO: 727          moltype = AA  length = 126
FEATURE                 Location/Qualifiers
REGION                  1..126
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..126
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 727
EVQLLESGGG LVQPGGSLRL SCAASGFTFS SYAMSWVRQA PGKGLEWVSA ISGSGGSTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDS PFLLDDYYYY YYMDVWGKGT  120
TVTVSS                                                             126

SEQ ID NO: 728          moltype = AA  length = 126
FEATURE                 Location/Qualifiers
REGION                  1..126
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..126
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 728
EVQLLESGGG LVQPGGSLRL SCAASGFTFR NYAMSWVRQA PGKGLEWVSA ISGSGDTTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCAKDS PFLLDDYYYY YYMDVWGKGT  120
TVTVSS                                                             126

SEQ ID NO: 729          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 729
QVQLVQSGAE VKKPGSSVKV SCKASGGTFN SYAISWVRQA PGQGLEWMGG IIPGFGTANY  60
AQKFQGRVTI TADESTSTAY MELSSLRSED TAVYYCARKN EEDGGFDHWG QGTLVTVSS   119

SEQ ID NO: 730          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
REGION                  1..118
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..118
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 730
QVQLVESGGG LVQPGGSLRL SCAASGFTFS DYYMHWVRQA PGKGLEWVSV ISGSGSNTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARLY AQFEGDFWGQ GTLVTVSS    118

SEQ ID NO: 731          moltype = AA  length = 116
```

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FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 731
QVQLVQSGAE VKKPGESLKI SCKGSGYSFS TYWISWVRQM PGKGLEWMGK IYPGDSYTNY  60
SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGY GIFDYWGQGT LVTVSS      116

SEQ ID NO: 732         moltype = AA  length = 116
FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 732
EVQLVQSGAE VKKPGESLRI SCKGSGYSFS TYWISWVRQM PGKGLEWMGK IYPGDSYTNY  60
SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGY GIFDYWGQGT LVTVSS      116

SEQ ID NO: 733         moltype = AA  length = 116
FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 733
QVQLVQSGAE VKKPGESLKI SCKGSGYSFS TYWISWVRQM PGKGLEWMGK IYPGDSYTNY  60
SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGY GIFDYWGQGT LVTVSS      116

SEQ ID NO: 734         moltype = AA  length = 116
FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..116
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 734
EVQLVQSGAE VKKPGESLRI SCKGSGYSFS TYWISWVRQM PGKGLEWMGK IYPGDSYTNY  60
SPSFQGQVTI SADKSISTAY LQWSSLKASD TAMYYCARGY GIFDYWGQGT LVTVSS      116

SEQ ID NO: 735         moltype = AA  length = 119
FEATURE                Location/Qualifiers
REGION                 1..119
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..119
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 735
QVQLVQSGAE VKKPGASVKL SCKASGYTFS SYWMHWVRQA PGQGLEWIGE INPGNGHTNY  60
NEKFKSRVTM TRDTSTSTAY MELSSLRSED TAVYYCARSF KTARAFAYWG QGTLVTVSS   119

SEQ ID NO: 736         moltype = AA  length = 122
FEATURE                Location/Qualifiers
REGION                 1..122
                       note = Description of Artificial Sequence: Synthetic
                        polypeptide
source                 1..122
                       mol_type = protein
                       organism = synthetic construct
SEQUENCE: 736
EVQLVESGGG LVKPGGSLRL SCAASGFTFS TFTMNWVRQA PGKGLEWVSS ISSKSTYIEY  60
ADSFKVRFTI SRDNAKNSLY LQMNSLRAED TAVYYCARYG AKNFLNWVFD YWGQGTLVTV  120
SS                                                                 122

SEQ ID NO: 737         moltype = AA  length = 116
FEATURE                Location/Qualifiers
REGION                 1..116
                       note = Description of Artificial Sequence: Synthetic
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                         polypeptide
source                  1..116
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 737
QVQLVQSGAE VKKPGASVKV SCKASGYTFA GFEMHWVRQA PGQGLEWMGA IDPKTGGTDY  60
NQKFKDRVTM TRDTSISTAY MELSRLRSDD TAVYYCARDL GYFDVWGQGT LVTVSS      116

SEQ ID NO: 738          moltype = AA  length = 120
FEATURE                 Location/Qualifiers
REGION                  1..120
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
VARIANT                 1
                        note = X can be any amino acid
source                  1..120
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 738
XVQLVQSGAE VKKPGASVKV SCKASGYTFT GYYMHWVRQA PGQGLEWMGW INPNSGGTNY  60
AQKFQGRVTM TRDTSISTAY MELSRLRSDD TAVYYCAREP GYYGSGLDYW GQGTLVTVSS  120

SEQ ID NO: 739          moltype = AA  length = 123
FEATURE                 Location/Qualifiers
REGION                  1..123
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..123
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 739
EVQLVESGGG LVQPGGSLRL SCAASGFSLS TGGVGVGWIR QAPGKGLEWL ALIDWADDKY  60
YSPSLKSRLT ISRDNSKNTL YLQLNSLRAE DTAVYYCARG GSDTVIGDWF AYWGQGTLVT  120
VSS                                                                123

SEQ ID NO: 740          moltype = AA  length = 118
FEATURE                 Location/Qualifiers
REGION                  1..118
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..118
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 740
EVQLLESGGG LVQPGGSLRL SCAASGFNFG YSYMSWVRQA PGKGLEWVSS IGSTSSHTYY  60
ADSVKGRFTI SRDNSKNTLY LQMNSLRAED TAVYYCARVY SSPGIDYWGQ GTLVTVSS    118

SEQ ID NO: 741          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 741
QVQLVQSGAE VVKPGASVKL SCKASGYTFS SYWMHWVKQA PGQVLEWIGE INPGNGHTNY  60
NEKFKSKATL TVDKSASTAY MELSSLRSED TAVYYCARSF TTARAFAYWG QGTLVTVSS   119

SEQ ID NO: 742          moltype = AA  length = 119
FEATURE                 Location/Qualifiers
REGION                  1..119
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..119
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 742
QVQLVQSGAE VKKPGASVKV SCKASGYTFS SYWMHWVRQA PGQRLEWMGE INPGNGHTNY  60
SQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARSF TTARAFAYWG QGTLVTVSS   119

SEQ ID NO: 743          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
REGION                  1..111
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
```

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```
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 743
NVMLTQPHSV SESPGKTVTI SCKLNSGNIG SYYVHWYQQR PGSSPTTMIY RDDKRPDGVP  60
DRFSGSIDSS SNSASLTISG LKTEDEADYY CHSYDSTITP VFGGGTKLTV L          111

SEQ ID NO: 744          moltype = AA  length = 111
FEATURE                 Location/Qualifiers
REGION                  1..111
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..111
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 744
QVVLTQPKSV STSLKSTVKL SCKLNSGNIG SYYVHWYQQH AGRSPTTMIY RDDKRPDGVP  60
DRFSGSIDSS SNSAFLTINN VQTEDDAIYF CHSYDSTITP VFGGGTKLTV L          111

SEQ ID NO: 745          moltype = AA  length = 112
FEATURE                 Location/Qualifiers
REGION                  1..112
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..112
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 745
DVVLTQTPSI LSATIGQSVS ISCRSSQSLL DSDGNTYLYW FLQRPGQSPQ RLIYLVSNLG  60
SGVPNRFSGS GSGTDFTLKI SGVEAEDLGI YYCMQPTHAP YTFGAGTKLE LK         112

SEQ ID NO: 746          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 746
DIVMTQSPAF LSVTPGEKVT ITCRASQTIS DYLHWYQQKP DQAPKLLIKY ASQSISGIPS  60
RFSGSGSGTD FTFTISSLEA EDAATYYCQD GHSFPPTFGQ GTKLEIK              107

SEQ ID NO: 747          moltype = AA  length = 113
FEATURE                 Location/Qualifiers
REGION                  1..113
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..113
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 747
DIVMTQSPDS LAVSLGERAT INCKSSQSLL SSGNQKNYLA WYQQKPGQPP KLLIYYASTR  60
QSGVPDRFSG SGSGTDFTLT ISSLQAEDVA VYYCLQYDRY PFTFGQGTKL EIK        113

SEQ ID NO: 748          moltype = AA  length = 109
FEATURE                 Location/Qualifiers
REGION                  1..109
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..109
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 748
EIVLTQSPAT LSLSPGERAT LSCRASQSVS SYLAWYQQKP GQAPRLLIYD ASNRATGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQQ RSNWPPALTF GGGTKVEIK            109

SEQ ID NO: 749          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 749
```

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DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ GHLFPITFGG GTKVEIK                107

SEQ ID NO: 750          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 750
DIQMTQSPSS VSASVGDRVT ITCRASQGIS SWLAWYQQKP GKAPKLLIYA ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ GHLFPITFGG GTKVEIK                107

SEQ ID NO: 751          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 751
DIELTQPPSV SVAPGQTARI SCSGDNLGDY YASWYQQKPG QAPVLVIYDD SNRPSGIPER  60
FSGSNSGNTA TLTISGTQAE DEADYYCQTW DGTLHFVFGG GTKLTVL                107

SEQ ID NO: 752          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 752
DIELTQPPSV SVAPGQTARI SCSGDNIGSK YVSWYQQKPG QAPVLVIYSD SERPSGIPER  60
FSGSNSGNTA TLTISGTQAE DEADYYCQSW DGSISRVFGG GTKLTVL                107

SEQ ID NO: 753          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 753
DIELTQPPSV SVAPGQTARI SCSGDNIGDQ YAHWYQQKPG QAPVVVIYQD KNRPSGIPER  60
FSGSNSGNTA TLTISGTQAE DEADYYCATY TGFGSLAVFG GGTKLTVL               108

SEQ ID NO: 754          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 754
SYELTQPPSV SVSPGQTASI TCSGDNIGDQ YAHWYQQKPG QSPVLVIYQD KNRPSGIPER  60
FSGSNSGNTA TLTISGTQAM DEADYYCATY TGFGSLAVFG GGTKLTVL               108

SEQ ID NO: 755          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 755
DIELTQPPSV SVAPGQTARI SCSGDNIGDQ YAHWYQQKPG QAPVVVIYQD KNRPSGIPER  60
FSGSNSGNTA TLTISGTQAE DEADYYCSTY TFVGFTTVFG GGTKLTVL               108
```

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SEQ ID NO: 756          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 756
SYELTQPPSV SVSPGQTASI TCSGDNIGDQ YAHWYQQKPG QSPVLVIYQD KNRPSGIPER  60
FSGSNSGNTA TLTISGTQAM DEADYYCATY TGFGSLAVFG GGTKLTVL               108

SEQ ID NO: 757          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 757
DIVMTQSPAF LSVTPGEKVT ITCRASQTIS DYLHWYQQKP DQAPKLLIKY ASQSISGIPS  60
RFSGSGSGTD FTFTISSLEA EDAATYYCQD GHSWPPTFGQ GTKLEIK                107

SEQ ID NO: 758          moltype = AA  length = 110
FEATURE                 Location/Qualifiers
REGION                  1..110
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..110
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 758
QSALTQPPSA SGSPGQTVTI SCTGTRYDVG YYEYVSWYQQ HPGKAPKLMI YETSKRLSGV  60
PDRFSGSKSG NTASLTVSGL QAEDEADYYC SSYRYEHQVS FGGGTKLTVL             110

SEQ ID NO: 759          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 759
DIQMTQSPSS LSASVGDRVT ITCRASQDIR SNLNWYQQKP GGAVKLLIYY TSRLHSGVPS  60
RFSGSGSGTD YTLTISSLQP EDFATYFCQQ SEKLPRTFGG GTKVEIR                107

SEQ ID NO: 760          moltype = AA  length = 109
FEATURE                 Location/Qualifiers
REGION                  1..109
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..109
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 760
SYVLTQPPSV SVAPGETARI TCGGDDIGDK RVHWYQKKPD QAPVLVVYED RYRPSGIPER  60
ISGSNSGNTA TLTLSRVEAG DEADYYCQVW DSSSDHPGVF GGGTQLIIL              109

SEQ ID NO: 761          moltype = AA  length = 106
FEATURE                 Location/Qualifiers
REGION                  1..106
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..106
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 761
DIQLTQSPSS LSASVGDRVT ITCRASQSIG SYLAWYQQKP GKAPKLLIYD ASNLETGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ GYYLWTFGQG TKVEIK                 106
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SEQ ID NO: 762          moltype = AA  length = 108
FEATURE                 Location/Qualifiers
REGION                  1..108
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..108
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 762
DIQMTQSPSS LSASVGDRVT ITCRASQSIG STLNWYQQKP GKAPKLLIYG ASSLQSGVPS  60
RFSGSGSGTD FTLTISSLQP EDFATYYCQQ YYTWVPFTFG QGTKLEIK               108

SEQ ID NO: 763          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 763
DIVMTQSPAT QSVSPGERVT LSCRASQTIS DYLHWYQQKP GQSPRLLIKY ASQSISGIPS  60
RFSGSGSGSD FTLTISSVEP EDFGVYYCQD GHSFPPTFGG GTKLEIK                107

SEQ ID NO: 764          moltype = AA  length = 107
FEATURE                 Location/Qualifiers
REGION                  1..107
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..107
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 764
DIVMTQSPPT LSLSPGERVT LSCRASQSIS DYLHWYQQKP GQSPRLLIKY ASQSISGIPA  60
RFSGSGSGTD FTLTISSLEP EDFAVYYCQD GHSFPPTFGG GTKVEIK                107

SEQ ID NO: 765          moltype = AA  length = 267
FEATURE                 Location/Qualifiers
REGION                  1..267
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..267
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 765
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS GGGGSGGGGS GGGGSGGGGS SYVLTQPPSV SVAPGQTARI TCGGNNIGYK  180
NVHWYQQKPG QAPVLVVYDD SDRPSGIPER FSGSNSGNTA TLTISRVEAG DEADYYCQVW  240
DHSSDHVVFG GGTKLTVLGG GHHHHHH                                      267

SEQ ID NO: 766          moltype = AA  length = 267
FEATURE                 Location/Qualifiers
REGION                  1..267
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..267
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 766
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVLGG GGSGGGGSGG  120
GGSGGGGSQV QLQQWGAGLL KPSETLSLTC AVYGGSFSGY YWSWIRQPPG KGLEWIGEIN  180
HRGSINYNNY NPSLKSRVTI SVDTSKNQFS LKLSSVTAAD TAVYYCAKPL RPHCINGVCY  240
SGDAFDIWGQ GTMVTVASGG GHHHHHH                                      267

SEQ ID NO: 767          moltype = AA  length = 262
FEATURE                 Location/Qualifiers
REGION                  1..262
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..262
                        mol_type = protein
                        organism = synthetic construct
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SEQUENCE: 767
QVQLQQWGAG LLKPSETLSL TCAVYGGSFS GYYWSWIRQP PGKGLEWIGE INHRGSINYN  60
NYNPSLKSRV TISVDTSKNQ FSLKLSSVTA ADTAVYYCAK PLRPHCINGV CYSGDAFDIW  120
GQGTMVTVAS GGGGSGGGGS GGGGSSYVLT QPPSVSVAPG QTARITCGGN NIGYKNVHWY  180
QQKPGQAPVL VVYDDSDRPS GIPERFSGSN SGNTATLTIS RVEAGDEADY YCQVWDHSSD  240
HVVFGGGTKL TVLGGGHHHH HH                                         262

SEQ ID NO: 768          moltype = AA  length = 262
FEATURE                 Location/Qualifiers
REGION                  1..262
                        note = Description of Artificial Sequence: Synthetic
                         polypeptide
source                  1..262
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 768
SYVLTQPPSV SVAPGQTARI TCGGNNIGYK NVHWYQQKPG QAPVLVVYDD SDRPSGIPER  60
FSGSNSGNTA TLTISRVEAG DEADYYCQVW DHSSDHVVFG GGTKLTVLGG GGSGGGGSGG  120
GGSQVQLQQW GAGLLKPSET LSLTCAVYGG SFSGYYWSWI RQPPGKGLEW IGEINHRGSI  180
NYNNYNPSLK SRVTISVDTS KNQFSLKLSS VTAADTAVYY CAKPLRPHCI NGVCYSGDAF  240
DIWGQGTMVT VASGGGHHHH HH                                         262

SEQ ID NO: 769          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 4
                        note = X can be F or L
VARIANT                 5
                        note = X can be S or N
VARIANT                 8
                        note = X can be Y or H
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
VARIANT                 6
                        note = X can be D or G
SEQUENCE: 769
GGSXXXYXWS                                                       10

SEQ ID NO: 770          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 2
                        note = X can be I or V
VARIANT                 5
                        note = X can be A, Q, R or S
VARIANT                 8
                        note = X can be I or T
VARIANT                 9
                        note = X can be N or S
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 770
EXNHXGSXXY NNYNPSLKS                                             19

SEQ ID NO: 771          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 4
                        note = X can be L or F
VARIANT                 9
                        note = X can be I or T
VARIANT                 14
                        note = X can be Y or S
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 771
AKPXRPHCXN GVCXSGDAFD I                                          21
```

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SEQ ID NO: 772          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 1
                        note = X can be G or R
VARIANT                 7
                        note = X can be S , T or Y
VARIANT                 8
                        note = X can be K or M
VARIANT                 9
                        note = X can be N or S
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 772
XGNNIGXXXV H                                                      11

SEQ ID NO: 773          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 5
                        note = X can be H or S
VARIANT                 6
                        note = X can be E, R or S
VARIANT                 10
                        note = X can be L or V
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 773
QVWDXXSDHX V                                                      11

SEQ ID NO: 774          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 2
                        note = X can be I or V
VARIANT                 5
                        note = X can be A,Q, R or S
VARIANT                 8
                        note = X can be I or T
VARIANT                 9
                        note = X can be N or S
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 774
EXNHXGSXXY NPSLKS                                                 16

SEQ ID NO: 775          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 4
                        note = X can be F or L
VARIANT                 5
                        note = X can be S or N
VARIANT                 8
                        note = X can be Y or H
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 775
GGSXXDYXWS                                                        10

SEQ ID NO: 776          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 2
                        note = X can be I or V
VARIANT                 5
```

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```
                        note = X can be R or S
VARIANT                 8
                        note = X can be I or T
VARIANT                 9
                        note = X can be N or S
source                  1..16
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 776
EXNHXGSXXY NPSLKS                                                    16

SEQ ID NO: 777          moltype = AA  length = 19
FEATURE                 Location/Qualifiers
REGION                  1..19
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 2
                        note = X can be I or V
VARIANT                 5
                        note = X can be R or S
VARIANT                 8
                        note = X can be I or T
VARIANT                 9
                        note = X can be N or S
source                  1..19
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 777
EXNHXGSXXY NNYNPSLKS                                                 19

SEQ ID NO: 778          moltype = AA  length = 21
FEATURE                 Location/Qualifiers
REGION                  1..21
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 4
                        note = X can be L or F
VARIANT                 14
                        note = X can be Y or S
source                  1..21
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 778
AKPXRPHCTN GVCXSGDAFD I                                              21

SEQ ID NO: 779          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 7
                        note = X can be S or T
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 779
GGNNIGXKNV H                                                         11

SEQ ID NO: 780          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Artificial Sequence: Synthetic peptide
VARIANT                 1
                        note = X can be G or R
VARIANT                 7
                        note = X can be  S, T or Y
VARIANT                 8
                        note = X can be K or M
VARIANT                 9
                        note = X can be N or I
source                  1..11
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 780
XGNNIGXXXV H                                                         11

SEQ ID NO: 781          moltype = AA  length = 10
FEATURE                 Location/Qualifiers
REGION                  1..10
```

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```
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..10
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 781
GGSGGGGSGG                                                        10

SEQ ID NO: 782          moltype = AA  length = 20
FEATURE                 Location/Qualifiers
REGION                  1..20
                        note = Description of Artificial Sequence: Synthetic peptide
source                  1..20
                        mol_type = protein
                        organism = synthetic construct
SEQUENCE: 782
GGGGSGGGGS GGGGSGGGGS                                             20

SEQ ID NO: 783          moltype = AA  length = 13
FEATURE                 Location/Qualifiers
REGION                  1..13
                        note = Description of Unknown: EphA2 sequence
source                  1..13
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 783
TEPPKVRLEG RST                                                    13

SEQ ID NO: 784          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA2 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 784
KKGDS                                                             5

SEQ ID NO: 785          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphA2 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 785
TYLVQVQ                                                           7

SEQ ID NO: 786          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphA2 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 786
AGSKVHEFQT L                                                      11

SEQ ID NO: 787          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphA1 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 787
AESLSGLSLR LVKKEP                                                 16

SEQ ID NO: 788          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA1 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 788
LNQDE                                                             5
```

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```
SEQ ID NO: 789          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphA1 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 789
TYIVRVR                                                                     7

SEQ ID NO: 790          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphA1 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 790
PFSPDHEFRT S                                                                11

SEQ ID NO: 791          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphA3 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 791
AAPSPVLTIK KDRTSR                                                           16

SEQ ID NO: 792          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA3 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 792
KQEQE                                                                       5

SEQ ID NO: 793          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphA3 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 793
IYVFQIR                                                                     7

SEQ ID NO: 794          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphA3 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 794
TNSRKFEFET S                                                                11

SEQ ID NO: 795          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphA4 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 795
AAPSSIALVQ AKEVTR                                                           16

SEQ ID NO: 796          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA4 sequence
source                  1..5
```

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```
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 796
KDQNE                                                              5

SEQ ID NO: 797           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
REGION                   1..7
                         note = Description of Unknown: EphA4 sequence
source                   1..7
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 797
SYVFHVR                                                            7

SEQ ID NO: 798           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Unknown: EphA4 sequence
source                   1..11
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 798
DFSEPLEVTT N                                                       11

SEQ ID NO: 799           moltype = AA  length = 16
FEATURE                  Location/Qualifiers
REGION                   1..16
                         note = Description of Unknown: EphA5 sequence
source                   1..16
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 799
AAPSPVTNVK KGKIAK                                                  16

SEQ ID NO: 800           moltype = AA  length = 4
FEATURE                  Location/Qualifiers
REGION                   1..4
                         note = Description of Unknown: EphA5 sequence
source                   1..4
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 800
KDQE                                                               4

SEQ ID NO: 801           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
REGION                   1..7
                         note = Description of Unknown: EphA5 sequence
source                   1..7
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 801
VYVFQIR                                                            7

SEQ ID NO: 802           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Unknown: EphA5 sequence
source                   1..11
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 802
VFSRRFEFET T                                                       11

SEQ ID NO: 803           moltype = AA  length = 16
FEATURE                  Location/Qualifiers
REGION                   1..16
                         note = Description of Unknown: EphA6 sequence
source                   1..16
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 803
DAPSLIGMMR KDWASQ                                                  16
```

-continued

```
SEQ ID NO: 804          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA6 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 804
KEHEQ                                                                   5

SEQ ID NO: 805          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphA6 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 805
KYVFHIR                                                                 7

SEQ ID NO: 806          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphA6 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 806
GYSQKFEFET G                                                            11

SEQ ID NO: 807          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphA7 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 807
AAPSQVSGVM KERVLQ                                                       16

SEQ ID NO: 808          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphA7 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 808
KDQRE                                                                   5

SEQ ID NO: 809          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphA7 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 809
VYVFQIR                                                                 7

SEQ ID NO: 810          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphA7 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 810
NYSPRLDVAT L                                                            11

SEQ ID NO: 811          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphA8 sequence
source                  1..16
```

-continued

```
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 811
AAPSQVVVIR QERAGQ                                              16

SEQ ID NO: 812         moltype = AA  length = 5
FEATURE                Location/Qualifiers
REGION                 1..5
                       note = Description of Unknown: EphA8 sequence
source                 1..5
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 812
KDKEM                                                          5

SEQ ID NO: 813         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Unknown: EphA8 sequence
source                 1..7
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 813
RYVFQVR                                                        7

SEQ ID NO: 814         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Unknown: EphA8 sequence
source                 1..11
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 814
RFSQAMEVET G                                                   11

SEQ ID NO: 815         moltype = AA  length = 16
FEATURE                Location/Qualifiers
REGION                 1..16
                       note = Description of Unknown: EphA10 sequence
source                 1..16
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 815
GAPWEEGEIR RDRVEP                                              16

SEQ ID NO: 816         moltype = AA  length = 5
FEATURE                Location/Qualifiers
REGION                 1..5
                       note = Description of Unknown: EphA10 sequence
source                 1..5
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 816
KGQSE                                                          5

SEQ ID NO: 817         moltype = AA  length = 7
FEATURE                Location/Qualifiers
REGION                 1..7
                       note = Description of Unknown: EphA10 sequence
source                 1..7
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 817
RYVFQIR                                                        7

SEQ ID NO: 818         moltype = AA  length = 11
FEATURE                Location/Qualifiers
REGION                 1..11
                       note = Description of Unknown: EphA10 sequence
source                 1..11
                       mol_type = protein
                       organism = unidentified
SEQUENCE: 818
SFNPSIEVQT L                                                   11
```

-continued

```
SEQ ID NO: 819          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphB1 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 819
AAPSTVPIMH QVSATM                                                  16

SEQ ID NO: 820          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphB1 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 820
KEHNE                                                              5

SEQ ID NO: 821          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphB1 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 821
VYVVQVR                                                            7

SEQ ID NO: 822          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphB1 sequence
source                  1..11
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 822
KFSGKMCFQT L                                                       11

SEQ ID NO: 823          moltype = AA  length = 16
FEATURE                 Location/Qualifiers
REGION                  1..16
                        note = Description of Unknown: EphB2 sequence
source                  1..16
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 823
AAPSAVSIMH QVSRTV                                                  16

SEQ ID NO: 824          moltype = AA  length = 5
FEATURE                 Location/Qualifiers
REGION                  1..5
                        note = Description of Unknown: EphB2 sequence
source                  1..5
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 824
KELSE                                                              5

SEQ ID NO: 825          moltype = AA  length = 7
FEATURE                 Location/Qualifiers
REGION                  1..7
                        note = Description of Unknown: EphB2 sequence
source                  1..7
                        mol_type = protein
                        organism = unidentified
SEQUENCE: 825
IYVFQVR                                                            7

SEQ ID NO: 826          moltype = AA  length = 11
FEATURE                 Location/Qualifiers
REGION                  1..11
                        note = Description of Unknown: EphB2 sequence
source                  1..11
```

-continued

```
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 826
RYSGKMYFQT M                                                11

SEQ ID NO: 827           moltype = AA  length = 16
FEATURE                  Location/Qualifiers
REGION                   1..16
                         note = Description of Unknown: EphB3 sequence
source                   1..16
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 827
AAPSEVPTLR LHSSSG                                           16

SEQ ID NO: 828           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
source                   1..7
                         mol_type = protein
                         organism = Rattus sp.
SEQUENCE: 828
SKVHEFQ                                                     7

SEQ ID NO: 829           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
REGION                   1..7
                         note = Description of Unknown: EphB3 sequence
source                   1..7
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 829
RYVVQVR                                                     7

SEQ ID NO: 830           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
                         note = Description of Unknown: EphB3 sequence
source                   1..11
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 830
QYSRPAEFET T                                                11

SEQ ID NO: 831           moltype = AA  length = 16
FEATURE                  Location/Qualifiers
REGION                   1..16
                         note = Description of Unknown: EphB4 sequence
source                   1..16
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 831
EVPPAVSDIR VTRSSP                                           16

SEQ ID NO: 832           moltype = AA  length = 5
FEATURE                  Location/Qualifiers
REGION                   1..5
                         note = Description of Unknown: EphB4 sequence
source                   1..5
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 832
KGAEG                                                       5

SEQ ID NO: 833           moltype = AA  length = 7
FEATURE                  Location/Qualifiers
REGION                   1..7
                         note = Description of Unknown: EphB4 sequence
source                   1..7
                         mol_type = protein
                         organism = unidentified
SEQUENCE: 833
SYLVQVR                                                     7

SEQ ID NO: 834           moltype = AA  length = 11
FEATURE                  Location/Qualifiers
REGION                   1..11
```

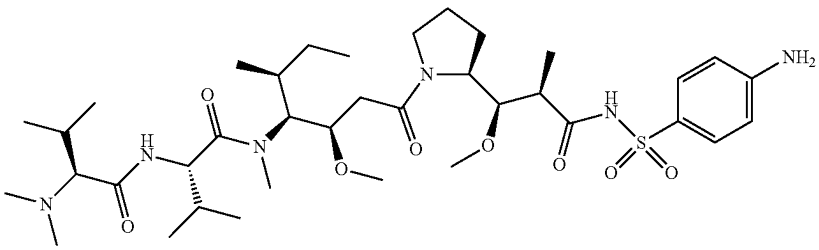
-continued

```
                          note = Description of Unknown: EphB4 sequence
source                    1..11
                          mol_type = protein
                          organism = unidentified
SEQUENCE: 834
PFGQEHHSQT Q                                                            11

SEQ ID NO: 835            moltype = AA  length = 16
FEATURE                   Location/Qualifiers
REGION                    1..16
                          note = Description of Unknown: EphB6 sequence
source                    1..16
                          mol_type = protein
                          organism = unidentified
SEQUENCE: 835
EVPSAVPVVH QVSRAS                                                       16

SEQ ID NO: 836            moltype = AA  length = 5
FEATURE                   Location/Qualifiers
REGION                    1..5
                          note = Description of Unknown: EphB6 sequence
source                    1..5
                          mol_type = protein
                          organism = unidentified
SEQUENCE: 836
QAEDE                                                                   5

SEQ ID NO: 837            moltype = AA  length = 7
FEATURE                   Location/Qualifiers
REGION                    1..7
                          note = Description of Unknown: EphB6 sequence
source                    1..7
                          mol_type = protein
                          organism = unidentified
SEQUENCE: 837
IYGFQVR                                                                 7

SEQ ID NO: 838            moltype = AA  length = 11
FEATURE                   Location/Qualifiers
REGION                    1..11
                          note = Description of Unknown: EphB6 sequence
source                    1..11
                          mol_type = protein
                          organism = unidentified
SEQUENCE: 838
PYGGKVYFQT L                                                            11

SEQ ID NO: 839            moltype = AA  length = 7
FEATURE                   Location/Qualifiers
source                    1..7
                          mol_type = protein
                          organism = Homo sapiens
SEQUENCE: 839
SKVHEFQ                                                                 7

SEQ ID NO: 840            moltype = AA  length = 7
FEATURE                   Location/Qualifiers
source                    1..7
                          mol_type = protein
                          organism = Mus sp.
SEQUENCE: 840
SKVHEFQ                                                                 7

SEQ ID NO: 841            moltype = AA  length = 7
FEATURE                   Location/Qualifiers
source                    1..7
                          mol_type = protein
                          organism = Macaca sp.
SEQUENCE: 841
SKVHEFQ                                                                 7
```

What is claimed is:

1. An immunoconjugate comprising an antibody that binds to ephrin receptor A2 (EphA2) and a cytotoxic agent, wherein the cytotoxic agent is an auristatin analogue, and wherein the antibody is conjugated to the auristatin analogue.

2. The immunoconjugate of claim **1**, wherein the auristatin analogue has a formula of:



3. The immunoconjugate of claim **2**, wherein the linker has a formula of

$$\#\!-\!Z\!-\!\!\left[Str\right]_s\!-\!AA_1\!-\!\left[AA_2\right]_m\!-\!\left[X\right]_o\!-\!\% \qquad (V)$$

wherein:

Z is a linking group that joins the linker to a target group on the antibody;

Str is a stretcher:

AA_1 and AA_2 are each independently an amino acid, wherein AA_1-$[AA_2]_r$ forms a protease cleavage site;

X is a self-immolative group;

s is 0 or 1;

m is 1, 2, 3 or 4;

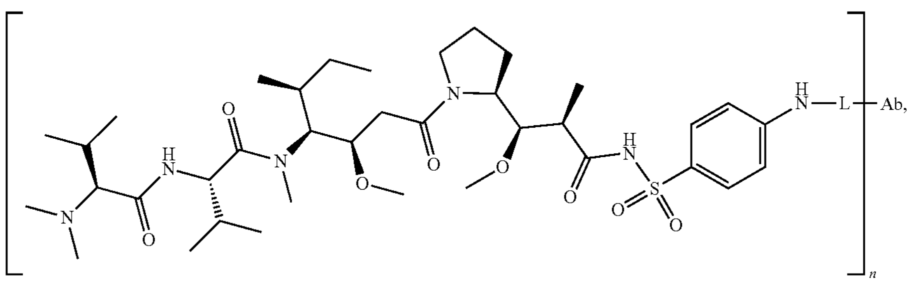
o is 0, 1 or 2;

is the point of attachment to the antibody, and

% is the point of attachment to the auristatin analogue.

4. The immunoconjugate of claim **1**, wherein the immunoconjugate has a formula of:

(I)



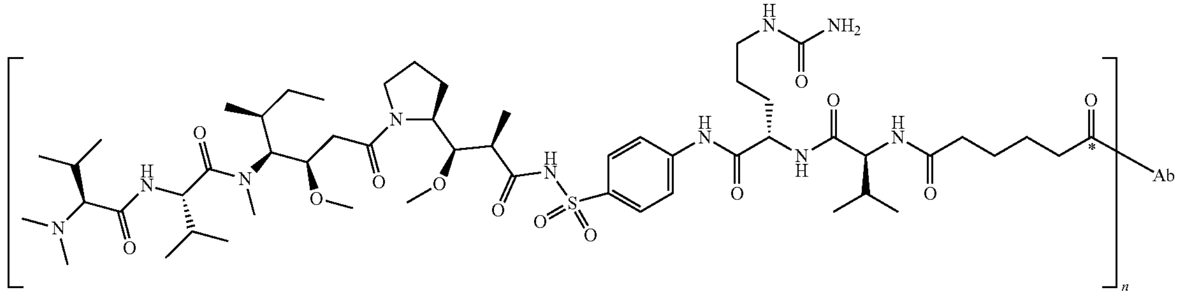
wherein:

L is a cleavable linker;

n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and

Ab is the antibody.

5. The immunoconjugate of claim **1**, wherein the immunoconjugate has a formula of:



wherein n is the drug-to-antibody ratio (DAR) and is an integer from 1 to 12, and

Ab is the antibody.

6. The immunoconjugate of any one of claims **1-5**, wherein the antibody of the immunoconjugate binds to an epitope of EphA2 located in the FN2 domain of EphA2, wherein the FN2 domain has a sequence of

```
                                        (SEQ ID NO: 95)
TEPPKVRLEGRSTTSLSVSWSIPPPQQSRVWKYEVTYRKKGDSNSYNVR
RTEGFSVTLDDLAPDTTYLVQVQALTQEGQGAGSKVHEFQTLSPEGSG
N.
```

7. The immunoconjugate of claim **6** wherein the epitope comprises at least one, at least two, at least three, at least four, at least five, or at least six, at least seven, at least eight, or at least nine, or all ten of amino acid residues Leu444, Arg447, Lys476, Gln506, Ser519, Lys520, Val521, Glu523, Phe524, and Gln525 of SEQ ID NO: 94.

8. The immunoconjugate of any of claims **1-7**, wherein the antibody of the immunoconjugate has one or more properties selected from the group consisting of:

a) activates the ephrin A1-EphA2 signaling axis at least 95%, 90%, 85%, or 80% less than ephrin-A1 activates the ephrin A1-EphA2 signaling axis,

b) binds preferentially to a tumor tissue than normal tissue,

c) has an EC_{50} for activation of the ephrinA1-EphA2 signaling axis that is at least 10-, 20-, 50-, 100-, 200-, 500-fold less potent than the EC_{50} of the antibody

d) has ADCP activity, and

e) has ADCC activity.

9. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

a heavy chain variable region comprising:

an HCDR1 comprising a sequence (SEQ ID NO:1), or a variant HCDR1 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence;

an HCDR2 comprising a sequence (SEQ ID NO:12), or a variant HCDR2 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; and

an HCDR3 comprising a sequence (SEQ ID NO:23), or a variant HCDR3 in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence;

a light chain variable region comprising:

an LCDR1 comprising a sequence (SEQ ID NO:34), or a variant LCDR1 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence;

an LCDR2 comprising a sequence (SEQ ID NO:45), or variant LCDR2 in which 1, 2, or 3 amino acid is substituted relative to the sequence; and

an LCDR3 comprising a sequence (SEQ ID NO:56), or a variant LCDR3 in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence.

10. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

a heavy chain variable region comprising:

an HCDR1 sequence set forth in any one of SEQ ID NOS:1-11 and 201-265 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence;

an HCDR2 sequence set forth in any one of SEQ ID NOS:12-22 and 266-330 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence; and

an HCDR3 sequence set forth in any one of SEQ ID NOS:23-33 and 331-395 or a variant thereof in which 1, 2, 3, 4, or 5 amino acids are substituted relative to the sequence;

a light chain variable region comprising:

an LCDR1 sequence set forth in any one of SEQ ID NOS:34-44 and 396-460 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence;

an LCDR2 sequence set forth in any one of SEQ ID NOS:45-55 and 461-525 or a variant thereof in which 1, 2, or 3 amino acid is substituted relative to the sequence; and

an LCDR3 sequence set forth in any one of SEQ ID NOS:56-66 and 526-590 or a variant thereof in which 1, 2, 3, 4, or 5 amino acid is substituted relative to the sequence.

11. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

(a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, and X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769);

(b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNNYNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 770);

(c) an HCDR3 sequence of $AKPX_1RPHCX_2NGVCX_3SGDAFDI$, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S (SEQ ID NO: 771);

(d) an LCDR1 sequence of $X_1GNNIGX_2X_3X_4VH$, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or S (SEQ ID NO: 772); (e) an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and

(f) an LCDR3 sequence of QVWDX1 X2SDHX3V, where X1 is H or S, X2 is E, R, or S, and X3 is L or V (SEQ ID NO: 773).

12. The immunoconjugate of claim any one of claims **1-8**, wherein the antibody comprises:

(a) an HCDR1 sequence of $GGSX_1X_2X_3YX_4WS$ where X_1 is F or L, X_2 is S or N, and X_3 is D or G, and X_4 is Y or H (SEQ ID NO: 769);

(b) an HCDR2 sequence of $EX_1NHX_2GSX_3X_4YNPSLKS$, where X_1 is I or V, X_2 is A, Q, R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 774);

(c) an HCDR3 sequence of $AKPX_1RPHCX_2NGVCX_3SGDAFDI$, where X_1 is L or F, X_2 is I or T; and X_3 is Y or S (SEQ ID NO: 771);

(d) an LCDR1 sequence of $X_1GNNIGX_2X_3X_4VH$, where X_1 is G or R, X_2 is S, T, or Y, X_3 is K or M, and X_4 is N or S (SEQ ID NO: 772);

(e) an LCDR2 sequence of DDSDRPS (SEQ ID NO: 45); and

(f) an LCDR3 sequence of $QVWDX_1X_2SDHX_3V$, where X_1 is H or S, X_2 is E, R, or S, and X_3 is L or V (SEQ ID NO: 773).

13. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

(a) an HCDR1 sequence of $GGSX_1X_2DYX_3WS$ where X_1 is F or L, X_2 is S or N, and X_3 is Y or H (SEQ ID NO: 775);

(b) an HCDR2 sequence of EX_1NHX_2 GS X_3 X_4YNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 776); or an HCDR2 sequence of EX_1NHX_2 GS X_3 X_4YNNYNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 777); and

(c) an HCDR3 sequence of AKP X_1RPHCTNGVCX_2SGDAFDI, where X_1 is L or F and X_2 is Y or S (SEQ ID NO: 778);

(d) an LCDR1 sequence of GGNNIGX_1KNVH, where X_1 is S or T (SEQ ID NO: 779);

(e) an LCDR2 sequence DDSDRPS (SEQ ID NO: 45); and

(f) an LCDR3 sequence QVWDSSSDHLV (SEQ ID NO: 56).

14. The immunoconjugate of any of claims **1-13**, wherein the V_H of the antibody comprises an amino acid sequence having at least 95% identity to (SEQ ID NO: 67); and the V_L comprises an amino sequence having at 95% identity to (SEQ ID NO: 78).

15. The immunoconjugate of any of claims **2-13**, wherein the V_H of the antibody comprises an amino acid sequence having at least 95% identity to a V_H of any one of SEQ ID NOS:67-77 and 591-655; and the V_L comprises an amino sequence having at 95% identity to a V_L of any one of SEQ ID NOS:78-88 and 656-720.

16. The immunoconjugate of any of claims **1-15**, wherein the antibody comprises:

a V_H region comprising amino acid sequence SEQ ID NO:67 and a V_L region comprising amino acid sequence SEQ ID NO:78;

a V_H region comprising amino acid sequence SEQ ID NO:68 and a V_L region comprising amino acid sequence SEQ ID NO:79;

a V_H region comprising amino acid sequence SEQ ID NO:69 and a V_L region comprising amino acid sequence SEQ ID NO:80;

a V_H region comprising amino acid sequence SEQ ID NO:70 and a V_L region comprising amino acid sequence SEQ TD NO:81;

a V_H region comprising amino acid sequence SEQ ID NO:71 and a V_L region comprising amino acid sequence SEQ ID NO:82;

a V_H region comprising amino acid sequence SEQ ID NO:72 and a V_L region comprising amino acid sequence SEQ ID NO:83;

a V_H region comprising amino acid sequence SEQ ID NO:73 and a V_L region comprising amino acid sequence SEQ ID NO:84;

a V_H region comprising amino acid sequence SEQ ID NO:74 and a V_L region comprising amino acid sequence SEQ ID NO:85;

a V_H region comprising amino acid sequence SEQ ID NO:75 and a V_L region comprising amino acid sequence SEQ ID NO:86;

a V_H region comprising amino acid sequence SEQ ID NO:76 and a V_L region comprising amino acid sequence SEQ ID NO:87;

a V_H region comprising amino acid sequence SEQ ID NO:77 and a V_L region comprising amino acid sequence SEQ ID NO:88,

a V_H region comprising amino acid sequence SEQ ID NO:624 and a V_L region comprising amino acid sequence SEQ ID NO:689,

a V_H region comprising amino acid sequence SEQ ID NO:652 and a V_L region comprising amino acid sequence SEQ ID NO:717, or

a V_H region comprising amino acid sequence SEQ ID NO:607 and a V_L region comprising amino acid sequence SEQ ID NO:672.

17. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

(1) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein:

(a) the V_H region has at least 70% identity to SEQ ID NO:67; and comprises a CDR1 of SEQ ID NO:1, or the CDR1 of SEQ ID NO:1 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:12, or the CDR2 of SEQ ID NO:12 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:23 or the CDR3 of SEQ ID NO:23 in which 1, 2, 3, 4, or 5 are substituted; and

(b) the V_L region has at least 70% identity to SEQ ID NO: 78, and comprises a CDR1 of SEQ ID NO:34 or the CDR1 of SEQ ID NO:34 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:45, or the CDR2 of SEQ ID NO:45 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:56 or the CDR3 of SEQ ID NO:56 in which 1, 2, 3, 4, or 5 are substituted.

(2) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein:

(a) the V_H region has at least 70% identity to SEQ ID NO:652; and comprises a CDR1 of SEQ ID NO:262, or the CDR1 of SEQ ID NO:262 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:327, or the CDR2 of SEQ ID NO:327 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:392 or the CDR3 of SEQ ID NO:392 in which 1, 2, 3, 4, or 5 are substituted; and

(b) the V_L region has at least 70% identity to SEQ ID NO: 717, and comprises a CDR1 of SEQ ID NO:457 or the CDR1 of SEQ ID NO:457 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:522, or the CDR2 of SEQ ID NO:522 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:587 or the CDR3 of SEQ ID NO:587 in which 1, 2, 3, 4, or 5 are substituted; or

(3) a heavy chain variable (V_H) region and a light chain variable (V_L) region, wherein:

(a) the V_H region has at least 70% identity to SEQ ID NO:607; and comprises a CDR1 of SEQ ID NO:217, or the CDR1 of SEQ ID NO:217 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:282, or the CDR2 of SEQ ID NO:282 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:347 or the CDR3 of SEQ ID NO:347 in which 1, 2, 3, 4, or 5 are substituted; and

(b) the V_L region has at least 70% identity to SEQ ID NO: 672, and comprises a CDR1 of SEQ ID NO:412 or the CDR1 of SEQ ID NO:412 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR2 of SEQ ID NO:477, or the CDR2 of SEQ ID NO:477 in which 1, 2, 3, 4, or 5 amino acids are substituted; a CDR3 of SEQ ID NO:542 or the CDR3 of SEQ ID NO:542 in which 1, 2, 3, 4, or 5 are substituted

18. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises:

a heavy chain variable (V_H) region and a light chain variable (V_L) region,

(i) wherein the V_H region has at least 70% identity to SEQ ID NO:67 and comprises

(a) an HCDR1 sequence of $GGSX_1X_2DYX_3WS$ where X_1 is F or L, X_2 is S or N, and X_3 is Y or H (SEQ ID NO: 775);

(b) an HCDR2 sequence of EX_1NHX_2 GS X_3 X_4YNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 is N or S (SEQ ID NO: 776); or an HCDR2 sequence of EX_1NHX_2 GS X_3 X_4YNNYNPSLKS, where X_1 is I or V, X_2 is R or S, X_3 is I or T, and X_4 N or S (SEQ ID NO: 777); and

(c) an HCDR3 sequence of AKP X_1RPHCTNGVCX_2SGDAFDI, where X_1 is L or F and X_2 is Y or S (SEQ ID NO: 778); and

(ii) wherein the V_L region has at least 70% identity to SEQ ID NO:78 and comprises

(d) an LCDR1 sequence of GGNNIGX_1KNVH, where X_1 is S or T (SEQ ID NO: 779);

(e) an LCDR2 sequence DDSDRPS (SEQ ID NO: 45); and

(f) an LCDR3 sequence QVWDSSSDHLV (SEQ ID NO: 56).

19. The immunoconjugate of any one of claim **1-18**, wherein the antibody comprises an HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702, or a variant thereof in which at least one, two, three, four, five, or all six of the CDRs contain 1 or 2 amino acid substitutions compared to the corresponding CDR.

20. The immunoconjugate of any one of claims **1-18**, wherein the antibody comprises a heavy chain variable region and a light chain variable region of an antibody designated as AB-008873; AB-009805; AB-009806; AB-009807; AB-009808; AB-009812; AB-009813; AB-009814; AB-009815; AB-009816; AB-009817, AB-010141, AB-010142, AB-010143, AB-010144, AB-010145, AB-010146, AB-010147, AB-010148, AB-010149, AB-010150, AB-010151, AB-010152, AB-010357, AB-010358, AB-010359, AB-010360, AB-010361, AB-010362, AB-010363, AB-010364, AB-010365, AB-010366, AB-010367, AB-010661, AB-010662, AB-010663, AB-010664, AB-010665, AB-010666, AB-010667, AB-010668, AB-010669, AB-010670, AB-010671, AB-010672, AB-010673, AB-010674, AB-010675, AB-010676, AB-010677, AB-010678, AB-010679, AB-010680, AB-010681, AB-010682, AB-010683, AB-010684, AB-010685, AB-010686, AB-010687, AB-010688, AB-010689, AB-010690, AB-010691, AB-010692, AB-010693, AB-010694, AB-010695, AB-010696, AB-010697, AB-010698, AB-010699, AB-010700, AB-010701, or AB-010702, or a variant thereof, wherein the variant comprises a heavy chain variable region having a sequence that is at least 95% identical to that of the corresponding heavy chain variable region and a light chain variable region having a sequence that is at least 95% identical to the corresponding light chain variable region.

21. The immunoconjugate of any one of claims above, wherein at least 1 or 2 of the substitutions in the antibody are conservative substitutions; at least 50% of the substitutions are conservative substitutions; or all of the substitutions are conservative substitutions.

22. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises a V_H region of any one of SEQ ID NO 67-77 and 591-655, and/or a V_L region of any one of SEQ ID NO: 78-88 and 656-720, or an antibody comprising a V_H region with at least 80% identity to any one of SEQ ID NO 67-77 and 591-655 and a V_L region having at least 80% identity to any one of SEQ ID NO: 78-88 and 656-720, with variations to the corresponding V_H or V_L regions present only in Framework regions.

23. The immunoconjugate of any one of claims **1-8**, wherein the antibody comprises an HCDR1 of any one of SEQ ID NOS:1-11, and 201-265, an HCDR2 of any one of SEQ ID NOS:12-22 and 266-330, an HCDR3 of any one of SEQ ID NOS:23-33 and 331-395, an LCDR1 of any one of SEQ ID NOS:34-44 and 396-460, an LCDR2 of any one of SEQ ID NOS:45-55 and 461-525, an LCDR3 of any one of SEQ ID NOS:56-66 and 526-590,

wherein the FW regions in the V_H region of the antibody are at least 80% identical to the FW regions present in the V_H region of SEQ ID NO 67-77 and 591-655, and

wherein the FW regions in the V_L region of the antibody are at least 80% identical to the FW regions present in the V_L region of SEQ ID NO: 78-88 and 656-720.

24. The immunoconjugate of any one of claims **1-23**, wherein the antibody is a non-natural antibody.

25. The immunoconjugate of any one of claims **1-21**, wherein the antibody is a bispecific antibody comprising two antigen binding fragments, one binding to EphA2, and the other binds to a different antigen.

26. The immunoconjugate of claim **25**, wherein the different antigen is 4-1bb or CD3.

27. A pharmaceutical composition comprising an immunoconjugate of any one of claims **1** to **26** and a pharmaceutically acceptable carrier.

28. A method of inducing an immune response and/or treating cancer, the method comprising administering an immunoconjugate of any one of claims **1** to **26**.

29. The method of claim **28**, wherein the immune response comprises an antibody-dependent cellular cytotoxicity (ADCC) and/or antibody dependent cellular phagocytosis (ADCP).

30. The method of claim **28** or **29**, wherein the immunoconjugate is administered intravenously.

31. A method of treating a cancer expresses EphA2, the method comprising administering the immunoconjugate of any one of claims **1-26** to a patient having the cancer.

32. The method of claim **31**, wherein the cancer is a gastric cancer, ovarian cancer, soft tissue sarcoma, lung cancer, head and neck cancer, uterine cancer, breast cancer, colorectal cancer, esophageal cancer, stomach cancer, kidney cancer, melanoma, liver cancer, bladder cancer, or testicular cancer.

33. The method of claim **32**, wherein the cancer is non-small cell lung cancer, triple negative breast cancer, colorectal cancer, ovarian cancer or melanoma.

34. The method of any one of claims **31**-**33**, wherein the immunoconjugate is administered intravenously.

35. The method of any one of claims **31** to **34**, further comprising administering chemotherapy and/or radiation therapy.

36. The method of any one of claims **31** to **35**, further comprising administering an agent that targets an immunological checkpoint antigen.

37. The method of claim **36**, wherein the agent is a monoclonal antibody.

38. The method of claim **37**, wherein the monoclonal antibody blocks PD-1 ligand binding to PD-1.

39. The method of claim **38**, wherein the monoclonal antibody is an anti-PD-1 antibody.

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